Equipment Communication Specifications for LG Display AP4 ToE MELSEC Type Ver 1.1

한글 버전



LG CNS

- Revision History -

Version	Date	Person	Contents
1.0	2020.01.07	Cha Jae Gook	Version 1.0 for LG Display AP4 ToE
1.1	2020.03.09	Cha Jae Gook	Version 1.1 for LG Display AP4 ToE

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- Revision List -

※ 아래 변경 목록은 단순참고용으로 상세내역은 이 문서의 본문을 참조해야 한다.

0. From Ver P4 ToE 1.2 to Ver AP4 ToE 1.0

No	Part or Section	Ver P4 ToE 1.2 Page	Ver AP4 ToE 1.0 Page	Contents & Description
1	2.6.3 Stage Load Request GLSCODE	23	23	A Cut, B Cut GLSCODE Division
2	4.1.6 Glass Type	30	30	Glass Type Data Recover (ToE Ver 1.0)
3	6.3 Recipe Body Data Request	43	43	Recipe Body Data Address Change
4	6.11 Glass APD Report	55	55	Glass APD Report Sequence 수정 및 설명 추가
5	6.21.1 Unload Glass Data Report (Process Inline)	66	66	A Cut, B Cut Unload Glass Data Area Division
6	6.21.2 Unload Glass Data Report (Stand Alone)	67	67	A Cut, B Cut Stage Unload Glass Data Area Division
7	6.22 Glass Data Send	68	68	A Cut, B Cut Stage Glass Data Area Division
8	2.1.2 Stage Load Request GLSCODE (Stand Alone)	203	203	A Cut, B Cut Stage Load Request GLSCODE Area Division
9	2.1.3 Stage Unload Request GLSCODE (Stand Alone)	204	204	A Cut, B Cut Stage Unload Request GLSCODE Area Division
10	2.2 Glass Data Send	205	205	A Cut, B Cut Send Glass Data Area Division
11	2.4 Unload Glass Data Report	208	208	A Cut, B Cut Unload Glass Data Area Division
12	6.30 Pair Glass Data Change Report		131	- Sequence Add

1. From Ver AP4 ToE 1.0 to Ver AP4 ToE 1.1

No	Part or Section	Ver AP4 ToE 1.0 Page	Ver AP4 ToE 1.1 Page	Contents & Description
1	6.31 Link Test		133	- Sequence Add

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- Explanation -

1. PLC Address 표기

본 사양서에 표기된 PLC Address는 ECS 및 EQP 등의 시작 Address로부터 상대적인 Address로 표기된 것으로 아래의 같이 구분한다.

▶ ECS Address 표시(B[@s?]#, W[@s?]#)

@ : ECS Link Setting Area (ex) B[2s]0 : Setting 2 Area)

s : ECS Start Address

? : ECS Start Address에서 10(Hex) 단위의 주소

: 10(Hex)로 나눈 나머지 Address

ex) B[sA]3 (s : 0H라고 할 때) : 0H + (10H * AH) + 3 ➔ B0A3

▶ EQP(Loader) Address 표시(B[@n?]#, W[@n?]#)

@ : EQP(Loader) Link Setting Area (ex) B[2n]0 : Setting 2 Area)

n : EQP (Loader) Start Address

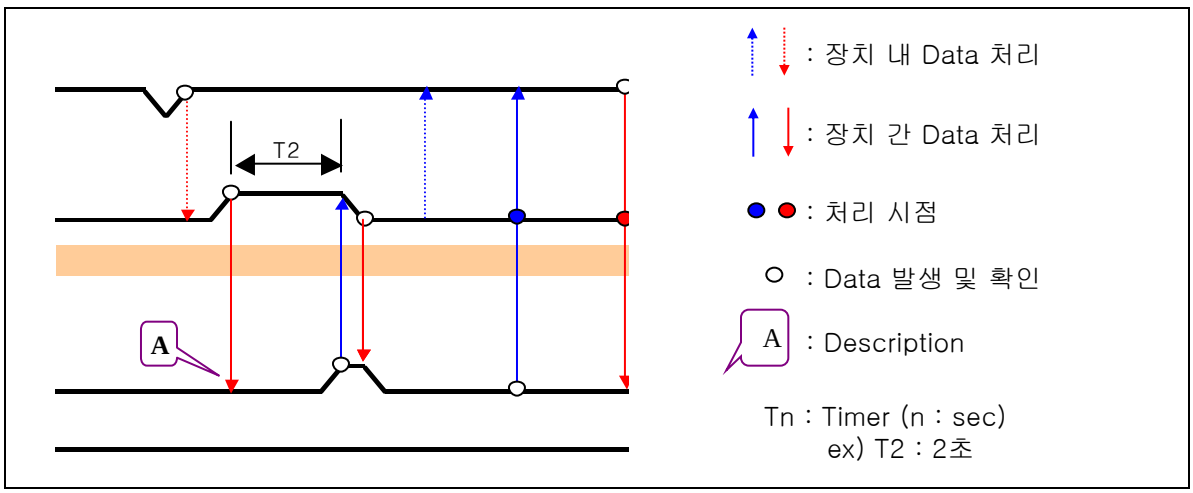
? : EQP(Loader) Start Address에서 10(Hex) 단위의 주소

: 10(Hex)로 나눈 나머지 Address

ex) B[n]3 (n : 0H라고 할 때) : 0H + (10H * 0H) + 3 ➔ B003

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2. Time Chart



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Chapter Ⅰ. ECS-Equipment MELSEC Standard Specification

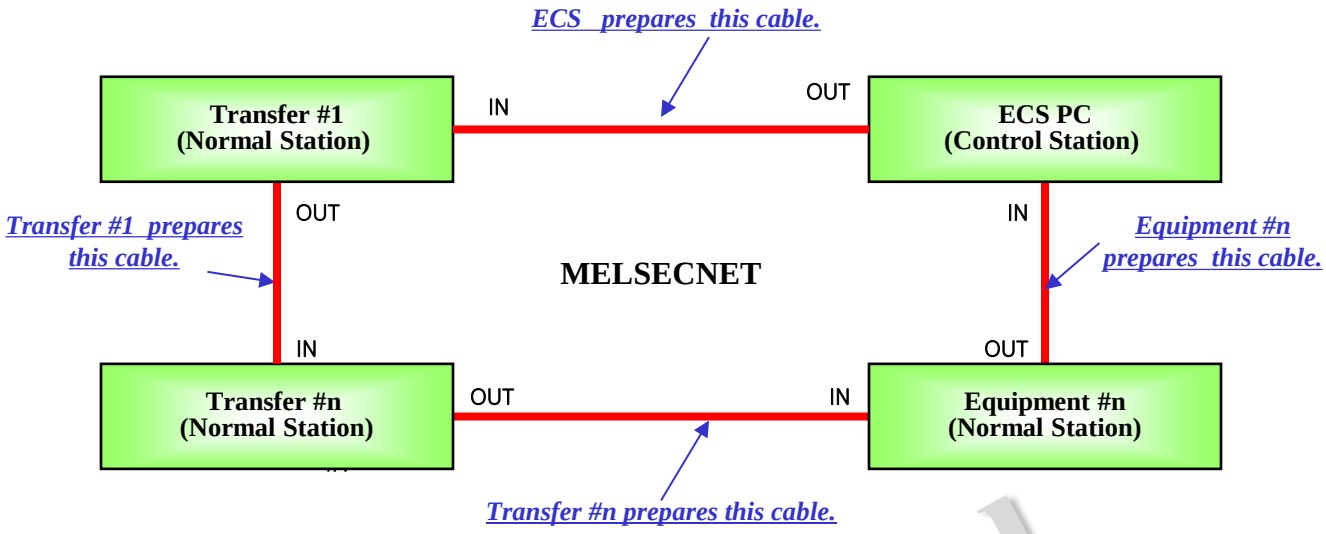
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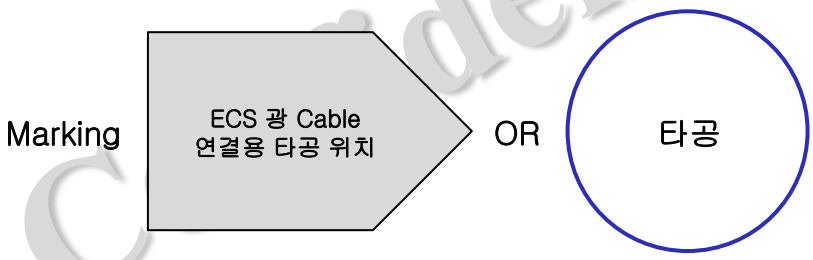
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1. System Format

1.1 Glass Processing Network



주1) ECS PC를 기준으로 OUT -> IN 방향으로 광Cable을 각 업체에서 준비한다.
주2) 광Cable 양끝 단에 "장치명-OUT" 및 "장치명-IN" 등과 같이 식별 가능하게 명기한다.
주의) 장비 업체는 ECS PC 현장 설치 시점에 장비 측 콘트롤러 하부에 ECS PC와 광 케이블 연결을 위해 아래와 같이 타공 또는 마킹을 완료 해야 함.
미완료 시 장비 측에서 ECS PC로 광 케이블링을 실시해야 함.



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▶ MELSECNET 통신 사양

항 목	사 양
보드 형명	PLC Type : QJ71GP21-SX or QJ71GP21S-SX PC Type : Q80BD-J71GP21-SX or Q80BD-J71GP21S-SX
MELSECNET Mode	MELSECNET/G Mode
통신 속도	1G bps
케이블 종별	Optical fiber cable (Multi-mode fiber)
통신 방식	토큰 링 방식
전송로 형식	2중 루프
네트워크의 접속대수	120국(Control Station : 1, Normal Station : 119)
Default Network No/ Group No	Network No : 1, Group No : 0
기타 네트워크 사양	기타 사양은 MELSECNET의 사양에 준함

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1.2 Codes and ID

1.2.1 CST ID (Cassette ID)
CST ID 는 개별적인 Cassette를 나타내는 고유한 이름이다.

1.2.2 LOT ID
한 Cassette 내에 적재되어 있는 Glass(Panel)에 대한 논리적인 ID로서 일부 장비를 제외하고는 1-Cassette 1-LOT을 기본으로 한다.

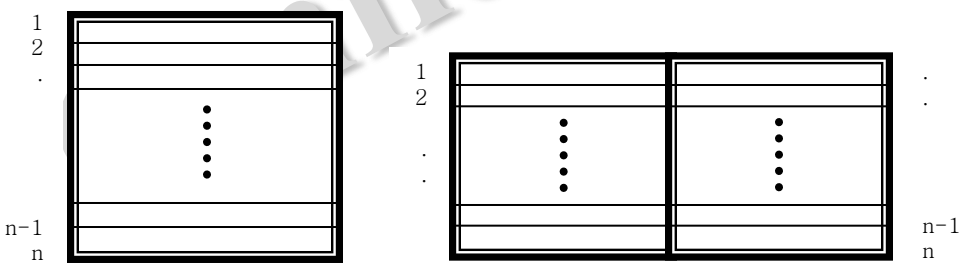
1.2.3 Equipment Recipe Number
장비에서 공정을 진행하기 위해 작성되는 Process Program에 대한 Number를 말한다.

1.2.4 GLS ID (PNL ID)
Glass(Panel)마다의 고유한 이름이다. ECS로부터 Download 받거나, OCR(Optical Character Reader)이 장착된 장비는 Glass(Panel)로부터 OCR을 통해 읽어 들인다.

1.2.5 장비는 입력 받은 CST ID, LOT ID, GLS ID, PNL ID, PPID 등의 문자열을 확인한다.
만약 ID에 Space나 제어 불가능한 문자열이 포함되어 있거나 문자열 길이가 맞지 않을 경우, Error로 처리하고 재입력을 요구한다.

1.2.6 SLOT ID
SLOT ID 는 Cassette의 Glass(Panel)가 들어가는 고유의 위치를 나타낸다. SLOT ID의 부여 기준은 아래. 그림과 같이 Cassette의 전면에서 보았을 때 상단으로부터 하단으로 1번부터 오름차순으로 부여한다.

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1.2.7 PORT ID
PORT ID 는 장비에서 각각의 PORT를 나타내는 고유한 번호이다.

1.2.8 Unit ID
UNIT ID는 각 개별 장비의 유일한 번호이다. Unit ID 는 LPL에 의해 지정되고 추후 상세 항목에 대한 협의를 필요로 한다.

1.2.9 Sub Unit ID
Sub UNIT ID 는 각 개별장비의 상세 부분별로 지정되는 유일한 번호이다. 이것은 UNIT ID와 같을 수도 있고 다를 수도 있다. 추후 상세 항목에 대한 협의를 필요로 한다.

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1.2.10 Different ID Code (<DICODE>)
0 : 전달 받은 Glass ID와 장비의 OCR에서 읽어 들인 Glass ID가 정확히 일치 할때 반환되는 값
1 : 전달 받은 Glass ID와 장비의 OCR에서 읽어 들인 Glass ID가 상이할 때 반환되는 값.

1.2.11 CST Port Status (<PORTSTATUS>)
0 : PORT에 Cassette가 존재하지 않을 경우
1 : PORT에 Cassette가 존재할 경우

1.2.12 Stage ID (<STAGEID>)
장비에서 Glass가 투입/취출 되는 영역을 나타내는 유일한 번호.
장비에 Stage가 1개만 존재할 경우 Stage ID를 1로 한다.

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2. Equipment Status Specifications

2.1 Operation Mode

2.1.1 Online State

장비의 상태를 데이터 영역에 Update 중임을 나타내는 것으로 항시 On한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

2.1.2 Dummy Status

장비가 Dummy Glass를 요구할 때 설정한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

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2.1.3 Unit Auto State

Glass의 자동 반송이 가능한 상태를 나타내는 것으로 자동 반송일 경우 On한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto State	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

2.1.4 Unit Operation Mode

장비의 특이 운영 모드를 나타내는 것으로 해당 Mode의 Bit만 설정한다.
(Unit Operation Mode는 추후 협의한다.)

B [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special	Unit Operation. Mode #1	Unit Operation. Mode #2	Unit Operation. Mode #3													

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2.2 Equipment Status Data

2.2.1 OCR(VCR) Status

OCR 또는 VCR의 상태를 나타내는 것으로 OCR(VCR)이 Normal 상태일 경우 'On'으로 Abnormal일 상태 경우 'OFF'로 설정한다.
OCR(VCR)이 2개 이상 존재할 경우 협의에 의해 상태를 결정한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

2.2.2 Operation Cycle Stop

작업자에 의해 투입중지가 요청될 때 설정하는 것으로 투입 중인 Glass는 정상 반송하며 새로운 Glass는 요청하지 않는다. (대상장비는 협의하여 결정함)

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

(참조. 8.2 Chemical Change Sequence)

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2.2.3 PD Down

장비 내에 처리 또는 반송중인 Glass가 존재하지 않으며 장비의 상태가 PD Down일 경우에 On 시킨다.
Run, Idle, Down 및 PD Down은 한 Bit만 반드시 On 되어야 하며 동시에 On또는 Off 될 수 없다
Run, Idle, Down 및 PD Down 상태는 라인의 특성에 따라 추후 협의하여 결정한다.

B [n5]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special	Unit Status (Experiment)	Unit Status (Development)	Unit Status (Monitoring)	Unit Status (Process Down)												

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2.2.4 Run

장비 내에 처리 또는 반송중인 Glass가 존재하며 장비의 상태가 Down이 아닐 경우에 On 시킨다.
Run, Idle, Down 및 PD Down은 한 Bit만 반드시 On 되어야 하며 동시에 On또는 Off 될 수 없다
Run, Idle, Down 및 PD Down 상태는 라인의 특성에 따라 추후 협의하여 결정한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

2.2.5 Idle

장비 내에 처리 또는 반송중인 Glass가 존재하지 않으며 장비의 상태가 Down이 아닐 경우에 On 시킨다.
Run, Idle 및 Down은 한 Bit만 반드시 On 되어야 하며 동시에 On또는 Off 될 수 없다
Run, Idle 및 Down 상태는 라인의 특성에 따라 추후 협의하여 결정한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

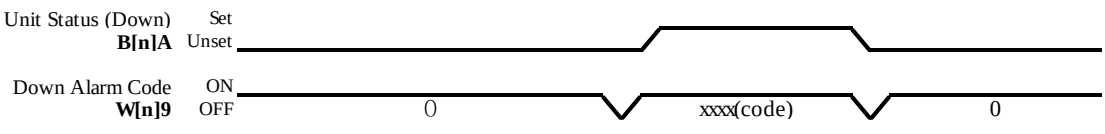
2.2.6 Down

장비에 이상이 발생하였거나 Glass를 처리 또는 반송하지 못할 경우에 Down Alarm Code를 설정한 후 Bit를 On 시킨다.
Run, Idle, Down 및 PD Down은 한 Bit만 반드시 On 되어야 하며 동시에 On또는 Off 될 수 없다
Run, Idle, Down 및 PD Down 상태는 라인의 특성에 따라 추후 협의하여 결정한다.

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status							Ionizer Status (0~15)	Ionizer Status (16~31)	LOT Judge Code	Down Alarm Code		Current Equip. Recipe Number	Glass Count in Unit	Put Into Possible Count	Light Alarm Code	Heavy Alarm Code

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2.2.7 Glass In Processing

장비 내에 1장 이상의 Glass를 Processing 중일 경우 상태 Bit를 'On' 시킨다

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

2.2.8 Glass Exist In Unit

장비 내에 1장 이상의 Glass가 존재할 경우 Bit를 'On' 시킨다

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

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2.2.9 Previous Transfer Enable

Glass의 Buffering이 가능한 장비에서 사용하는 것으로 하류 장비로 반송 지연이 예상될 때 상류 Buffering 가능 장비에게 투입 제한을 요구하는 것으로 상태 Bit를 Off시킨다.
사용 여부는 추후 협의하여 결정하여 결정한다.
(8.3 Glass 투입 제한 및 Glass Buffering 참조)

B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

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2.2.10 Current Equipment Recipe Number

장비에 현재 처리중인 Equipment Recipe Number를 나타내며 Recipe가 설정되어 있지 않을 경우 데이터를 Clear한다.

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status							Ionizer Status (0~15)	Ionizer Status (16~31)	LOT Judge Code	Down Alarm Code		Current Equip. Recipe Number	Glass Count in Unit	Put Into Possible Count	Light Alarm Code	Heavy Alarm Code

2.2.11 Glass Count In Unit

장비에 존재하는 Glass 개수를 나타낸다.

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status							Ionizer Status (0~15)	Ionizer Status (16~31)	LOT Judge Code	Down Alarm Code		Current Equip. Recipe Number	Glass Count in Unit	Put Into Possible Count	Light Alarm Code	Heavy Alarm Code

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2.2.12 Put Into Possible Count

장비에 투입 가능한 Glass 개수를 나타낸다. (Buffer Slot 포함)

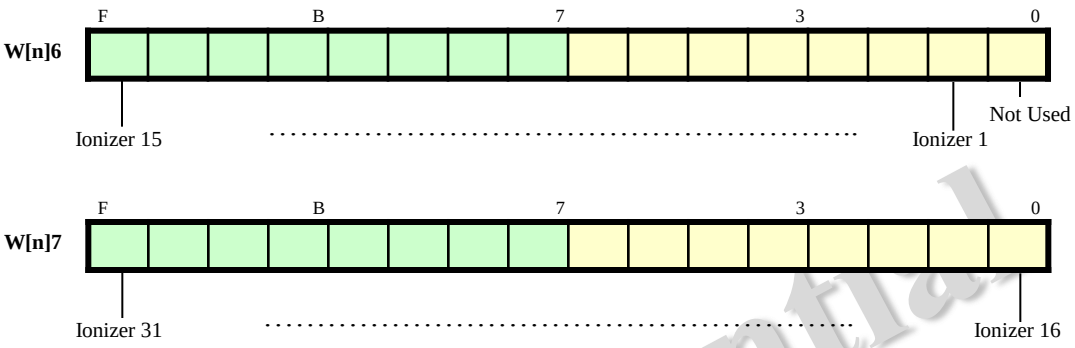
W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status							Ionizer Status (0~15)	Ionizer Status (16~31)	LOT Judge Code	Down Alarm Code		Current Equip. Recipe Number	Glass Count in Unit	Put Into Possible Count	Light Alarm Code	Heavy Alarm Code

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2.2.13 Ionizer Status

Word의 각 Bit는 Ionizer의 상태를 나타내는 것으로 Ionizer가 작동이 가능한 상태일 때 Bit를 On 한다.
(사용 여부는 추후 협의하여 결정함)

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special							Ionizer Status (0~15)	Ionizer Status (16~31)	LOT Judge Code	Down Alarm Code		Current Equip. Recipe Number	Glass Count in Unit	Put Into Possible Count	Light Alarm Code	Heavy Alarm Code



2.2.14 Glass Call

이전 장비로부터 Glass를 요구하는 것으로 Glass 요구 시 상태 Bit를 On시킨다.
(사용 여부는 추후 협의하여 결정함)

B [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special			Glass Call	Chemical Change												

2.2.15 Chemical Change

장비는 화학약품을 변경할 때, chemical change를 보고한다. 사용 여부는 추후 협의하여 결정함
(참조. 8.2 Chemical Change Sequence)

B [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special			Glass Call	Chemical Change												

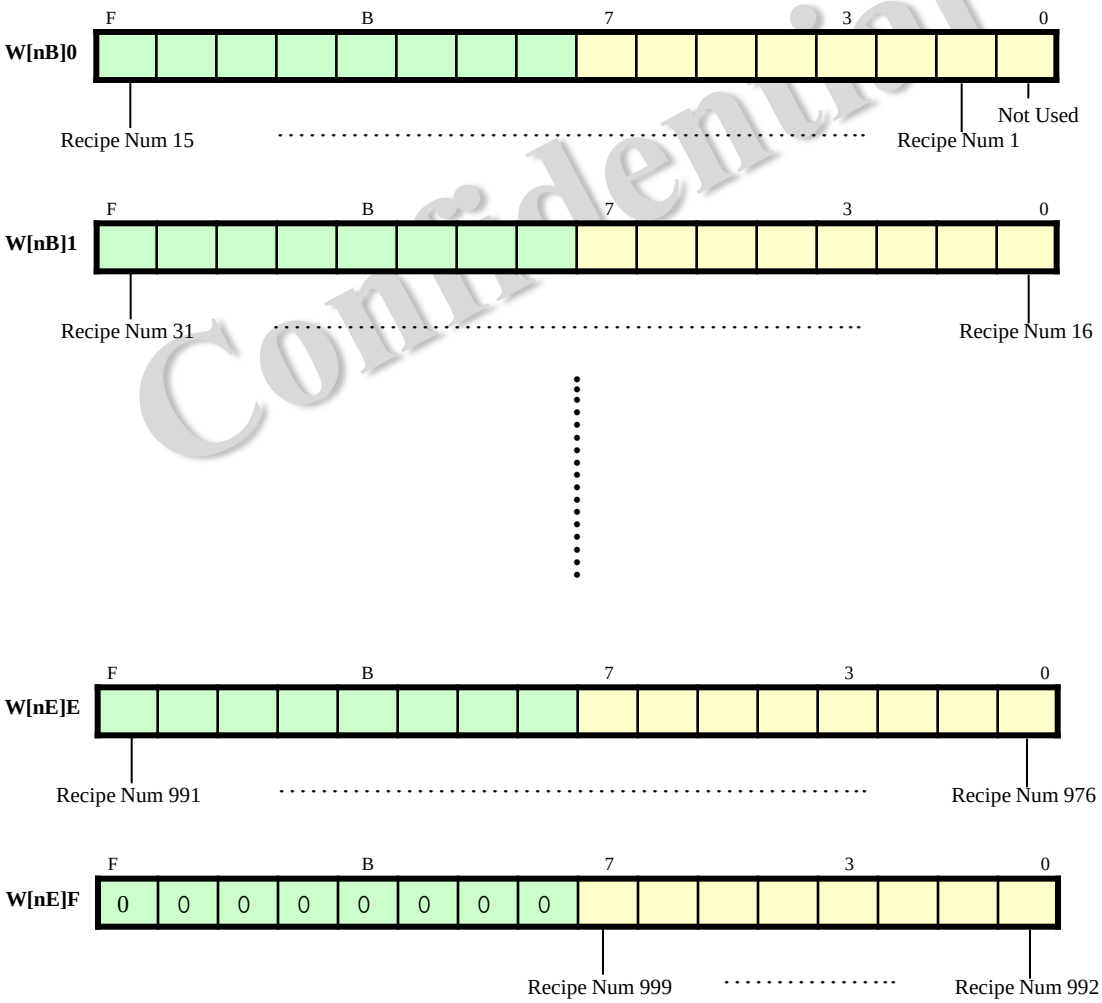
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2.2.16 Equipment Recipe Number List

등록되어 있는 모든 Equipment Recipe Number에 해당 Bit를 항시 On한다.

W [nB]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
EQP Recipe Number List	EQP Recipe Num. 1 ~ 15	EQP Recipe Num. 16 ~ 31	EQP Recipe Num. 32~47	EQP Recipe Num. 48~63	EQP Recipe Num. 64~79	EQP Recipe Num. 80 ~ 95	EQP Recipe Num. 96~111	EQP Recipe Num. 112~127	EQP Recipe Num. 128~143	EQP Recipe Num. 144~159	EQP Recipe Num. 160~175	EQP Recipe Num. 176~191	EQP Recipe Num. 192~207	EQP Recipe Num. 208~223	EQP Recipe Num. 224~239	EQP Recipe Num. 240~255
	~															
	EQP Recipe Num. 768~783	EQP Recipe Num. 784~799	EQP Recipe Num. 800~815	EQP Recipe Num. 816~831	EQP Recipe Num. 832~847	EQP Recipe Num. 848~863	EQP Recipe Num. 864~879	EQP Recipe Num. 880~895	EQP Recipe Num. 896~911	EQP Recipe Num. 912~927	EQP Recipe Num. 928~943	EQP Recipe Num. 944~959	EQP Recipe Num. 960~975	EQP Recipe Num. 976~991	EQP Recipe Num. 992~999	



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2.3 Glass Monitoring Data

2.3.1 Glass Position Data

장비의 Sub Unit간의 Glass 이동이 발생할 때 보고하는 것으로 장비는 ECS와의 Handshake 없이 Glass의 이동이 발생하였을 때마다 데이터를 Update한다.

< Glass Exist in Equipment >
장비의 존재 여부를 나타내는 것으로 Glass가 존재할 경우 상태 Bit를 'On'시킨다.

B [n8]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Position Data	Position 1Glass Exist	Position 2Glass Exist	Position 3Glass Exist	Position 4Glass Exist	Position 5Glass Exist	Position 6Glass Exist	Position 7Glass Exist	Position 8Glass Exist	Position 9Glass Exist	Position 10Glass Exist	Position 11Glass Exist	Position 12Glass Exist	Position 13Glass Exist	Position 14Glass Exist	Position 15Glass Exist	Position 16Glass Exist

B [n8]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Position Data	Position 1Glass Exist	Position 2Glass Exist	Position 3Glass Exist	Position 4Glass Exist	Position 5Glass Exist	Position 6Glass Exist	Position 7Glass Exist	Position 8Glass Exist	Position 9Glass Exist	Position 10Glass Exist	Position 11Glass Exist	Position 12Glass Exist	Position 13Glass Exist	Position 14Glass Exist	Position 15Glass Exist	Position 16Glass Exist
	Position 17Glass Exist	Position 18Glass Exist	Position 19Glass Exist	Position 20Glass Exist	Position 21Glass Exist	Position 22Glass Exist	Position 23Glass Exist	Position 24Glass Exist	Position 25Glass Exist	Position 26Glass Exist	Position 27Glass Exist	Position 28Glass Exist	Position 29Glass Exist	Position 30Glass Exist	Position 31Glass Exist	Position 32Glass Exist

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< Glass Code in Equipment >
장비에 존재하는 Glass Code를 나타낸다.

W [n8]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Position Data	Position 1Glass Code	Position 2Glass Code	Position 3Glass Code	Position 4Glass Code	Position 5Glass Code	Position 6Glass Code	Position 7Glass Code	Position 8Glass Code	Position 9Glass Code	Position 10Glass Code	Position 11Glass Code	Position 12Glass Code	Position 13Glass Code	Position 14Glass Code	Position 15Glass Code	Position 16Glass Code

W [n8]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Position Data	Position 1Glass Code	Position 2Glass Code	Position 3Glass Code	Position 4Glass Code	Position 5Glass Code	Position 6Glass Code	Position 7Glass Code	Position 8Glass Code	Position 9Glass Code	Position 10Glass Code	Position 11Glass Code	Position 12Glass Code	Position 13Glass Code	Position 14Glass Code	Position 15Glass Code	Position 16Glass Code
	Position 17Glass Code	Position 18Glass Code	Position 19Glass Code	Position 20Glass Code	Position 21Glass Code	Position 22Glass Code	Position 23Glass Code	Position 24Glass Code	Position 25Glass Code	Position 26Glass Code	Position 27Glass Code	Position 28Glass Code	Position 29Glass Code	Position 30Glass Code	Position 31Glass Code	Position 32Glass Code

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2.4 TPD Report

Time based Process Data를 나타내며 최소 0.5초 이하로 Update하며 출력 Cycle은 추후 협의 하에 결정함.

2.4.1 TPD Report Number(- TPD Report Counter)

TPD Data의 보고를 위한 Number로 데이터는 0 → 32767 → -32768 → 0과 같이 Loop로 설정한다.
PLC 장비인 경우 PLC Scan Count(SD420) 값의 사용이 가능하다.

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	TPD Report Number	TPD Update Time														
TPD Data Report	~															
	TPD Data Report V48	TPD Data Report V49	TPD Data Report V50	TPD Data Report V51	TPD Data Report V52	TPD Data Report V53	TPD Data Report V54	TPD Data Report V55	TPD Data Report V56	TPD Data Report V57	TPD Data Report V58	TPD Data Report V59	TPD Data Report V60	TPD Data Report V61	TPD Data Report V62	TPD Report Number

2.4.2 TPD Update Time

TPD Data의 보고 주기를 나타내는 것으로 데이터는 msec 단위로 표시하며 PLC 장비인 경우 PLC Scan Time(SD520) 사용이 가능하다

W [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	TPD Report Number	TPD Update Time														

2.4.3 Glass Processing Step Number

TPD Data 보고 시 장비 내에 존재하는 Glass의 Processing Step Number 보고하는 것으로 데이터는 Glass Position Data와 동일한 방법으로 설정한다.
Glass Processing Step Number 의 보고 여부 및 시작 주소는 추후 협의에 의해 결정 한다

< Glass Processing Step Number >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Processing Step Data	Position 1 Proc Step	Position 2 Proc Step	Position 3 Proc Step	Position 4 Proc Step	Position 5 Proc Step	Position 6 Proc Step	Position 7 Proc Step	Position 8 Proc Step	Position 9 Proc Step	Position 10 Proc Step	Position 11 Proc Step	Position 12 Proc Step	Position 13 Proc Step	Position 14 Proc Step	Position 15 Proc Step	Position 16 Proc Step
	Position 17 Proc Step	Position 18 Proc Step	Position 19 Proc Step	Position 20 Proc Step	Position 21 Proc Step	Position 22 Proc Step	Position 23 Proc Step	Position 24 Proc Step	Position 25 Proc Step	Position 26 Proc Step	Position 27 Proc Step	Position 28 Proc Step	Position 29 Proc Step	Position 30 Proc Step	Position 31 Proc Step	Position 32 Proc Step
	Position 33 Proc Step	Position 34 Proc Step	Position 35 Proc Step	Position 36 Proc Step	Position 37 Proc Step	Position 38 Proc Step	Position 39 Proc Step	Position 40 Proc Step	Position 41 Proc Step	Position 42 Proc Step	Position 43 Proc Step	Position 44 Proc Step	Position 45 Proc Step	Position 46 Proc Step	Position 47 Proc Step	

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2.4.4 TPD Data

TPD Data 는 4.2.3 APD/Recipe Body Data Format과 동일한 방식으로 설정하며 TPD Report Number는 TPD Data에 따라 이동한다
TPD Data의 시작위치 및 정보는 추후 협의에 의해 결정 한다

Example 1 >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
TPD Data Report	TPD Data Report V1	TPD Data Report V2	TPD Data Report V3	TPD Data Report V4	TPD Data Report V5	TPD Data Report V6	TPD Data Report V7	TPD Data Report V8	TPD Data Report V9	TPD Data Report V10	TPD Data Report V11	TPD Data Report V12	TPD Data Report V13	TPD Data Report V14	TPD Data Report V15	TPD Data Report V16
	TPD Data Report V17	TPD Data Report V18	TPD Data Report V19	TPD Data Report V20	TPD Data Report V21	TPD Data Report V22	TPD Data Report V23	TPD Data Report V24	TPD Data Report V25	TPD Data Report V26	TPD Data Report V27	TPD Data Report V28	TPD Data Report V29	TPD Data Report V30	TPD Data Report V31	TPD Report Number

Example 2 >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
TPD Data Report	TPD Data Report V1	TPD Data Report V2	TPD Data Report V3	TPD Data Report V4	TPD Data Report V5	TPD Data Report V6	TPD Data Report V7	TPD Data Report V8	TPD Data Report V9	TPD Data Report V10	TPD Data Report V11	TPD Data Report V12	TPD Data Report V13	TPD Data Report V14	TPD Data Report V15	TPD Data Report V16
	TPD Data Report V17	TPD Data Report V18	TPD Data Report V19	TPD Data Report V20	TPD Data Report V21	TPD Data Report V22	TPD Data Report V23	TPD Data Report V24	TPD Data Report V25	TPD Data Report V26	TPD Data Report V27	TPD Data Report V28	TPD Data Report V29	TPD Data Report V30	TPD Data Report V31	TPD Data Report V32
	TPD Data Report V33	TPD Data Report V34	TPD Data Report V35	TPD Data Report V36	TPD Data Report V37	TPD Data Report V38	TPD Data Report V39	TPD Data Report V40	TPD Data Report V41	TPD Data Report V42	TPD Data Report V43	TPD Data Report V44	TPD Data Report V45	TPD Data Report V46	TPD Data Report V47	TPD Data Report V48
	TPD Data Report V49	TPD Data Report V50	TPD Data Report V51	TPD Data Report V52	TPD Data Report V53	TPD Data Report V54	TPD Data Report V55	TPD Data Report V56	TPD Data Report V57	TPD Data Report V58	TPD Data Report V59	TPD Data Report V60	TPD Data Report V61	TPD Data Report V62	TPD Data Report V63	TPD Report Number

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2.5 TAS Data

Tact Time Analysis Data를 나타내며 출력 Cycle은 추후 협의 하에 결정함.

2.5.1 TAS Data

TAS Data 는 4.2.3 APD/Recipe Body Data Format과 동일한 방식으로 설정하며 TAS Data의 시작위치 및 정보는 추후 협의에 의해 결정 한다

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
TAS Data Report	TAS Data Report V1	TAS Data Report V2	TAS Data Report V3	TAS Data Report V4	TAS Data Report V5	TAS Data Report V6	TAS Data Report V7	TAS Data Report V8	TAS Data Report V9	TAS Data Report V10	TAS Data Report V11	TAS Data Report V12	TAS Data Report V13	TAS Data Report V14	TAS Data Report V15	TAS Data Report V16
	TAS Data Report V17	TAS Data Report V18	TAS Data Report V19	TAS Data Report V20	TAS Data Report V21	TAS Data Report V22	TAS Data Report V23	TAS Data Report V24	TAS Data Report V25	TAS Data Report V26	TAS Data Report V27	TAS Data Report V28	TAS Data Report V29	TAS Data Report V30	TAS Data Report V31	

Example 1 >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Sub-Unit #1 TAS Data	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	Recipe Number	Reserve	Reserve	TAS Data Report V1	TAS Data Report V2	TAS Data Report V3	TAS Data Report V4	TAS Data Report V5
Sub-Unit #2 TAS Data	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	Recipe Number	Reserve	Reserve	TAS Data Report V1	TAS Data Report V2	TAS Data Report V3	TAS Data Report V4	TAS Data Report V5

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2.6 Glass Move Status

2.6.1 Glass Move Status (Process Inline)

Process Inline에서는 Interlock 사양에 포함되어 있으며 상,하류 장비에 따라 결정된다.
(참조. 7 Glass Move Sequence)

< Unload Request >

B [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unload Inter-lock	Unload Enable	Glass Data Report			Normal Status	Unload Request	Unload Comp. Confirm						Shutter Open	Shutter All Close		

< Load Request >

B [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Load Inter-lock	Load Enable	Glass Data Request			Normal Status	Load Request	Load Comp. Confirm						Shutter Open	Shutter All Close		

< Exchange Request >

B [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Load Inter-lock	Load Enable	Glass Data Request			Normal Status	Load Request	Load Comp. Confirm						Shutter Open	Shutter All Close		
Unload Inter-lock	Unload Enable	Glass Data Report			Normal Status	Unload Request	Unload Comp. Confirm					Unload Wait	Shutter Open	Shutter All Close		

< Unload Wait >

B [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unload Inter-lock	Unload Enable	Glass Data Report			Normal Status	Unload Request	Unload Comp. Confirm					Unload Wait	Shutter Open	Shutter All Close		

< Load Wait >

B [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Load Inter-lock	Load Enable	Glass Data Request			Normal Status	Load Request	Load Comp. Confirm					Load Wait	Shutter Open	Shutter All Close		

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2.6.2 Glass Move Status (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 반송 장치로부터 완료 신호가 수신될 때까지 On 한다.

< Glass Move Status >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Receive Ack	Stage #1 Glass Move Statue	Stage #1 Load Request GLSCODE													

Item	내 용
Glass Move Status	Bit 0 : Cancel Request (CR) Bit 1 : Unload Request (UR) Bit 2 : Load Request (LR) Bit 3 : Exchange Request (ER) Bit 4 : Unload Wait Request (UW) Bit 5 : Load Wait Request (LW)

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2.6.3 Stage Load Request GLSCODE (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 장비 Stage로 투입을 요청하는 Glass의 GLSCODE를 나타내는 것으로 투입 완료 시 0으로 설정된다.

※ 장비는 A Cut, B Cut 지정된 Stage에 구분하여 GLSCODE를 Write 하여야 한다.

< Glass Move Status >

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Receive Ack	Stage #1 Glass Move Statue	Stage #1 Load Request GLSCODE_A	Stage #1 Load Request GLSCODE_B												

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2.7 Accessory Unit

2.7.1 Accessory Unit Status

Turn Unit 또는 Single Stage와 같은 주변 장치에 대한 상태를 나타내는 것으로 상태 변경시 해당 Bit를 설정한다.
(사용 유무 및 Accessory Unit은 협의하여 결정함)

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Accessory Unit	Accessory Unit #1 Status	Accessory Unit #2 Status	Accessory Unit #3 Status	Accessory Unit #4 Status	Accessory Unit #5 Status	Accessory Unit #6 Status	Accessory Unit #7 Status	Accessory Unit #8 Status	Accessory Unit #9 Status	Accessory Unit #10 Status	Accessory Unit #11 Status	Accessory Unit #12 Status	Accessory Unit #13 Status	Accessory Unit #14 Status	Accessory Unit #15 Status	Accessory Unit #16 Status

Name	Description
Accessory Unit Status	Bit 0 : Accessory Unit Run Bit 1 : Accessory Unit Down

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2.7.2 Accessory Unit Down Alarm Code

Accessory Unit에 이상(Down)이 발생하였을 경우에 Down Alarm Code를 설정한 후 Accessory Unit Down을 설정한다. (사용 유무 및 Accessory Unit은 협의하여 결정함)

W [n6]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Accessory Unit	Accessory Unit #1 Down Alarm	Accessory Unit #2 Down Alarm	Accessory Unit #3 Down Alarm	Accessory Unit #4 Down Alarm	Accessory Unit #5 Down Alarm	Accessory Unit #6 Down Alarm	Accessory Unit #7 Down Alarm	Accessory Unit #8 Down Alarm	Accessory Unit #9 Down Alarm	Accessory Unit #10 Down Alarm	Accessory Unit #11 Down Alarm	Accessory Unit #12 Down Alarm	Accessory Unit #13 Down Alarm	Accessory Unit #14 Down Alarm	Accessory Unit #15 Down Alarm	Accessory Unit #16 Down Alarm

2.7.3 Accessory Unit Glass Exist & Glass Code

Accessory Unit의 Glass 정보는 2.3 Glass Monitoring Data로 표시한다.

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3. ECS Status Specifications

3.1 ECS Status Data

3.1.1 Glass Recovery

ECS가 Glass Recovery 요구 시 장비는 처리중인 Glass는 처리 후 반송하고 처리전인 Glass는 다음 장비로 반송한다.(사용여부는 협의에 의해 결정함.)

B [s]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Status			Glass Recovery													

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4. Data Definition

4.1 Glass Data Structure

Glass 실처리 데이터 보고 및 Glass Unload 또는 Load 될 때 전달하거나 전달 받은 Glass 정보로 기본 데이터는 48 word로 구성되어 있다.

※ Process Inline Glass Data Structure (LOT Information/Glass Information)

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	LOT ID(1,2)	LOT ID (3,4)	LOT ID (5,6)	LOT ID (7,8)	LOT ID (9,10)	LOT ID (11,12)	LOT ID (13,14)	LOT ID (15,16)	Processing Code (1,2)	Processing Code (3,4)	Processing Code (5,6)	Processing Code (7,8)	LOT Specific Data (1)	LOT Specific Data (2)	LOT Specific Data (3)	LOT Specific Data (4)
	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

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※ Process Inline Glass Data Structure (Main Glass Information/ Sub Glass Information)

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Sub Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Main Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Main Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

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※ Stand Alone Glass Data Structure (LOT Information/Glass Information)

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	LOT ID(1,2)	LOT ID (3,4)	LOT ID (5,6)	LOT ID (7,8)	LOT ID (9,10)	LOT ID (11,12)	LOT ID (13,14)	LOT ID (15,16)	Processing Code (1,2)	Processing Code (3,4)	Processing Code (5,6)	Processing Code (7,8)	LOT Specific Data (1)	LOT Specific Data (2)	LOT Specific Data (3)	LOT Specific Data (4)
	EQP Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

※ Stand Alone Glass Data Structure (Main Glass Information/ Sub Glass Information)

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Sub Glass Data	EQP Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Main Glass Data	EQP Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)
Main Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

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4.1.1 LOTID

LOT의 고유 ID로 최대 16자까지 ASCII Code 구성되어 있다
사용 가능한 ASCII Code는 alphabet(41h ~ 5Ah) 및 Numeric(30h ~ 39h) 문자이며 Space(20h)는 첫 문자 및 문자와 문자 사이를 제외하고 사용 가능하나 무시된다.

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	LOT ID(1,2)	LOT ID (3,4)	LOT ID (5,6)	LOT ID (7,8)	LOT ID (9,10)	LOT ID (11,12)	LOT ID (13,14)	LOT ID (15,16)	Processing Code (1,2)	Processing Code (3,4)	Processing Code (5,6)	Processing Code (7,8)	LOT Specific Data (1)	LOT Specific Data (2)	LOT Specific Data (3)	LOT Specific Data (4)

Example) LOTID : “ABCDE”

W[n1]0	W[n1]1	W[n1]2	W[n1]3	W[n1]4	W[n1]5	W[n1]6	W[n1]7
4241h	4443h	45h	0h	0h	0h	0h	0h

Example) LOTID : “ABCDE ” → 마지막 Space는 무시되고 “ABCDE”로 인식함.

W[n1]0	W[n1]1	W[n1]2	W[n1]3	W[n1]4	W[n1]5	W[n1]6	W[n1]7
4241h	4443h	2045h	0h	0h	0h	0h	0h

Space 무시

4.1.2 Processing Code

ASCII Code이며 최대 8자까지 가능하며 ASCII Code는 Numeric(30h ~ 39h)으로 구성된다.

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	LOT ID(1,2)	LOT ID (3,4)	LOT ID (5,6)	LOT ID (7,8)	LOT ID (9,10)	LOT ID (11,12)	LOT ID (13,14)	LOT ID (15,16)	Processing Code (1,2)	Processing Code (3,4)	Processing Code (5,6)	Processing Code (7,8)	LOT Specific Data (1)	LOT Specific Data (2)	LOT Specific Data (3)	LOT Specific Data (4)

Example) Processing Code : “12345”

W[n1]8	W[n1]9	W[n1]A	W[n1]B
3231h	3433h	35h	0h

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4.1.3 Lot Specific Data

LOT의 특수한 데이터를 나타내는 것으로 라인의 특성에 따라 추후 협의하여 결정한다

W [n1]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	LOT ID(1,2)	LOT ID (3,4)	LOT ID (5,6)	LOT ID (7,8)	LOT ID (9,10)	LOT ID (11,12)	LOT ID (13,14)	LOT ID (15,16)	Processing Code (1,2)	Processing Code (3,4)	Processing Code (5,6)	Processing Code (7,8)	LOT Specific Data (1)	LOT Specific Data (2)	LOT Specific Data (3)	LOT Specific Data (4)

Example)
Operator ID : "1234"

W[n1]C	W[n1]D
3231h	3433h

Example)
Operator ID : "1234", MASK ID : "ABCD"

W[n1]C	W[n1]D	W[n1]E	W[n1]F
3231h	3433h	4241h	4443h

LOT Specific Data(4) <= Production Type
- Unit Status(Run, Idle, Down, Experiment, Development , Monitoring)를 아래와 같이 보고 하여야 함.

0 Bit : LOT의 종듀 2^0
1 Bit : LOT의 종듀 2^1
2 Bit : LOT의 종듀 2^2

0 : 양산(Production)
1 : 실험(Experiment)
2 : 개발(Development)
4 : 모니터링(Monitoring)

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B [n]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Status	Online State	Dummy Status				Unit Auto Mode	OCR Status	Operation Cycle Stop	Unit Status (Run)	Unit Status (Idle)	Unit Status (Down)	Glass In Processing	Glass Exist In Unit	Previous Transfer Enable	Light Alarm Report	Heavy Alarm Report

B [n5]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Unit Special	Unit Status (Experiment)	Unit Status (Development)	Unit Status (Monitoring)	Unit Status (Process Down)	Unit Status (Model Change)	Unit Status (Layer Change)										

Unit Special Status의 Run, Idle, Down, Experiment, Development, Monitoring, Process Down, Model Change, Layer Change는 한 Bit만 반드시 On 되어야 하며 동시에 On또는 Off 될 수 없다.

4.1.4 Equipment Recipe Number

장비에서 공정을 진행하기 위해 작성되는 Process Program에 대한 Number를 말한다. (Stand Alone)

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	EQP Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

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4.1.5 Host Recipe Number

해당 공정을 진행하기 위한 대표 Recipe Number 이며 장비는 ECS로 장비에 해당되는 Equipment Recipe Number를 요청한다. (Process Inline)

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

4.1.6 Glass Type

Glass의 추가적인 정보를 나타내며 각 장비에 따라 사용여부가 결정된다.

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

0 Bit : Glass size code 2^0
1 Bit : Glass size code 2^1
2 Bit : Glass size code 2^2
3 Bit : Glass size code 2^3

The detail use is determined after discussing.

4 Bit : Glass thickness code 2^0
5 Bit : Glass thickness code 2^1
6 Bit : Glass thickness code 2^2

The detail use is determined after discussing.

7 Bit : Glass Shop Bit(1 : TFT, 0 : CF) , CELL Shop Only
8 Bit : Pass Glass
9 Bit : Dummy Glass (1 : Dummy)
A Bit : Reserve
B Bit : Glass ID Update (1 : Updated)
C Bit : OCR(VCR) Not Used (1 : Not Used)
D Bit : Scraped Glass(1 : Scraped)
E Bit : Sample Code (1 : Sampling)
F Bit : Glass Data Updated (By Operator)

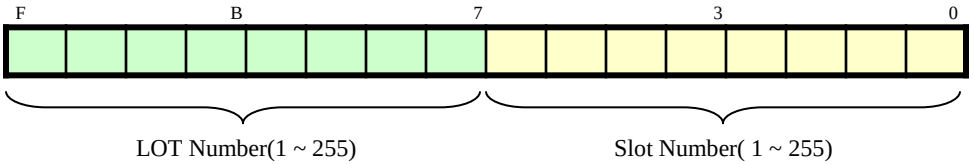
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4.1.7 GLSCODE

Glass를 구분하는 고유의 코드로 Lot 및 Slot Number로 구성되어 있다.

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)



4.1.8 GLSID

Glass의 고유 ID로 최대 16자까지 ASCII Code 구성되어 있다
사용 가능한 ASCII Code는 alphabet(41h ~ 5Ah) 및 Numeric(30h ~ 39h) 문자이며 Space(20h)는 첫 문자 및 문자와 문자 사이를 제외하고 사용 가능하나 무시된다. (LOTID 예제 참조)

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

4.1.9 GLSJUDGE

Glass의 판정코드로 최대 2자까지 ASCII Code 구성되어 있다
사용 가능한 ASCII Code는 alphabet(41h ~ 5Ah) 문자이며 Space(20h)는 첫 문자 제외하고 사용 가능하나 무시된다. (LOTID 예제 참조)

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

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4.1.10 Glass Specific Data

Glass의 특수한 데이터를 나타내는 것 사용 여부는 협의에 의해 결정함.

W [n2]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Report Glass Data	Host Recipe Number	Glass Type	GLS CODE	Glass ID (1,2)	Glass ID (3,4)	Glass ID (5,6)	Glass ID (7,8)	Glass ID (9,10)	Glass ID (11,12)	Glass ID (13,14)	Glass ID (15,16)	GLS JUDGE	Glass Specific Data (1)	Glass Specific Data (2)	Glass Specific Data (3)	Glass Specific Data (4)

Example)

Glass Special Data(1)

0 Bit : Destination code 2^0
1 Bit : Destination code 2^1

2 Bit : Sorting type 2^0
3 Bit : Sorting type 2^1
4 Bit : Sorting type 2^2
5 Bit : Sorting type 2^3

6 Bit : DICODE

7 Bit : Glass Order 2^0
8 Bit : Glass Order 2^1

9 Bit : Direction code(1 : Reverse), CELL Only

A Bit : Glass turn information code 2^0
B Bit : Glass turn information code 2^1] 0 : 0°, 1 : 90°, 2 : 180°, 3 : -90°

C Bit : Reserve
D Bit : Reserve
E Bit : Reserve
F Bit : Reserve

Glass Special Data(4)

0 Bit ~ 6 Bit : Reserve

7 Bit : Pair Glass — 0 : Unpair, 1 : Pair

8 Bit : Glass Cut 20
9 Bit : Glass Cut 21] 0 : A Cut, 1 : B Cut
2 : A Cut Dummy, 3 : B Cut Dummy

A Bit ~ F Bit : Reserve

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4.1.11 Glass Add Data

Glass의 추가적인 정보를 나타내며 사용 여부는 협의에 의해 결정함.

W [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

Example)

Glass Add Data(1)

C Bit： Routing no.1 Processing Complete (CELL Rubbing Only)
D Bit： Routing no.2 Processing Complete (CELL Rubbing Only)
E Bit： Routing no.3 Processing Complete (CELL Rubbing Only)
F Bit： Routing no.4 Processing Complete (CELL Rubbing Only)

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4.1.12 Previous Unit Processing Data

상류 장비의 처리 데이터로 각 Bit의 사용여부는 협의에 의해 결정한다.

W [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

Example)

Previous Unit Processing Data(1)

0 Bit： Coater Processing
1 Bit： Reserve
2 Bit： Reserve
3 Bit： Reserve
4 Bit： Reserve
5 Bit： Reserve
6 Bit： Reserve
7 Bit： Reserve
8 Bit： Reserve
9 Bit： Reserve
A Bit： Reserve
B Bit： Reserve
C Bit： Reserve
D Bit： Reserve
E Bit： Reserve
F Bit： Reserve

Previous Unit Processing Data(2)

0 Bit： DHB Processing
1 Bit： Reserve
2 Bit： Reserve
3 Bit： Reserve
4 Bit： DHB Processing Code 2^0
5 Bit： DHB Processing Code 2^1
6 Bit： DHB Processing Code 2^2
7 Bit： DHB Processing Code 2^3
8 Bit： PRB Processing
9 Bit： Reserve
A Bit： Reserve
B Bit： Reserve
C Bit： PRB Processing Code 2^0
D Bit： PRB Processing Code 2^1
E Bit： PRB Processing Code 2^2
F Bit： PRB Processing Code 2^3

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4.1.13 Glass Processing Status

각 장비의 처리 결과를 나타내는 것으로 처리 결과가 NG일 경우 상태 Bit를 'On'시킨다.

W [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)

Glass Processing Status(1)

0 Bit : Equipment #1 glass processing status
1 Bit : Equipment #2 glass processing status
2 Bit : Equipment #3 glass processing status
3 Bit : Equipment #4 glass processing status
4 Bit : Equipment #5 glass processing status
5 Bit : Equipment #6 glass processing status
6 Bit : Equipment #7 glass processing status
7 Bit : Equipment #8 glass processing status
8 Bit : Equipment #9 glass processing status
9 Bit : Equipment #10 glass processing status
A Bit : Equipment #11 glass processing status
B Bit : Equipment #12 glass processing status
C Bit : Equipment #13 glass processing status
D Bit : Equipment #14 glass processing status
E Bit : Equipment #15 glass processing status
F Bit : Equipment #16 glass processing status

Glass Processing Status(2)

0 Bit : Equipment #17 glass processing status
1 Bit : Equipment #18 glass processing status
2 Bit : Equipment #19 glass processing status
3 Bit : Equipment #20 glass processing status
4 Bit : Equipment #21 glass processing status
5 Bit : Equipment #22 glass processing status
6 Bit : Equipment #23 glass processing status
7 Bit : Equipment #24 glass processing status
8 Bit : Equipment #25 glass processing status
9 Bit : Equipment #26 glass processing status
A Bit : Equipment #27 glass processing status
B Bit : Equipment #28 glass processing status
C Bit : Equipment #29 glass processing status
D Bit : Equipment #30 glass processing status
E Bit : Equipment #31 glass processing status
F Bit : Equipment #32 glass processing status

Glass Processing Status(3)

0 Bit : Equipment #33 glass processing status
1 Bit : Equipment #34 glass processing status
2 Bit : Equipment #35 glass processing status
3 Bit : Equipment #36 glass processing status
4 Bit : Equipment #37 glass processing status
5 Bit : Equipment #38 glass processing status
6 Bit : Equipment #39 glass processing status
7 Bit : Equipment #40 glass processing status
8 Bit : Equipment #41 glass processing status
9 Bit : Equipment #42 glass processing status
A Bit : Equipment #43 glass processing status
B Bit : Equipment #44 glass processing status
C Bit : Equipment #45 glass processing status
D Bit : Equipment #46 glass processing status
E Bit : Equipment #47 glass processing status
F Bit : Equipment #48 glass processing status

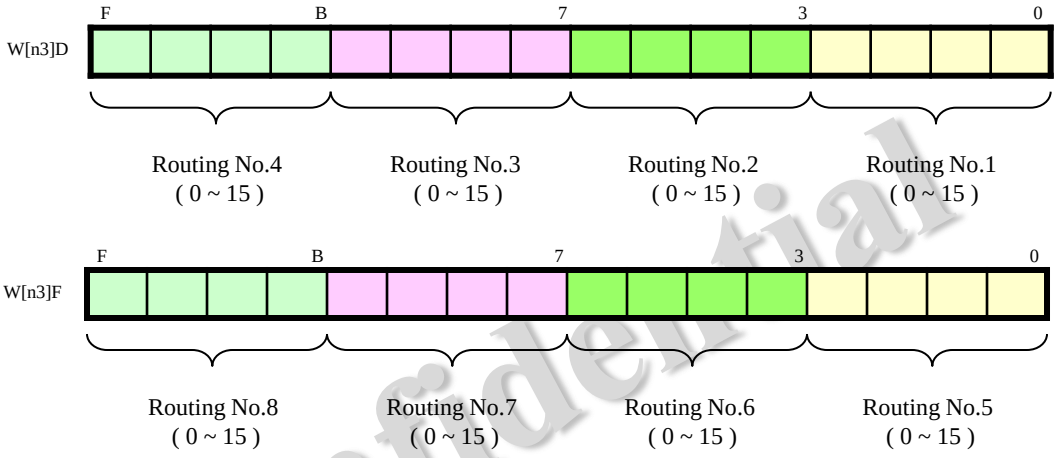
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4.1.14 Glass Routing Data

장비 사이의 Routing 정보로 사용 여부는 협의에 의해 결정함.

W [n3]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Glass Add Data	Glass Add Data (1)	Glass Add Data (2)	Glass Add Data (3)	Glass Add Data (4)	Glass Add Data (5)	Glass Add Data (6)	Previous Unit Processing Data(1)	Previous Unit Processing Data(2)	Previous Unit Processing Data(3)	Previous Unit Processing Data(4)	Glass Processing Status (1)	Glass Processing Status (2)	Glass Processing Status (3)	Glass Routing Data (1)	Glass Routing Data (2)	Glass Routing Data (3)



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4.2 Glass Processing and Report Data

Glass 처리 및 보고 시 필요한 추가적인 정보

4.2.1 Glass Panel Information

Panel과 Sub Panel 정보로 Judge Code는 협의에 의해 결정함.

Panel Info(1)
0 Bit : Panel 1 Judge Code 2^0
1 Bit : Panel 1 Judge Code 2^1
2 Bit : Panel 1 Judge Code 2^2
3 Bit : Panel 1 Judge Code 2^3
4 Bit : Panel 1 Judge Code 2^4

5 Bit : Panel 2 Judge Code 2^0
6 Bit : Panel 2 Judge Code 2^1
7 Bit : Panel 2 Judge Code 2^2
8 Bit : Panel 2 Judge Code 2^3
9 Bit : Panel 2 Judge Code 2^4

A Bit : Panel 3 Judge Code 2^0
B Bit : Panel 3 Judge Code 2^1
C Bit : Panel 3 Judge Code 2^2
D Bit : Panel 3 Judge Code 2^3
E Bit : Panel 3 Judge Code 2^4

F Bit : Panel 4 Judge Code 2^0

Panel Info(2)
0 Bit : Panel 4 Judge Code 2^1
1 Bit : Panel 4 Judge Code 2^2
2 Bit : Panel 4 Judge Code 2^3
3 Bit : Panel 4 Judge Code 2^4

4 Bit : Panel 5 Judge Code 2^0
5 Bit : Panel 5 Judge Code 2^1
6 Bit : Panel 5 Judge Code 2^2
7 Bit : Panel 5 Judge Code 2^3
8 Bit : Panel 5 Judge Code 2^4

9 Bit : Panel 6 Judge Code 2^0
A Bit : Panel 6 Judge Code 2^1
B Bit : Panel 6 Judge Code 2^2
C Bit : Panel 6 Judge Code 2^3
D Bit : Panel 6 Judge Code 2^4

E Bit : Panel 7 Judge Code 2^0
F Bit : Panel 7 Judge Code 2^1

Panel Info(3)
0 Bit : Panel 7 Judge Code 2^2
1 Bit : Panel 7 Judge Code 2^3
2 Bit : Panel 7 Judge Code 2^4
…

Panel Info(4)
…
Panel Info(5)
…
Panel Info(6)
…
Panel Info(7)
…
…
…
…
…
Panel Info(400)

Sub-Panel (Model) Info(1)
0 Bit : Sub-Panel 1 Judge Code 2^0
1 Bit : Sub-Panel 1 Judge Code 2^1
2 Bit : Sub-Panel 1 Judge Code 2^2
3 Bit : Sub-Panel 1 Judge Code 2^3
4 Bit : Sub-Panel 1 Judge Code 2^4

5 Bit : Sub-Panel 2 Judge Code 2^0
6 Bit : Sub-Panel 2 Judge Code 2^1
7 Bit : Sub-Panel 2 Judge Code 2^2
8 Bit : Sub-Panel 2 Judge Code 2^3
9 Bit : Sub-Panel 2 Judge Code 2^4

A Bit : Sub-Panel 3 Judge Code 2^0
B Bit : Sub-Panel 3 Judge Code 2^1
C Bit : Sub-Panel 3 Judge Code 2^2
D Bit : Sub-Panel 3 Judge Code 2^3
E Bit : Sub-Panel 3 Judge Code 2^4

F Bit : Sub-Panel 4 Judge Code 2^0

Sub-Panel Info(2)
0 Bit : Sub-Panel 4 Judge Code 2^1
1 Bit : Sub-Panel 4 Judge Code 2^2
2 Bit : Sub-Panel 4 Judge Code 2^3
3 Bit : Sub-Panel 4 Judge Code 2^4

4 Bit : Sub-Panel 5 Judge Code 2^0
5 Bit : Sub-Panel 5 Judge Code 2^1
6 Bit : Sub-Panel 5 Judge Code 2^2
7 Bit : Sub-Panel 5 Judge Code 2^3
8 Bit : Sub-Panel 5 Judge Code 2^4

9 Bit : Sub-Panel 6 Judge Code 2^0
A Bit : Sub-Panel 6 Judge Code 2^1
B Bit : Sub-Panel 6 Judge Code 2^2
C Bit : Sub-Panel 6 Judge Code 2^3
D Bit : Sub-Panel 6 Judge Code 2^4

E Bit : Sub-Panel 7 Judge Code 2^0
F Bit : Sub-Panel 7 Judge Code 2^1

Sub-Panel Info(3)
0 Bit : Sub-Panel 7 Judge Code 2^2
1 Bit : Sub-Panel 7 Judge Code 2^3
2 Bit : Sub-Panel 7 Judge Code 2^4
…

Sub-Panel Info(4)
…
Sub-Panel Info(5)
…
Sub-Panel Info(6)
…
Sub-Panel Info(7)
…
…
…
…
…
Sub-Panel Info(400)

< Panel Judge > < Sub Panel Judge >

Code	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Judge		G	N	P	M	R	H	I	A	Q	E	C	B	F	D	L

Code	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Judge		W	U	T	V	K	S	X	Y	J						

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4.2.2 Data Report CEID

Glass의 처리 및 보고 Data를 위한 데이터 구분 ID

CEID	Description
1	Glass APD Report
2	Scrap Glass Data Report
3	Recipe Body Data Report
7	Glass Data Change Report

<Data 보고 우선순위>

1. Recipe Body Data Report (Reference 6.3 Recipe Body Data Request)
2. Glass APD Report (Reference 6.11 Glass APD Report)
3. Scrap Glass Data Report (Reference 6.5 Scrap Glass Data Report)
4. Glass Data Change Report (Reference 6.7 Glass Data Change Report)

※ 장비에서 Glass APD를 한번에 보고하지 못할 시에는 처음 보고 시는 CEID를 Original Value(1)를 보고하고 그 이후에는 10을 더한 값을 보고한다 (이 시나리오는 추후 협의에 의해 결정함)

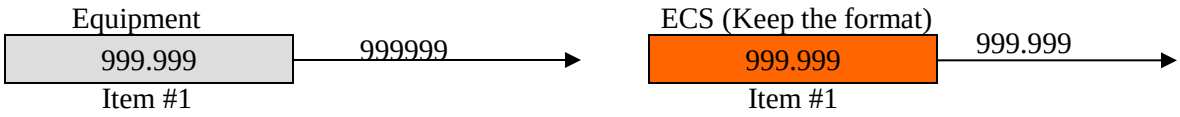
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4.2.3 APD/Recipe Body Data Format

장비의 APD와 Recipe Body는 아래와 같은 규칙에 따라 전송해야만 한다..

1) APD/Recipe Body의 Item 및 Data Format은 일정한 형식을 가진다.
2) Data Types 은 Numeric and Character 2가지로 한정한다.
3) 장비는 Data 보고 시 소수점을 제외한 Data를 전송하며 ECS는 원래의 소수점으로 복원한다.



4.2.4 다음 은 ECS와 장비사이의 Data Format 의 예이다.

Item No	Item ID	Item Format(Max value)	Data Type	Length(Word Unit)	Num. Of Decimal Place	Unit
1	MIN_CIJ	999.999	I4	2	3	
2	LOTEND TIME	YYYYMMDD_HHMMSS	A	8	nothing	
3	MAX_GAS	9999	I1	1	0	
4	MIN_GAS	0.200	I1	1	3	
5	MIN_AFC	0.01234	I4	2	5	
6	MAX_GG	0.1234E-10	I2	1	4	E-10

1) Item 1 : 장비는 ECS에게 '999999'로 Data 보고 시 ECS는 2 word로 읽고 '999.999'로 복원하는 것으로 정의한다. 이때, ECS는 Host로 이Data를 보고한다.
2) Item 2 : ECS는 고정 길이로 Data를 읽고 Data 변경 없이 Host로 보고한다.
3) Item 3 : Data타입은 Integer이며 최대값은 '9999', ECS는 Host로 이Data를 보고한다.
4) Item 4, 5 : 장비는 '1234'로 Data 보고 시 ECS는 1 word로 읽고 원Data복원하는 것으로 정의한다.
5) Item 6 : ECS는 '1234'로 전달받은 Data를 변환하여 '0.1234E-10'로 Host에게 보고한다.

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5. Required items of the counter specification document

대응 사양서에는 최소한 다음과 같은 사항을 포함해야 한다.

5.1 장비 제어계 기능 구성도

☞ 장비 내부의 제어 구성도

☞ Sub Unit 구성 및 그 구성도

5.2 Interface 전기적 특성이나 통신 Protocol

☞ MELSECNET 통신 사양

5.3 Normal Sequence 과 Memory MAP의 항목 및 영역의 사용 여부 표시

☞ 색이나 선을 이용하여 표시

☞ 단, 사용하지 않는 영역 및 항목은 삭제하지 않는다.

5.4 ECS와 통신 시 각 Status 및 Data에 대한 설명

☞ Alarm 발생/해제 시점과 그 조건 기술

☞ Equipment Status 보고 시점 정의 및 그 내용

☞ Glass Loading and Unloading 요청 시점 정의 및 그 내용

5.5 각 Normal Sequence 의 자세한 기술

☞ 적절한 그림(Chart)을 이용하여 자세히 기술한다.

5.6 각종 Code의 사용 유무 및 그 정의

5.7 Recipe Number에 대한 상세 기술

☞ Naming 규칙들(최대/ 최소 수), 등록된 개수 등

5.8 타 장비와의 Interlock Diagram

☞ Interlock chart 와 상세 설명

5.9 LOT/Glass APD Lists, Alarm Lists , Recipe Body Lists의 상세 기술

☞ LG에서 제공하는 문서 포맷에 맞추어 작성 후 제출한다.

☞ LG에서 추후 기준 Format 문서 송부함.

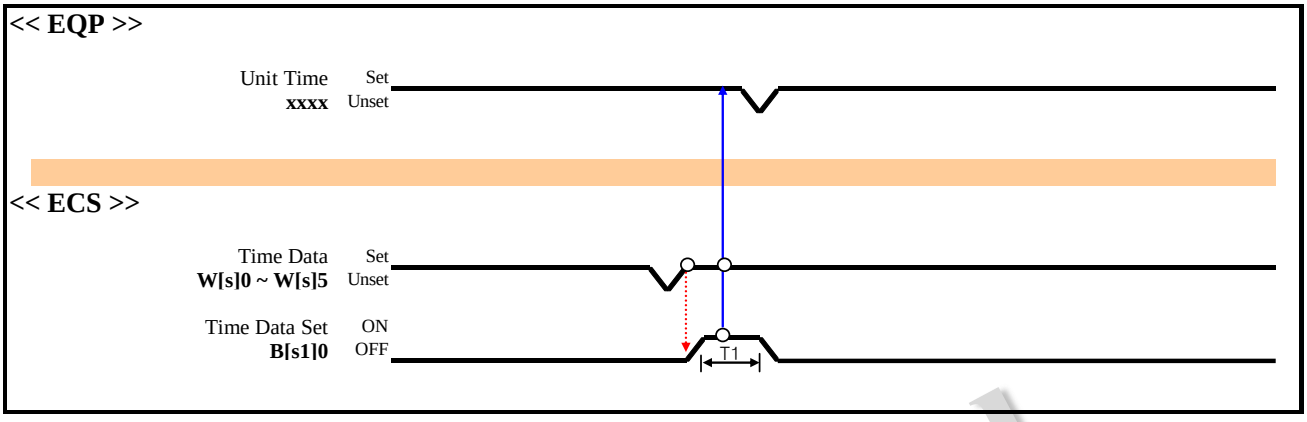
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6. Normal Sequence

6.1 Time Data Send

장비는 ECS로부터 Time Data을 Read하여 Time 동기화 한다. 이 항목은 필수 항목임.



< Time Data – ECS>

Name	Address	Format	Description
Time Setting (Year)	W[s]0	Integer	Current Time data Ex) “2003/ 9/ 9/ 20/ 27/ 6”
Time Setting (Month)	W[s]1	Integer	
Time Setting (Day)	W[s]2	Integer	
Time Setting (Hour)	W[s]3	Integer	
Time Setting (Minute)	W[s]4	Integer	
Time Setting (Second)	W[s]5	Integer	

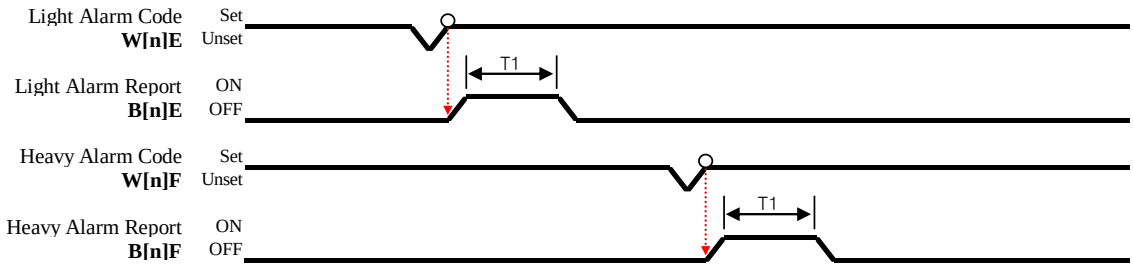
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6.2 Light/Heavy Alarm Report

장비에 Alarm이 발생 또는 해지 되었을 때 보고하는 것으로 Alarm 발생 및 해제는 Alarm Code로 구분하며 Alarm의 발생 및 해제 보고는 최소 1 Sec간은 유지시켜야만 한다.

<< EQP >>



※ 1초 trigger는 Parameter를 통해 변경 가능 하도록 해야만 한다.

< Alarm Code – EQP >

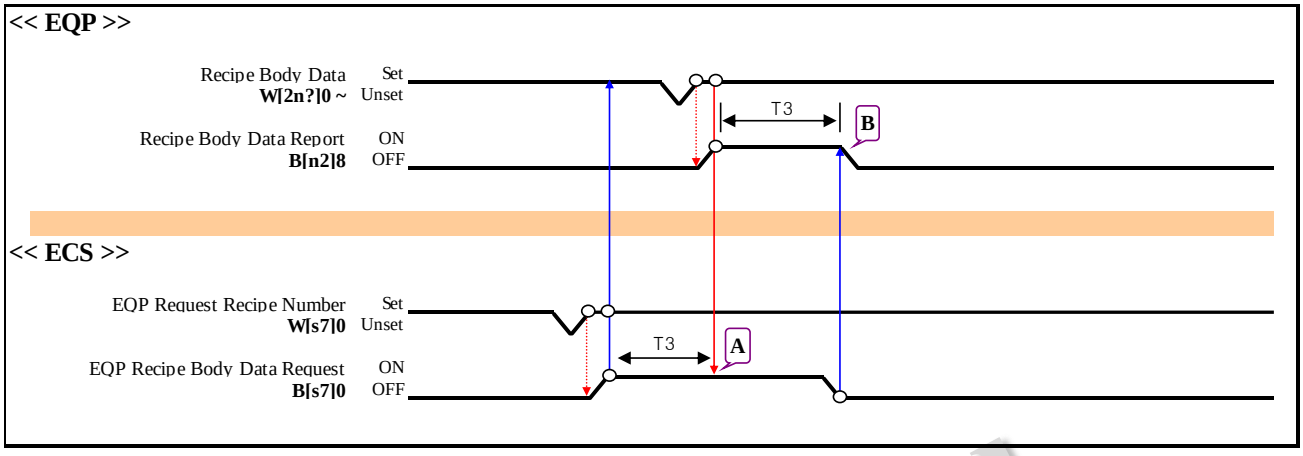
Name	Address	Format	Description
Light Alarm Code	W[n]E	Integer	Bit 0 ~ E : Alarm ID Bit F : Alarm Set (On) Alarm Reset (Off)
Heavy Alarm Code	W[n]F	Integer	Bit 0 ~ E : Alarm ID Bit F : Alarm Set (On) Alarm Reset (Off)

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6.3 Recipe Body Data Request

Recipe Number의 존재 여부 및 Recipe Body Data를 요청하는 것으로 Recipe Number는 존재하되 Recipe Body Data가 없을 시에는 CEID , Equipment Recipe Number 및 ACK Data 만을 보고한다.



A : ECS가 Recipe Body Data를 요구한 후 장비는 일정시간(Timeout - 3초) 이내에 Recipe Body Data를 보고 해야만 한다.

B : 장비는 일정시간(Timeout - 3 초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Recipe Body Data Request Confirm – ECS >

Name	Address	Format	Description
Request Recipe Number	W[s7]0	Integer	- Equipment Recipe Number

< Recipe Body Data Report – EQP >

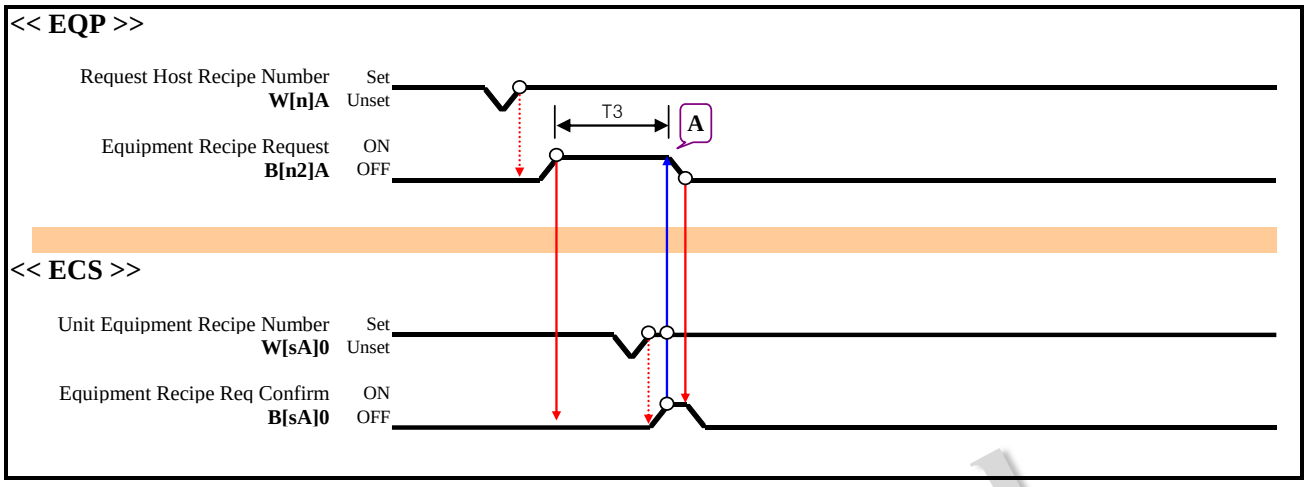
Name	Address	Format	Description
Data Report CEID	W[2n]0	Integer	3 : Recipe Body Data
Recipe Number	W[2n]1	Integer	- Equipment Recipe Number
Request Confirm ACK	W[2n]2	Integer	0 : Recipe Confirm OK 1 : Invalid Recipe Number 2 : Recipe Confirm Not Accept
Recipe Level	W[2n]3	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe
Recipe Version	W[2n]5 ~ W[2n]7	-	Recipe Version (The detail use is determined after discussing)
Unit #1 Equipment Specific Data V1 ~	W[2n1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Unit #1 Recipe Report V1 ~	W[2n2]0 ~	-	Recipe Body Data (The detail use is determined after discussing)
Unit #n Equipment Specific Data V1 ~	W[2n?]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Unit #n Recipe Report V1 ~	W[2n?]0 ~	-	Recipe Body Data (The detail use is determined after discussing)

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6.4 Equipment Recipe Request

상류 장비로부터 전달 받은 Glass Data의 Host Recipe Number를 ECS로 전달하여 Equipment Recipe Number를 요구한다. 이 시나리오는 추후 협의함



A : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Equipment Recipe Request 신호를 Off 시킨다.

※ ECS는 등록 되어 있지 않은 Host Recipe Number가 요청된 경우 Equipment Recipe Number 값을 0으로 설정하여 전달한다.
※ 장비는 이전 Glass의 LOT Number 또는 Host Recipe Number가 변경 되었을 경우 요청한다.

※ Timeout 는 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Equipment Recipe Data – EQP >

Name	Address	Format	Description
Request Host Recipe Number	W[n]A	Integer	Host Recipe Number

< Equipment Recipe Data – ECS >

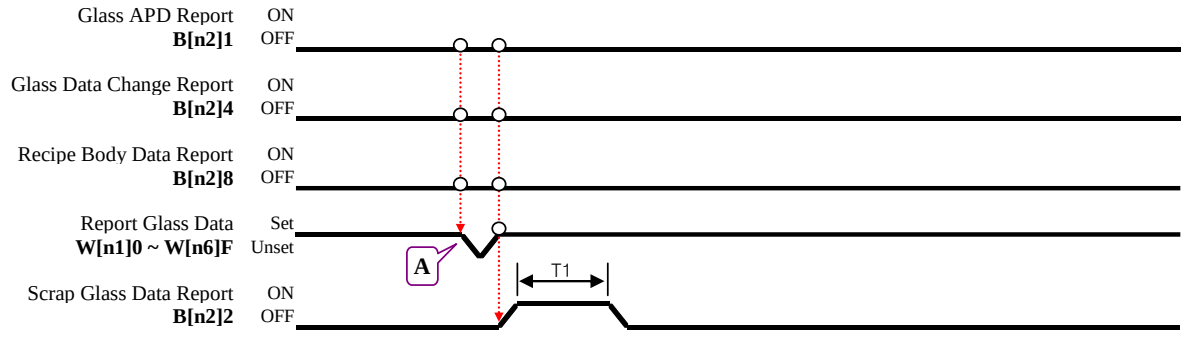
Name	Address	Format	Description
Unit Equipment Recipe Number	W[sA]0	Integer	Equipment Recipe Number

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6.5 Scrap Glass Data Report

장비는 processing중에 scrap된 Glass가 존재할 경우 Scrap Glass Data를 보고한다. 이 항목은 필수 항목임

<< **EQP** >>



A : 장비는 Glass가 scrap되었을 때 Glass APD Report, Glass Data Change Report 및 Recipe Body Data Report의 Trigger 가 Off 일 때 Report Glass Data를 보고한다.

※ Scrap Glass Data Report bit는 1초 동안 'ON' 하며 1초 trigger는 Parameter를 통해 변경 가능 하도록 해야만 한다.

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< Report Glass Data >

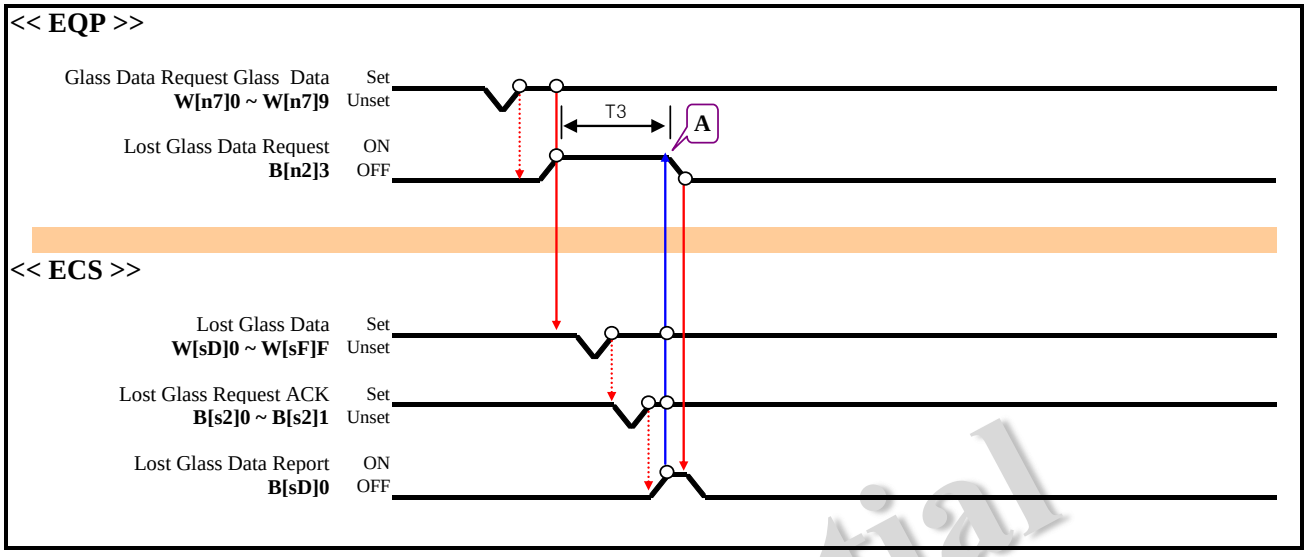
Name	Address	Format	Description
Glass Data	W[n1]0 ~ W[n3]F	-	Scrap Glass Data - Reference 4.1 Glass Data Structure
Data Report V1 <CEID>	W[n4]0	Integer	2 : Scrap Glass Data Report
Data Report V2~ V3	W[n4]1 ~ W[n4]2	ASCII	Scrap Code

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6.6 Lost Glass Data Request

장비가 해당 Glass Data을 소실하였을 때, Glass ID (or GLSCODE)를 사용하여 해당 Glass Data를 요청하는 경우 사용된다.
장비는 단지 수동 조작일 경우만 이 control sequence를 사용할 수 있다.



A : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Glass Recipe Request 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< Glass Information - EQP >

Name	Address	Format	Description
Glass Data Request Glass ID	W[n7]0 ~ W[n7]7	ASCII	- Requested Glass ID
Glass Data Request GLSCODE	W[n7]8	Integer	- Requested Glass Code
Glass Data Request Option	W[n7]9	Integer	- Request Option 0 : Glass ID 1 : GLSCODE

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< Lost Glass Data – ECS >

Name	Address	Format	Description
Lost Glass Data	W[nD]0 ~ W[nF]F	-	Glass Data - Reference 4.1 Glass Data Structure

< Lost Glass Request ACK - ECS >

Name	Address	Format	Description
Lost Glass Request ACK	B[s2]0 ~ B[s2]1	Bit	B[s2]0 : OK B[s2]1 : NG (Unknown Glass ID or GLSCODE)

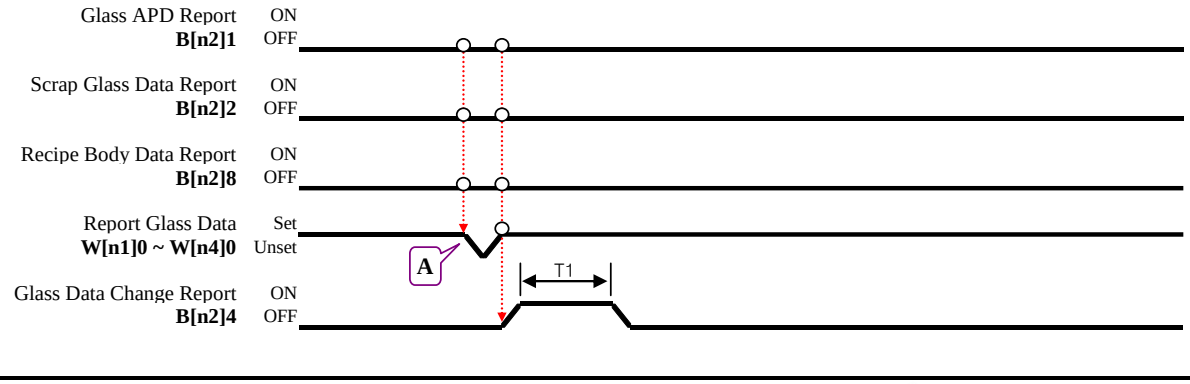
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6.7 Glass Data Change Report

장비는 Loading된 Glass Data를 Operator가 변경한 경우 Glass Data Change Report를 보고한다.
이 시나리오는 추후 협의함

<< EQP >>



A : 장비는 Scrap Glass Data Report, Glass APD Report 및 Recipe Body Data Report의 Trigger가 'Off'일 때 Report Glass Data를 보고한다.

※ Glass Data Change Report bit는 1초 동안 'ON' 하며 1초 trigger는 Parameter를 통해 변경 가능 하도록 해야만 한다.

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※ Glass Data 중 Operator에 의해 최소 변경 가능 항목

Name	Address	Format	Description
LOT ID	W[n1]0 ~ W[n1]7	ASCII (16 byte)	LOT ID
Processing Code	W[n1]8 ~ W[n1]B	ASCII (8 byte)	Processing Code
Host Recipe Number	W[n2]0	Integer	Host Recipe Number
GLSCODE	W[n2]2	Integer	Glass Code
Glass ID	W[n2]3 ~ W[n2]A	ASCII (16 byte)	Glass ID
Glass Judge	W[n2]B	ASCII	Glass Judge

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< Glass Data – EQP >

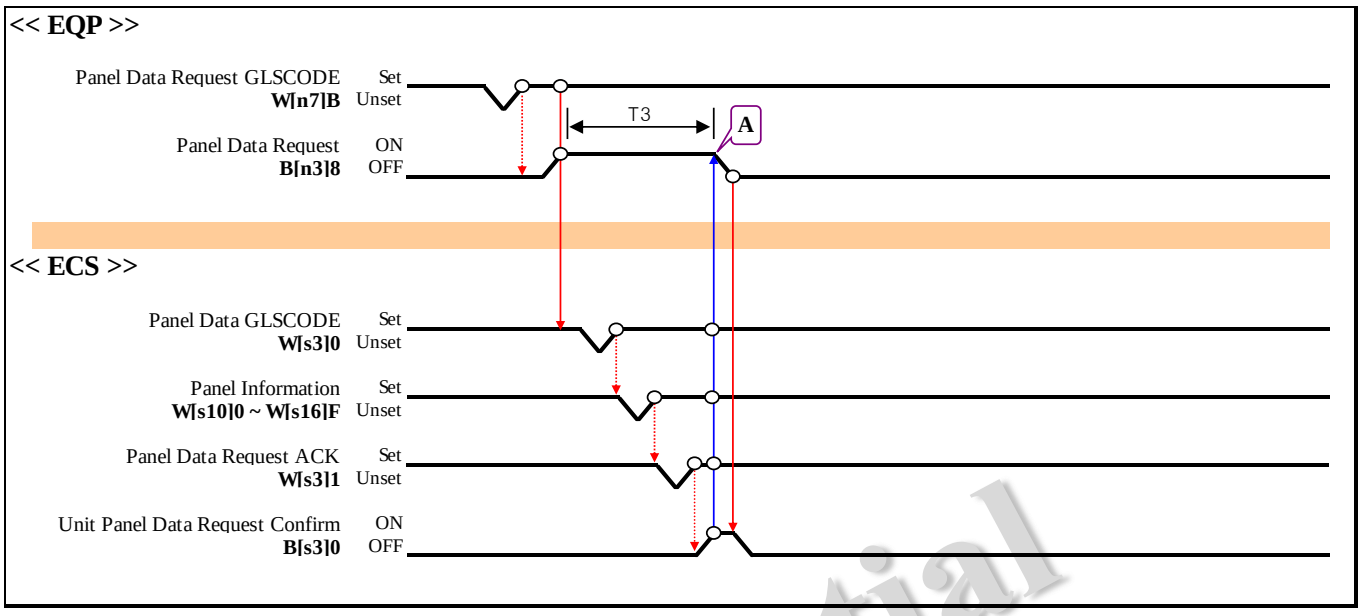
Name	Address	Format	Description
Glass Data	W[n1]0 ~ W[n3]F	-	Glass Data (Glass Type F bit : On) - Reference 4.1 Glass Data Structure
Data Report V1 <CEID>	W[n4]0	Integer	7 : Glass Data Change Report

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6.8 Panel Data Request

장비는 Panel 정보를 ECS에게 요구한다. (해당 장비에 한함)



A : 장비는 일정시간(Timeout - 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Panel Data Request 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Panel Data – EQP >

Name	Address	Format	Description
Panel Data Request GLSCODE	W[n7]B	Integer	Request GLSCODE

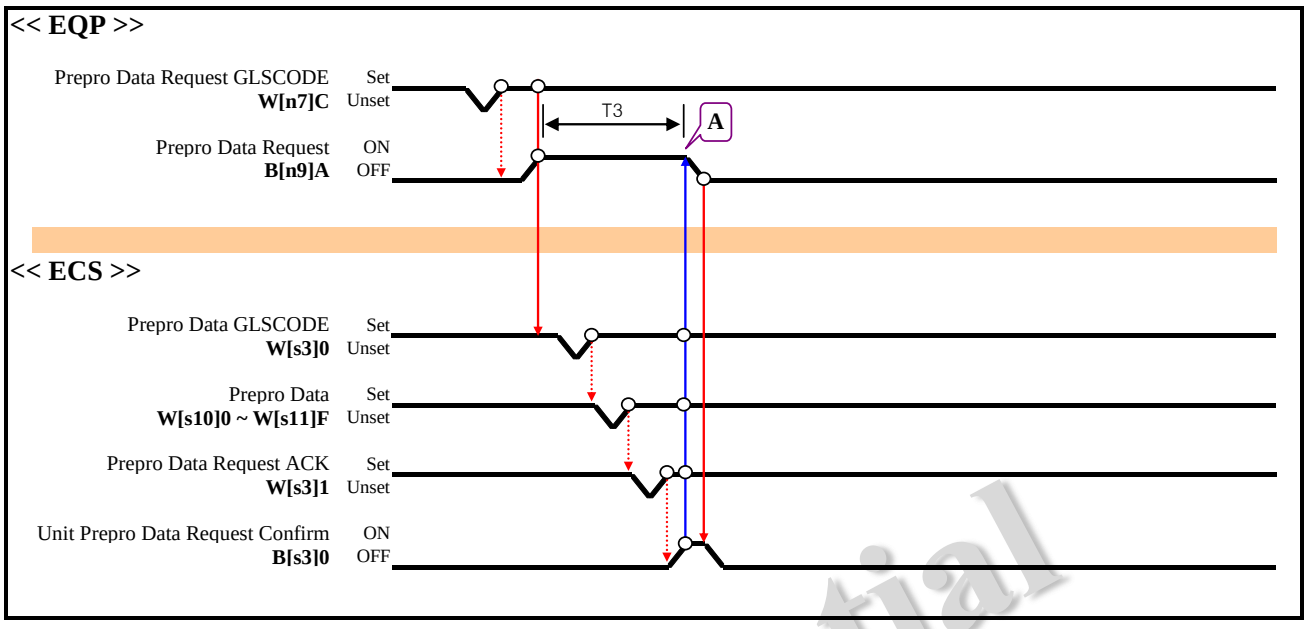
< Panel Data – ECS >

Name	Address	Format	Description
Panel Data GLSCODE	W[s3]0	Integer	Requested GLSCODE
Panel Data Request ACK	W[s3]1	Integer	0 : OK 1 : NG (Unknown Glass Code)
Panel Information	W[s10]0 ~ W[s13]F	Bit	W[s10]0 ~ W[s16]F : Panel Data W[s20]0 ~ W[s26]F : Sub Panel Data

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6.9 PREPRO Data Request

장비는 PREPRO Data를 ECS에게 요구한다. (해당 장비에 한함)



A : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 PREPRO Data Request 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< PREPRO Data – EQP >

Name	Address	Format	Description
PREPRO Data Request GLSCODE	W[n7]C	Integer	Requested GLSCODE

< PREPRO Data – ECS >

Name	Address	Format	Description
PREPRO Data GLSCODE	W[s3]0	Integer	Requested GLSCODE
PREPRO Data Request ACK	W[s3]1	Integer	0 : OK 1 : NG (Unknown Glass Code)
PREPRO Data	W[s10]0 ~ W[s11]F	ASCII	W[s10]0 ~ W[s10]F : PREPRO1 Data W[s11]0 ~ W[s11]F : PREPRO2 Data

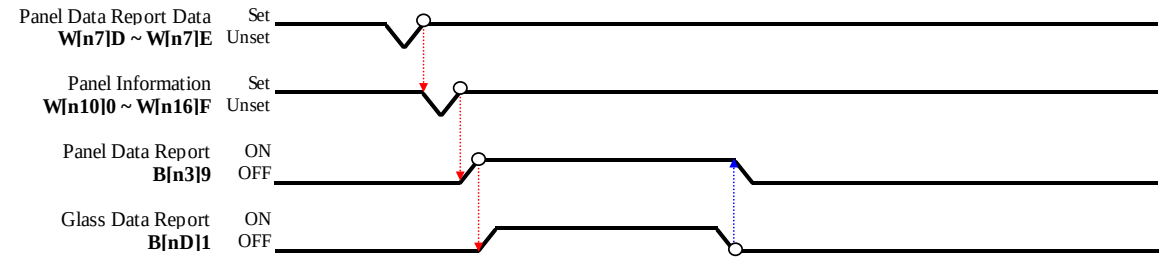
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6.10 Panel Data Report

장비는 Panel Data를 ECS에게 보고한다. (해당 장비에 한함)
반드시 Glass Data Report 이전에 설정하며 Glass Data Report의 Off 후 Panel Data Report를 Off시킨다.

<< EQP >>



※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< Panel Data – EQP >

Name	Address	Format	Description
Panel Data Report GLSCODE	W[n7]D	Integer	GLSCODE
Panel Data Report GLSJUDGE	W[n7]E	ASCII	Glass Judge
Panel Information	W[n10]0 ~ W[n13]F	Bit	W[n10]0 ~ W[n16]F : Panel Data W[n20]0 ~ W[n26]F : Sub Panel Data

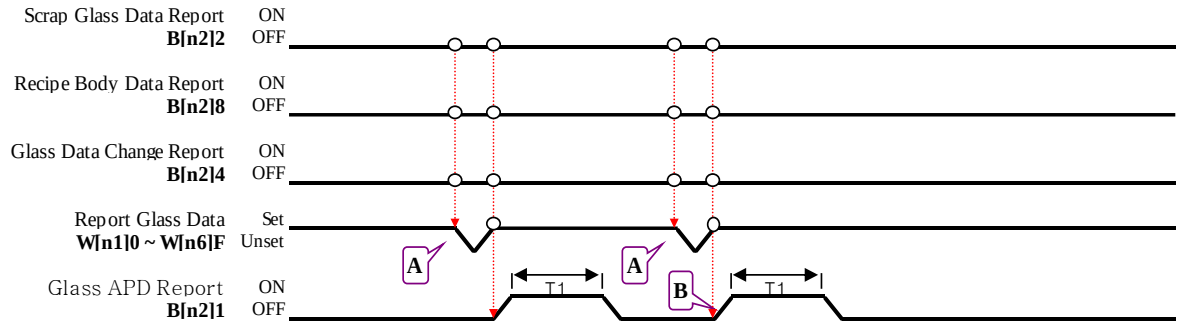
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6.11 Glass APD Report

장비는 처리가 완료된 Glass를 Unloading할 때 Glass APD를 보고한다. 이 항목은 필수 항목임.

<< EQP >>



A : 장비는 Scrap Glass Data Report, Glass Data Change Report 및 Recipe Body Data Report의 Trigger가 Off 일 때 Glass APD를 보고한다.
Glass APD Report trigger bit는 1초 동안 'ON' 한다.

B : A Cut, B Cut Pair Glass에 대해 연속적으로 Glass APD Report 시 1초 Delay를 반영한다.

※ 1초 trigger는 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Glass Data - EQP >

Name	Address	Format	Description
Glass Data	W[n1]0 ~ W[n3]F	-	Glass Data - Reference 4.1 Glass Data Structure
Data Report V1 <CEID>	W[n4]0	Integer	1 : Glass APD Report
Data Report	W[n4]1 ~ W[n6]F	-	Glass APD
Data Report2	W[n30]0 ~	-	Glass APD

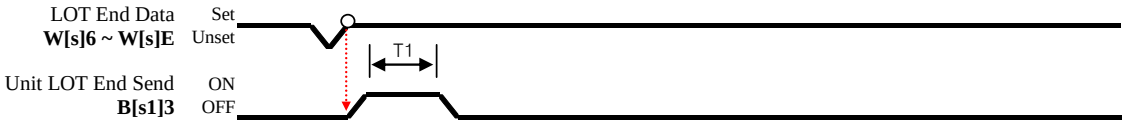
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6.12 LOT End Send

LOT의 처리가 완료되면 ECS는 LOT End를 알린다.

<< ECS >>



< LOT End Data – ECS >

Name	Address	Format	Description
LOT End LOT ID	W[s]6 ~ W[s]D	ASCII	-LOT ID
LOT End LOT Number	W[s]E	Integer	- LOT Number

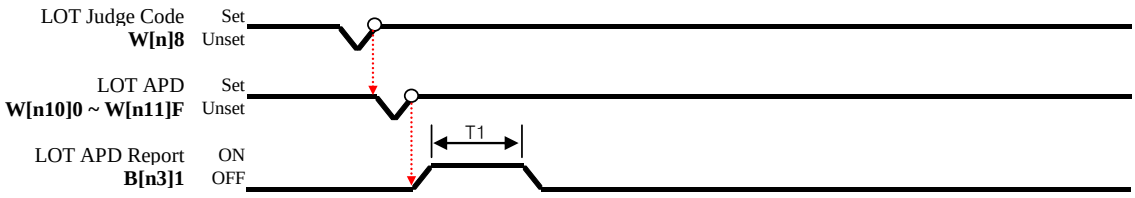
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6.13 LOT APD Report

장비는 LOT End를 수신한 후 LOT의 Actual Processing Data를 보고한다.
(이 시나리오의 사용 여부는 협의에 의해 결정함)

<< EQP >>



※ LOT APD Report bit는 1초 동안 'ON' 하며 1초 trigger는 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< LOT APD – EQP >

Name	Address	Format	Description
LOT Judge Code	W[n]8	ASCII	- LOT Judge Code
LOT Number	W[n10]0	Integer	Lot Number
LOT APD	W[n10]1 ~ W[n11]F		LOT APD

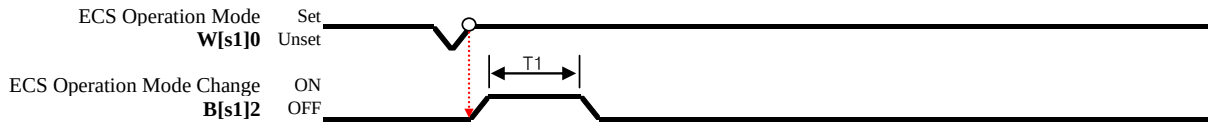
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6.14 ECS Operation Mode Change

ECS의 Operation Mode가 변경되었음을 알린다.

<< ECS >>



< ECS Operation Mode – Master >

Name	Address	Format	Description
ECS Operation Mode	W[s1]0	Integer	ECS Operation Mode

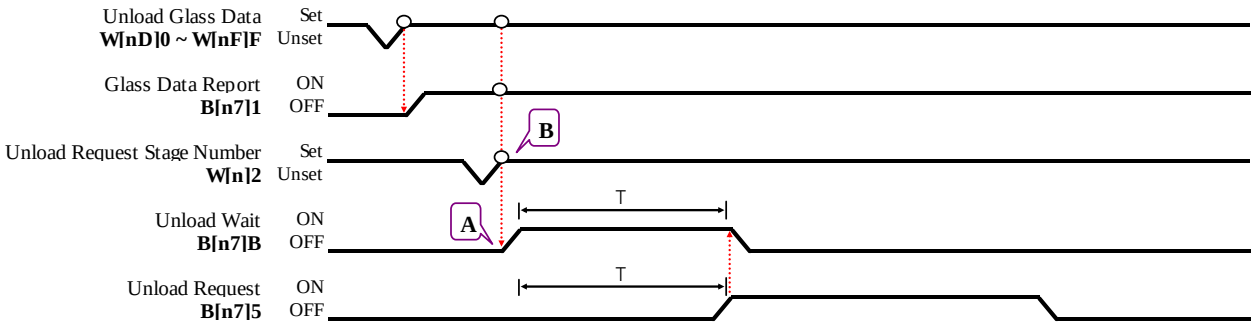
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6.15 Unload Wait

배출측 장치가 일정시간(T) 후에 Glass 배출 동작(Interlock)이 시작될 것을 수취측 장치에게 알려주는 것으로서 배출측 장치는 신호(Unload Wait)를 'ON' 한다. (Process Inline Only)
(이 시나리오의 사용 여부는 협의에 의해 결정함)

<< EQP >>



A : 배출측 장치는 Glass Data Report가 On일 경우에만 Unload Wait 를 ON으로 설정할 수 있다.
B : Unload Request Stage Number는 Multi Stage 장치에 한한다.

※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< Unload Glass Data - EQP >

Name	Address	Format	Description
Glass Data	W[nD]0 ~ W[nF]F	-	Unload Glass Data - Reference 4.1 Glass Data Structure

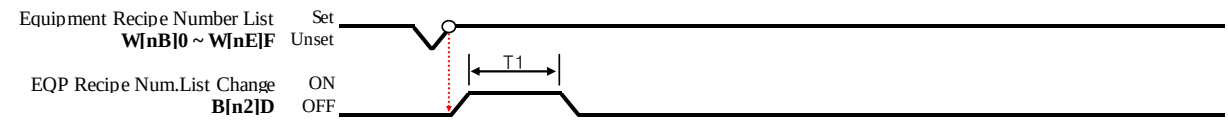
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6.16 Equipment Recipe Number List Change

장비에 등록되어 있는 Equipment Recipe Number List 변경되었을 때 보고한다.

<< EQP >>



※ EQP Recipe Num. List Change bit는 1초 동안 'ON' 하며 1초 trigger는 Parameter를 통해 변경 가능하도록 해야만 한다.

< Equipment Recipe Number List – Equipment >

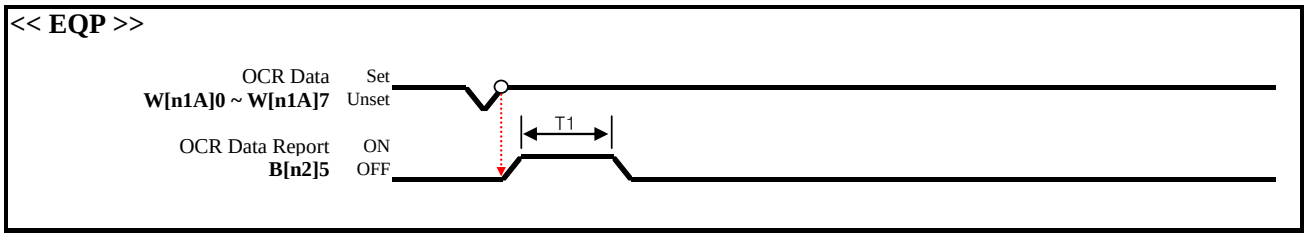
Name	Address	Format	Description
Equipment Recipe Number List	W[nB]0 ~ W[nE]F	Bit	Equipment Recipe Number List - Reference 2.2.16 Equipment Recipe Number List

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6.17 OCR Data Report

장비는 ECS로 OCR Data를 보고한다. (해당 장비에 한함)



※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< OCR Data – Equipment >

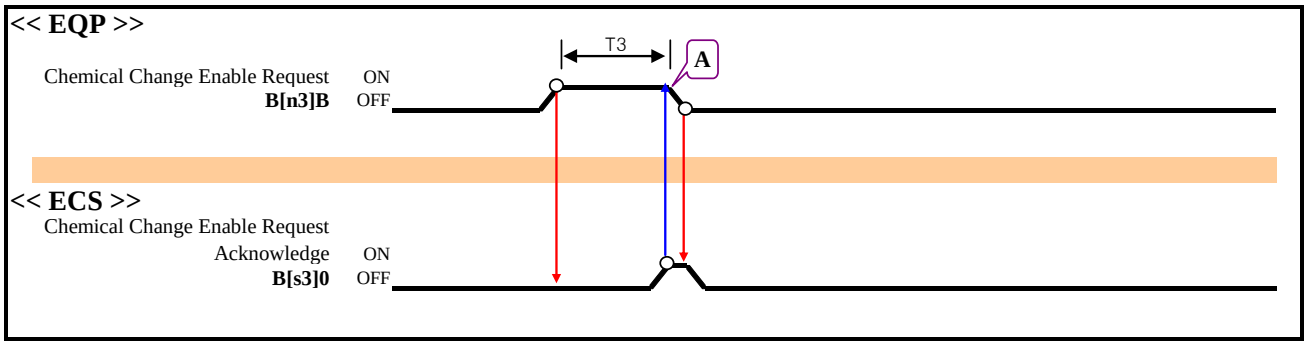
Name	Address	Format	Description
OCR Data	W[n1A]0~ W[n1A]7	ASCII	OCR Data

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6.18 Chemical Change Enable Request

ECS에 화학약품의 교환 가능 여부를 확인한다.
(Reference 8.2 Chemical Change Sequence)



A : 장비는 일정시간(3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Chemical Change Enable Request 신호를 Off 시킨다.

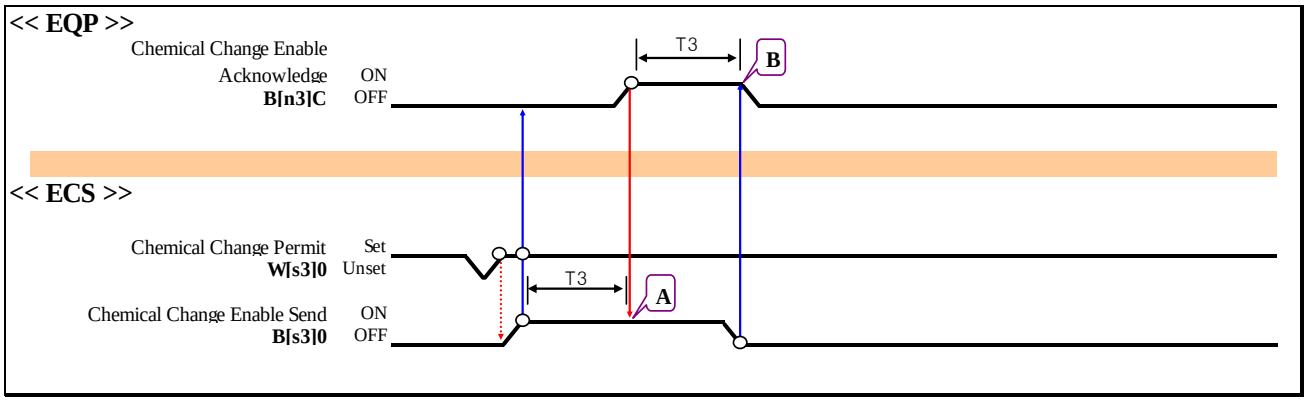
※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.

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6.19 Chemical Change Enable Send

ECS는 화학약품의 교환 가능 여부를 전달한다.
(Reference 8.2 Chemical Change Sequence)



A : ECS가 Chemical Change Enable Send한 후 장비는 3초 이내에 Chemical Change Enable Ack를 보고 해야만 한다.

B : 장비는 일정시간(3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.

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< Chemical Change Permit >

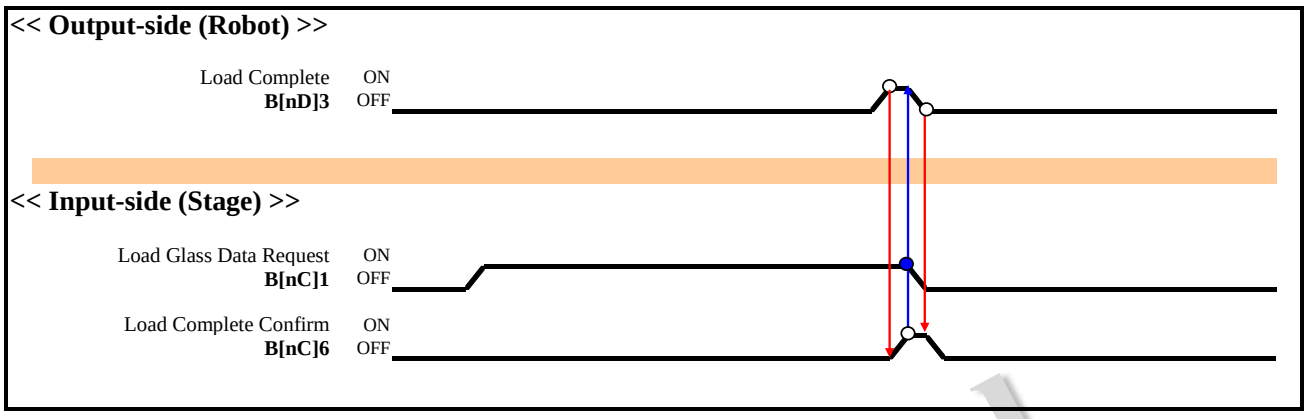
Name	Address	Format	Description
Chemical Change Permit	W[s3]0	Integer	0 : 즉시 가능 1 : 처리중인 Glass와 다른 LOT의 Glass가 투입 예정일 때

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6.20 Load Glass Data Request

6.20.1 Load Glass Data Request (Process Inline)

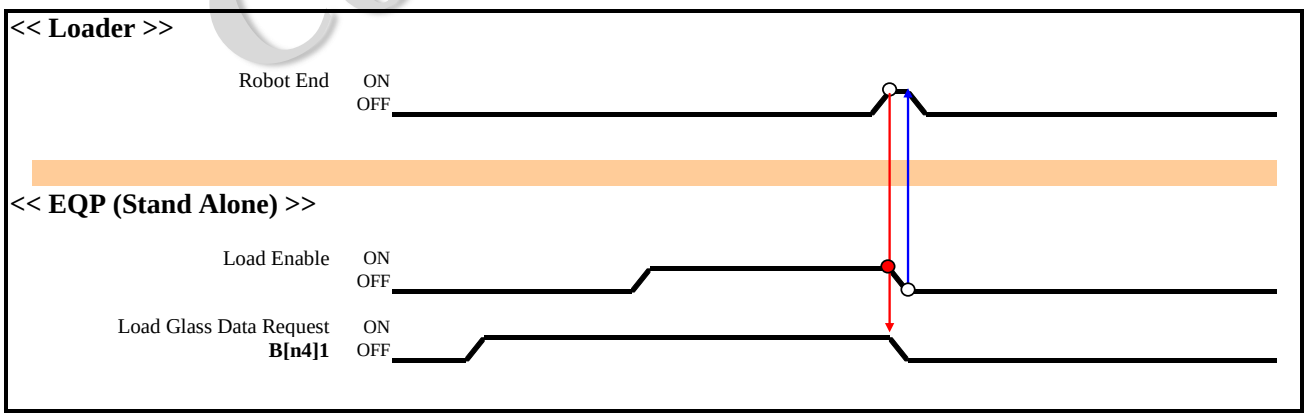
Process Inline의 장비간의 Interlock 신호 중 상류 장비로 Glass Data를 요구할 때 설정하는 것으로 상류 장비의 투입 완료에 의해 Off 된다. (**Interlock 사양서 참조**)



※ Process Inline Interlock Type(1 ~ 4)에 대한 예시로 장비간의 Interlock Type에 따라 다를 수 있음

6.20.2 Load Glass Data Request (Stand Alone)

Stand Alone 장비에서 ECS로 Glass Data를 요구할 때 설정하는 것으로 Loader(또는 반송 장치)의 투입 완료에 의해 Off된다.



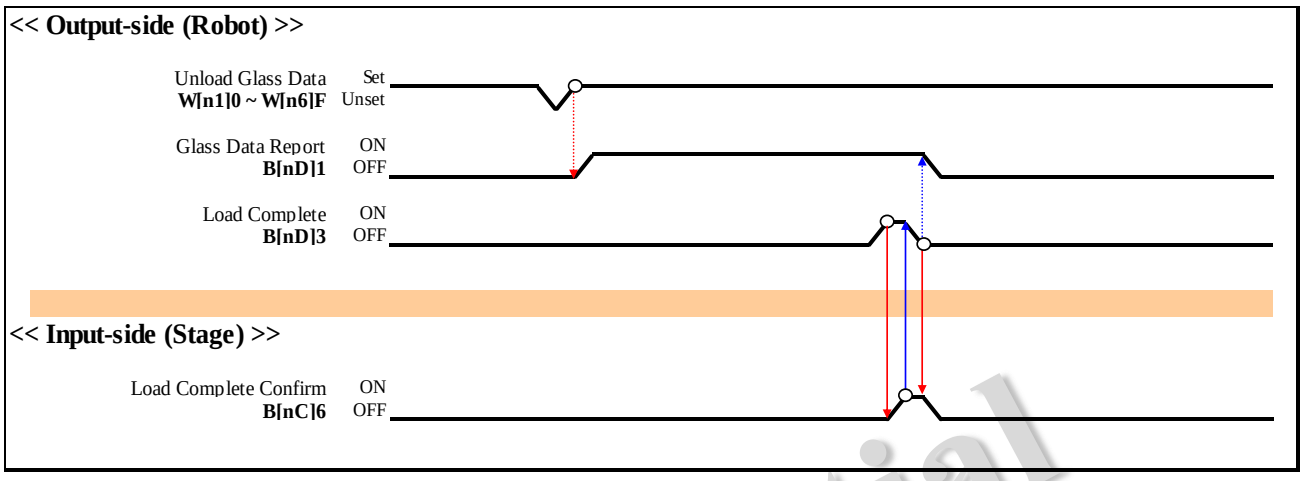
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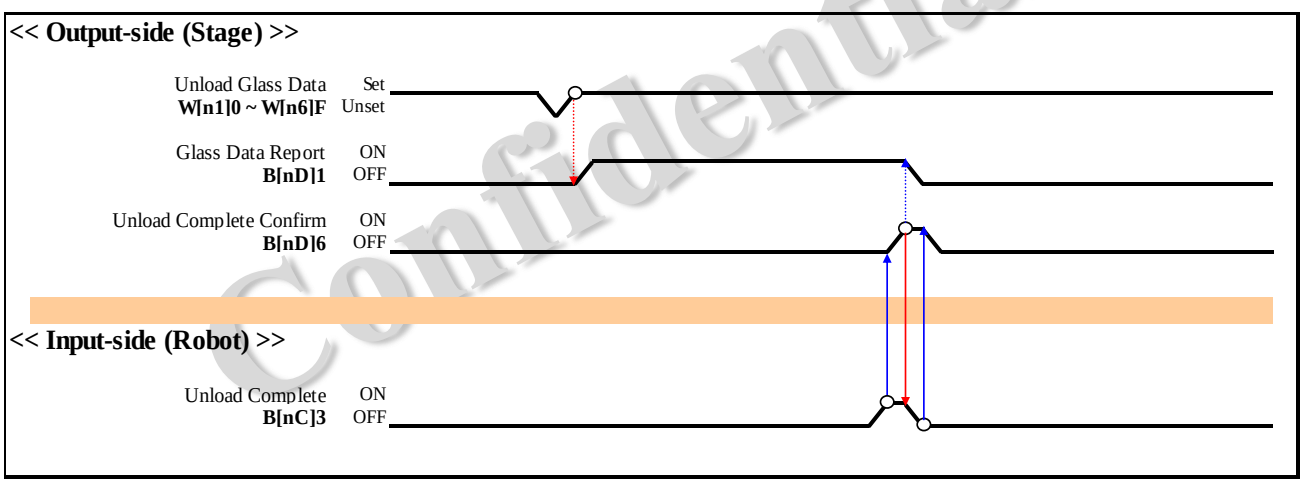
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6.21 Unload Glass Data Report

6.21.1 Unload Glass Data Report (Process Inline)

Process Inline의 장비간의 Interlock 신호 중 상류 장비에서 배출 Glass 대한 정보를 하류 장비로 전달하는 것으로 하류 장비의 투입 완료에 의해 Off 된다. (**Interlock 사양서 참조**)





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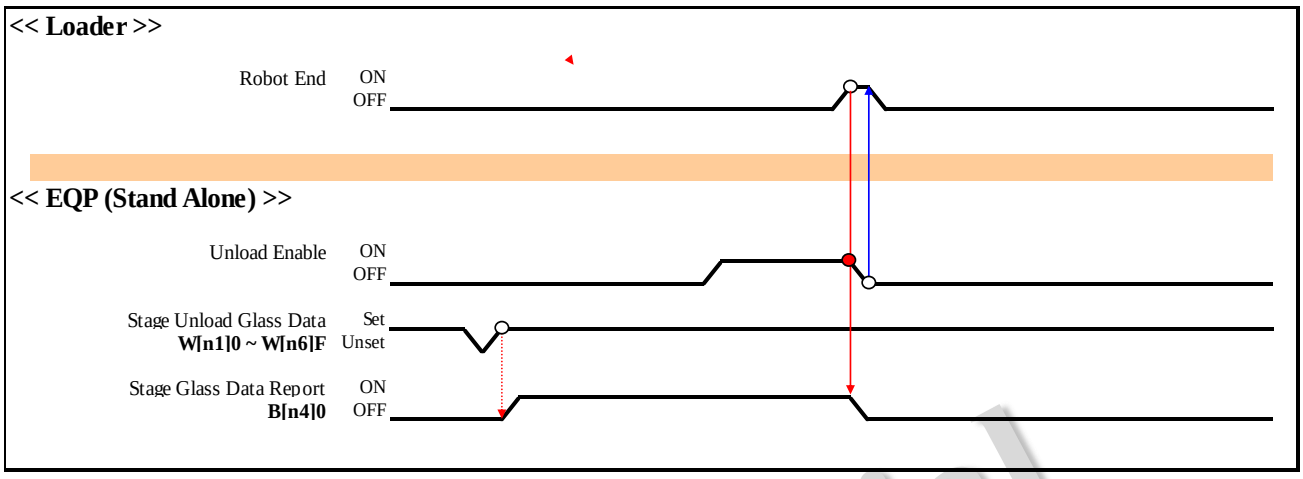
< Unload Glass Data >

Name	Address	Format	Description
Unload Glass Data	W[n1]0 ~ W[n6]F	-	Glass Data (Each Glass Data of A Cut, B Cut) - Reference 4.1 Glass Data Structure

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6.21.2 Unload Glass Data Report (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 배출 Glass에 대한 정보를 보고한다



< Stage Unload Glass Data >

Name	Address	Format	Description
Stage Unload Glass Data	W[n1]0 ~ W[n6]F	-	Glass Data (Each Glass Data of A Cut, B Cut) - Reference 4.1 Glass Data Structure

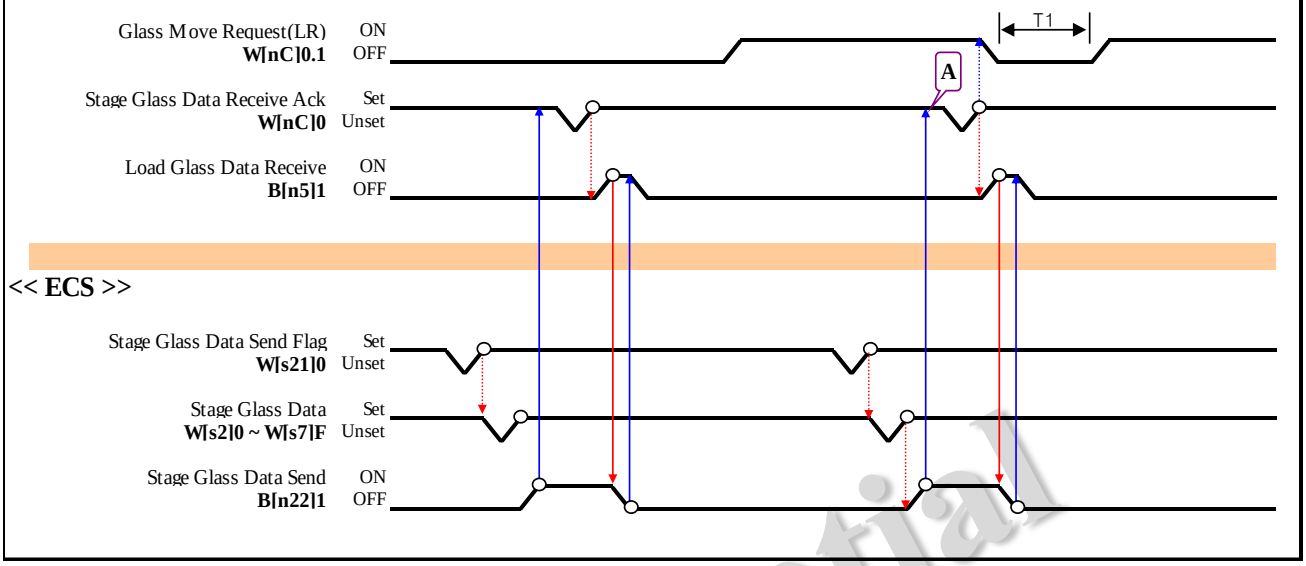
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6.22 Glass Data Send

Stand Alone 장비에서 사용하는 것으로 ECS가 장비로 투입될 Glass에 대한 정보를 전달한다.(Stand Alone)

<< EQP (Stand Alone) >>



※ A : 장치는 Glass의 반송이 이루어지지 전(Robot Busy or Conveyor Busy)에 해당 Glass 정보가 변경되었음 보고 받았을 경우 수신 가능 여부를 확인 한 후 “Glass Move Request (LR)”를 Off시킨다.
※ Glass Move Request (LR) Unset시 1초 이상 유지하며 Parameter를 통해 변경 가능하도록 해야만 한다.

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< Stage Glass Data - ECS >

Name	Address	Format	Description
Glass Data Send Flag	W[s21]0	Integer	0 : Glass Exist 1 : Glass Not Exist
Stage Glass Data	W[n2]0 ~ W[n7]F	-	Glass Data (Each Glass Data of A Cut, B Cut) - Reference 4.1 Glass Data Structure

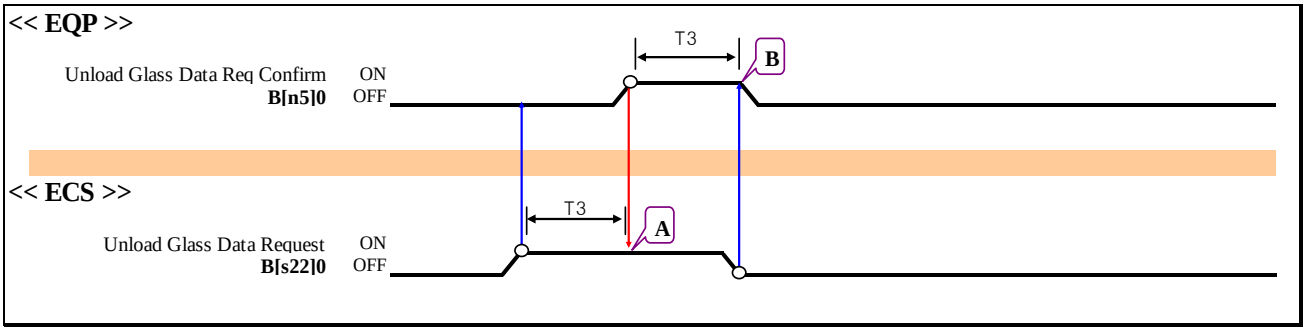
< Stage Glass Data Receive Ack - EQP >

Name	Address	Format	Description
Stage Glass Data Receive ACK	W[nC]0	Integer	0 : Accept 1 : Not Accept (Glass Processing) 2 : Not Accept (Unknown Recipe Number) 3 : Not Accept (Glass Data Error) Etc : Other Error

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6.23 Unload Glass Data Request

ECS가 장비로 Unload Glass Data를 요청하는 것으로 수신 시 작업자가 선택창을 통하여 Lost Glass Data Request 또는 Unload Glass Data Report를 선택 가능하게 한다.
Unload Glass Data Report 보고 시 상태가 On일 경우 1초간 Off 시킨 후 다시 On 시킨다.
이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)



A : ECS가 Unload Glass Data Request한 후 장비는 3초 이내에 Unload Glass Data Request Confirm를 보고 해야만 한다.

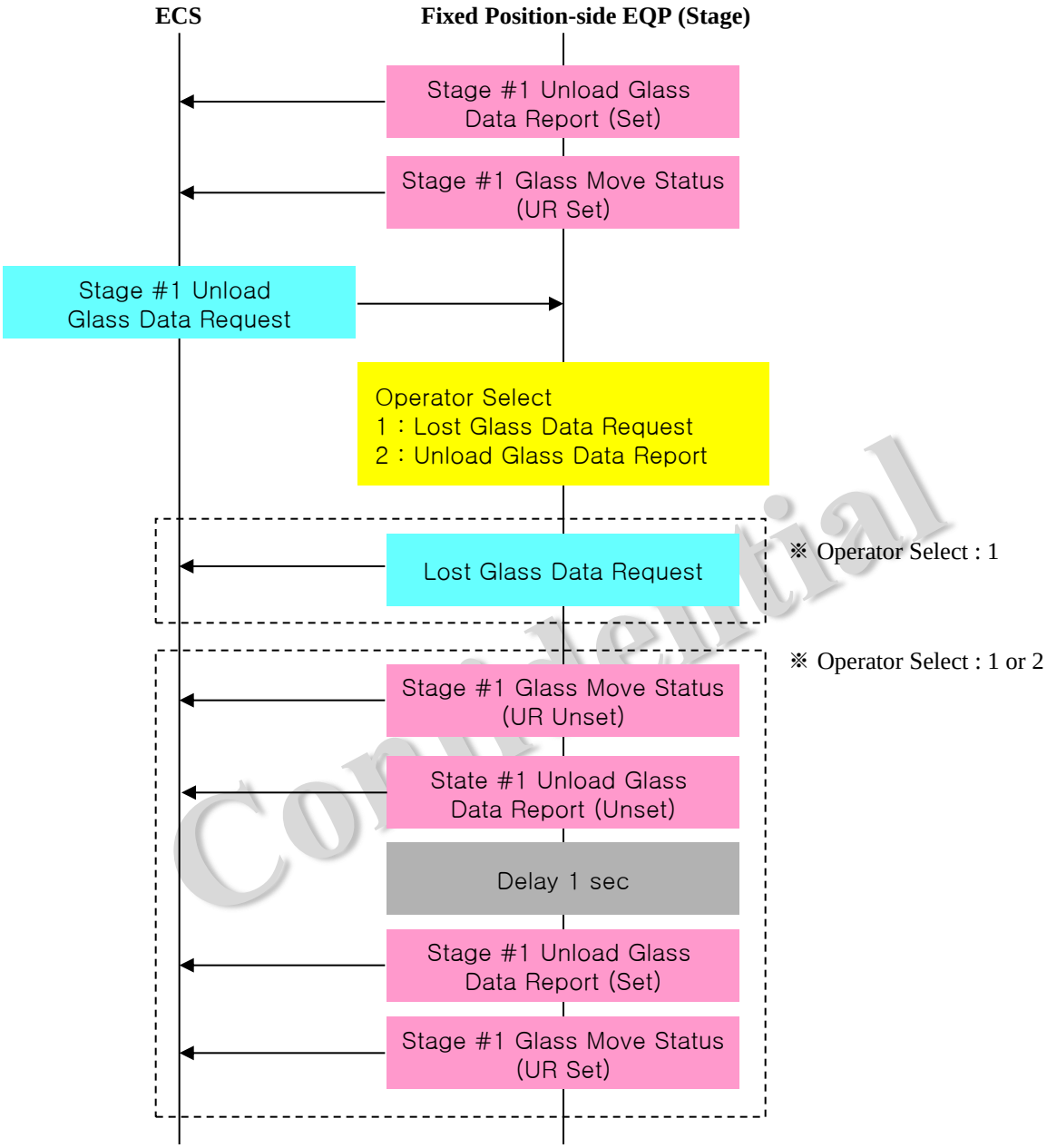
B : 장비는 일정시간(3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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※ Reference : Unload Glass Data Request



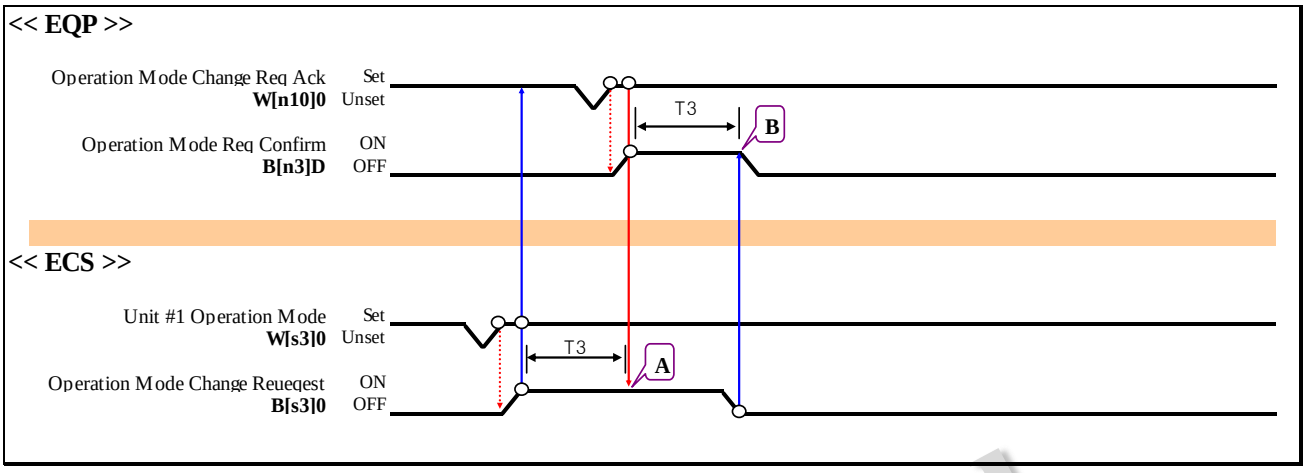
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Interlock | Operator | Report | Data Set | Internal

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6.24 Operation Mode Change Request

장비에게 Operation Mode 변경을 요구하는 것으로 장비는 자동으로 변경이 가능하면 Operation Mode를 변경하고 변경이 불 가능한 경우는 작업자를 호출 한다.



A : ECS가 Operation Mode Change를 요구한 후 장비는 3초 이내에 Operation Mode Request Confirm를 보고 해야만 한다.

B : 장비는 일정시간(T)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Operation Mode – ECS>

Name	Address	Format	Description
Operation Mode	W[s3]0	Integer	- Operation Mode 협의 후 결정함

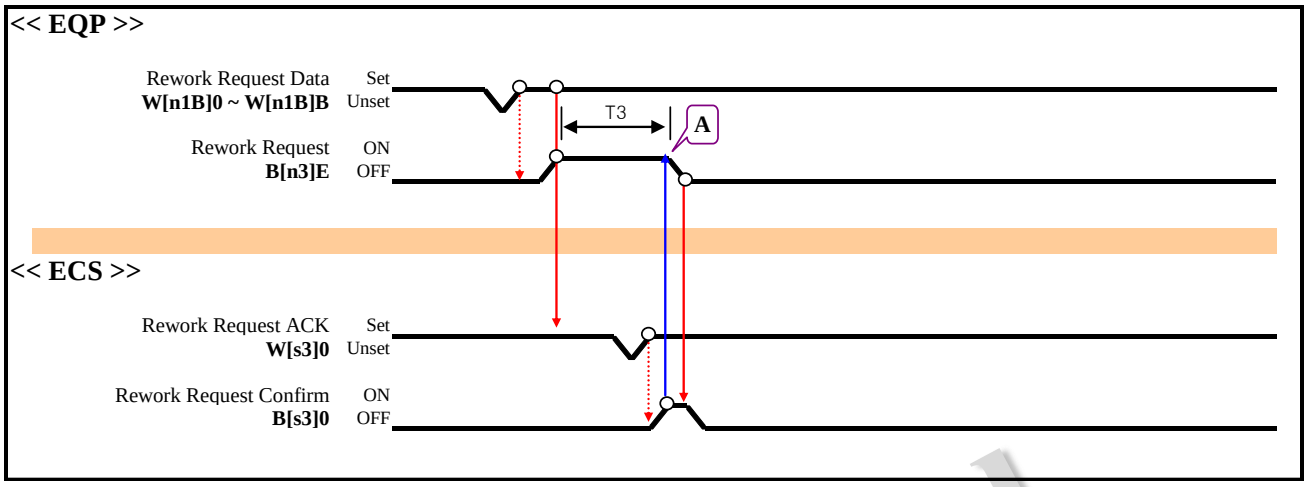
< Operation Mode Request Ack – EQP>

Name	Address	Format	Description
Operation Mode Req Ack	W[n10]0	Integer	0 : Accept 1 : Not Accept Etc : Other Error

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6.25 Rework Request

장비에서 특정 LOT에 대한 Rework을 요청한다. (Stand Alone)



A : 장비는 일정시간(T)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시키며 Rework Request 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Rework Request Data - EQP>

Name	Address	Format	Description
Rework LOT ID	W[n1B]0 ~ W[n1B]7	ASCII	Rework Request LOT ID
Rework Reserve Code	W[n1B]8	ASCII	Rework Reservation Code 'R' : Reservation of rework 'S' : Reservation start
Rework Select Slot	W[n1B]9 ~ W[n1B]B	Bit	Selected Slot (1 ~ 48) Bit On : Selected Bit Off : Not Selected Ex) '1111111110000000000'

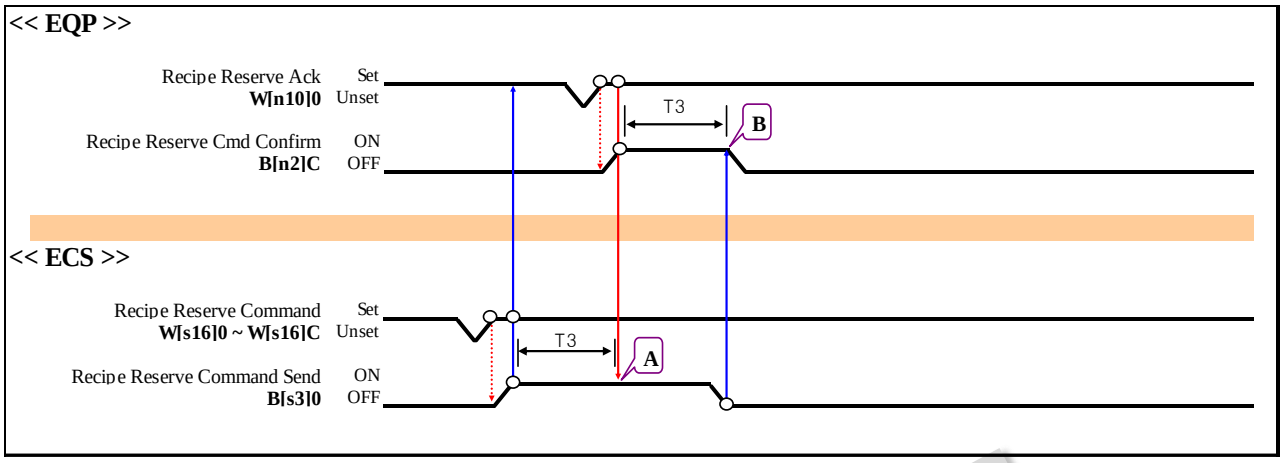
< Rework Request ACK - ECS >

Name	Address	Format	Description
Rework Request ACK	W[s3]0	Integer	0 : Accept 1 : Not Accept Etc : Other Error

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6.26 Recipe Reserve Command Send

장비로 Recipe Reserve Command를 전송한다. (해당 장비에 한함)



A : ECS가 "Recipe Reserve Command Sned"를 On한 후 장비는 3초 이내에 Recipe Reserve Command Confirm를 보고 해야만 한다.

B : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Recipe Reserve Command - ECS>

Name	Address	Format	Description
Recipe Reserve Code	W[s16]0	Integer	1 : Recipe Reservation 2 : Recipe Reservation Cancel
Recipe Reserve Unique Number	W[s16]1	Integer	Recipe Reserve Unique Number
Recipe Reserve LOTID	W[s16]2 ~ W[s16]9	ASCII	LOT ID
Recipe Reserve Equipment Recipe Number	W[s16]A	Integer	Equipment Recipe Number

< Recipe Reserve Ack – EQP >

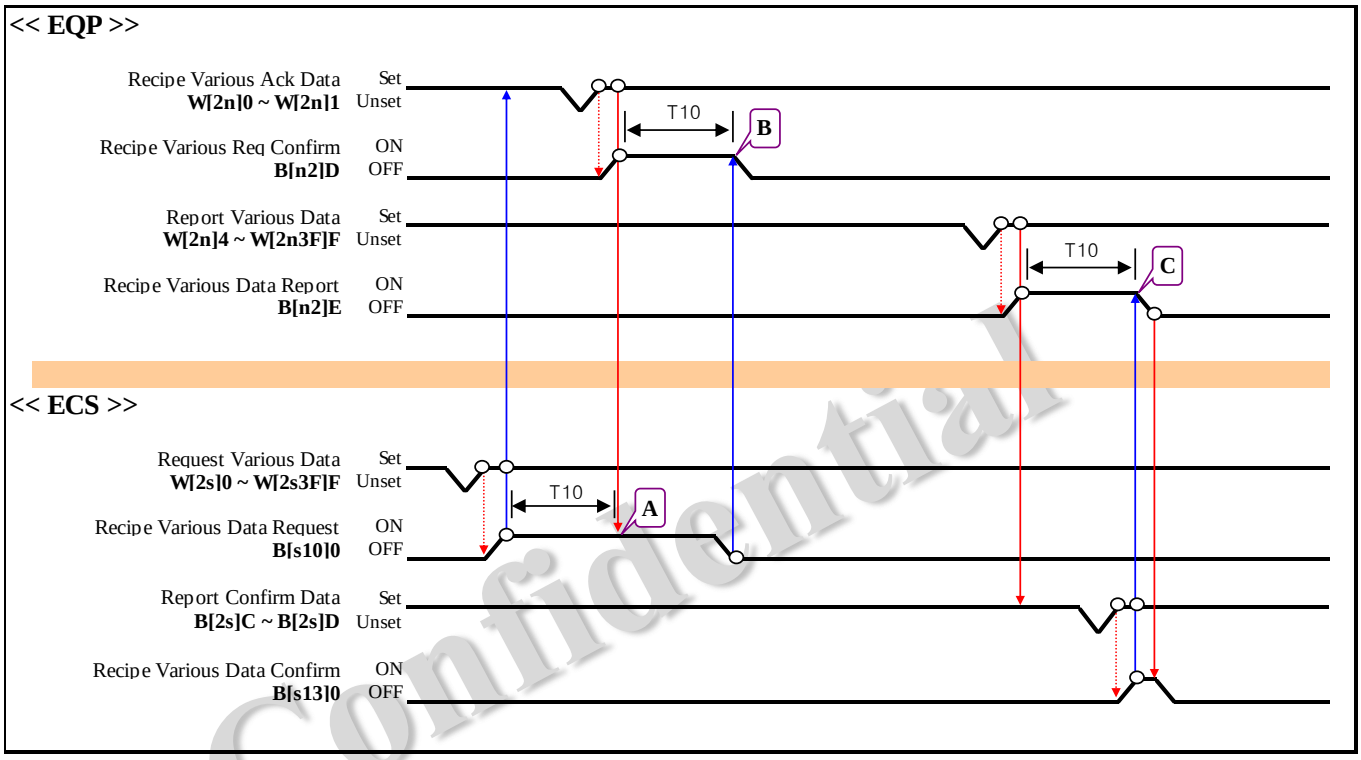
Name	Address	Format	Description
Recipe Reserve Ack	W[n10]0	Integer	0 : Accept 1 : Not Accept Etc : Other Error

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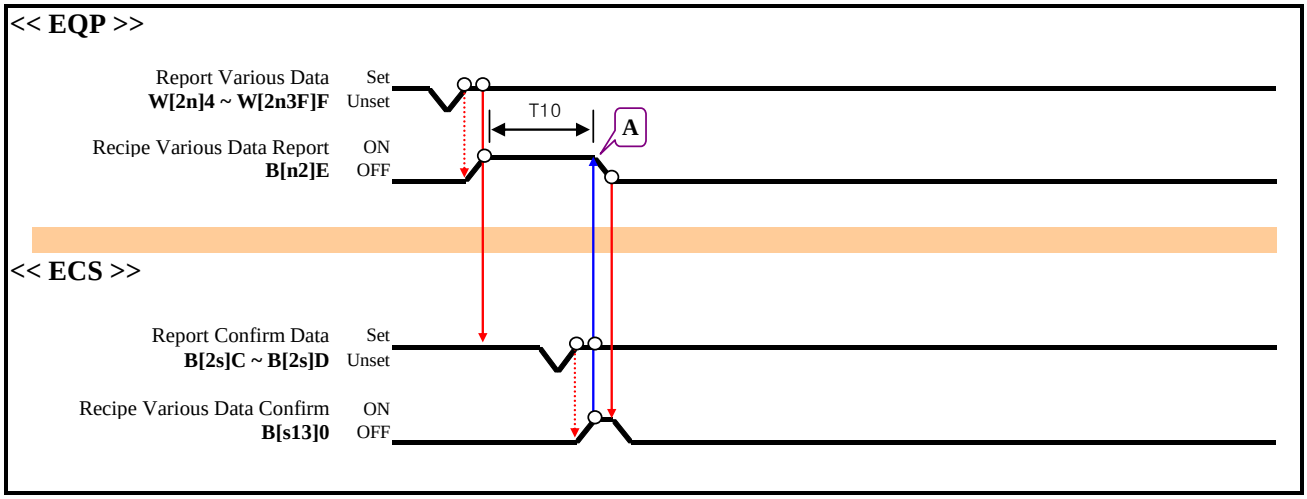
6.27 Recipe Various Data Request & Report

장비에 등록되어 있는 Recipe 정보 요청 하거나 Recipe 정보를 보고 받는 것으로 CEID에 따라 요청 및 보고 정보를 구분한다.
사용여부는 협의하여 결정한다.

※ <Recipe Various Data Request & Report> Time Chart



※ <Recipe Various Data Report> Time Chart

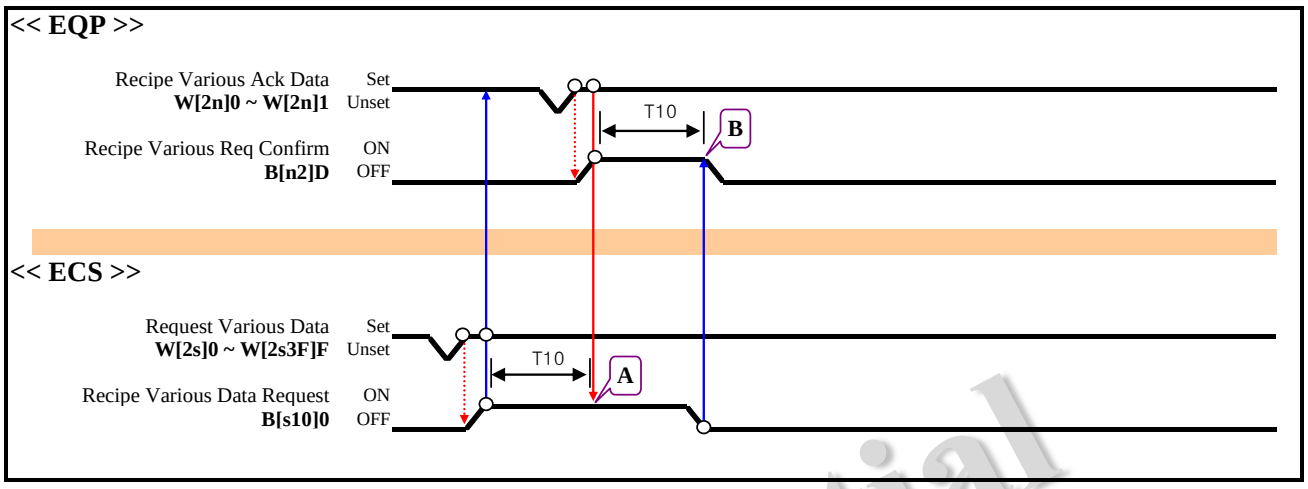


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6.27.1 Recipe Various Data Request

장비에 등록되어 있는 Recipe 정보 요청 또는 Recipe Command를 전달하는 것으로 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 4 : Recipe Command 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 8 : EQP Constant Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data - Recipe Parameter Data - Lower Recipe Number List ※ The detail use is determined after discussing.

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< Recipe Various ACK Data - EQP >

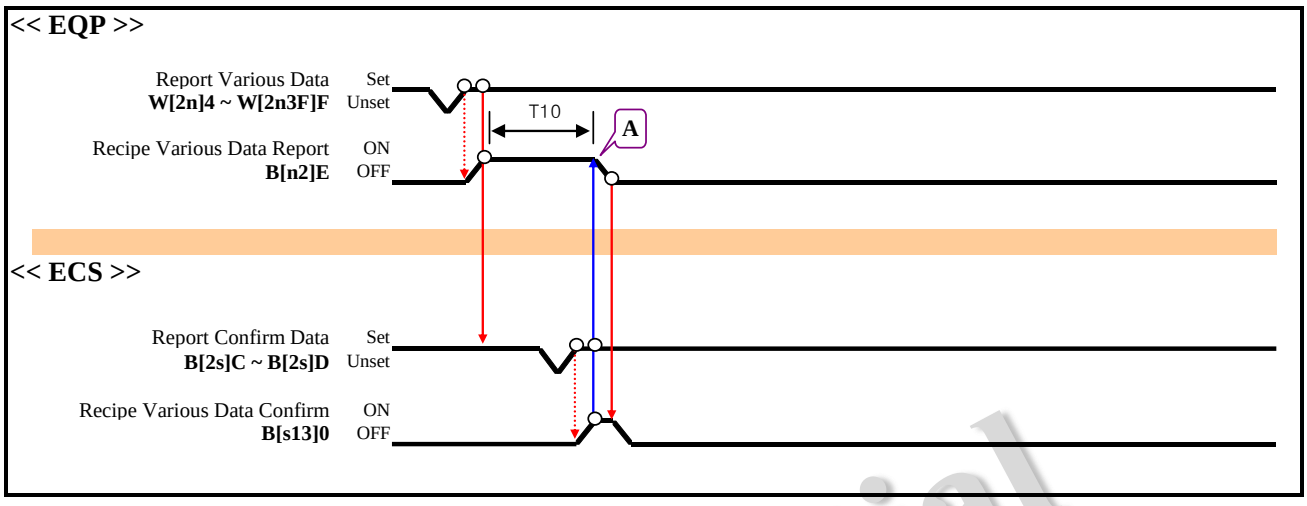
Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 4 : Recipe Command 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 8 : EQP Constant Recipe Command
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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6.27.2 Recipe Various Data Report

ECS의 요청 또는 장비에서 변경된 Recipe 정보를 보고하는 것으로 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 5 : Recipe Change Report 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 9 : EQP Constant Recipe Change Report
Recipe Report CEID Order	W[2n]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n]6	Integer	Recipe Total Count
Recipe Report Count	W[2n]7	Integer	Recipe Data Structure Count
Recipe Command	W[2n]8	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete
Recipe Number	W[2n]9	Integer	Recipe Number
Recipe Level	W[2n]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n]C ~ W[2n]E	-	Recipe Version
Recipe Report V1 ~	W[2n1]0 ~	-	- Equipment Specific Data - Recipe Report Data (The detail use is determined after discussing)

※ 보고하는 Recipe 데이터가 Report Various Data 영역을 초과할 경우 처음 보고 시는 CEID를 Original Value를 보고 하고 그 이후에는 10을 더한 값을 보고한다 (이 시나리오는 추후 협의에 의해 결정함)

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< Report Confirm Data – ECS >

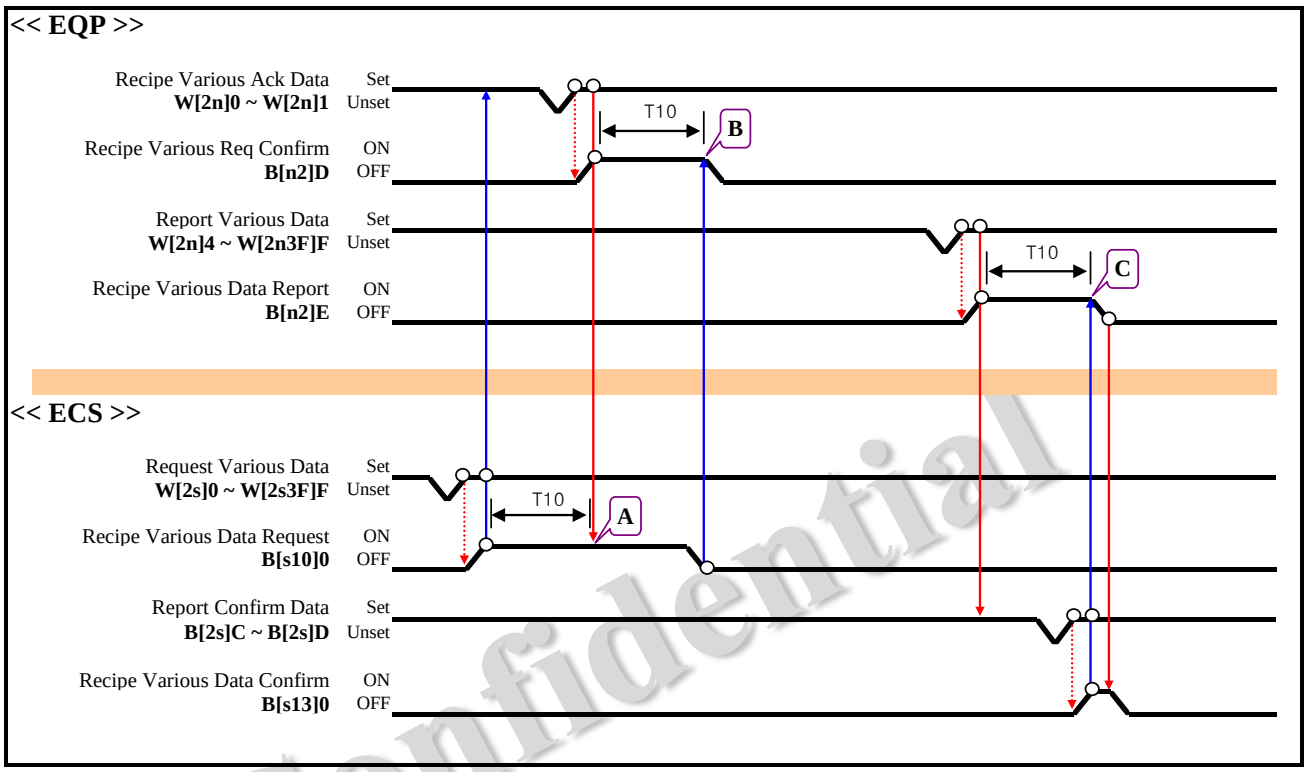
Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 5 : Recipe Change Report 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 9 : EQP Constant Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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6.27.3 Recipe List Request

장비에 등록되어 있는 Recipe 정보를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	1 : Recipe List
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data (The detail use is determined after discussing)

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< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	1 : Recipe List
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	1 : Recipe List
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	1 : Recipe List
Recipe Report CEID Order	W[2n]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n]6	Integer	Recipe Total Count
Recipe Report Count	W[2n]7	Integer	Recipe Data Structure Count
Recipe Report V1 ~	W[2s1]0 ~	-	- Equipment Specific Data - Recipe Report Structure ※ Reference Recipe Report Structure (The detail use is determined after discussing)

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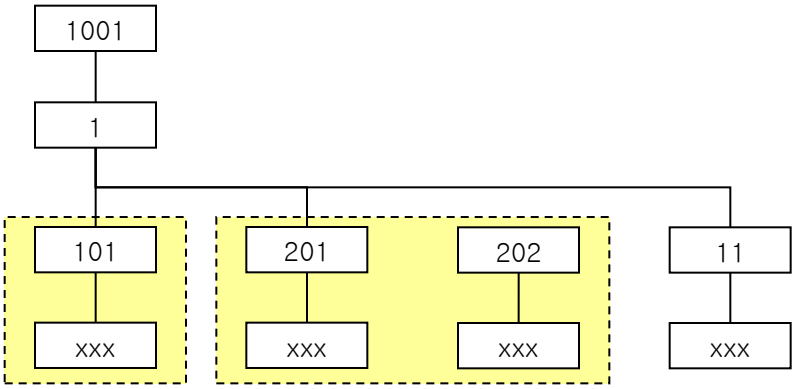
< Recipe Report Structure >

Index	Name	Length	Format	Description
1	Recipe Number	1 word	Integer	Recipe Number
2	Recipe Level	1 word	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
3	Recipe Type	1 word	ASCII	Type of recipe “M” : Machine Recipe “U” : Unit Recipe “1”, “2”, “3”, … : Special Recipe (The detail use is determined after discussing)
4	Recipe Version	3 word	-	Recipe version
5	Recipe Status	1 word	Integer	Recipe Status 1 : Running (using) 2 : Idle (Editable) 3 : Disable (locked)
6	Unit Number	1 word	Integer	Unit Number (The detail use is determined after discussing)

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※ Example 1 (Machine Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	1
Recipe Report Count	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

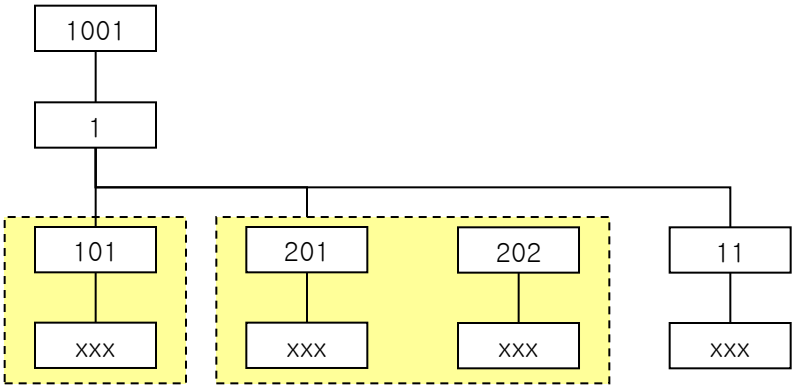
No	Name	Value
1	Recipe Number	1
2	Recipe Level	2 : Level 2 Recipe
3	Recipe Type	"M" : Machine Recipe
5	Recipe Status	2 : Idle (Editable)
6	Unit Number	0

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※ Example 2 (Special Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Unit Number	0

< Request ACK - EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	1
Recipe Report Count	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

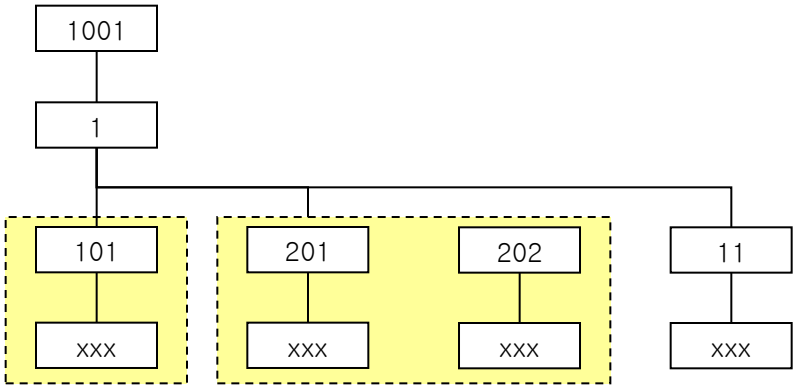
No	Name	Value 1
1	Recipe Number	11
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"1" : Special Recipe
5	Recipe Status	2 : Idle (Editable)
6	Unit Number	0

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※ Example 3 (Unit Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK - EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	2
Recipe Report Count	2
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

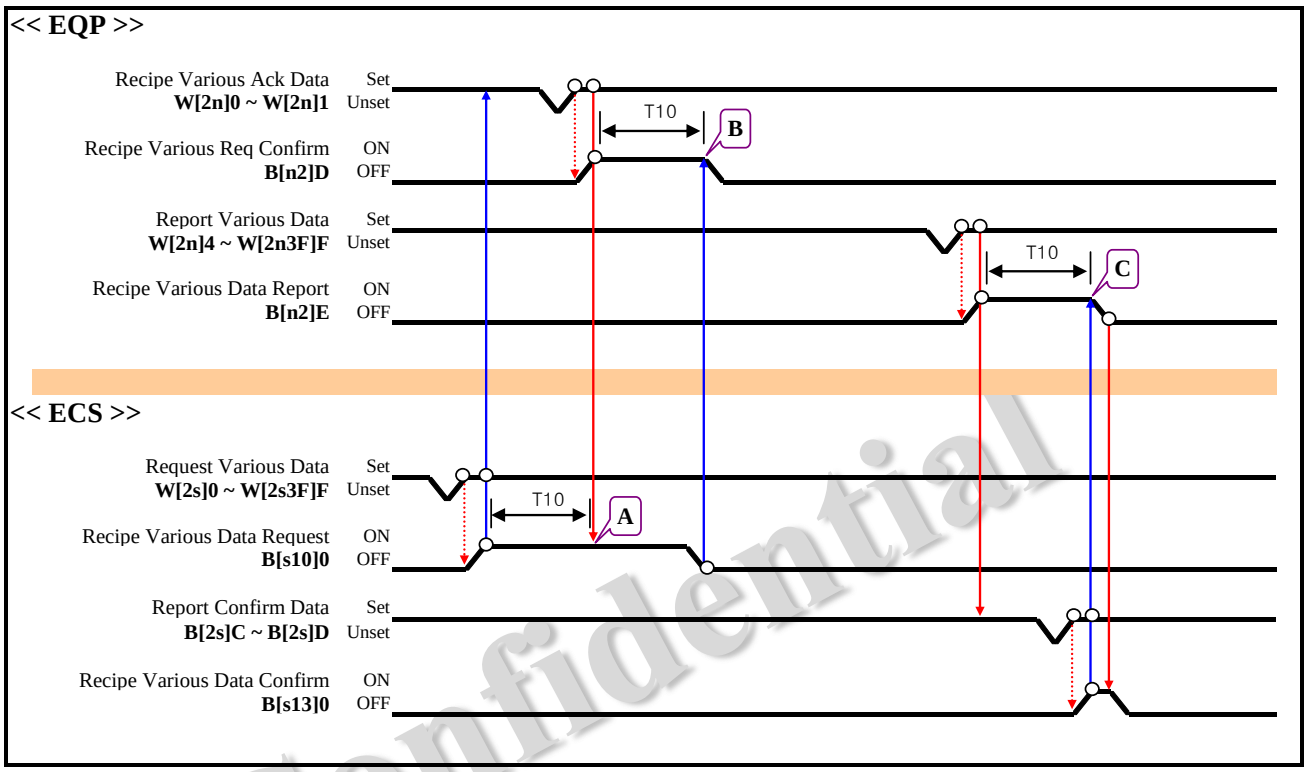
No	Name	Value 1	Value 2
1	Recipe Number	201	202
2	Recipe Level	3 : Level 3 Recipe	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe	"U" : Unit Recipe
5	Recipe Status	2 : Idle (Editable)	2 : Idle (Editable)
6	Unit Number	2	2

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6.27.4 Lower Recipe List Request

장비에 등록되어 있는 Recipe의 Lower Recipe List를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	2 : Lower Recipe List
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data (The detail use is determined after discussing)

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< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	2 : Lower Recipe List
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	2 : Lower Recipe List
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	2 : Lower Recipe List
Recipe Report CEID Order	W[2n]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n]6	Integer	Recipe Total Count
Recipe Report Count	W[2n]7	Integer	Recipe Data Structure Count
Recipe Number	W[2n]9	Integer	Requested Recipe Number
Recipe Report V1 ~	W[2s1]0 ~	-	- Equipment Specific Data - Recipe Report Structure ※ Reference Recipe Report Structure (The detail use is determined after discussing)

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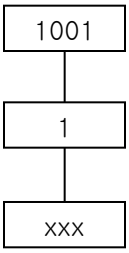
< Recipe Report Structure >

Index	Name	Length	Format	Description
1	Recipe Number	1 word	Integer	Recipe Number
2	Recipe Level	1 word	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
3	Recipe Type	1 word	ASCII	Type of recipe “M” : Machine Recipe “U” : Unit Recipe “1”, “2”, “3”, … : Special Recipe (The detail use is determined after discussing)
4	Recipe Version	3 word	-	Recipe version
6	Unit Number	1 word	Integer	Unit Number (The detail use is determined after discussing)

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※ Example 1 (2 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Lower Recipe List Request – ECS >

Name	Value
Request CEID	2 : Lower Recipe List
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

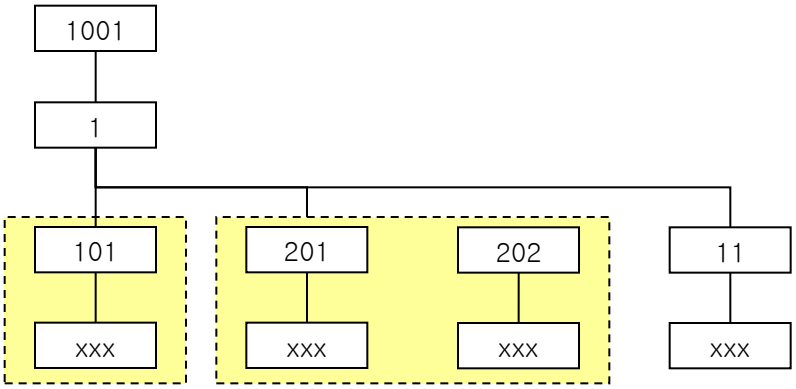
Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	5 : Recipe Number not found

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※ Example 2 (3 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Lower Recipe List Request – ECS >

Name	Value
Request CEID	2 : Lower Recipe List
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	2 : Lower Recipe List
Request Confirm ACK	0 : Accepted

< Lower Recipe List – EQP >

Name	Value
Recipe Report CEID	2 : Lower Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	3
Recipe Report Count	3
Recipe Number	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

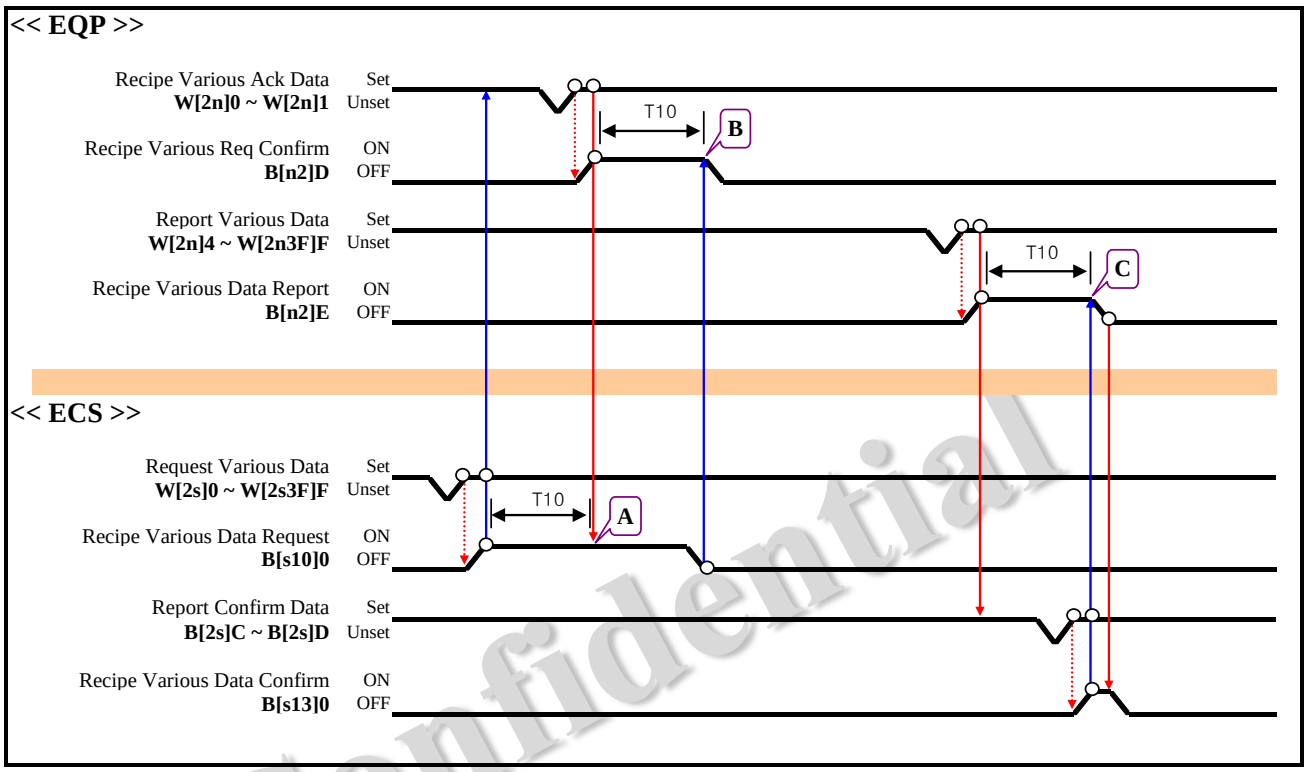
No	Name	Value 1	Value 2	Value 3
1	Recipe Number	101	201	11
2	Recipe Level	3 : Level 3 Recipe	3 : Level 3 Recipe	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe	"U" : Unit Recipe	"1" : Special Recipe
6	Unit Number	1	2	0

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6.27.5 Recipe Parameter Request

장비에 등록되어 있는 Recipe의 Parameter를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	3 : Recipe Parameter
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data (The detail use is determined after discussing)

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< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	3 : Recipe Parameter
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	3 : Recipe Parameter
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

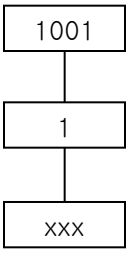
Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	3 : Recipe Parameter
Recipe Number	W[2n]9	Integer	Recipe Number
Recipe Level	W[2n]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n]C ~ W[2n]E	-	Recipe Version
Recipe Report V1 ~	W[2n1]0 ~	-	Recipe Parameter Data ※ Reference 4.2.3 APD/Recipe Body Data Format

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※ Example 1 (Machine Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

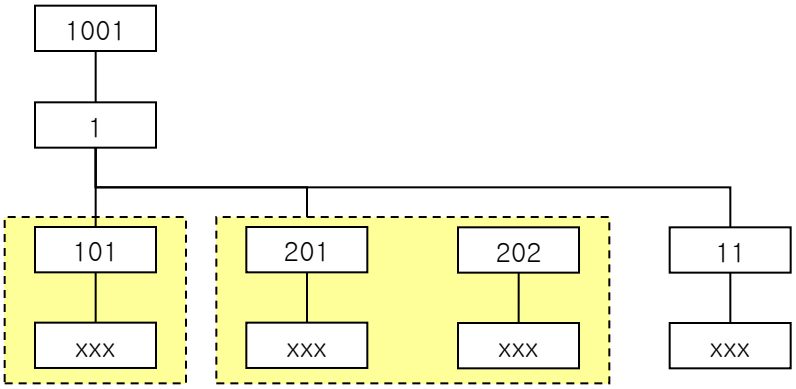
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	xxx

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※ Example 2 (Special Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	11
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

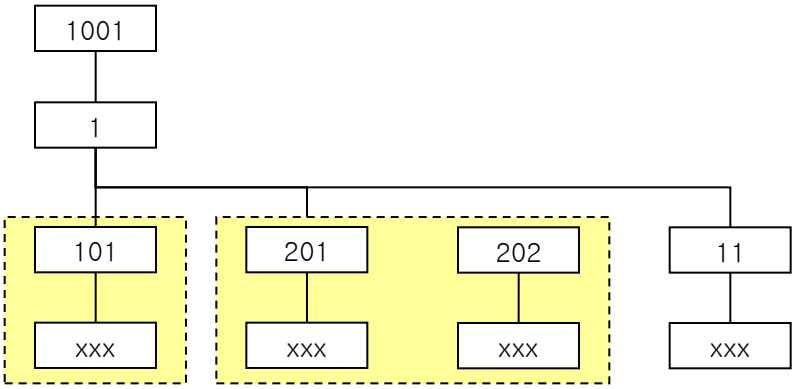
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	11
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Recipe Report V1 ~	xxx

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※ Example 3 (Unit Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

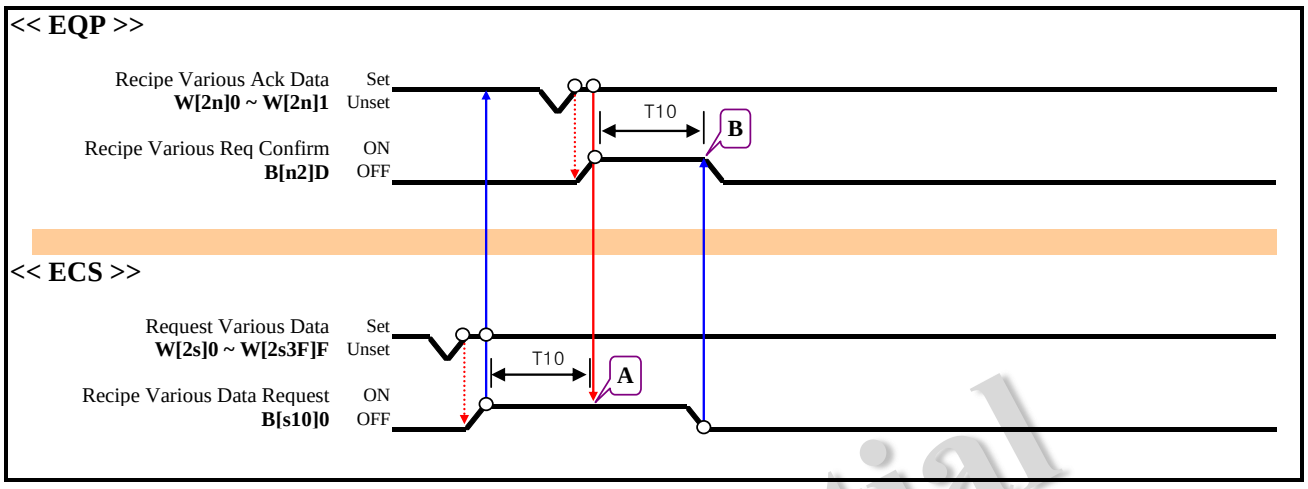
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Recipe Report V1	2 : Report Unit Number
Recipe Report V2 ~	xxx

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6.27.6 Recipe Command Send

장비에 Recipe의 Command를 전달하는 것으로 ECS는 “Recipe Various Data Request”로 요청 하며 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete 4 : Rename
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data - Lower Recipe Number List - Recipe Parameter Data (The detail use is determined after discussing)

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< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	4 : Recipe Command
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Built Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data -Recipe Parameter Data (The detail use is determined after discussing)

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< Change Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 2 : Change
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data -Recipe Parameter Data (The detail use is determined after discussing)

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< Delete Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 3 : Delete
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)

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< Rename Case – ECS >

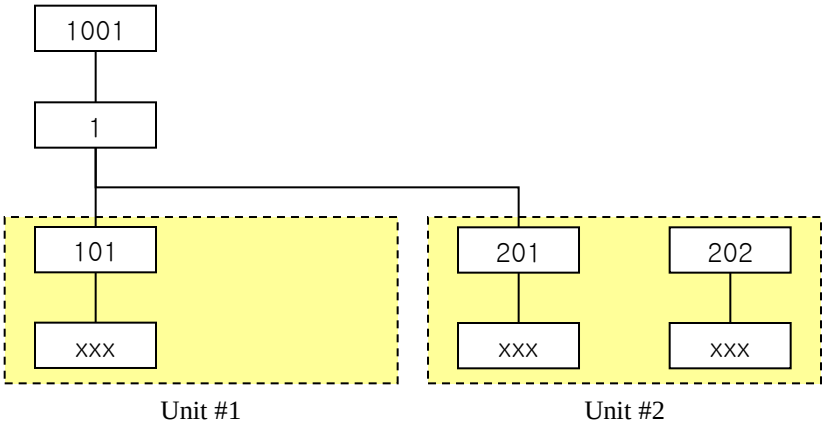
Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 4 : Rename
Recipe Number	W[2s]2	Integer	New Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1	W[2s1]0 ~	-	- From Recipe Number

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※ Example 1 (Equipment Sub-Recipe Built)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

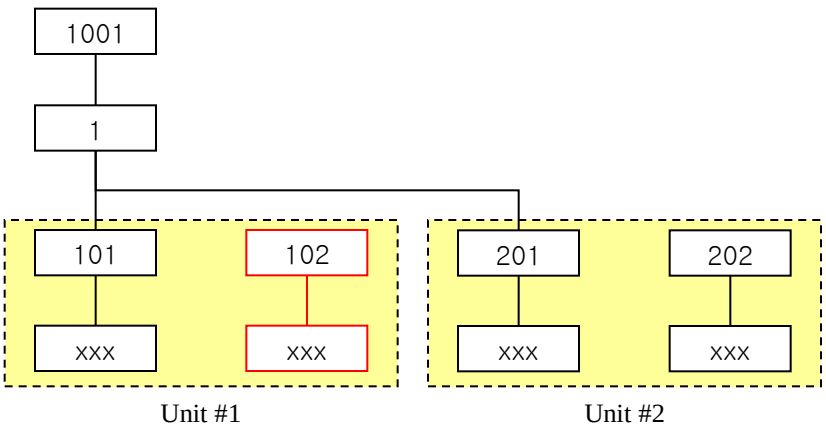
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	102
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U": Unit Recipe
Unit Number	1
Recipe Parameter V1 ~	xxx

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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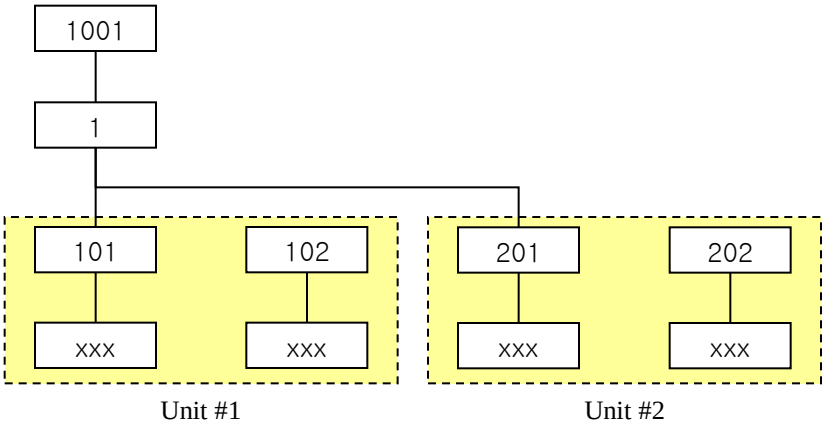
< Recipe Structure (After) >



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※ Example 2 (Equipment Recipe Built)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

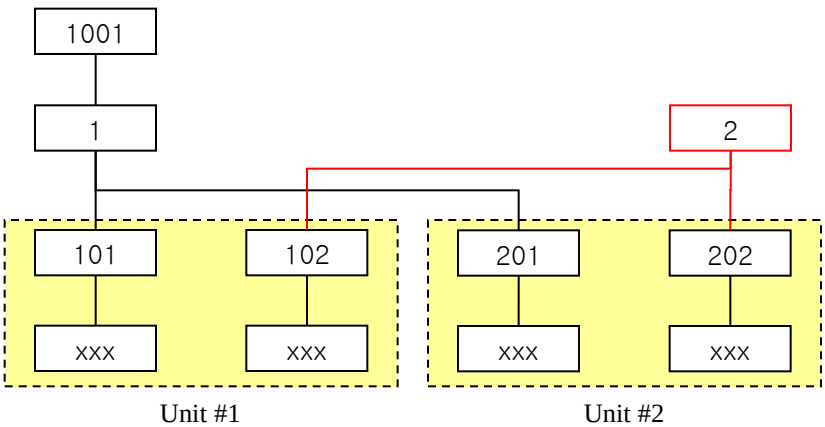
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	2
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1	102 : Unit #1 Sub-Recipe
Recipe Parameter V2	202 : Unit #2 Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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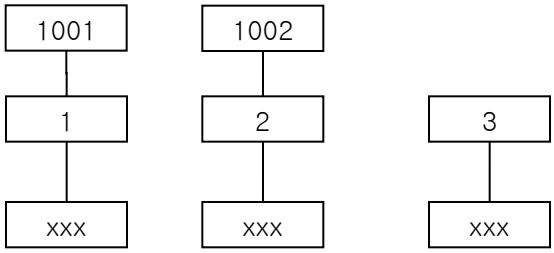
< Recipe Structure (After) >



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※ Example 3 (Equipment Recipe Built)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

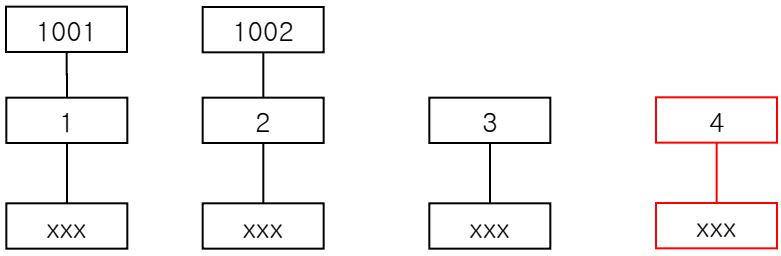
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	4
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	xxx

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

< Recipe Structure (After) >



Level 1 : Host Recipe

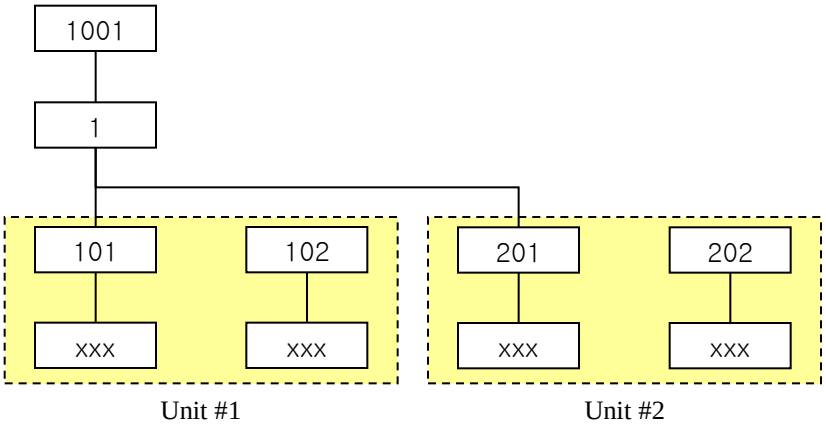
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 4 (Equipment Sub-Recipe Change)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

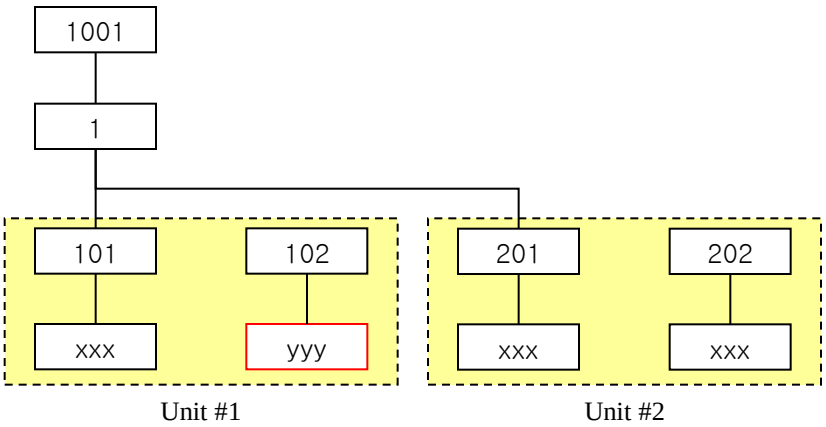
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	102
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" :Unit Recipe
Unit Number	1
Recipe Parameter V1 ~	yyy

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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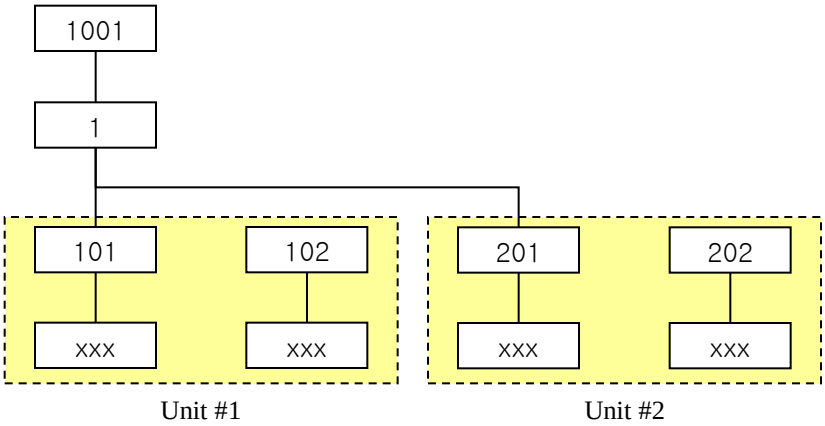
< Recipe Structure (After) >



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※ Example 5 (Equipment Recipe Change)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

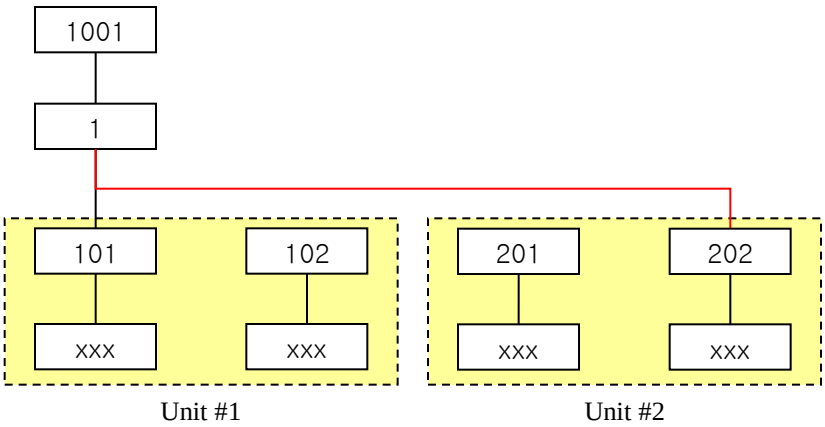
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1	101 : Unit #1 Sub-Recipe
Recipe Parameter V2	202 : Unit #2 Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

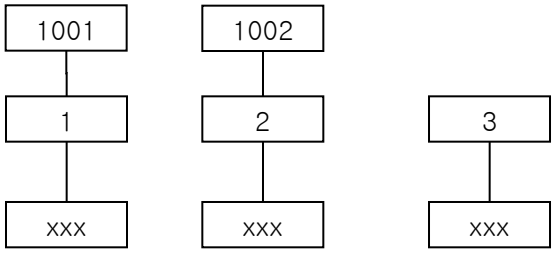
< Recipe Structure (After) >



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※ Example 6 (Equipment Recipe Change)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

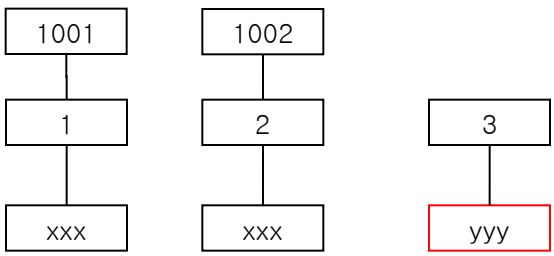
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	3
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	yyy

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

< Recipe Structure (After) >



Level 1 : Host Recipe

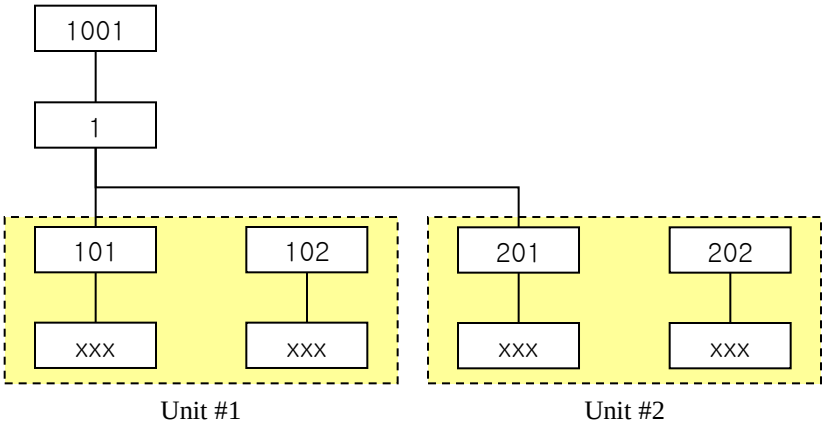
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 7 (Equipment Sub-Recipe Delete)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

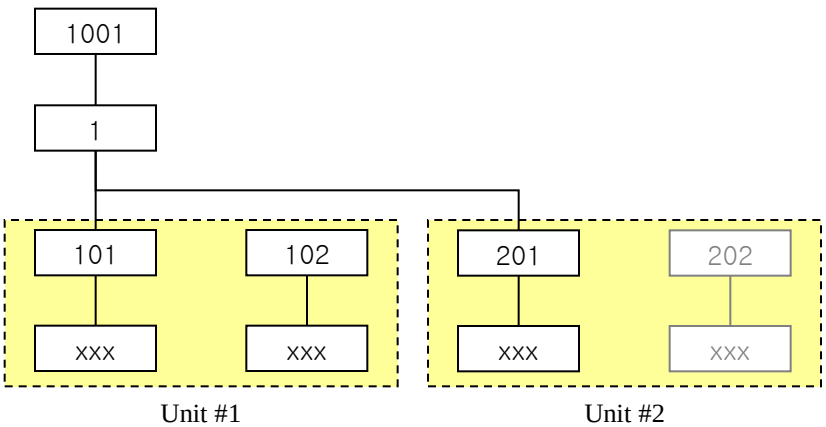
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	3 : Delete
Recipe Number	202
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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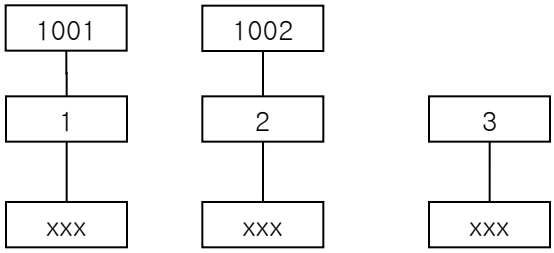
< Recipe Structure (After) >



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※ Example 8 (Equipment Recipe Delete)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

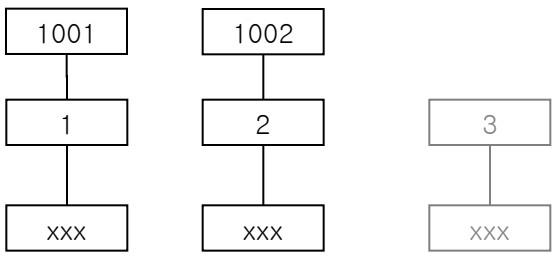
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	3 : Delete
Recipe Number	3
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

< Recipe Structure (After) >



Level 1 : Host Recipe

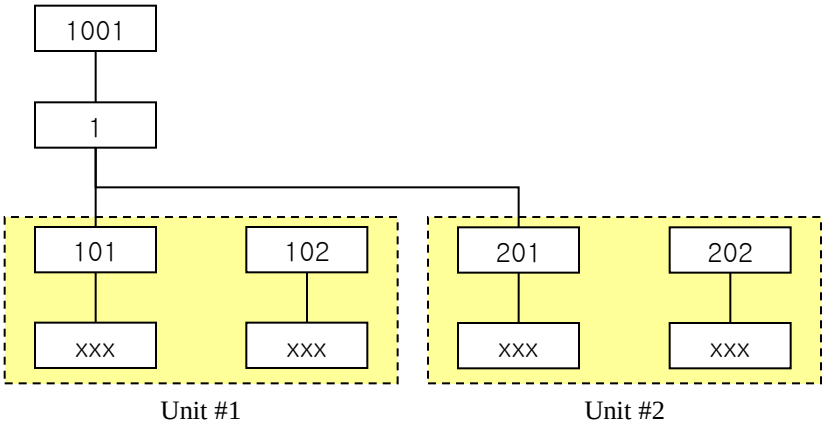
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 9 (Equipment Sub-Recipe Rename)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

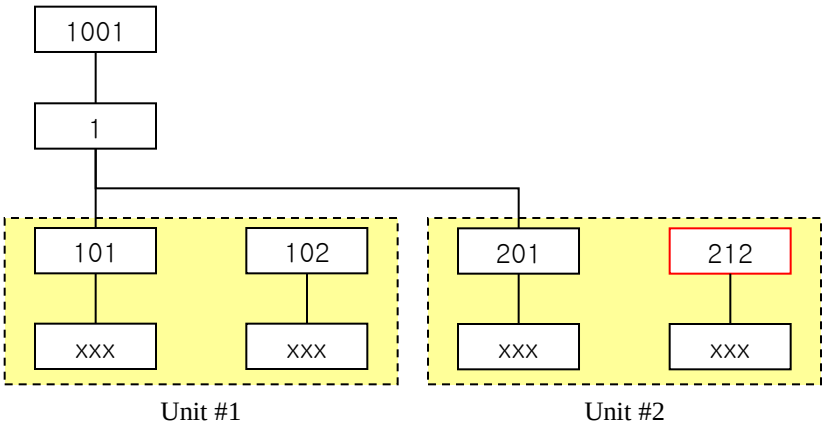
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	4 : Rename
Recipe Number	212
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2
Recipe Parameter V1	202 : From Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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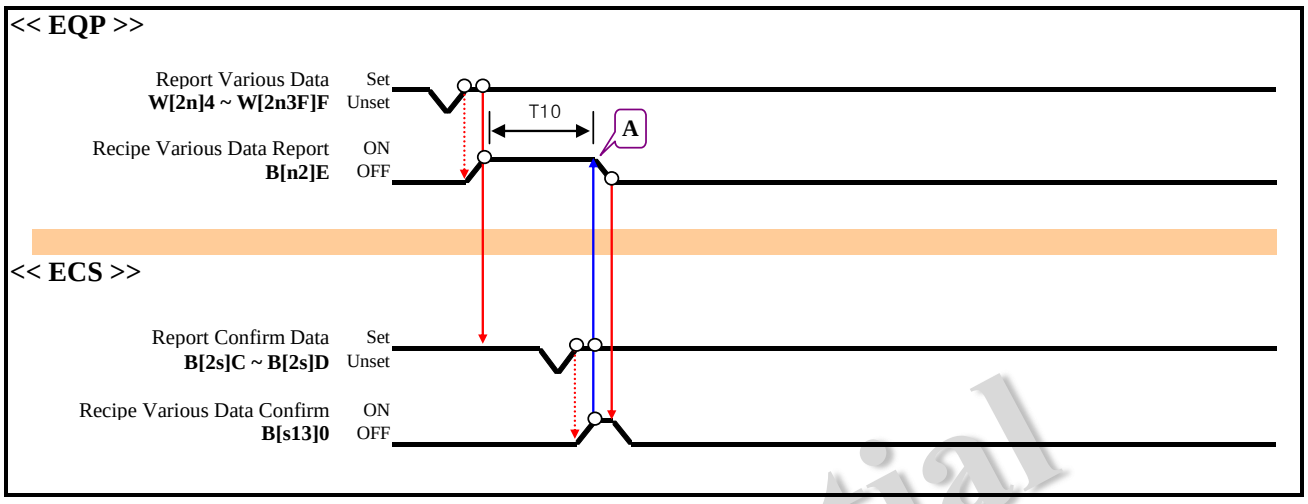
< Recipe Structure (After) >



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6.27.7 Recipe Change Report

작업자에 의해 장비에서 Recipe가 변경되었을 경우 보고하는 것으로 장비는 “Recipe Various Data Report”로 보고 하며 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	5 : Recipe Change Report
Recipe Command	W[2n]8	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete
Recipe Number	W[2n]9	Integer	Recipe Number
Recipe Level	W[2n]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n]C ~ W[2n]E	-	Recipe Version
Recipe Report V1 ~	W[2n1]0 ~	-	Equipment Specific Data Recipe Parameter Data ※ Reference 4.2.3 APD/Recipe Body Data Format

< Report Confirm Data – ECS >

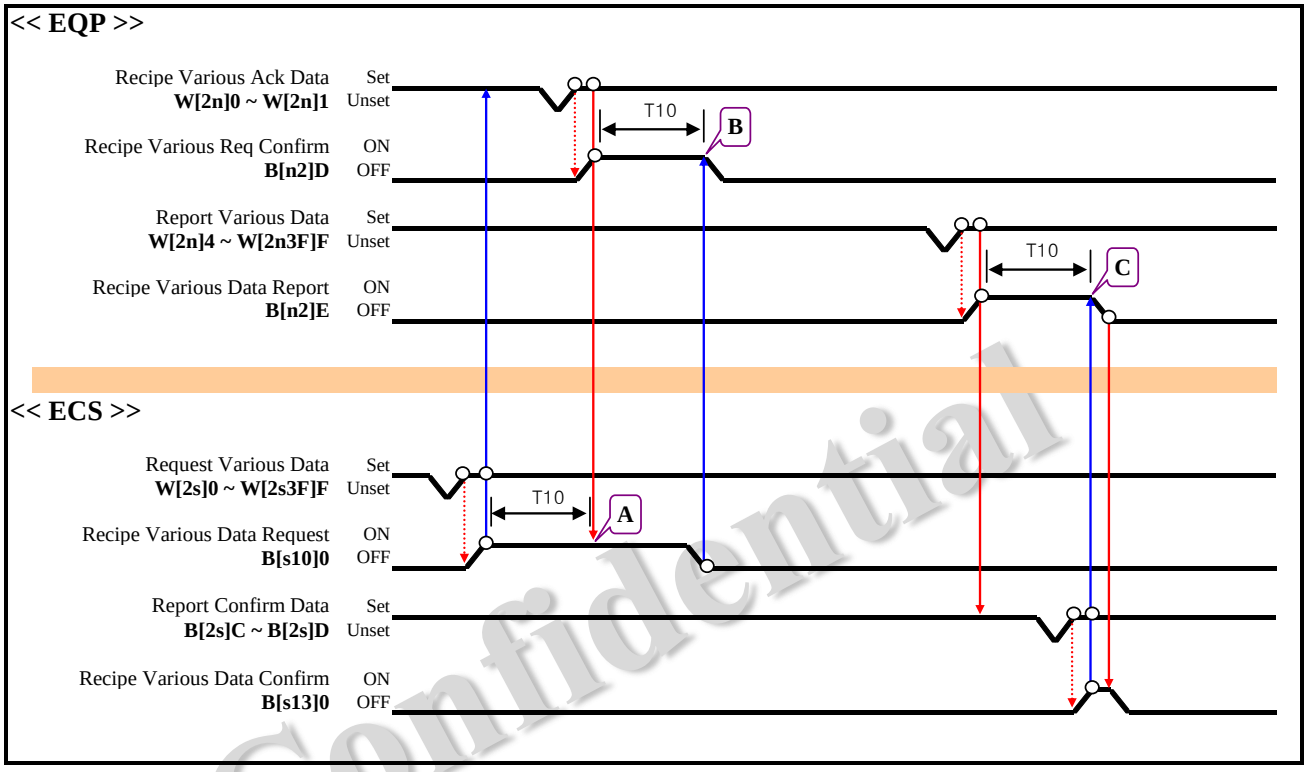
Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	5 : Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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6.27.8 Equipment Constant Recipe Parameter Request

장비에 등록되어 있는 Equipment Constant Recipe의 Parameter를 요청하는 것으로 ECS는 "Recipe Various Data Request"로 요청하며 장비는 "Recipe Various Data Report"로 보고한다.
(사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing.)
Recipe Number	W[2s]2	Integer	Not Used
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Not Used
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data (The detail use is determined after discussing)

< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

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< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

< Report Various Data - EQP >

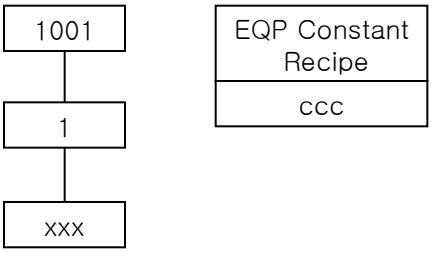
Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Recipe Number	W[2n]9	Integer	Not Used
Recipe Level	W[2n]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n]C ~ W[2n]E	-	Not Used
Recipe Report V1 ~	W[2n1]0 ~	-	Equipment Specific Data Recipe Parameter Data ※ Reference 4.2.3 APD/Recipe Body Data Format

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※ Example 1 (As Host Online)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

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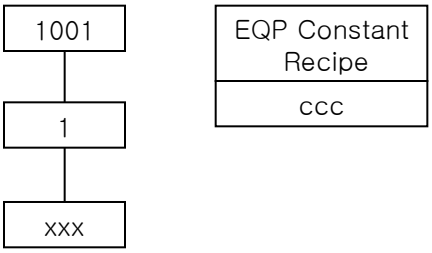
< Recipe Parameter – EQP >

Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 2 (As Host Request)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

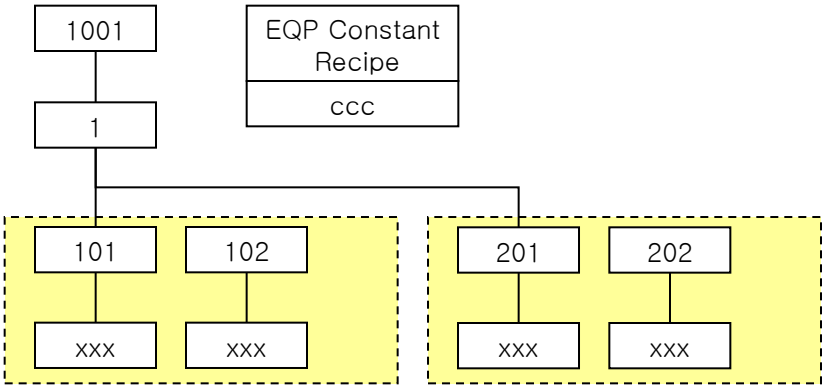
Name	Value
Recipe Report CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 3 (Machine Recipe)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

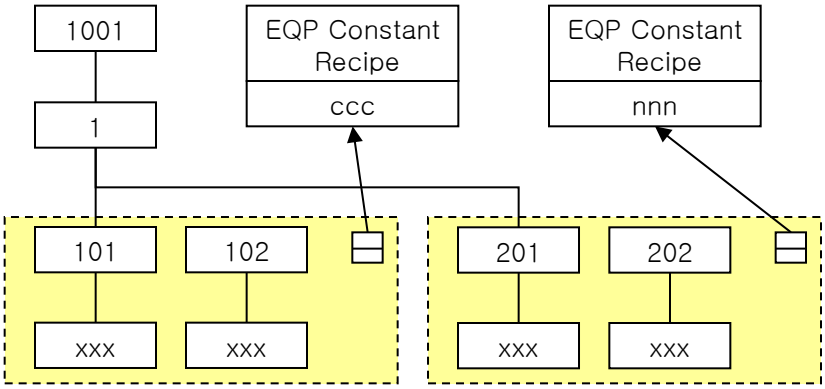
Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 4 (Unit Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

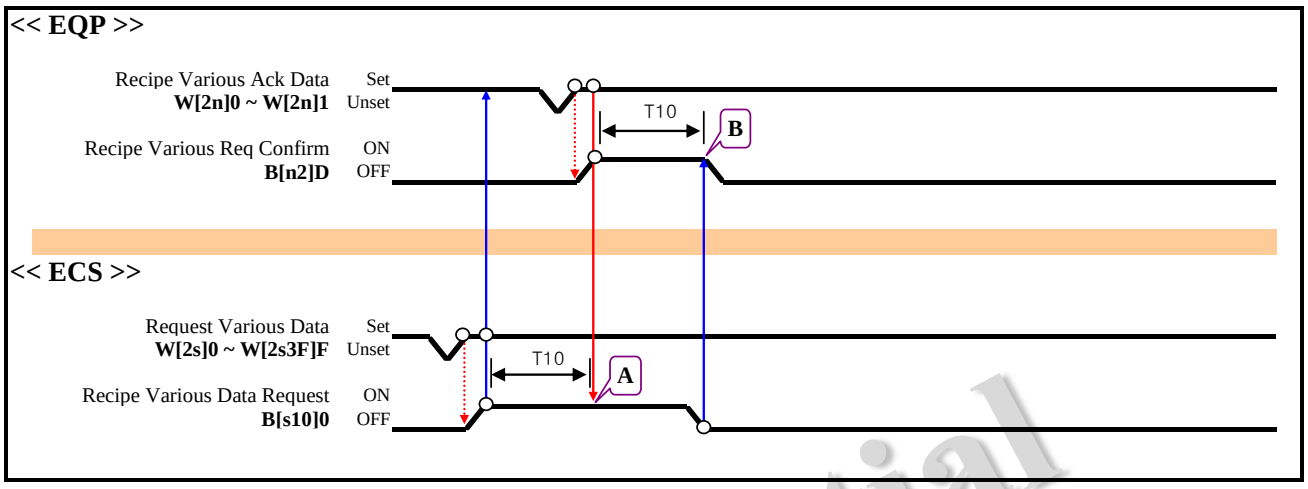
Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Recipe Report V1	2 : Report Unit Number
Recipe Report V2 ~	nnn

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6.27.9 Equipment Constant Recipe Command Send

장비에 Equipment Constant Recipe의 Command를 전달하는 것으로 ECS는 “Recipe Various Data Request”로 요청 하며 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	8 : EQP Constant Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 2 : Change
Recipe Number	W[2s]2	Integer	Not Used
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Not Used
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s1]0 ~	-	- Equipment Specific Data - Recipe Parameter Data ※ The detail use is determined after discussing.

< Recipe Various ACK Data - EQP >

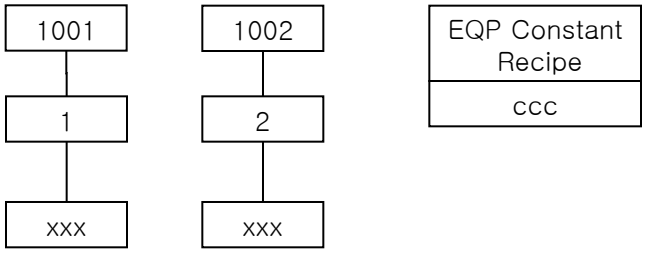
Name	Address	Format	Description
Request Confirm CEID	W[2n]0	Integer	8 : EQP Constant Recipe Command
Request Confirm ACK	W[2n]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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※ Example 1 (Machine Recipe)

< Recipe Structure (Before) >



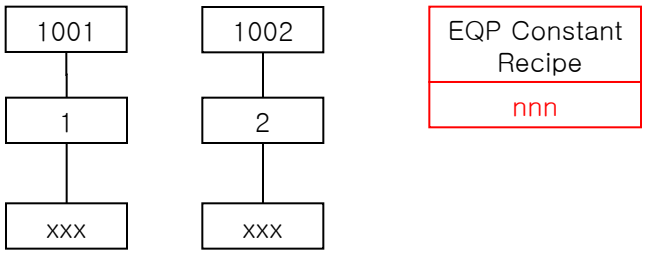
< Recipe Command Send – ECS >

Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

< Recipe Structure (After) >

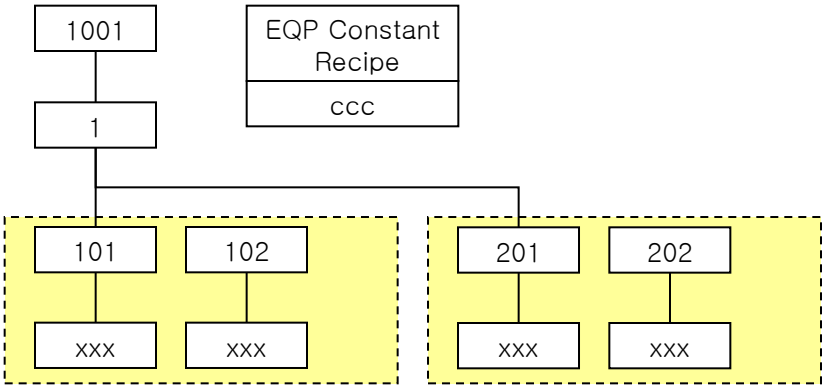


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※ Example 2 (Machine Recipe)

< Recipe Structure (Before) >



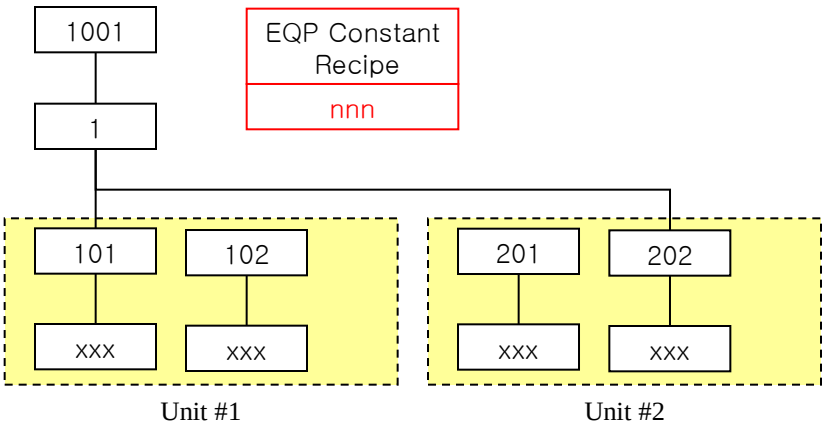
< Recipe Command Send – ECS >

Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

< Recipe Structure (After) >

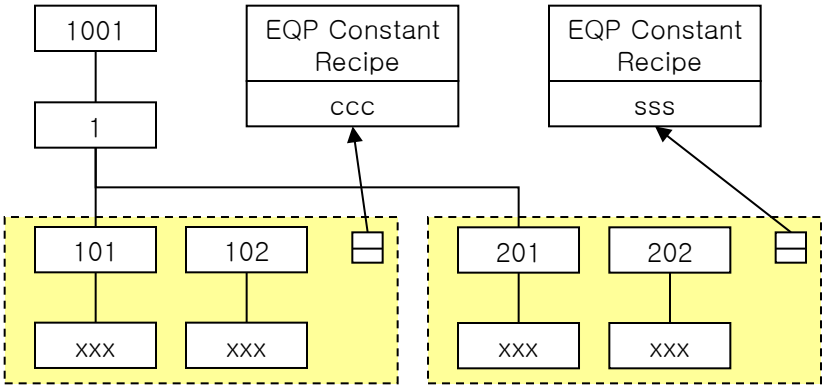


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※ Example 3 (Unit Recipe)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Command Send – ECS >

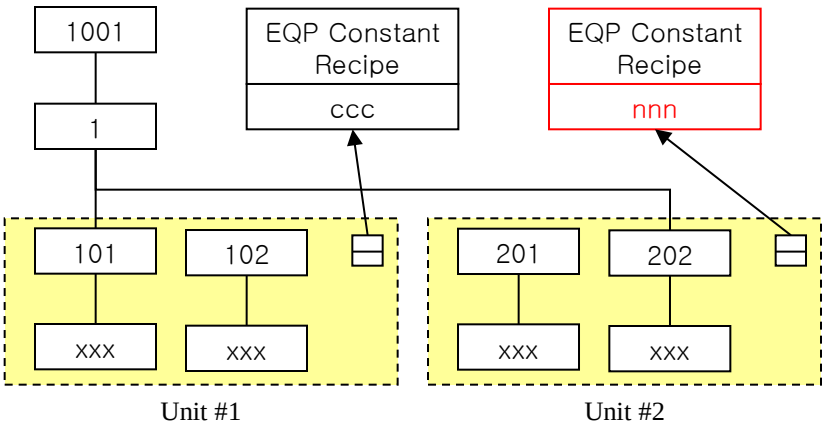
Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

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< Recipe Structure (After) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

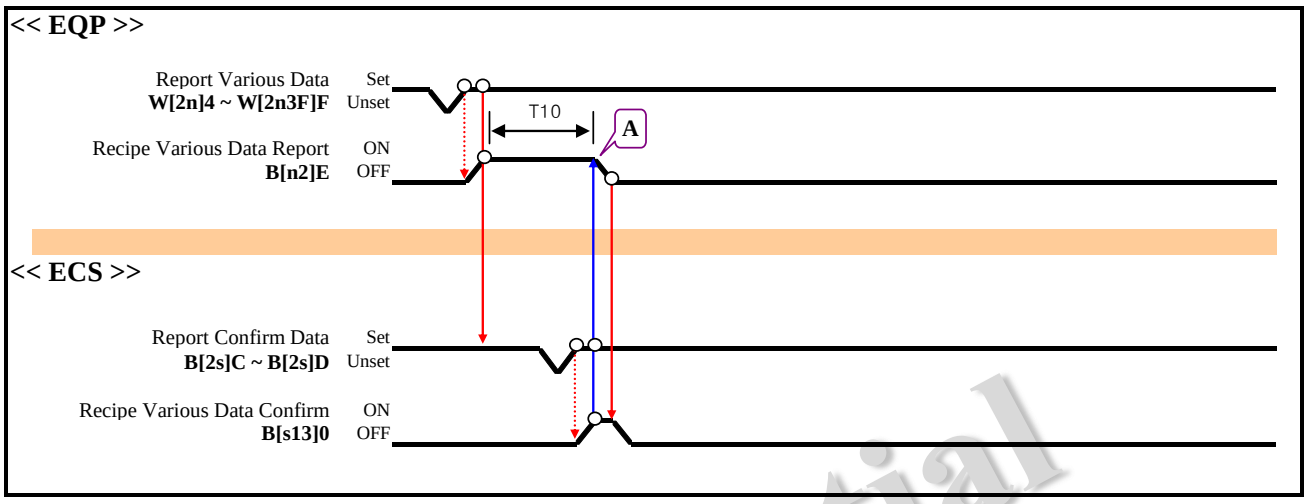
Level 3 : Equipment Sub-Recipe

Recipe Parameter

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6.27.10 Equipment Constant Recipe Change Report

작업자에 의해 장비에서 Equipment Constant Recipe가 변경되었을 경우 보고하는 것으로 장비는 "Recipe Various Data Report"로 보고 하며 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n]4	Integer	9 : EQP Constant Recipe Change Report
Recipe Command	W[2n]8	Integer	Recipe Command 2 : Change
Recipe Number	W[2n]9	Integer	Not Used
Recipe Level	W[2n]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n]C ~ W[2n]E	-	Not Used
Recipe Report V1 ~	W[2n1]0 ~	-	- Equipment Specific Data - Recipe Parameter Data ※ Reference 4.2.3 APD/Recipe Body Data Format

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< Report Confirm Data – ECS >

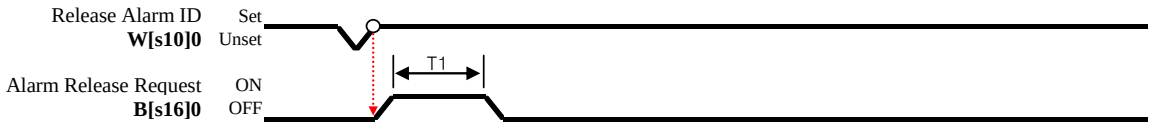
Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	9 : EQP Constant Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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6.28 Alarm Release Request

장비에서 발생한 Alarm 중에 특정 Alarm의 해제를 요청하는 것으로 Alarm이 해제 될 경우 "Light/Heavy Alarm Report"를 통해 보고한다. (사용여부는 협의하여 결정한다.)

<< ECS >>



< Release Alarm ID - ECS>

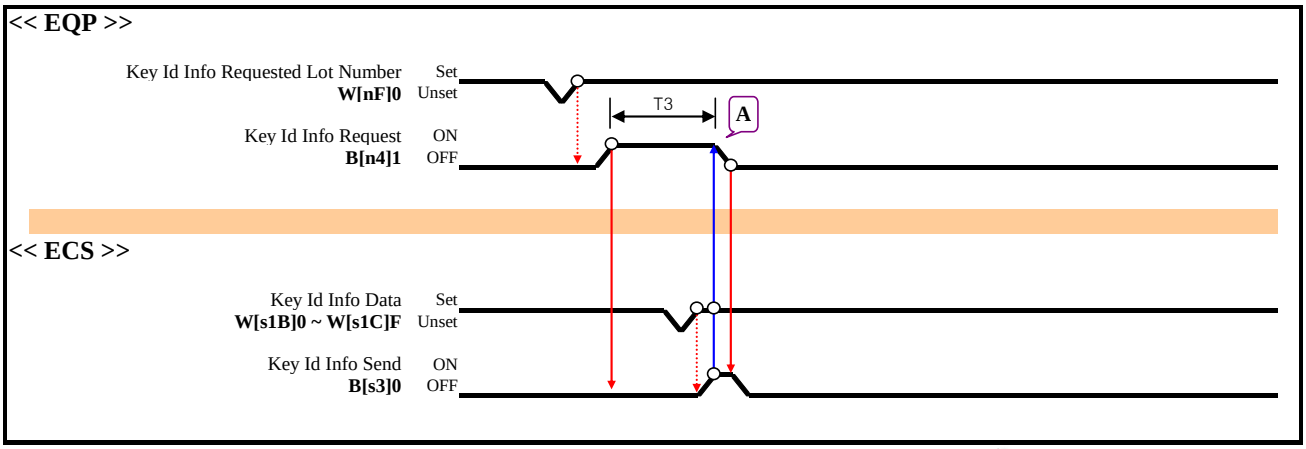
Name	Address	Format	Description
Release Alarm ID	W[s10]0	Integer	Heavy or Light Alarm ID ※ Reference 6.2 Light/Heavy Alarm Report

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6.29 Key ID Information Request

상류 장비로부터 전달 받은 Glass의 LOT Number가 이전 Glass의 LOT Number와 다를 경우 ECS로 LOT Number를 전달하여 Key ID 정보를 요구한다. 이 시나리오는 추후 협의함



A : 장비는 일정시간(Timeout - 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Key Id Info Request 신호를 Off 시킨다.

※ Timeout 는 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 장비는 이전 Glass의 LOT Number가 변경 되었을 경우 요청한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< Key ID Information – EQP >

Name	Address	Format	Description
Key ID Information Requested LOT Number	W[nF]0	Integer	LOT Number

< Key ID Information Data – ECS >

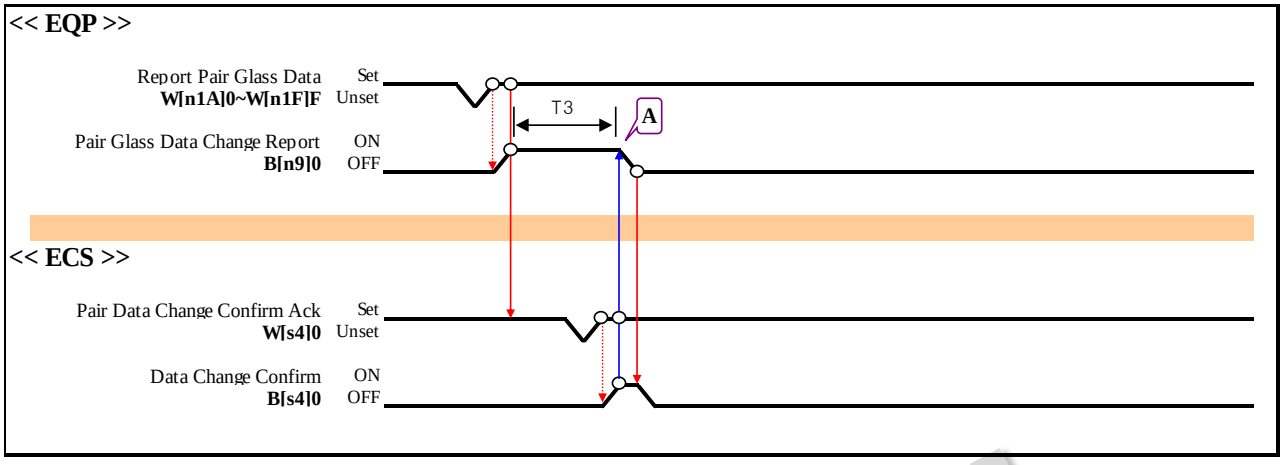
Name	Address	Format	Description
LOT ID	W[s1B]0~ W[s1B]7	ASCII	LOT ID
LOT START TIME	W[s1B]8~ W[s1B]E	ASCII	LOT Start Time Ex) 2008년 10월 02일 11시 27분 30초 “2008” / ”10” / ”02” “11” : ”27” : ”30”
Key ID Info Request Ack	W[s1B]F	Integer	0 : OK 1 : NG (Unknown LOT Number)
CST ID	W[s1C]0 ~ W[s1C]6	ASCII	CST ID
Key ID Information Option Data	W[s1C]7 ~ W[s1C]F	-	Optional Data ※ Optional Data는 협의에 의해 결정함.

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6.30 Pair Glass Data Change Report

장비에 Unpair Glass 2매 투입 후 Glass 배출 전 장비에서 ECS도 Pair Glass Data Change Report를 전송한다.
(해당 장비에 한함)



A : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 Off되지 않은 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별도 Handshake 또는 Data address가 다를 수 있다.

< Report Pair Glass Data - EQP>

Name	Address	Format	Description
A Cut Glass Data	W[n1A]0 ~ W[n1C]F	-	A Cut Glass Data - Reference 4.1 Glass Data Structure
B Cut Glass Data	W[n1D]0 ~ W[n1F]F	-	B Cut Glass Data - Reference 4.1 Glass Data Structure
A Cut GLSCODE	W[n1B]2	Integer	Glass Code
A Cut Glass Judge	W[n1B]B	ASCII	Glass Judge
A Cut Pair/Unpair	W[n1B]F.7	BIT	7 Bit On - Modify
A Cut Type	W[n1B]F.8 W[n1B]F.9	Integer	0 : A Cut, 1 : B Cut 2 : A Cut Dummy, 3 : B Cut Dummy
B Cut GLSCODE	W[n1E]2	Integer	Glass Code
B Cut Glass Judge	W[n1E]B	ASCII	Glass Judge
B Cut Pair/Unpair	W[n1B]F.7	BIT	7 Bit On - Modify
B Cut Type	W[n1E]F.8 W[n1E]F.9	Integer	0 : A Cut, 1 : B Cut 2 : A Cut Dummy, 3 : B Cut Dummy

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6.30 Pair Glass Data Change Report

< Pair Data Change Confirm Ack – ECS >

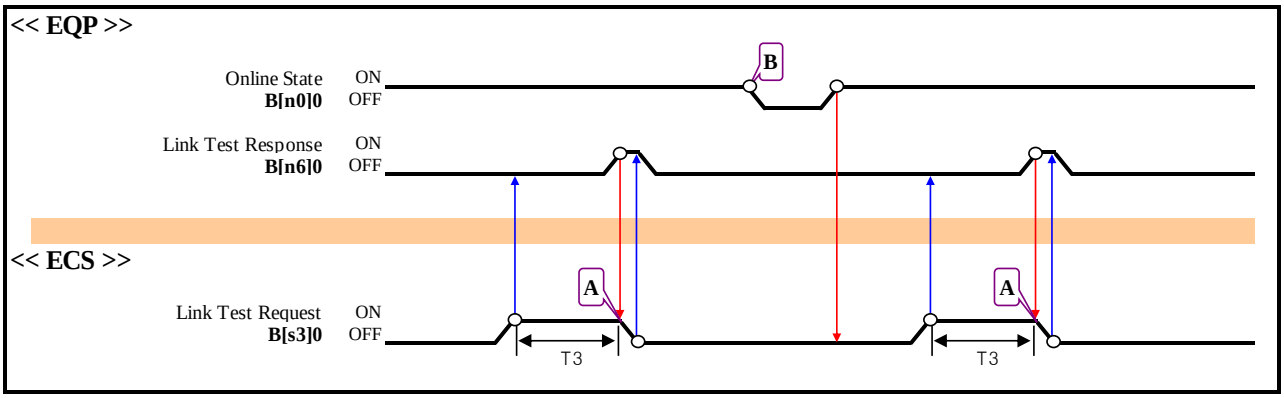
Name	Address	Format	Description
Pair Data Change Confirm Ack	W[s4]0	Integer	0 : Accept 1 : Not Accept (Unknown GLSCODE or Cut Type Miss Match)

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6.31 Link Test

ECS는 기동 중에 EQP도 주기적으도 Link Test를 실시하여 동신 상태를 체크 함.
(해당 장비에 한함)



A : ECS는 일정시간(Timeout – 3초)안에 EQP의 Trigger가 On되지 않은 경우 Heavy Alarm를 발생 및 EQP Status를 Down으도 변경 시키며 Link test Request 신호를 Off 시킨다.

B : EQP 조치 이후 Offline 변경 후 Online 변경 한다.

※ Link Test 주기는 협의 하에 결정한다.

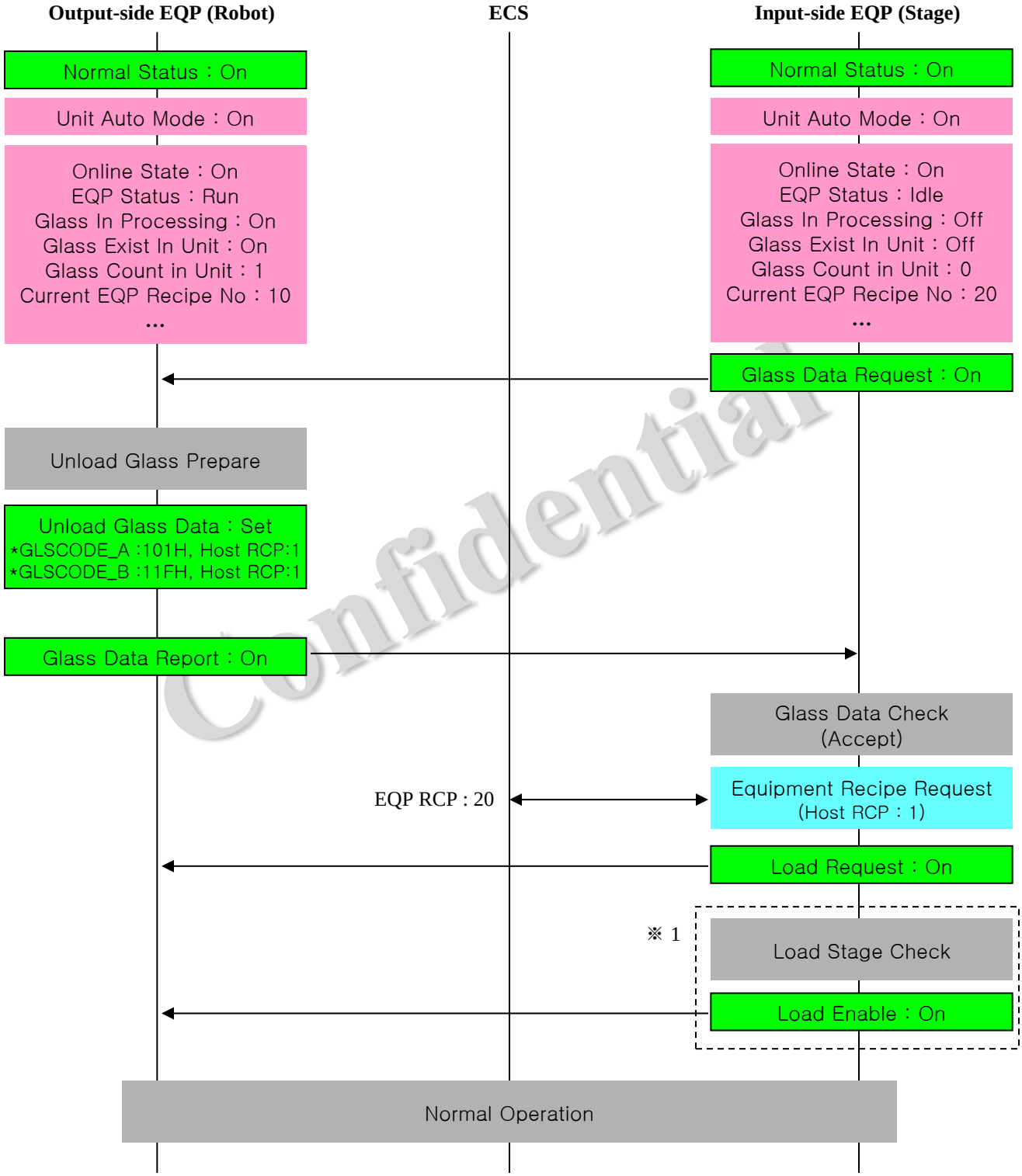
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7. Glass Move Sequence

7.1 Glass Move State (Load Request)

7.1.1 Load Glass Data Accepted (Process Inline, Robot → Stage)



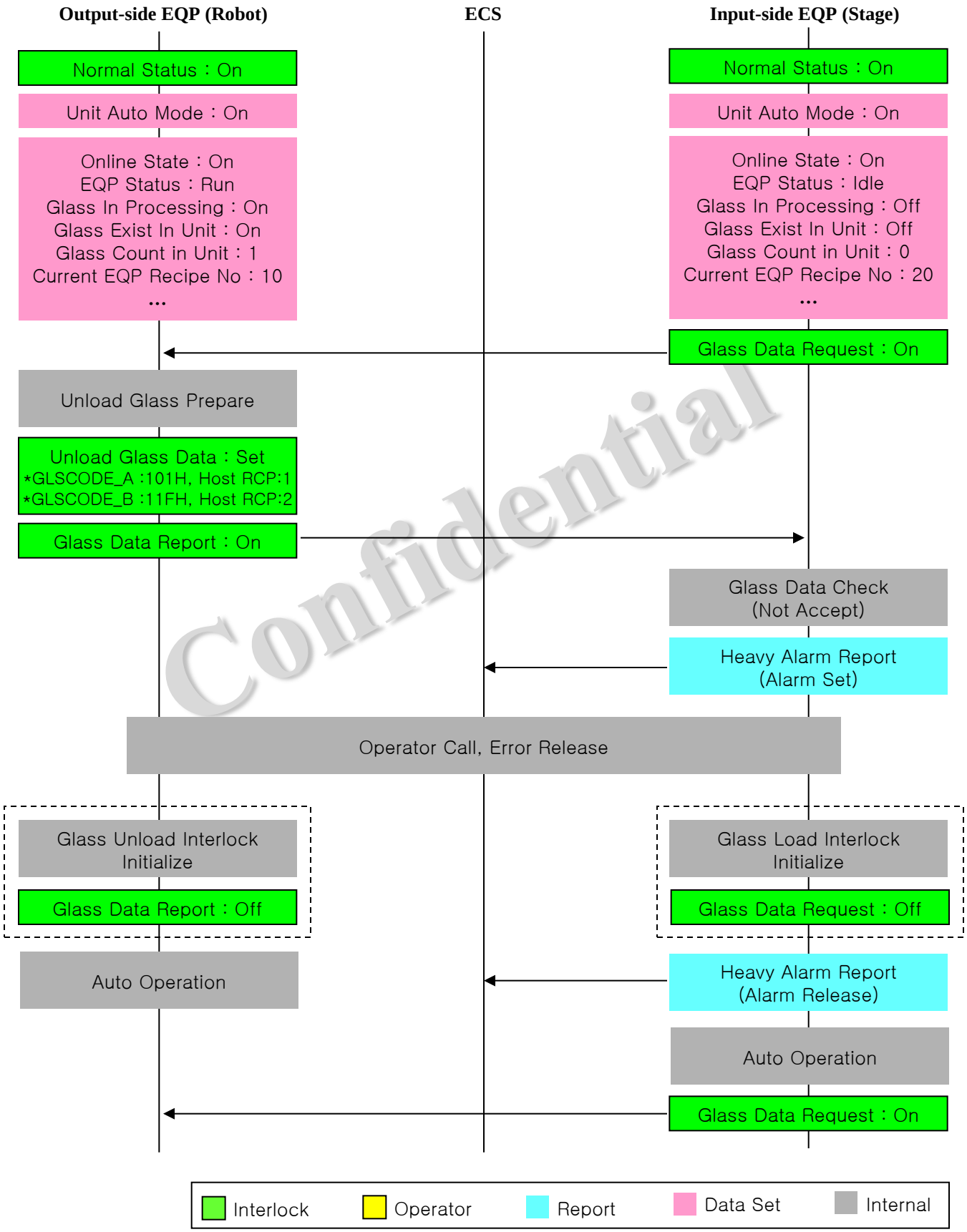
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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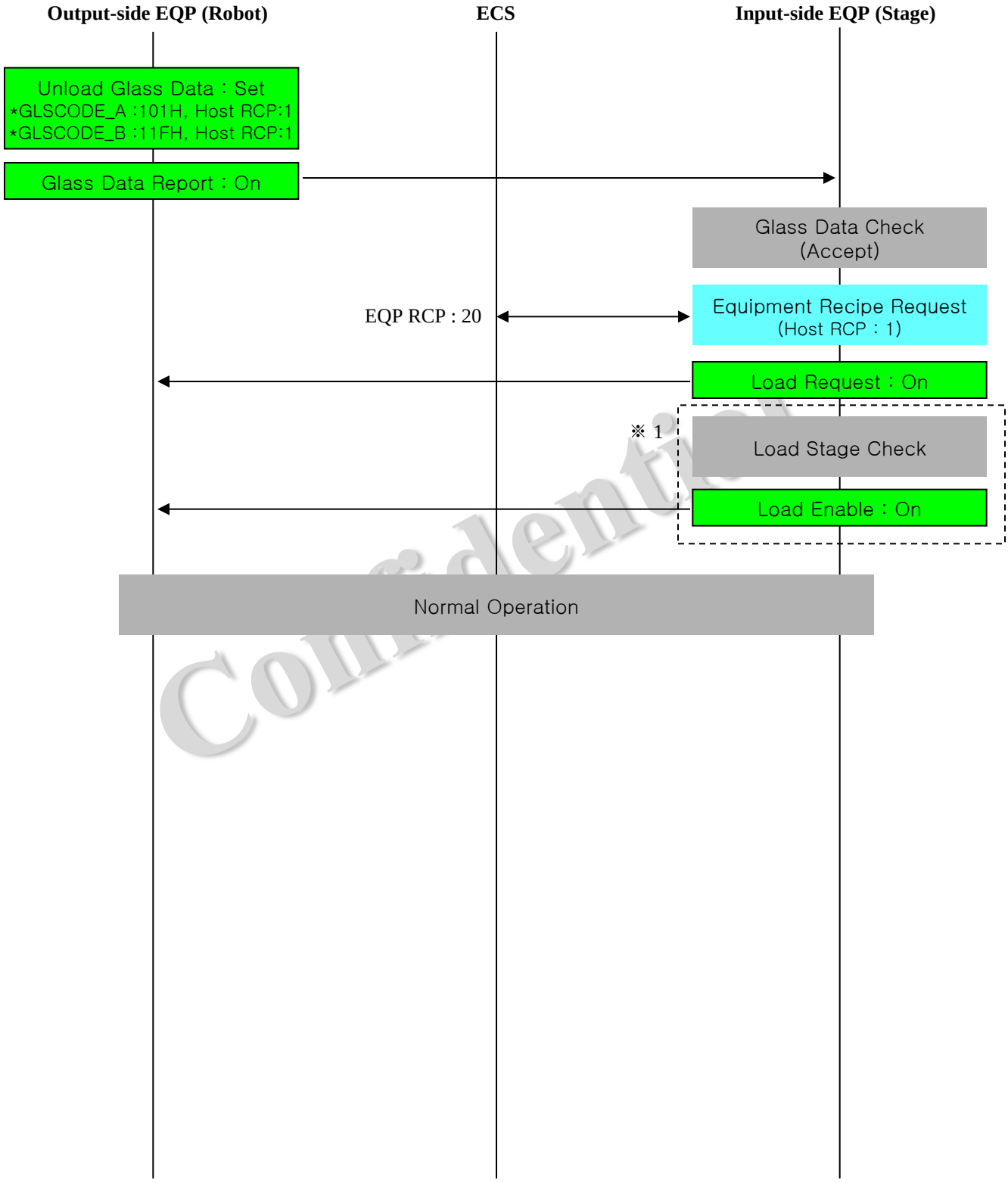
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7.1.2 Load Glass Data Not Accepted (Process Inline, Robot → Stage)



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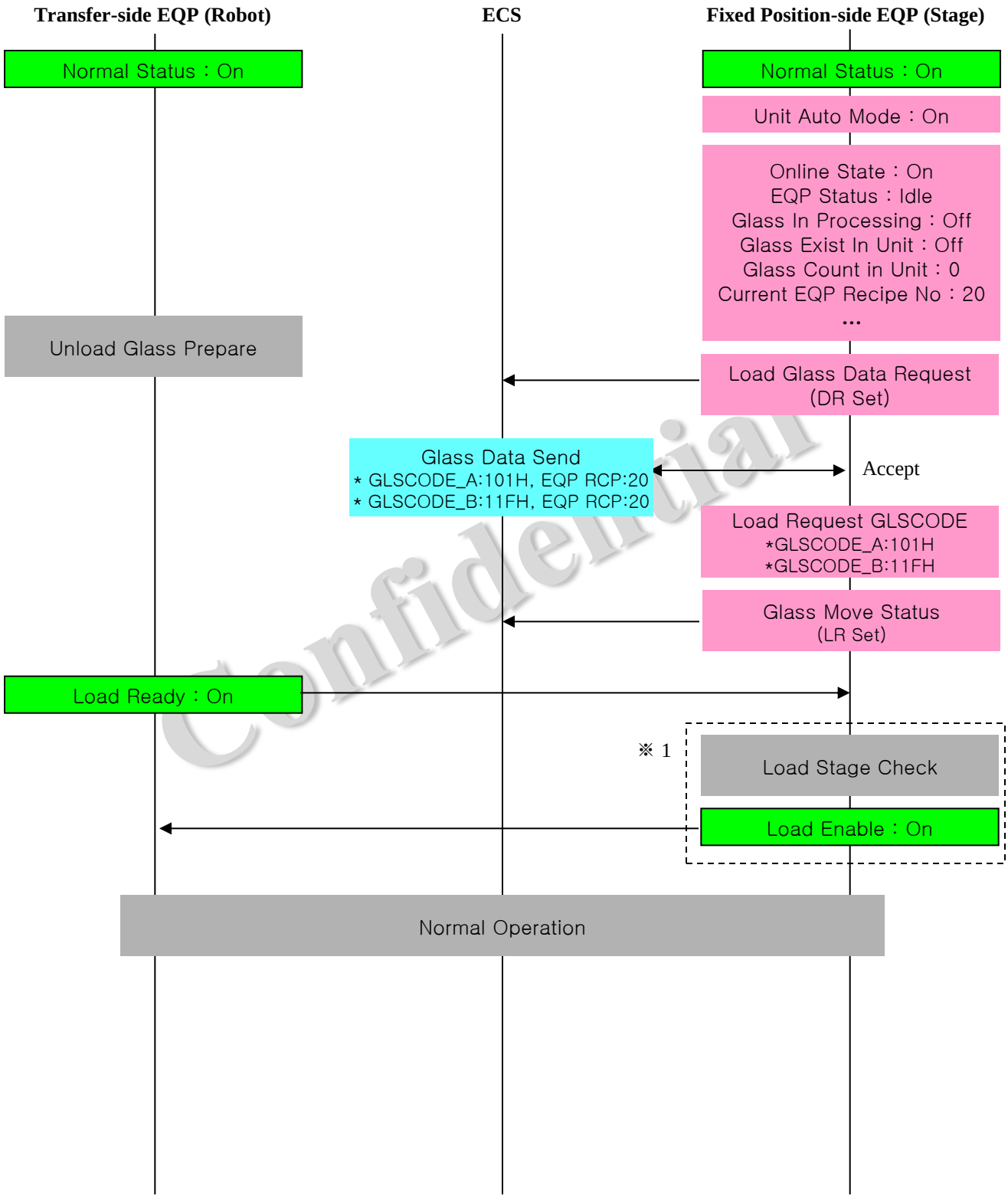
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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7.1.3 Load Glass Data Accepted (Stand Alone, Robot → Stage)



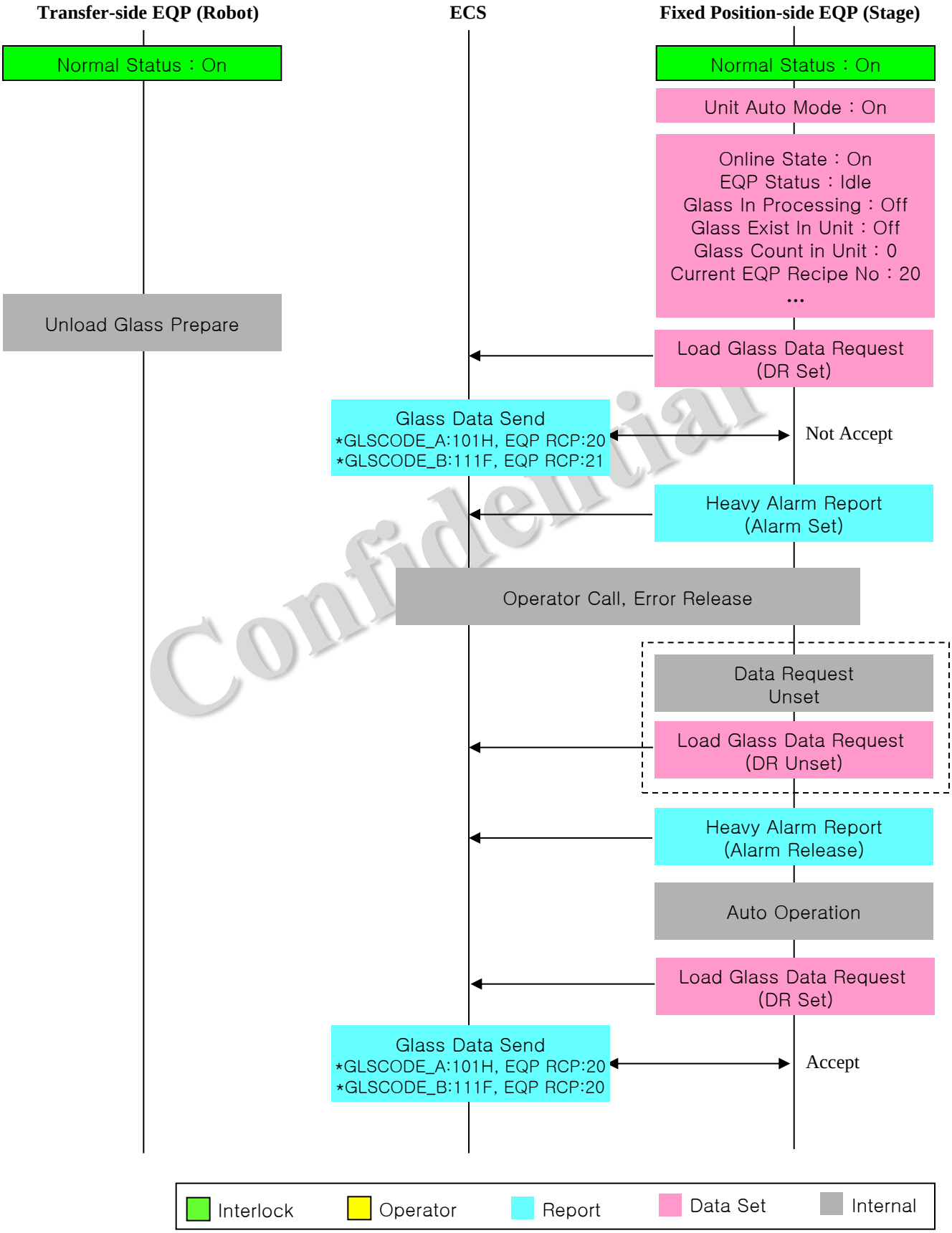
※ 1: To set before according to a physical Interlock condition.



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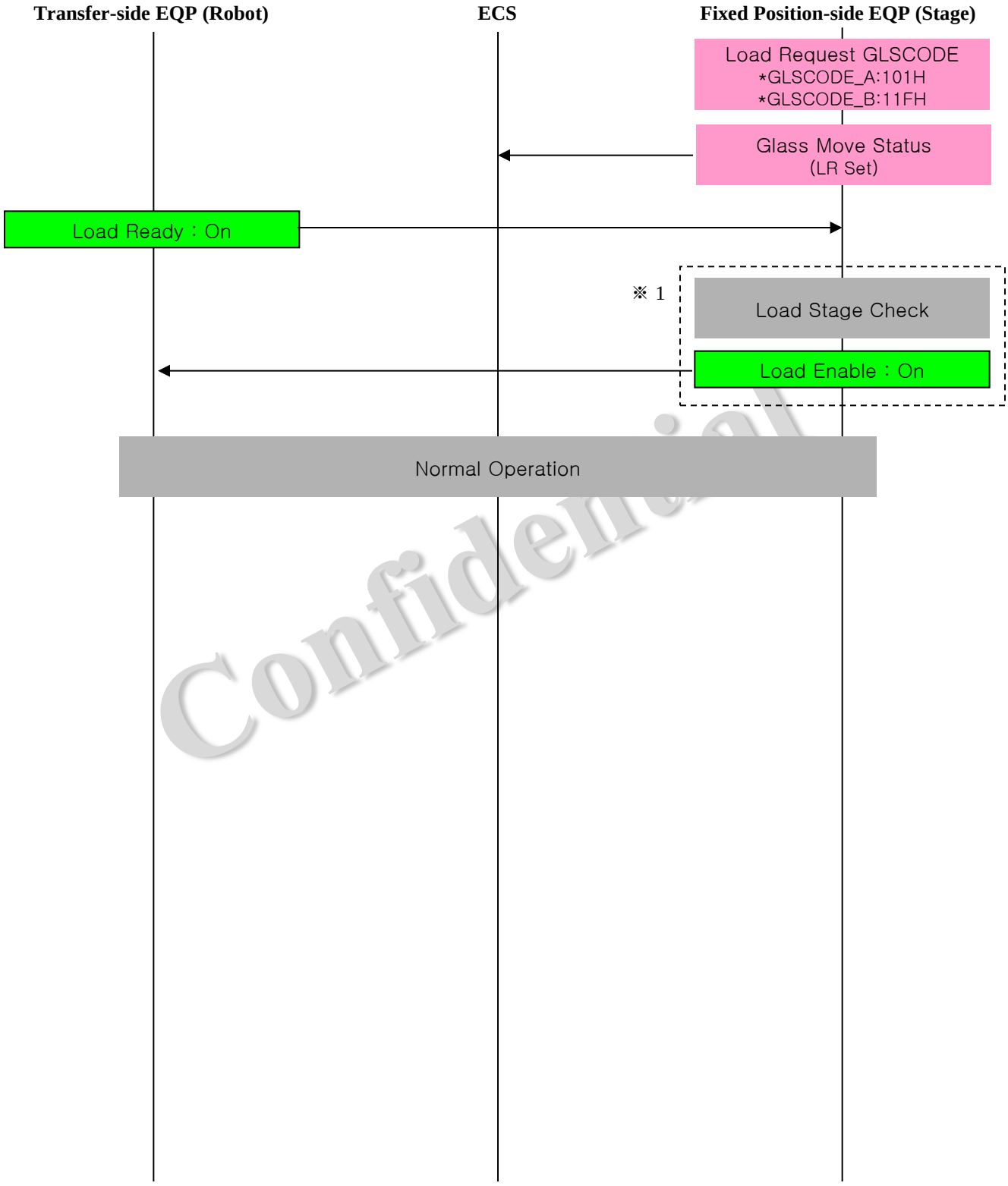
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7.1.4 Load Glass Data Not Accepted (Stand Alone, Robot → Stage)



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※ 1: To set before according to a physical Interlock condition.

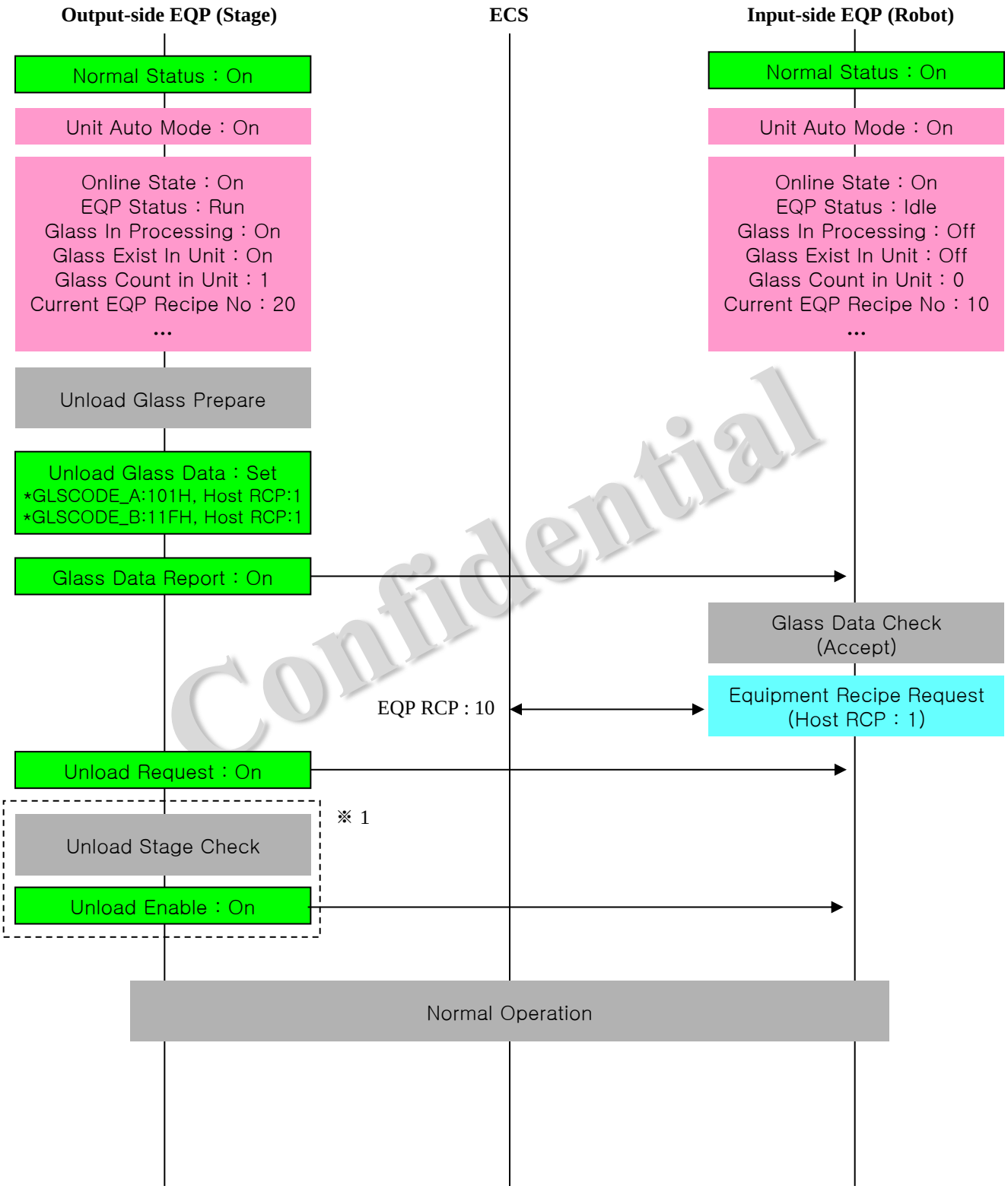
Interlock | Operator | Report | Data Set | Internal

Confidential

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7.2 Glass Move State (Unload Request)

7.2.1 Unload Glass Data Accepted (Process Inline, Stage → Robot)



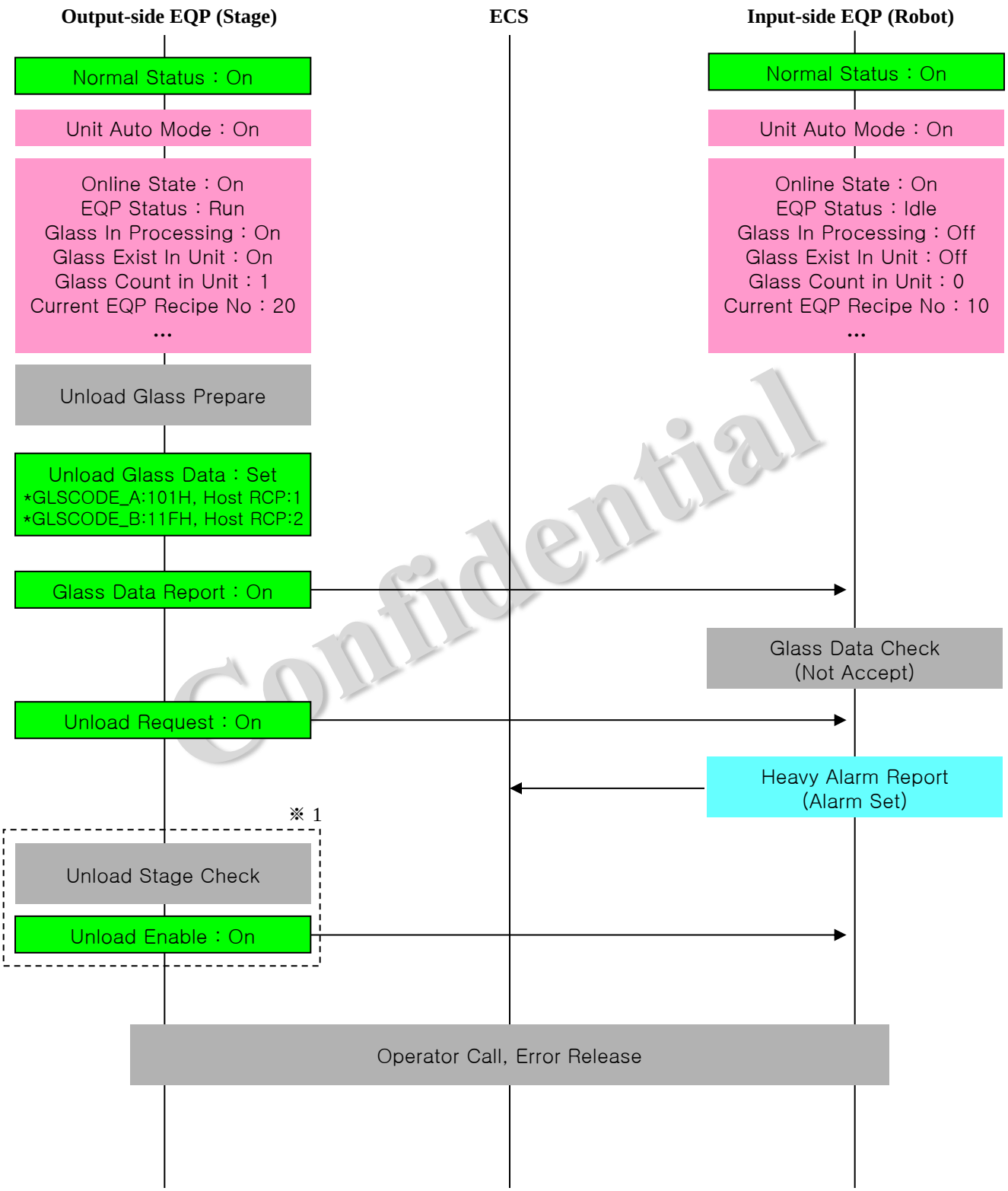
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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7.2.2 Unload Glass Data Not Accepted (Process Inline, Stage → Robot)

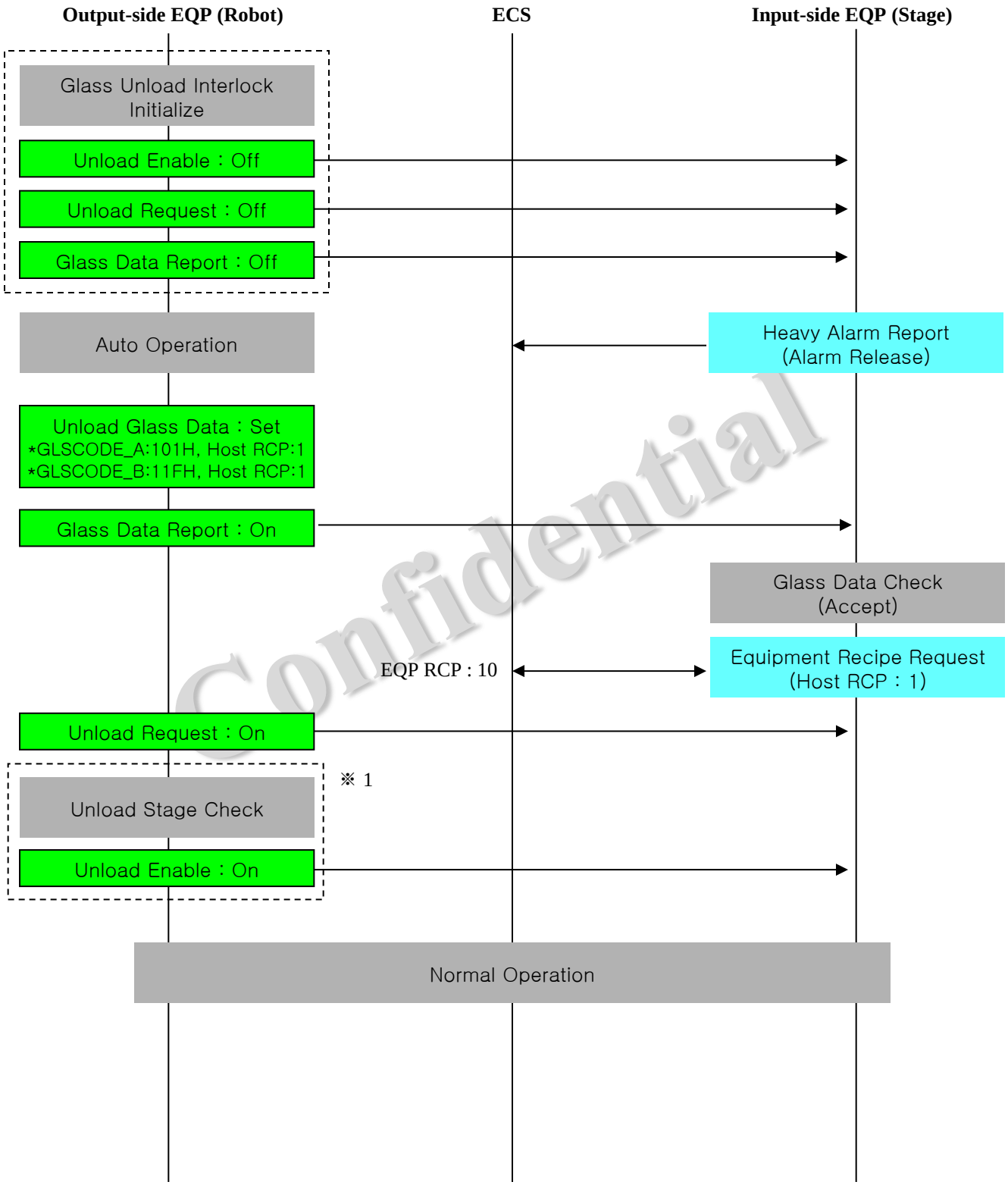


※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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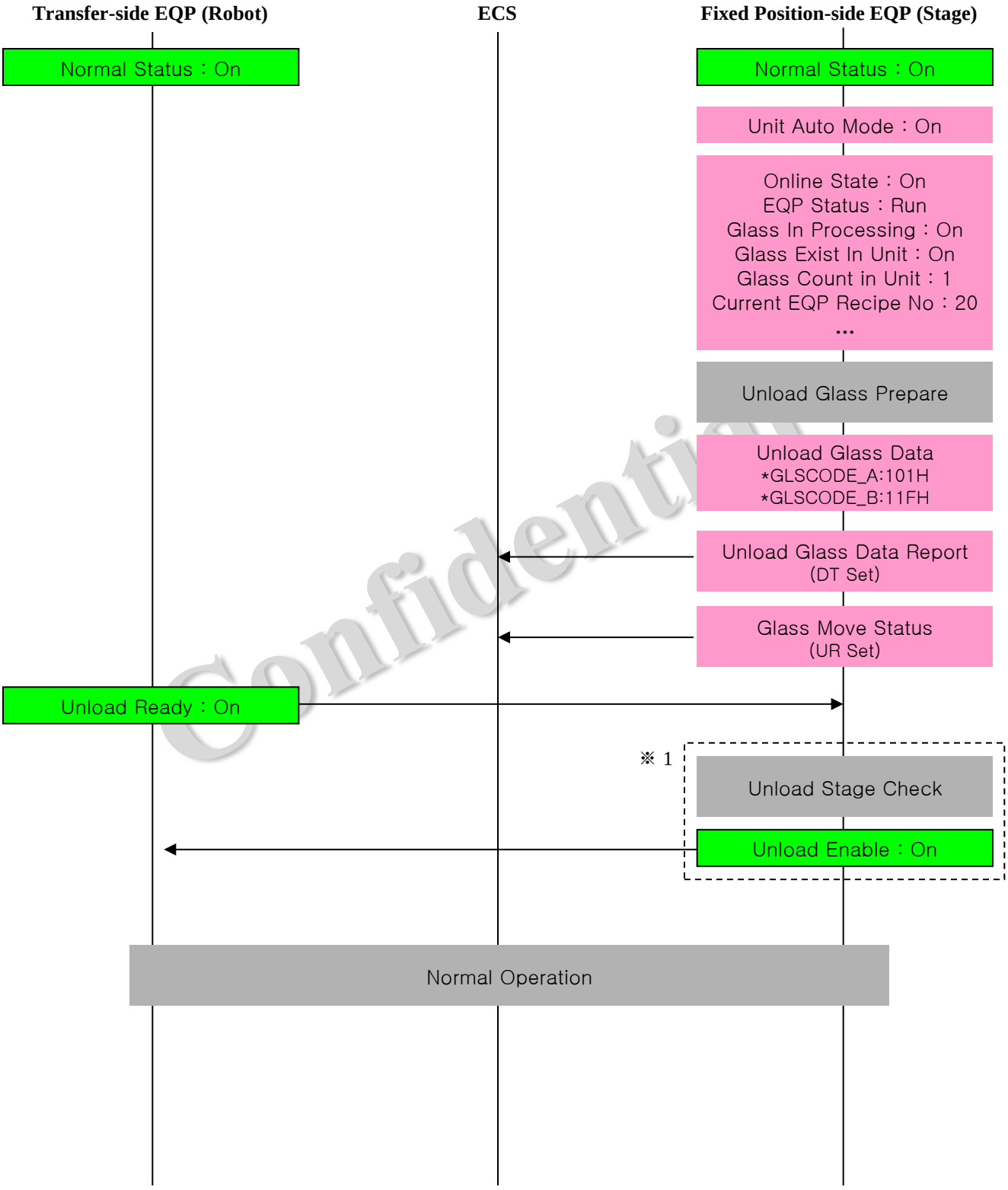
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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7.2.3 Unload Glass Data Accept (Stand Alone, Robot ← Stage)



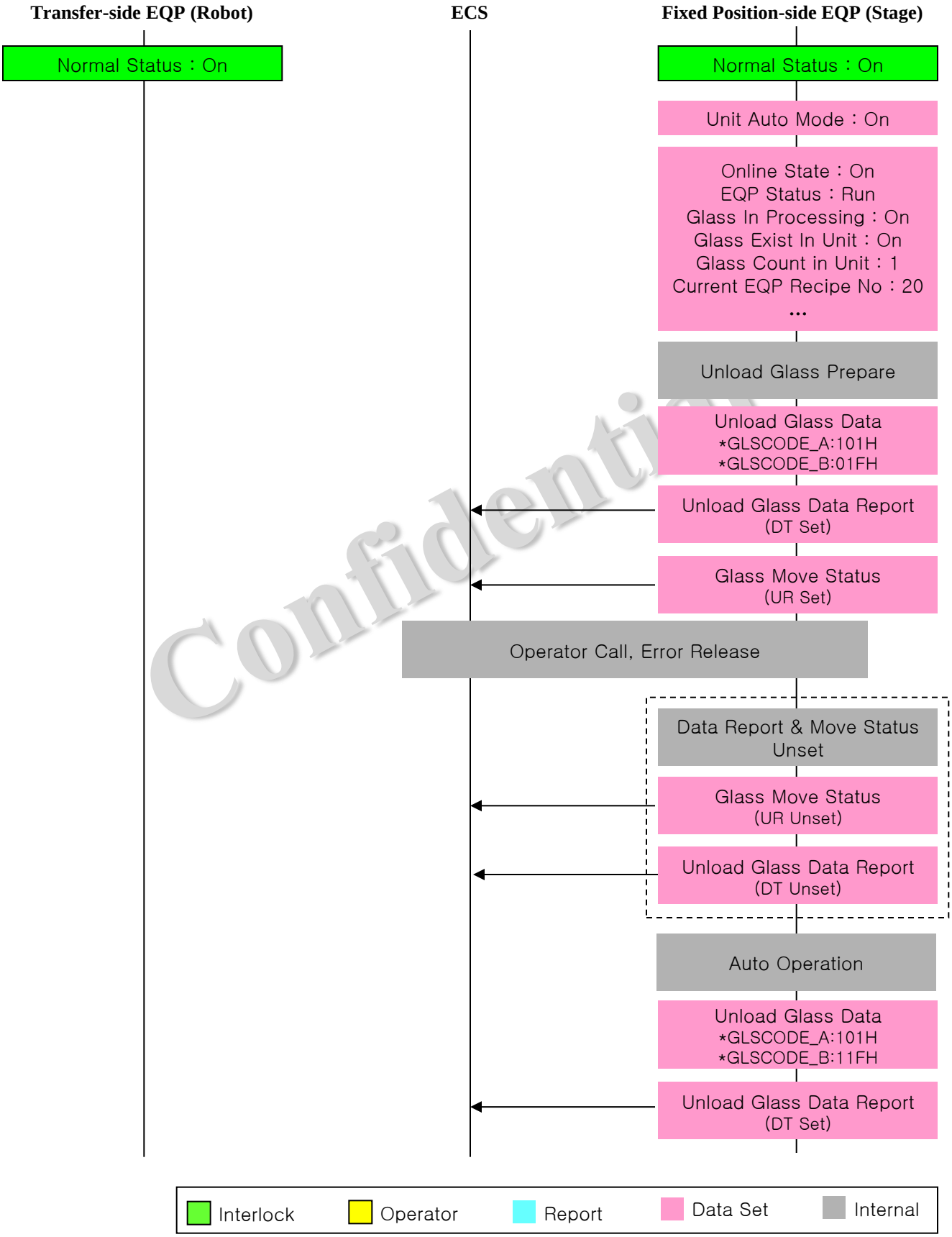
※ 1: To set before according to a physical Interlock condition.

Interlock Operator Report Data Set Internal

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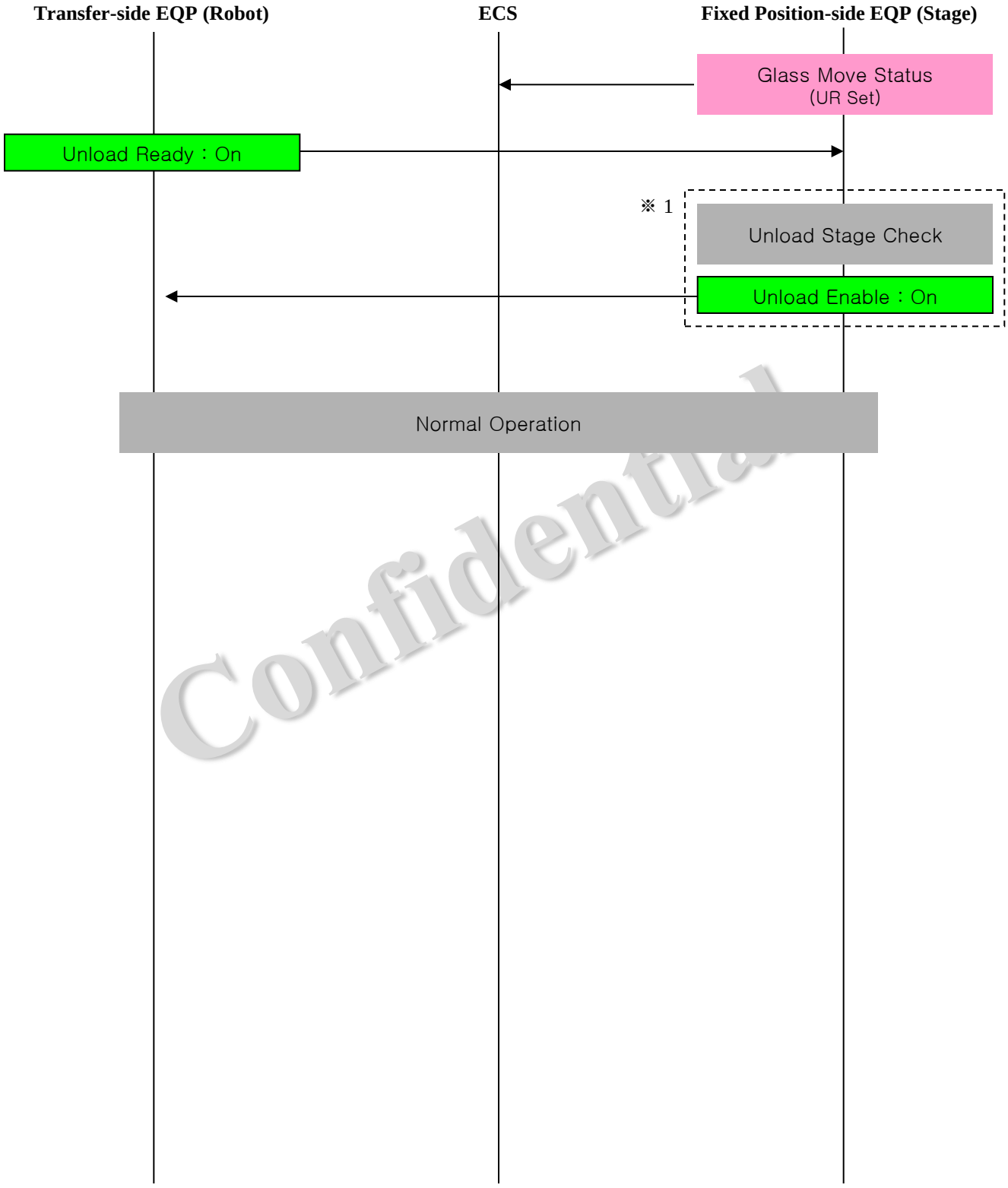
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7.2.4 Unload Glass Data Not Accept (Stand Alone, Robot ← Stage)



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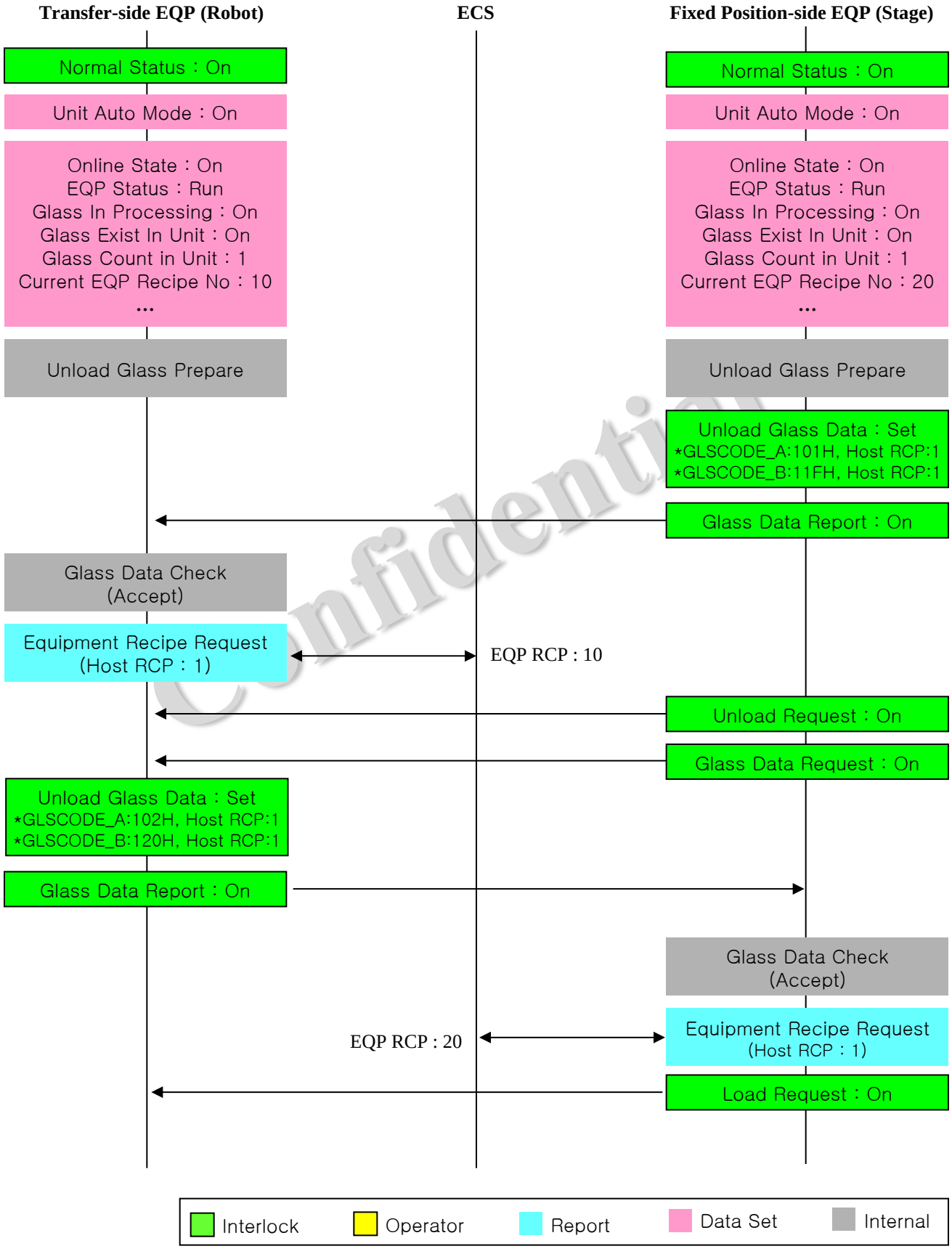
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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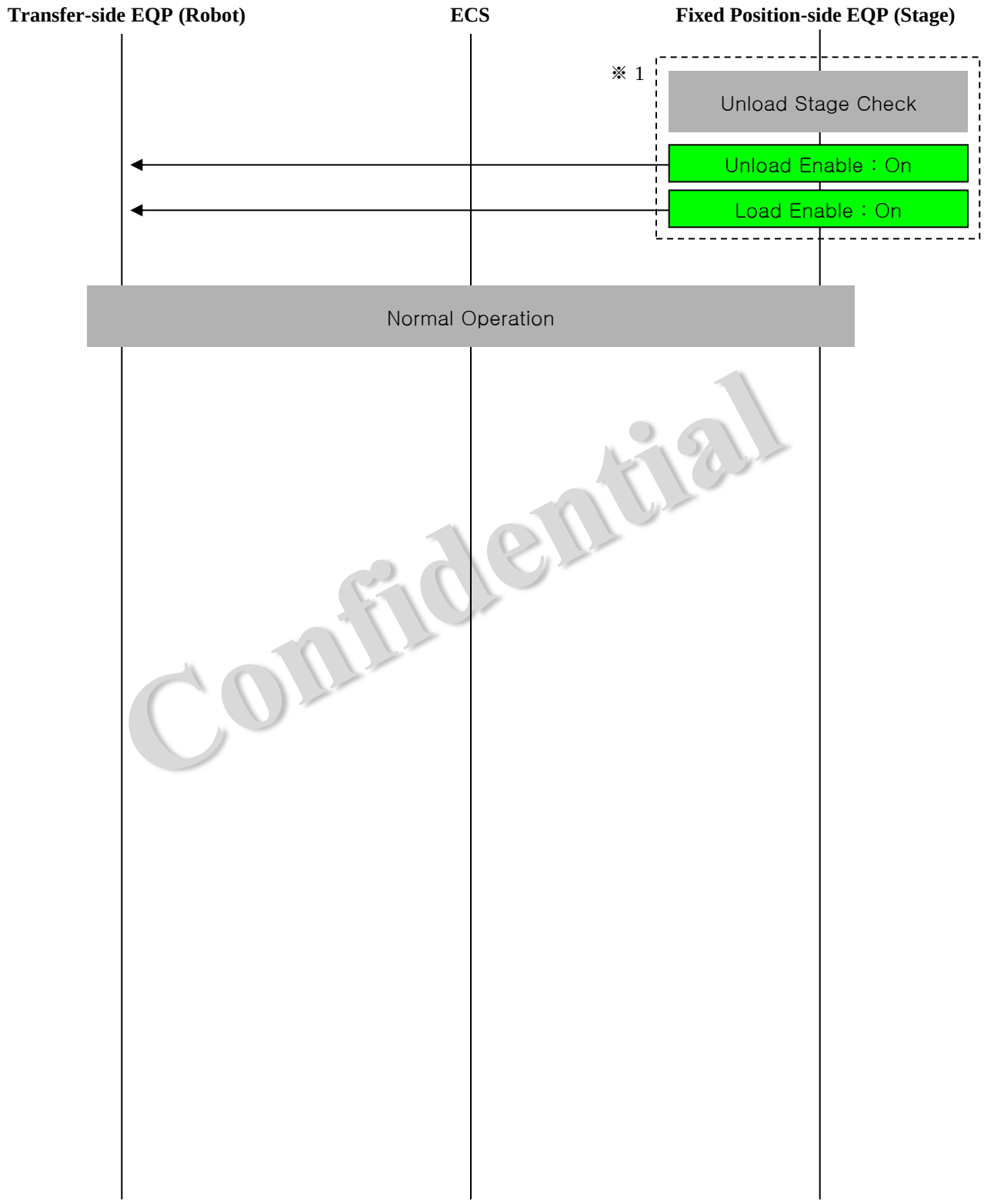
7.3 Glass Move State (Exchange Request)

7.3.1 Load & Unload Glass Data Accepted (Process Inline, Robot ↔ Stage)



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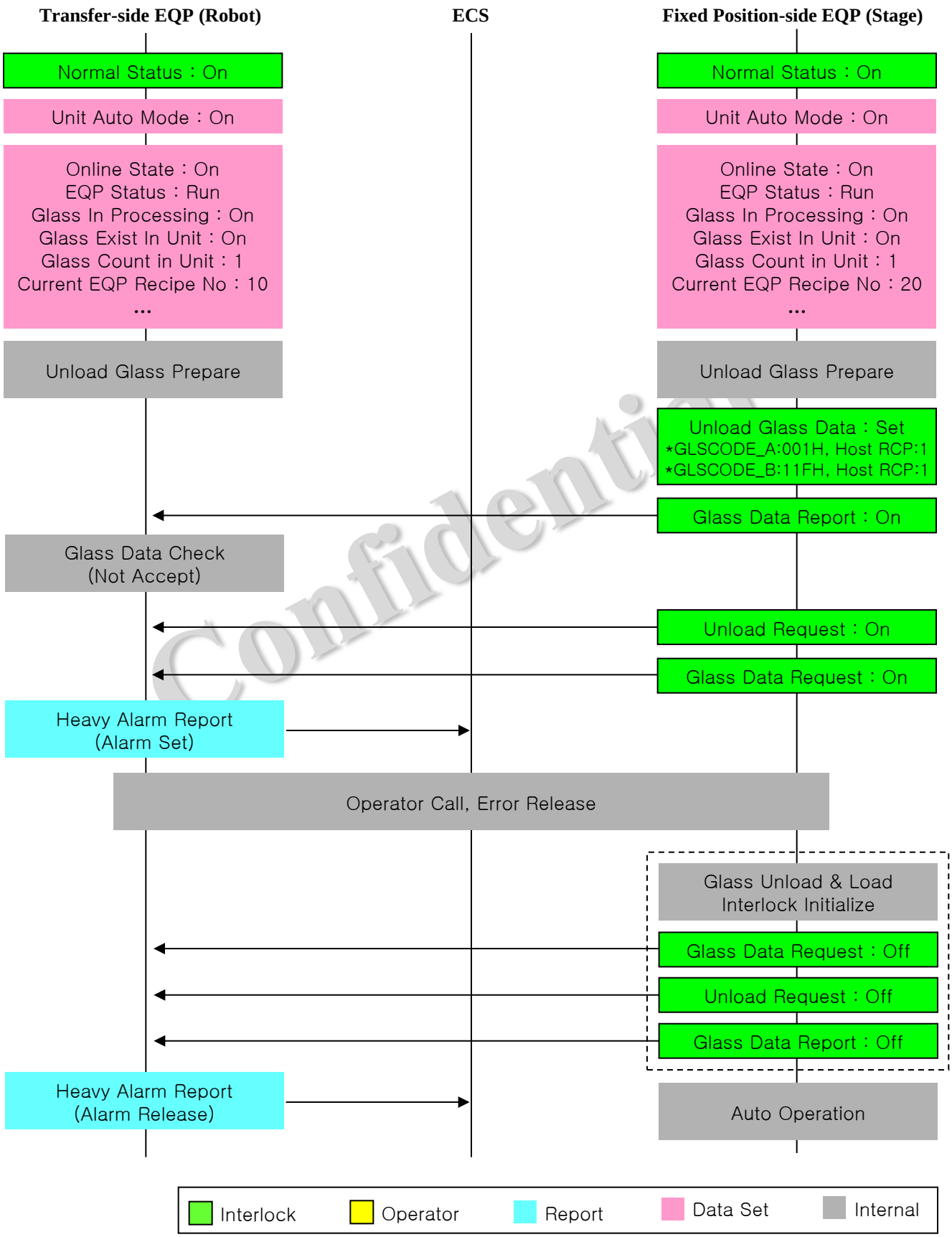
Confidential

※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

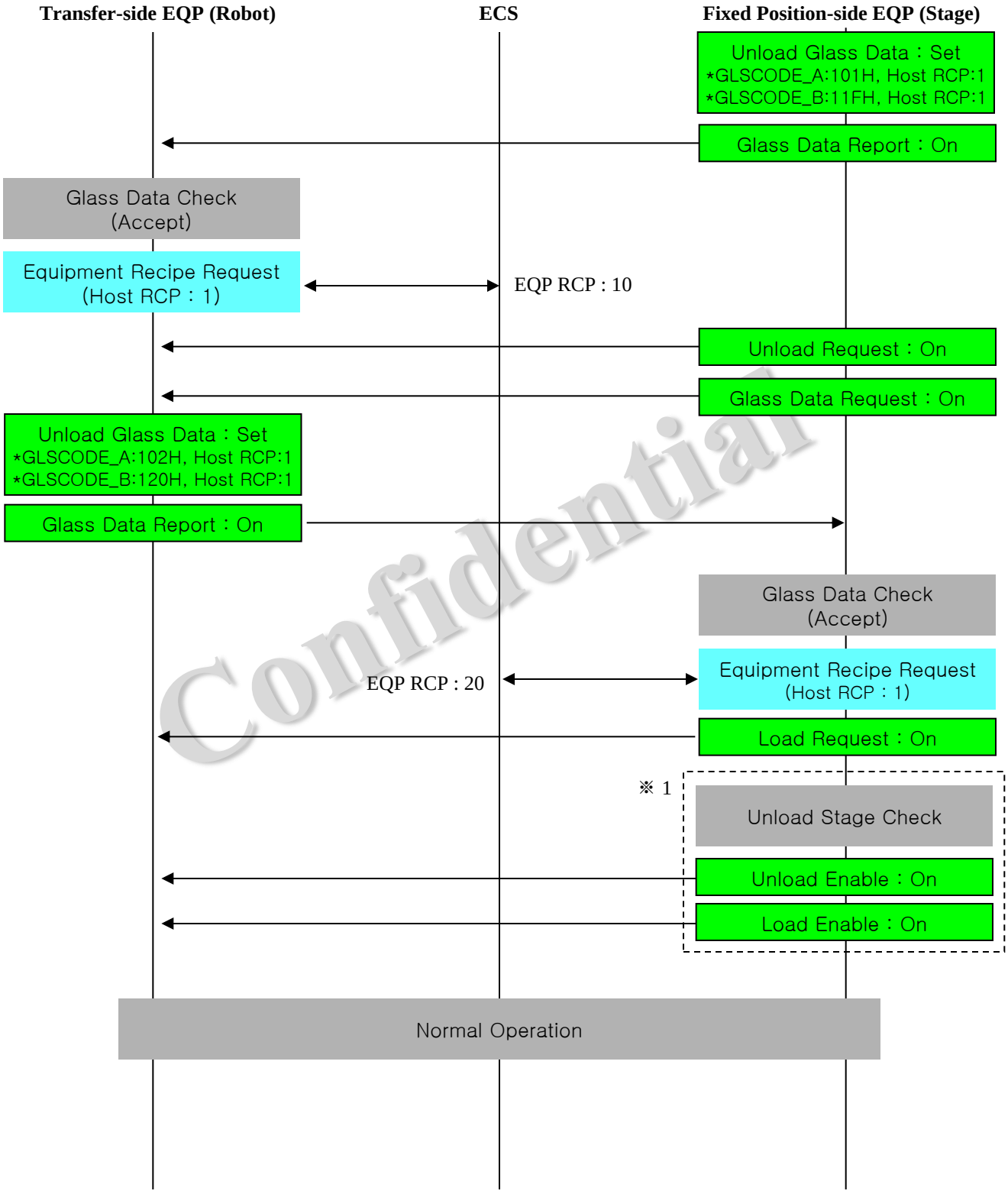
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7.3.2 Unload Glass Data Not Accepted (Process Inline, Robot ↔ Stage)



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※ 1: To set before according to a physical Interlock condition.

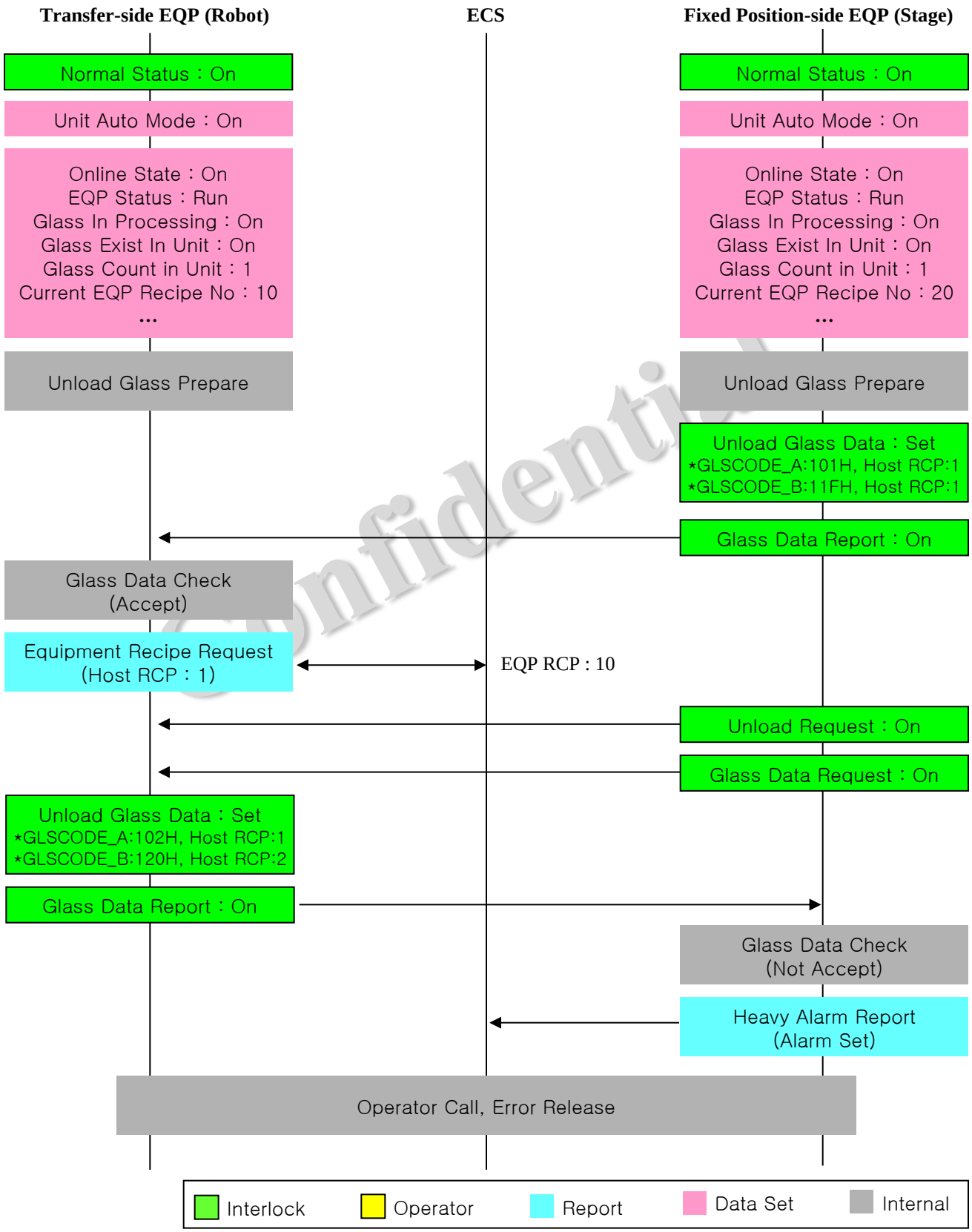
Interlock | Operator | Report | Data Set | Internal

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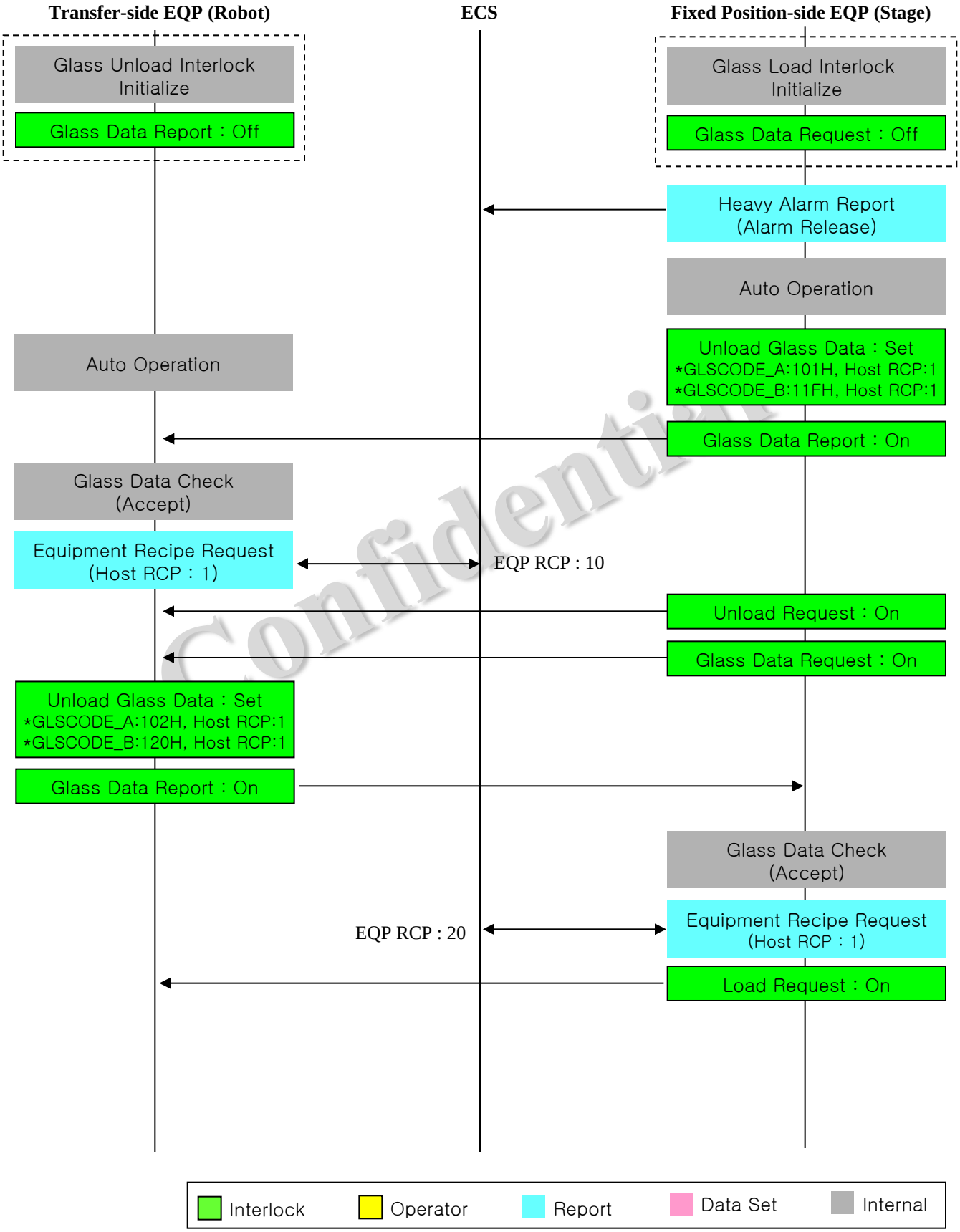
7.3.3 Load Glass Data Not Accepted (Process Inline, Robot ↔ Stage)

1) Glass Exchange Sequence after data not accept error release



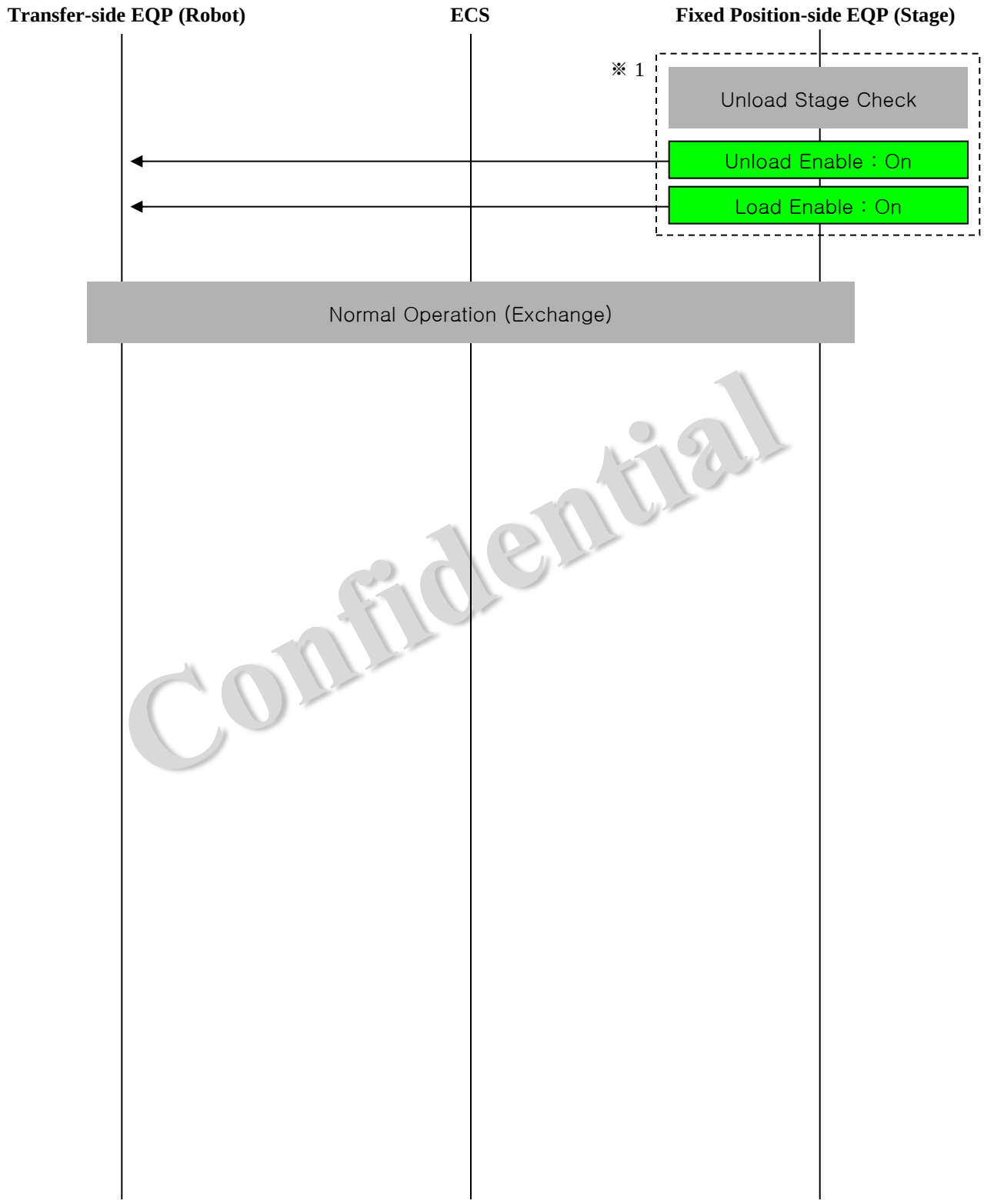
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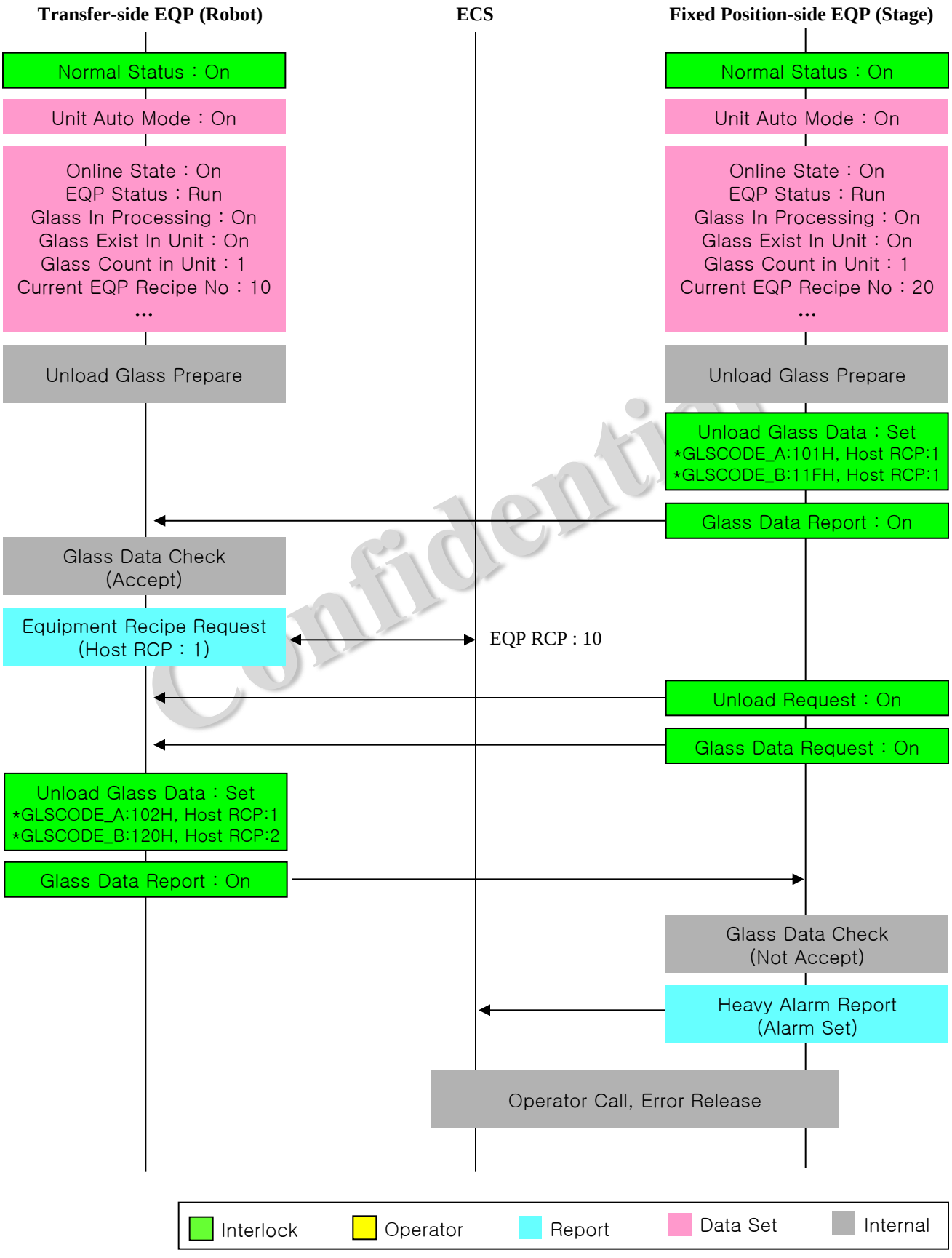
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※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

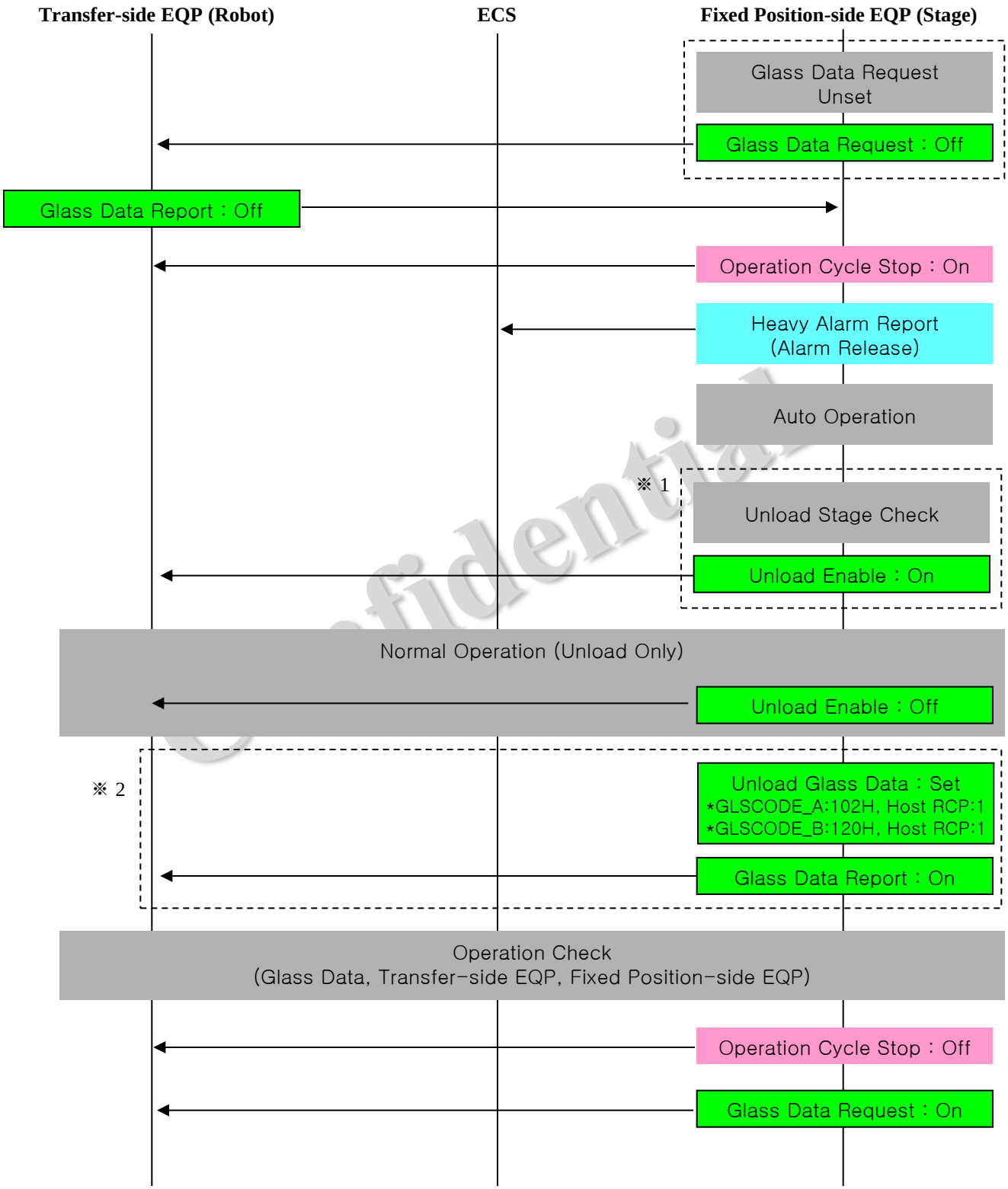
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2) Glass Unloading Sequence after data not accept error release



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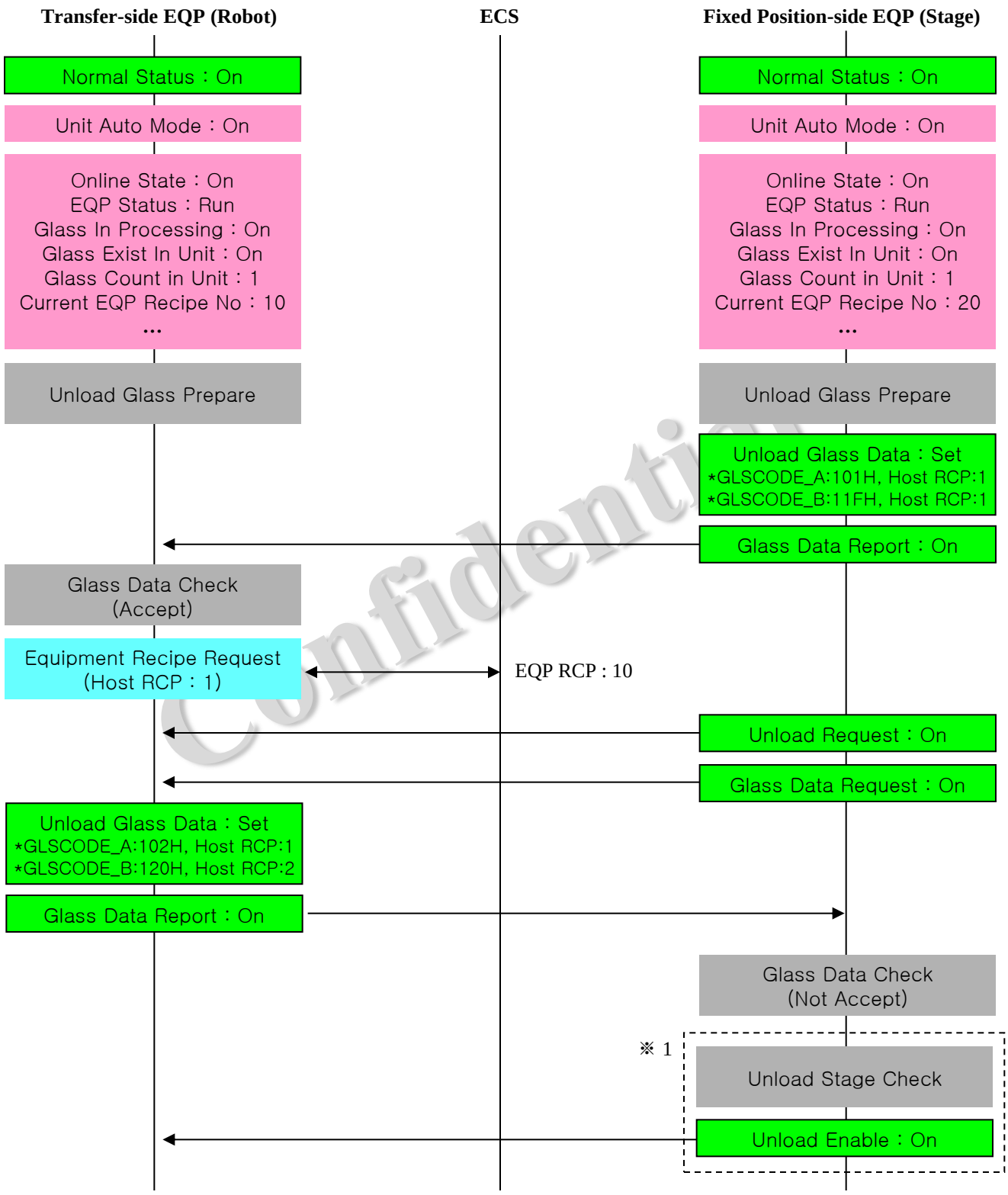


※ 1: To set before according to a physical Interlock condition.
※ 2 : If the next Unload Glass exists, starts Exchange Interlock

Interlock | Operator | Report | Data Set | Internal

Beyond Promise
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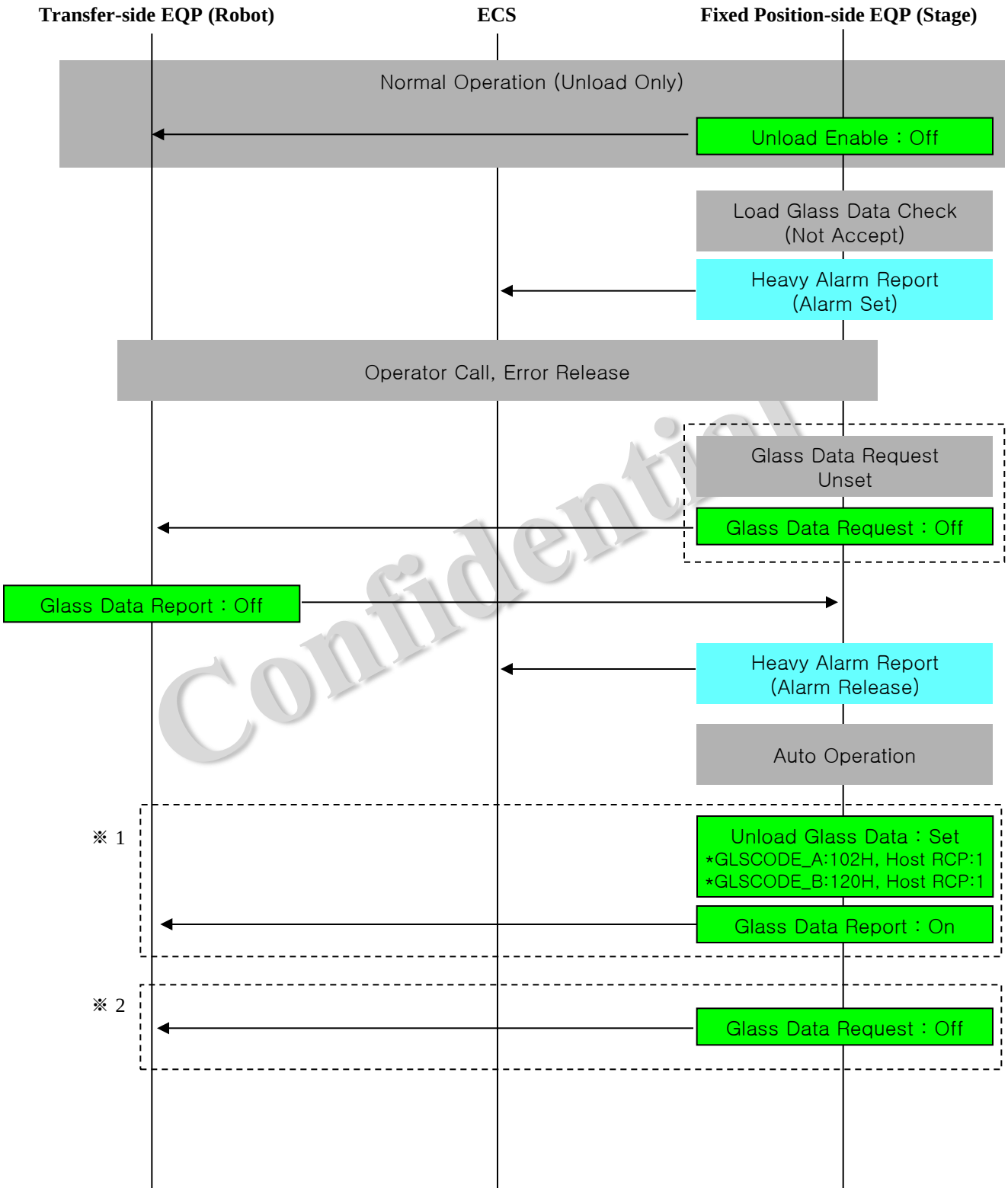
3) Alarm Report after Glass Unloading Sequence



※ 1: To set before according to a physical Interlock condition.

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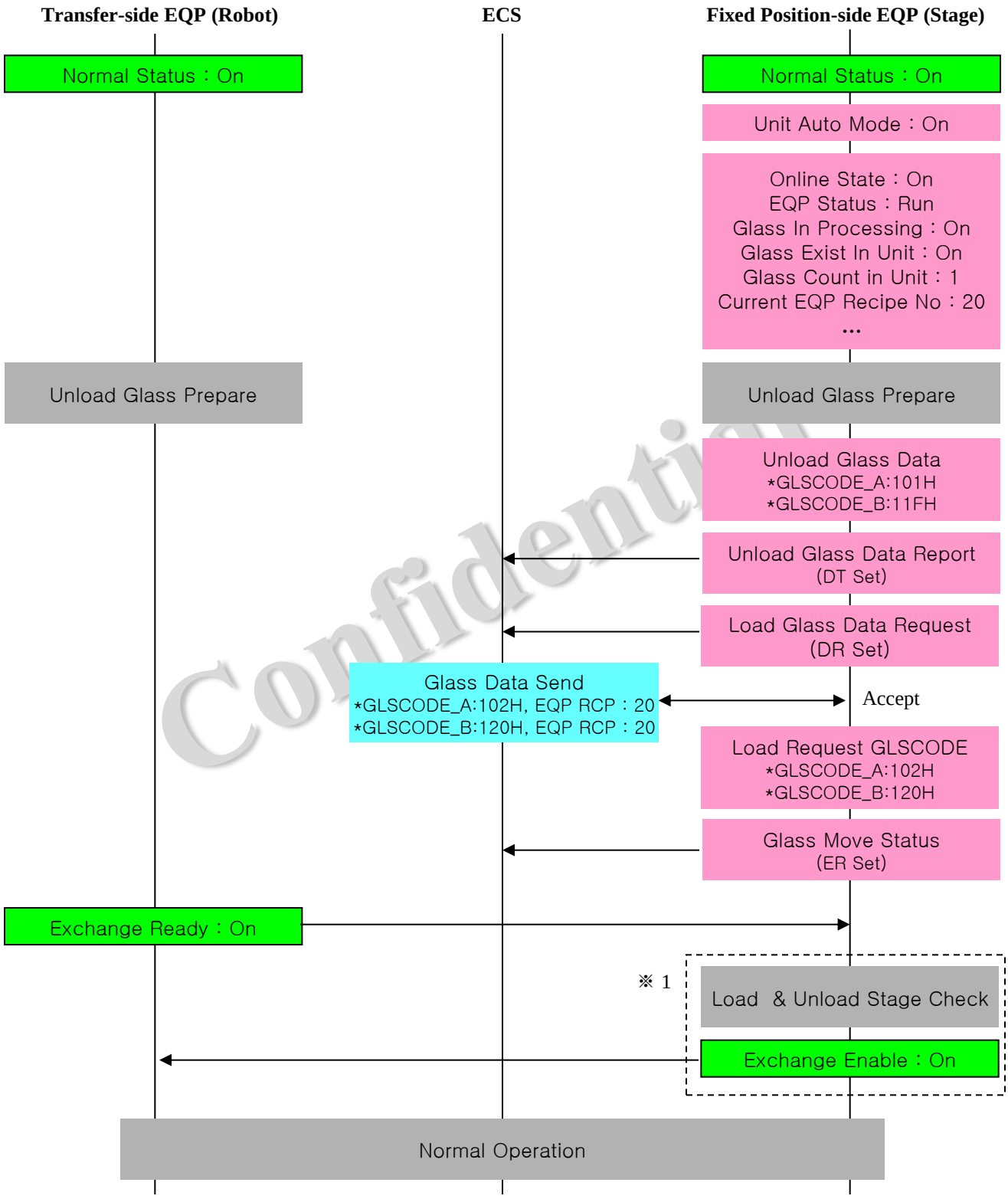
※ 1 : If the next Unload Glass exists, starts Exchange Interlock
※ 2 : If the next Unload Glass does not exist, starts Load Only Interlock

Interlock | Operator | Report | Data Set | Internal

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7.3.4 Load & Unload Glass Data Accepted (Stand Alone, Robot ↔ Stage)



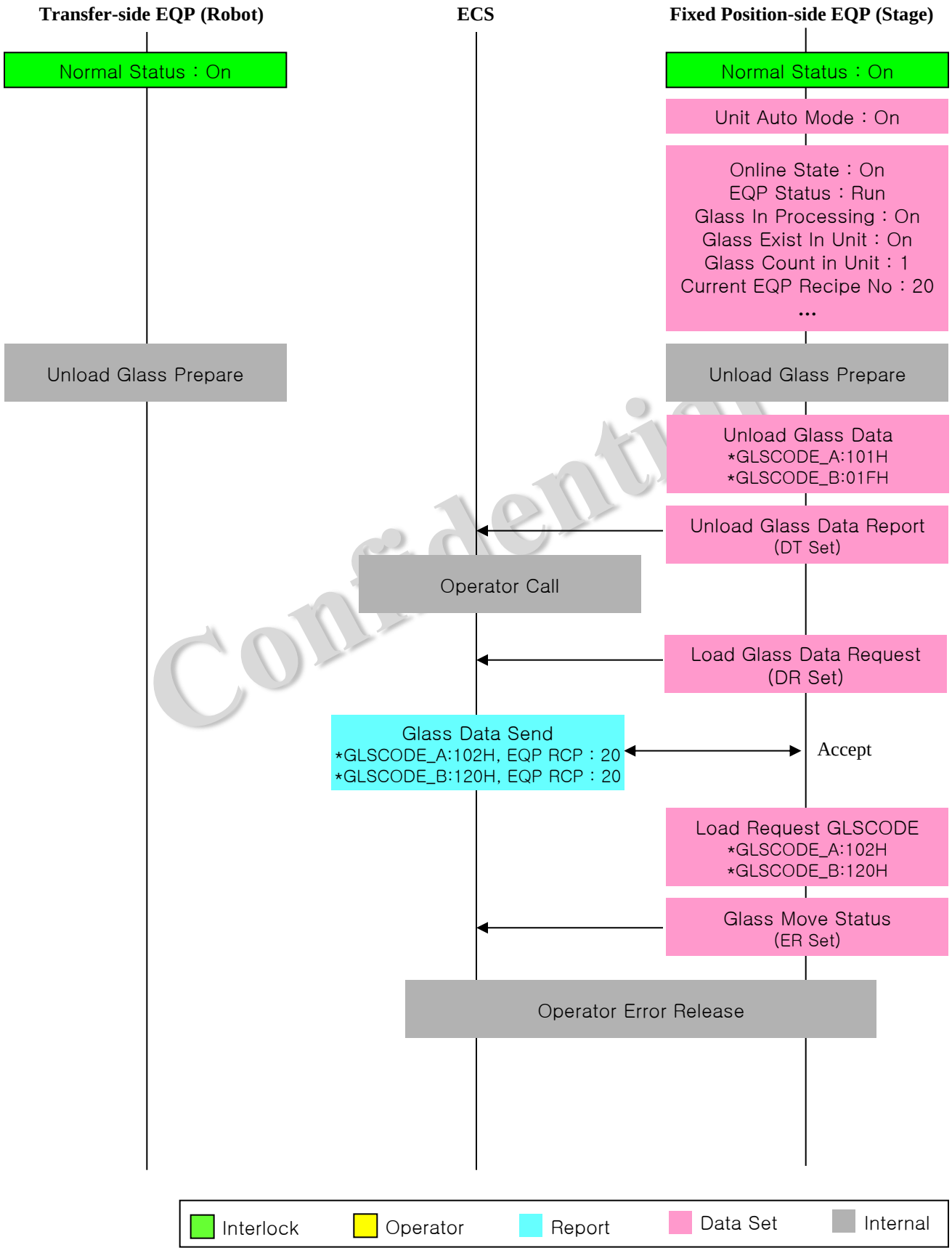
※ 1: To set before according to a physical Interlock condition.



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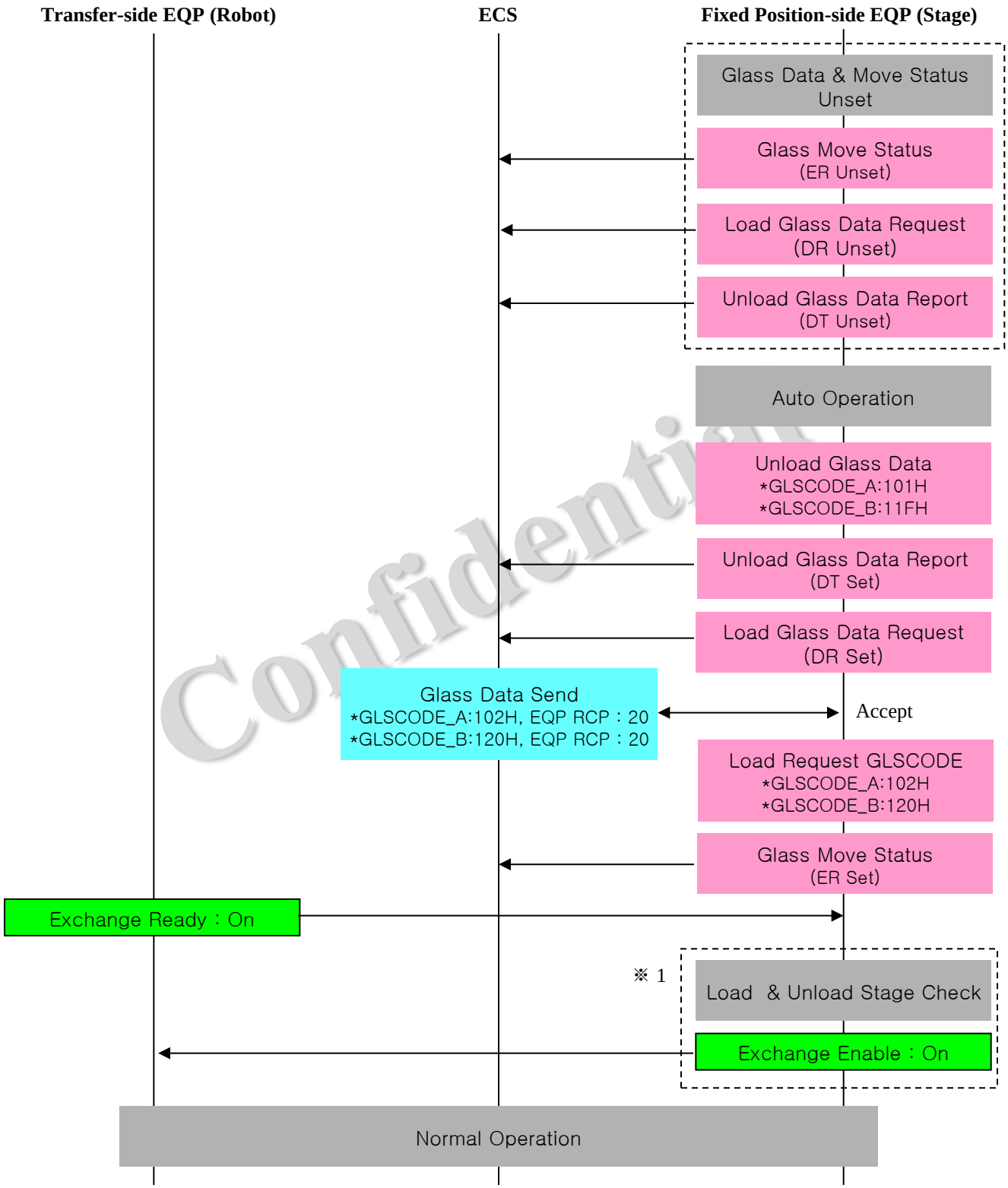
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7.3.5 Unload Glass Data Not Accepted (Stand Alone, Robot ↔ Stage)



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※ 1: To set before according to a physical Interlock condition.

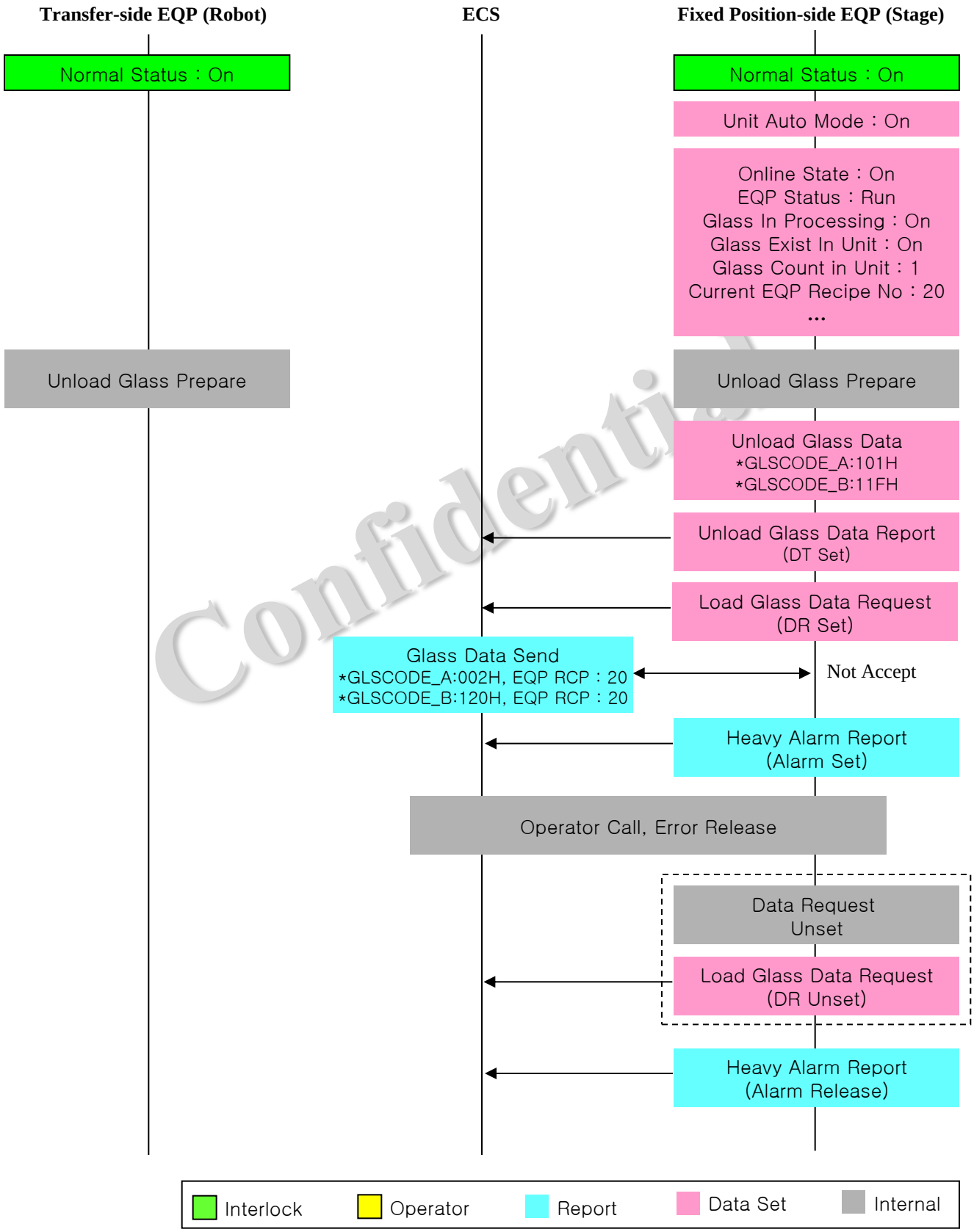
Interlock | Operator | Report | Data Set | Internal

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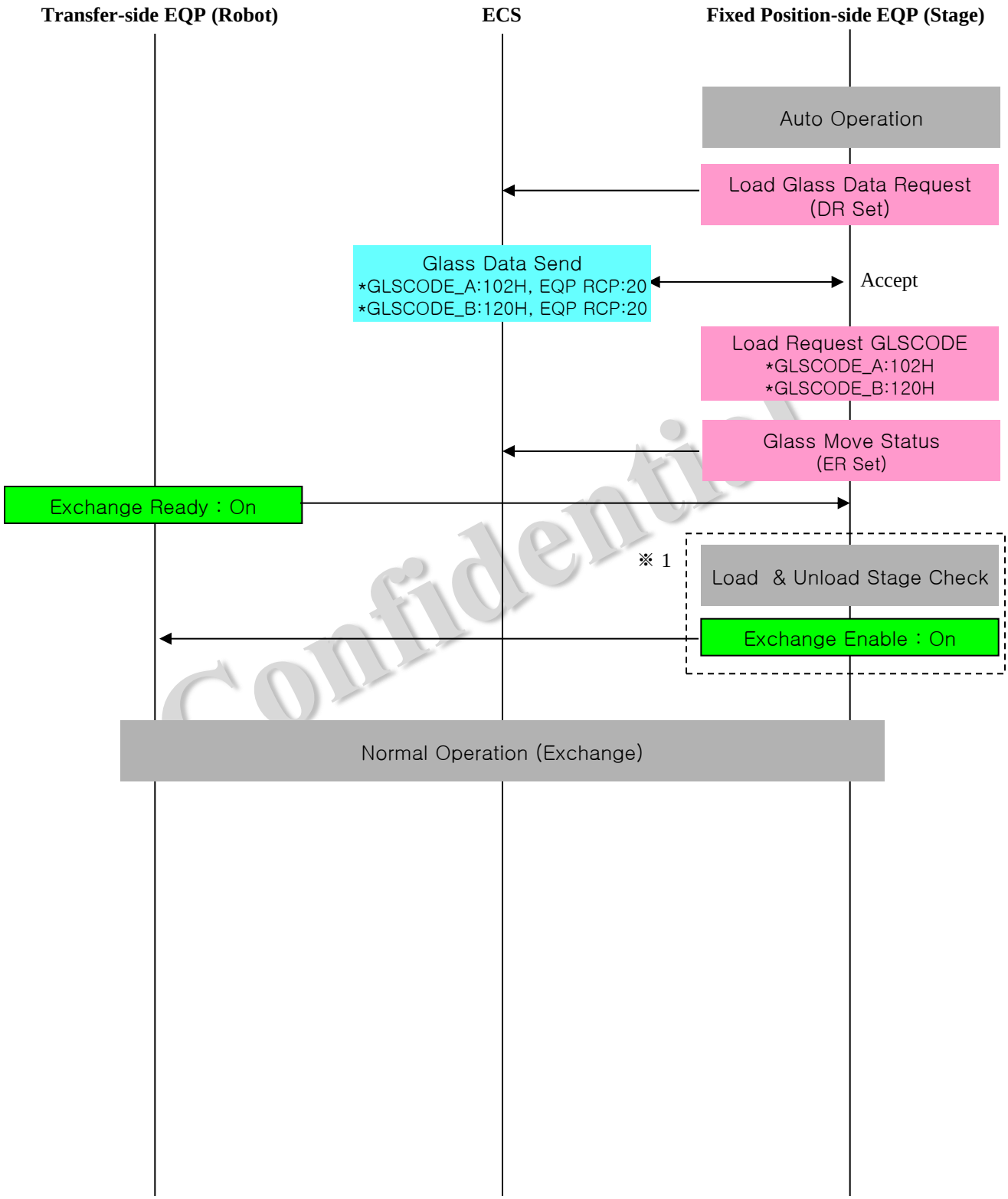
7.3.6 Load Glass Data Not Accepted (Stand Alone, Robot ↔ Stage)

1) Glass Exchange Sequence after data not accept error release



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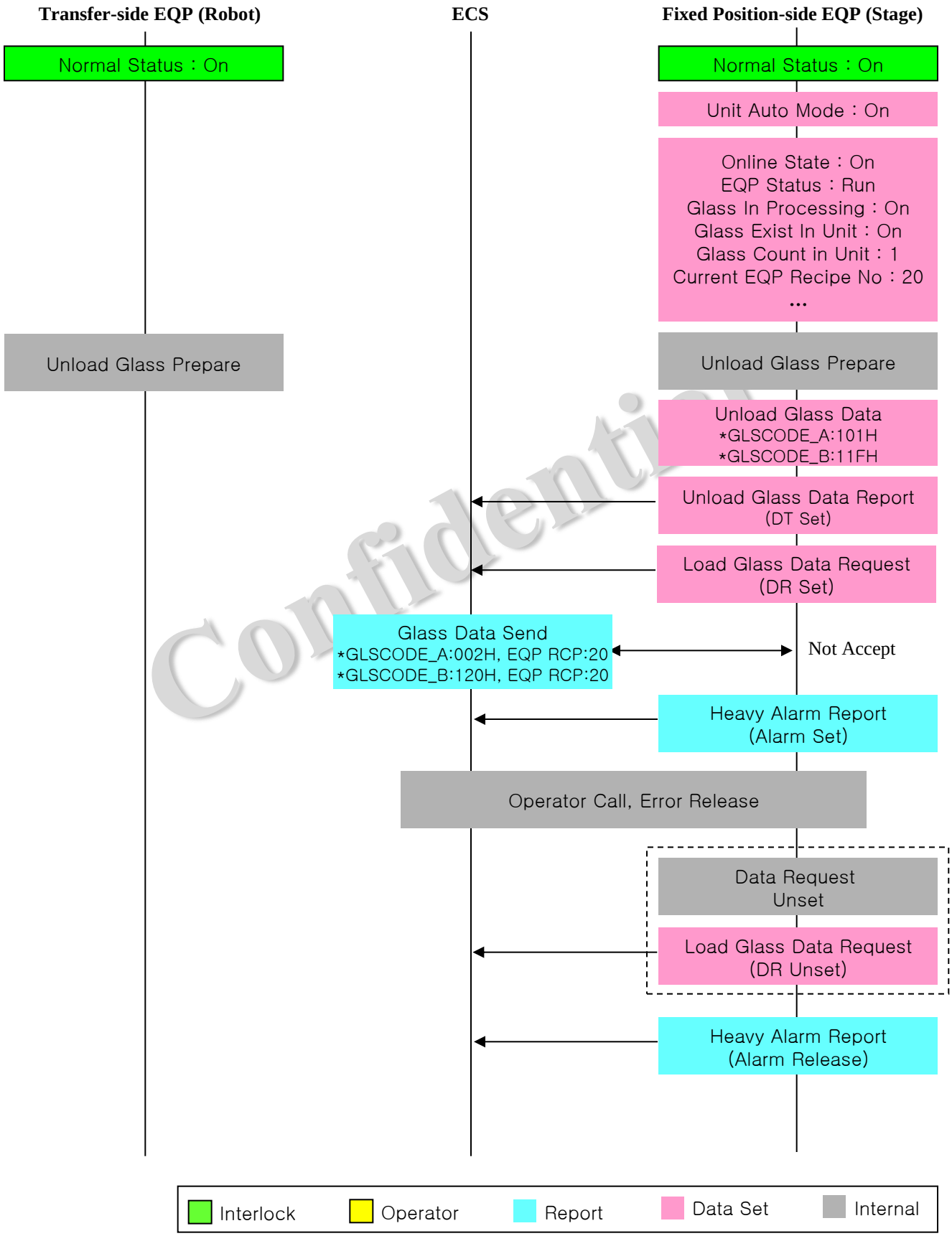
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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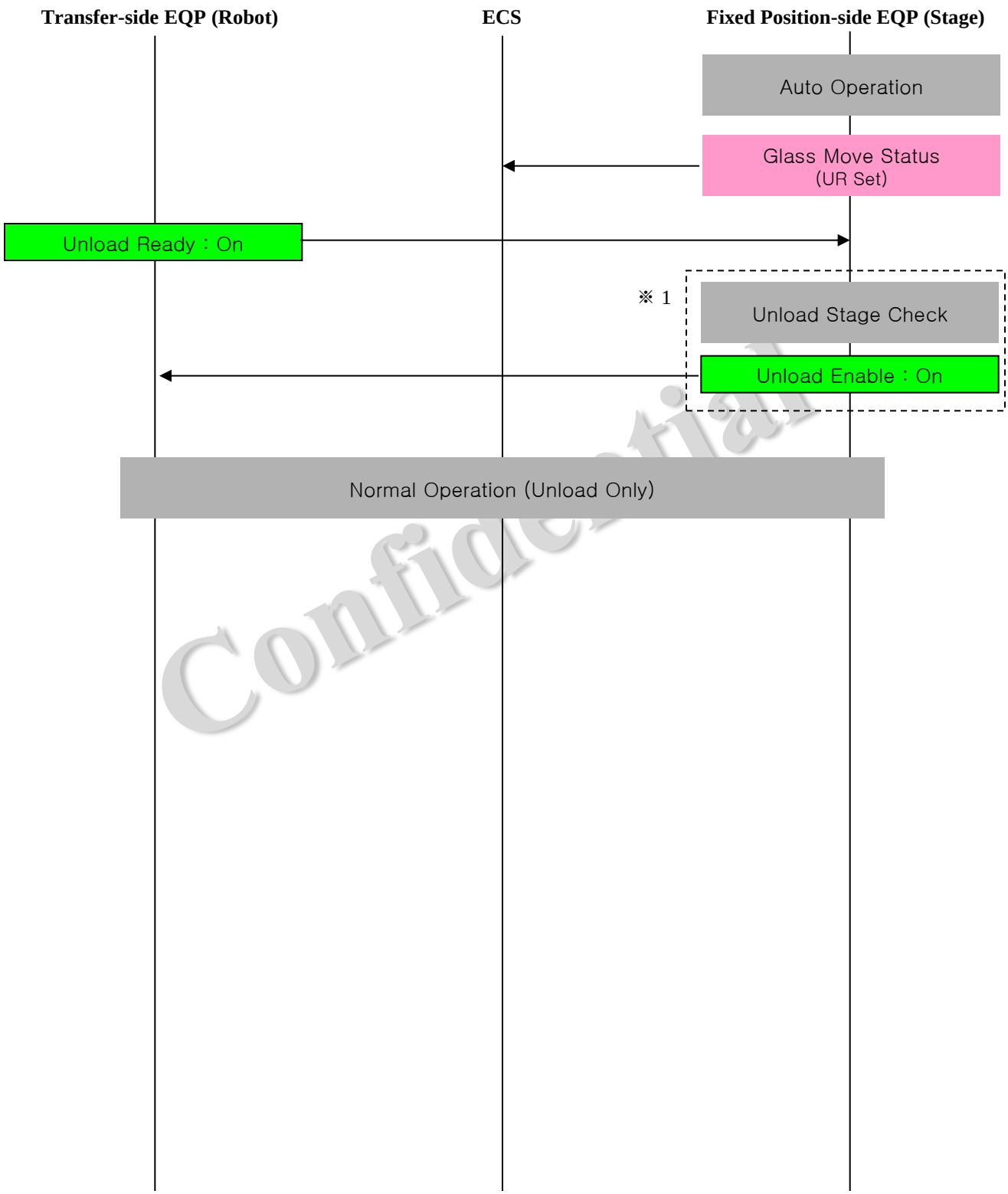
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2) Glass Unloading Sequence after data not accept error release



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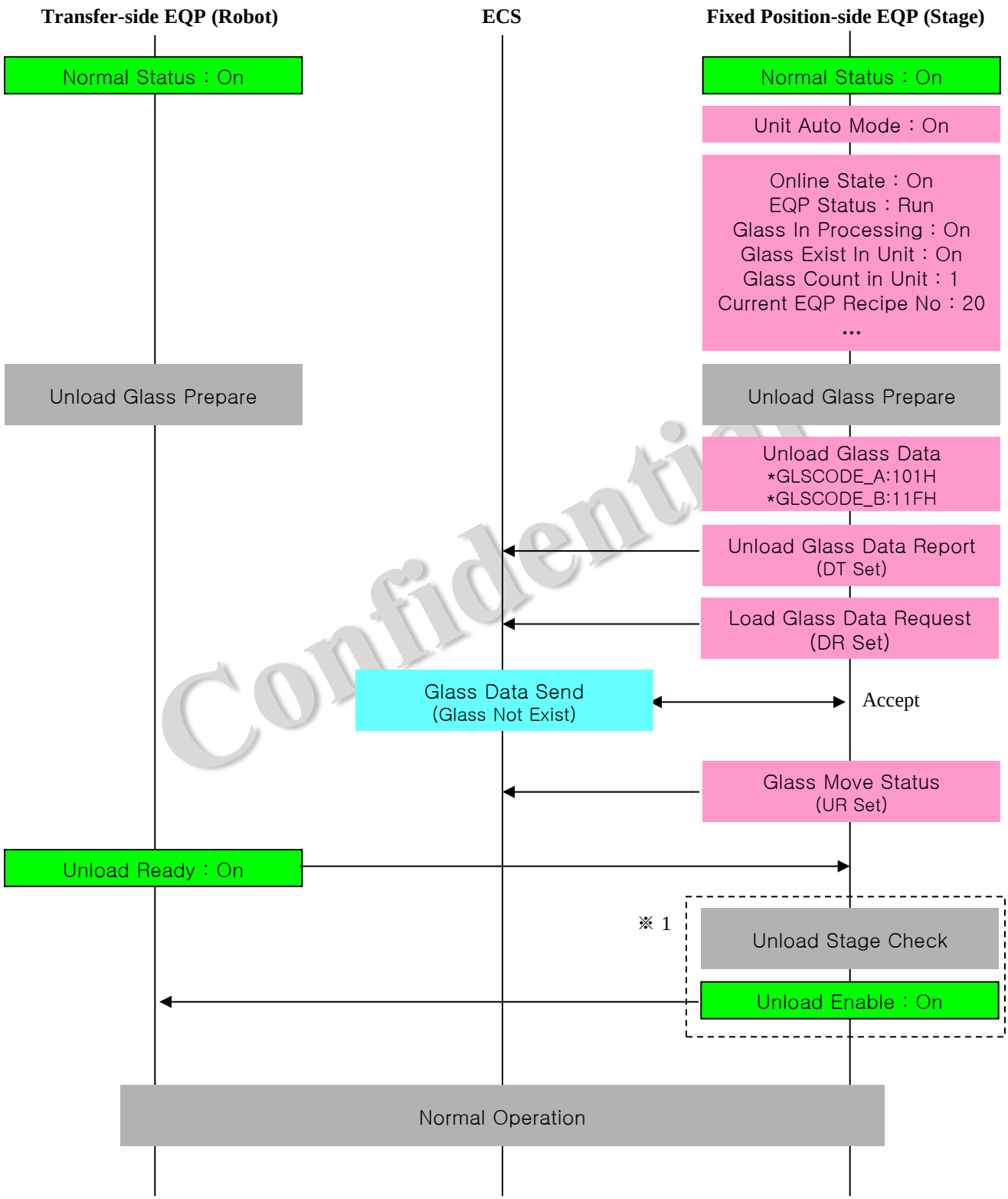
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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7.3.7 Load Glass Not Exist (Stand Alone, Robot ↔ Stage)



※ 1: To set before according to a physical Interlock condition.



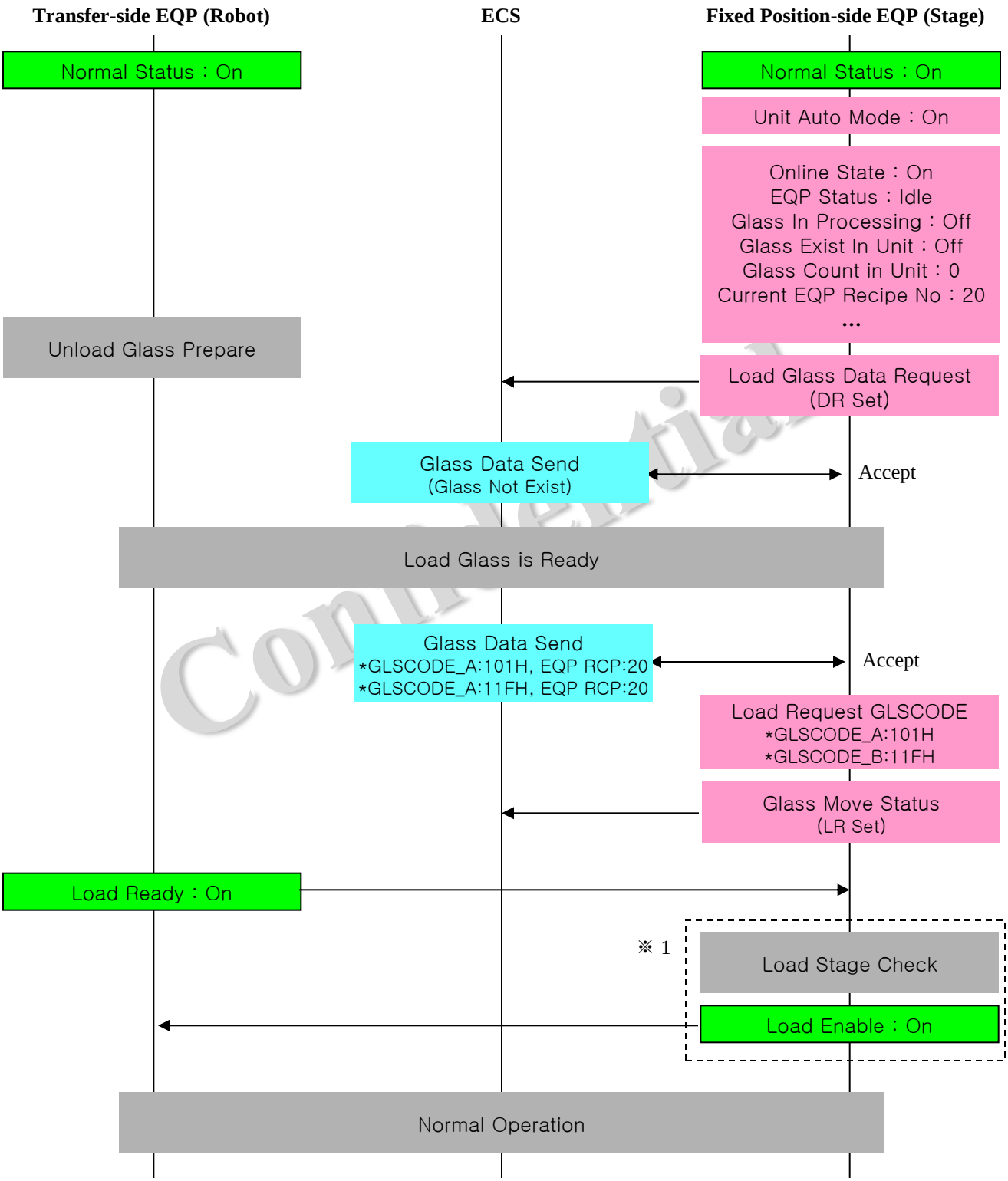
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7.4 Load Glass Data Change Case

7.4.1 Change From Not Exist to Exist (Stand Alone, Robot → Stage)

1) Unload Glass Not Exist



※ 1: To set before according to a physical Interlock condition.

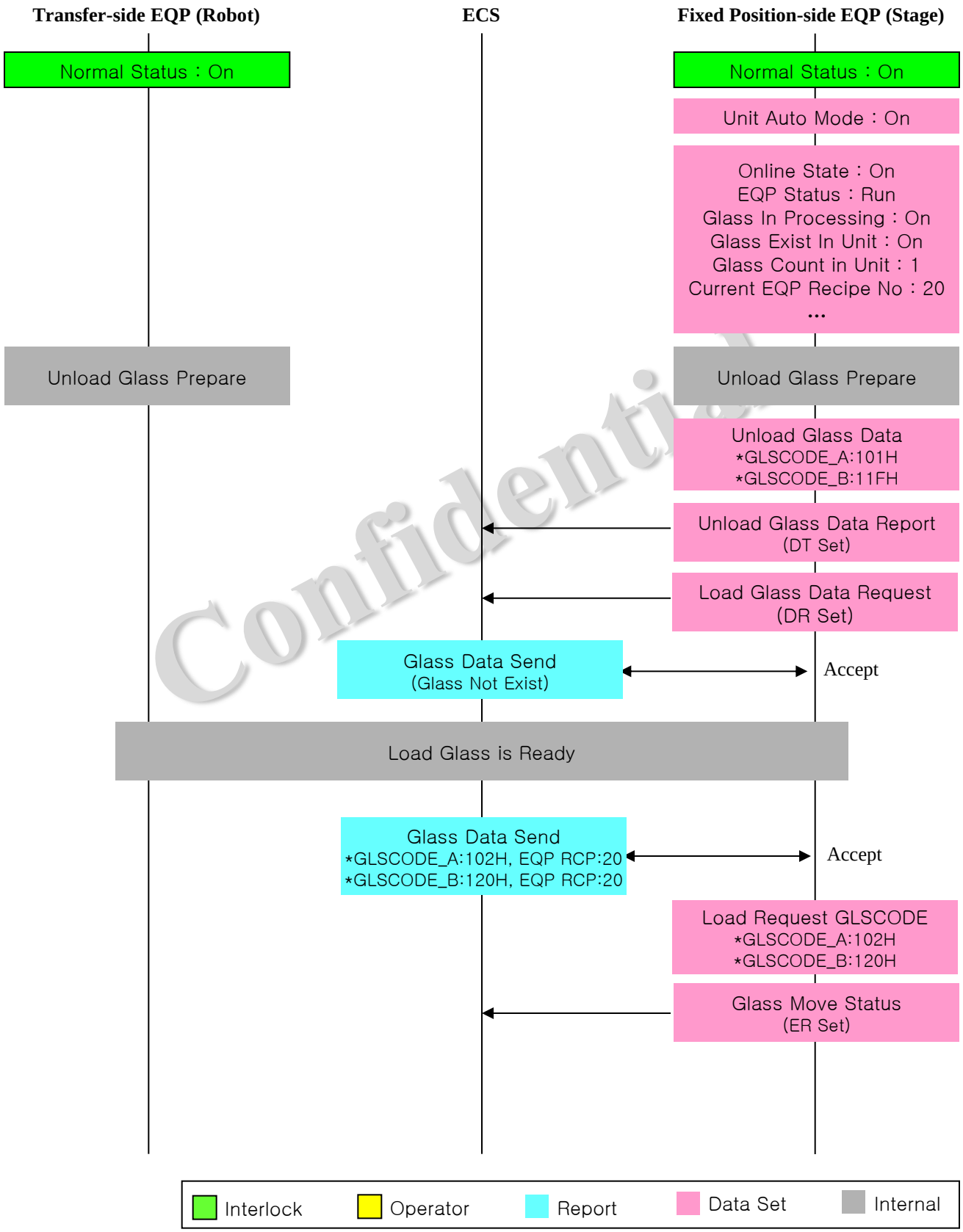
Interlock | Operator | Report | Data Set | Internal

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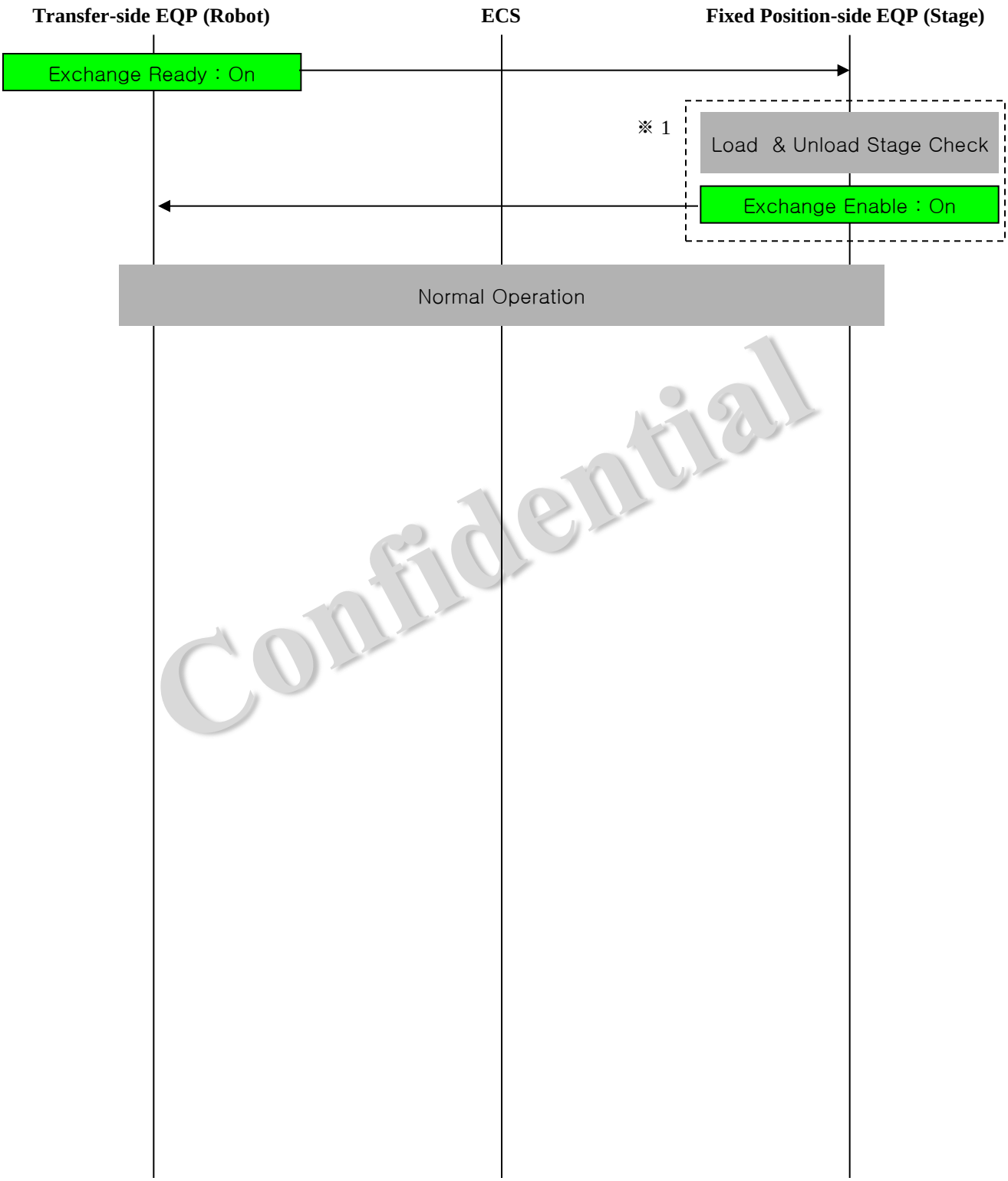
2) Unload Glass Exist

- Case ⓐ : Glass Data Send before 'UR'



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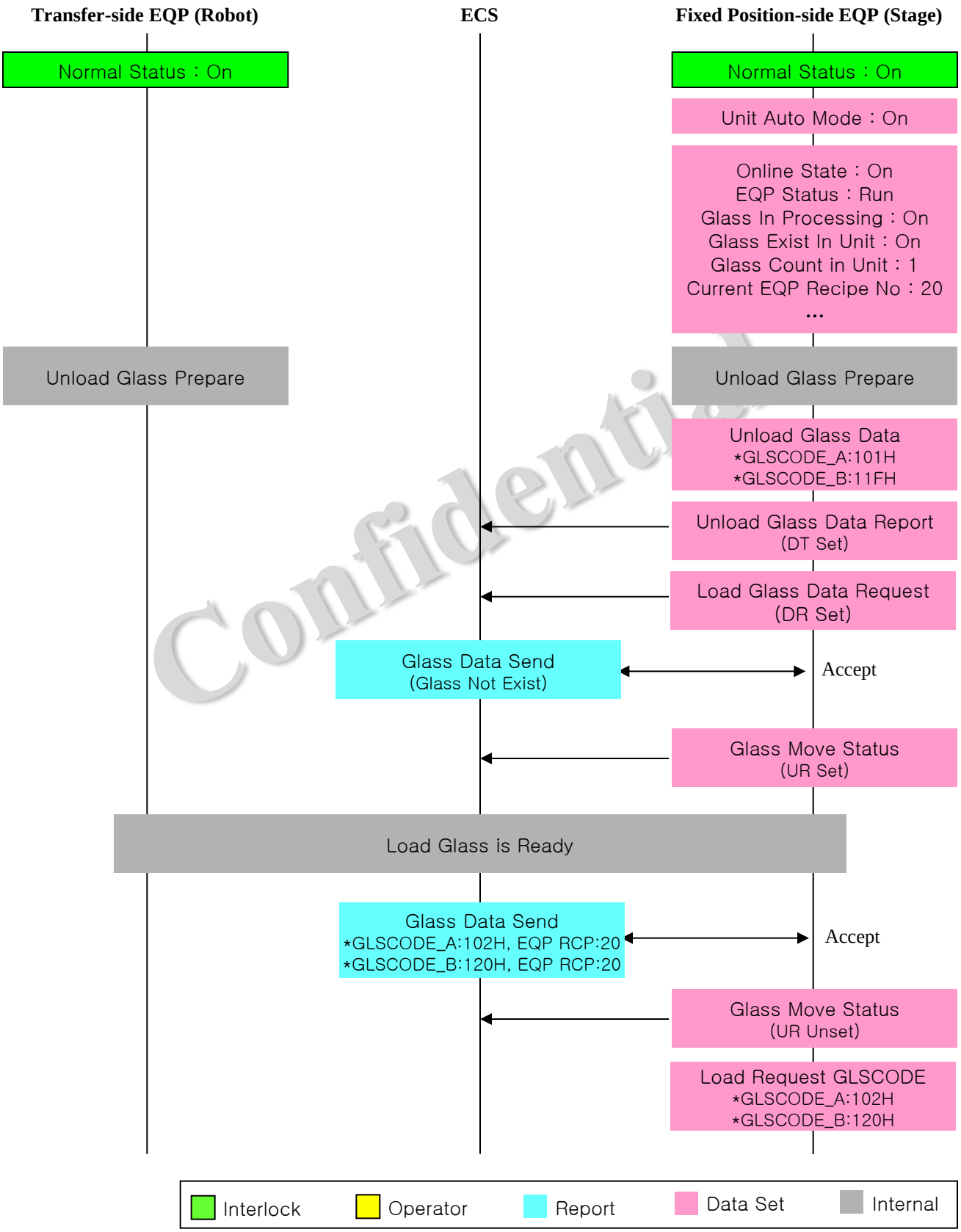
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※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

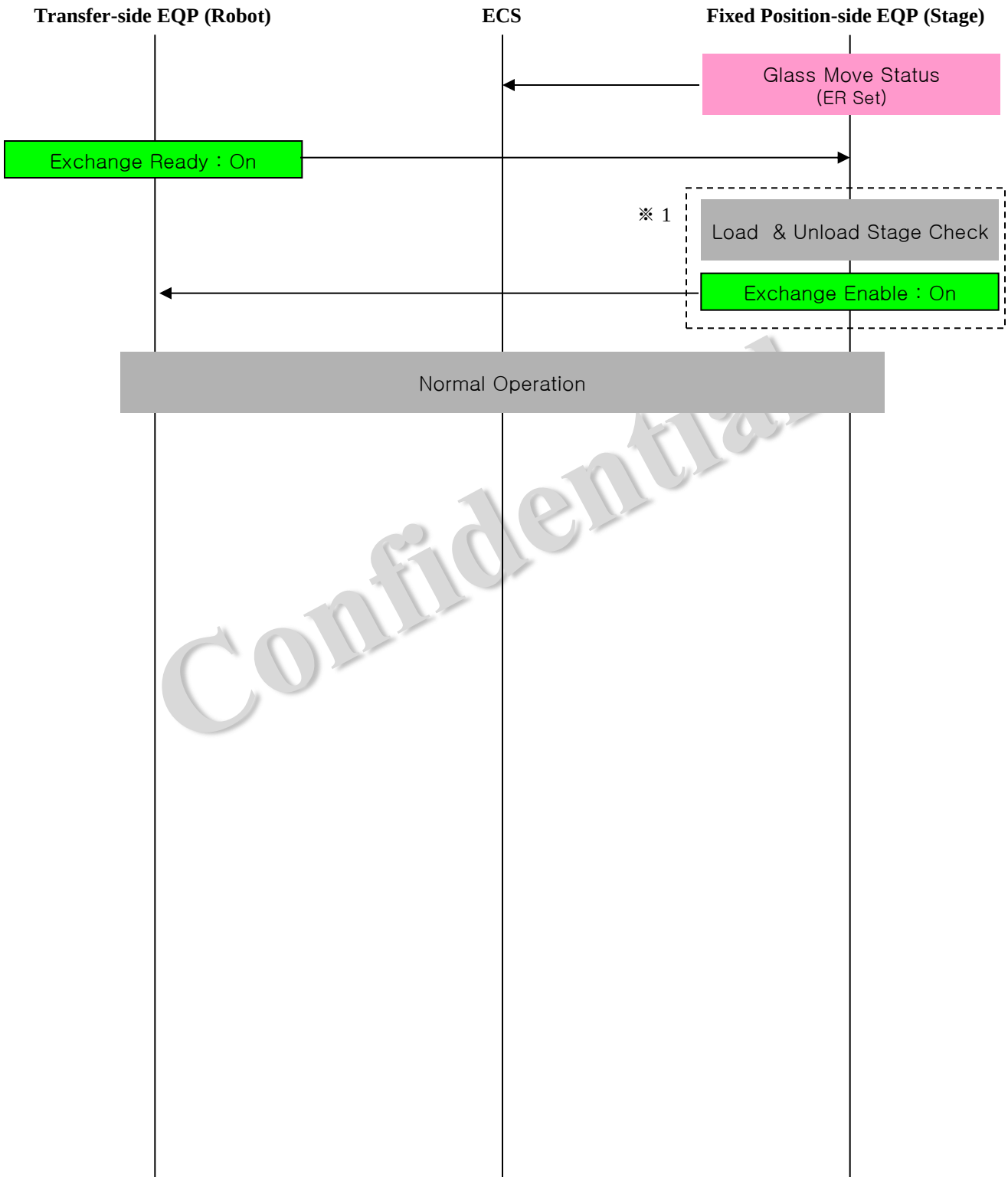
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- Case ⓑ : Glass Data Send after 'UR'



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※ 1: To set before according to a physical Interlock condition.

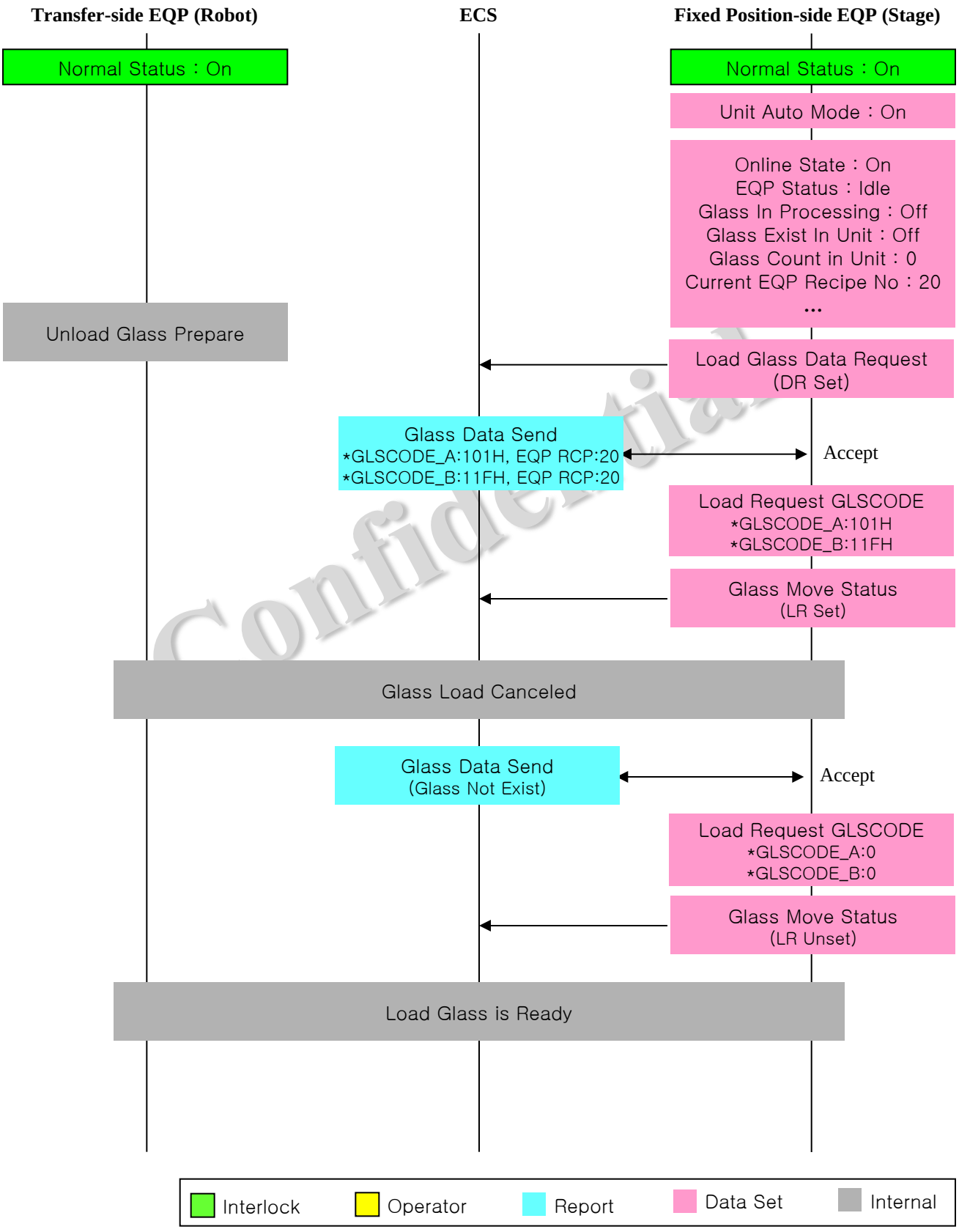
Interlock | Operator | Report | Data Set | Internal

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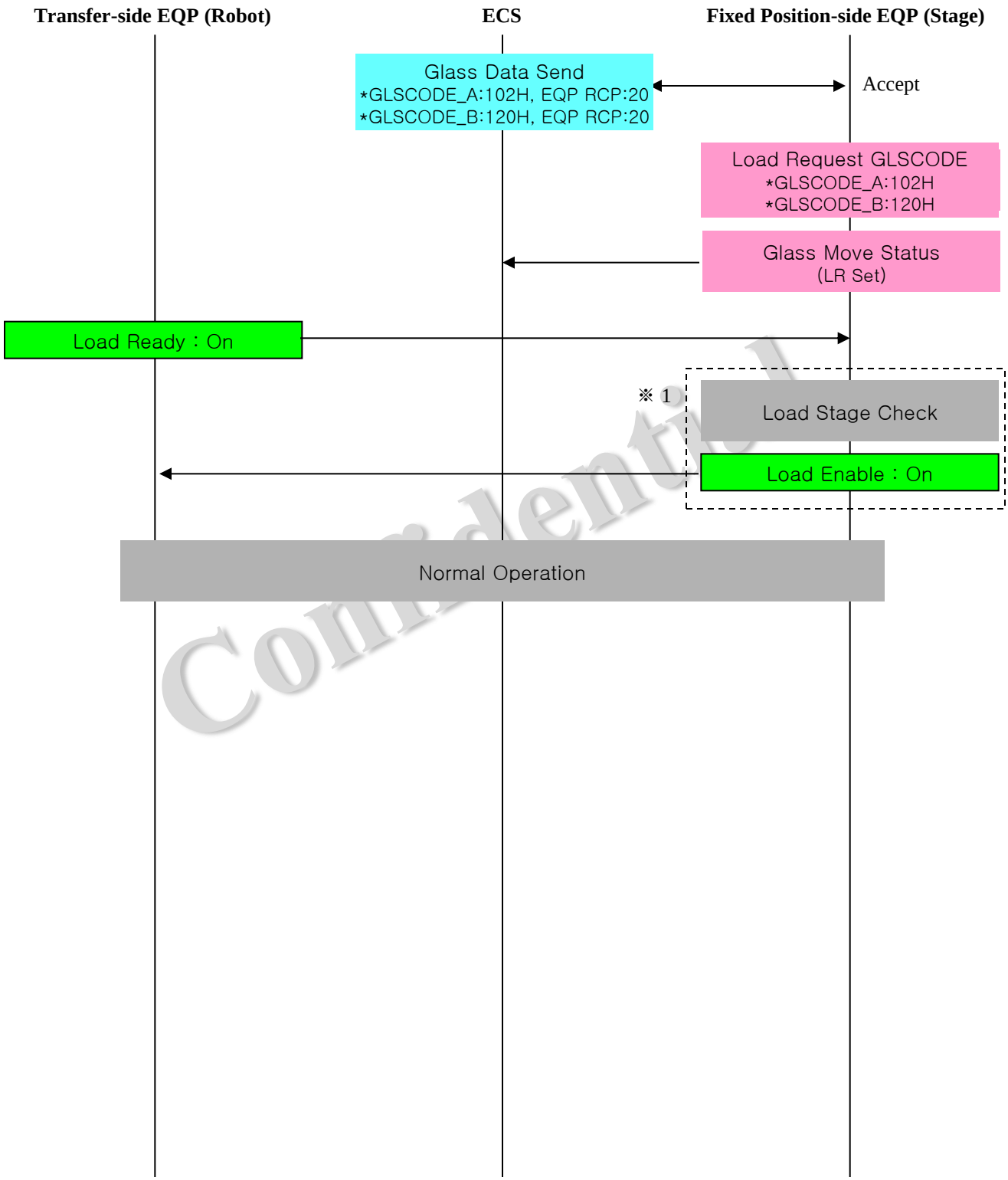
7.4.2 Change From Exist to Not Exist (Stand Alone, Robot → Stage)

1) Unload Glass Not Exist



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※ 1: To set before according to a physical Interlock condition.

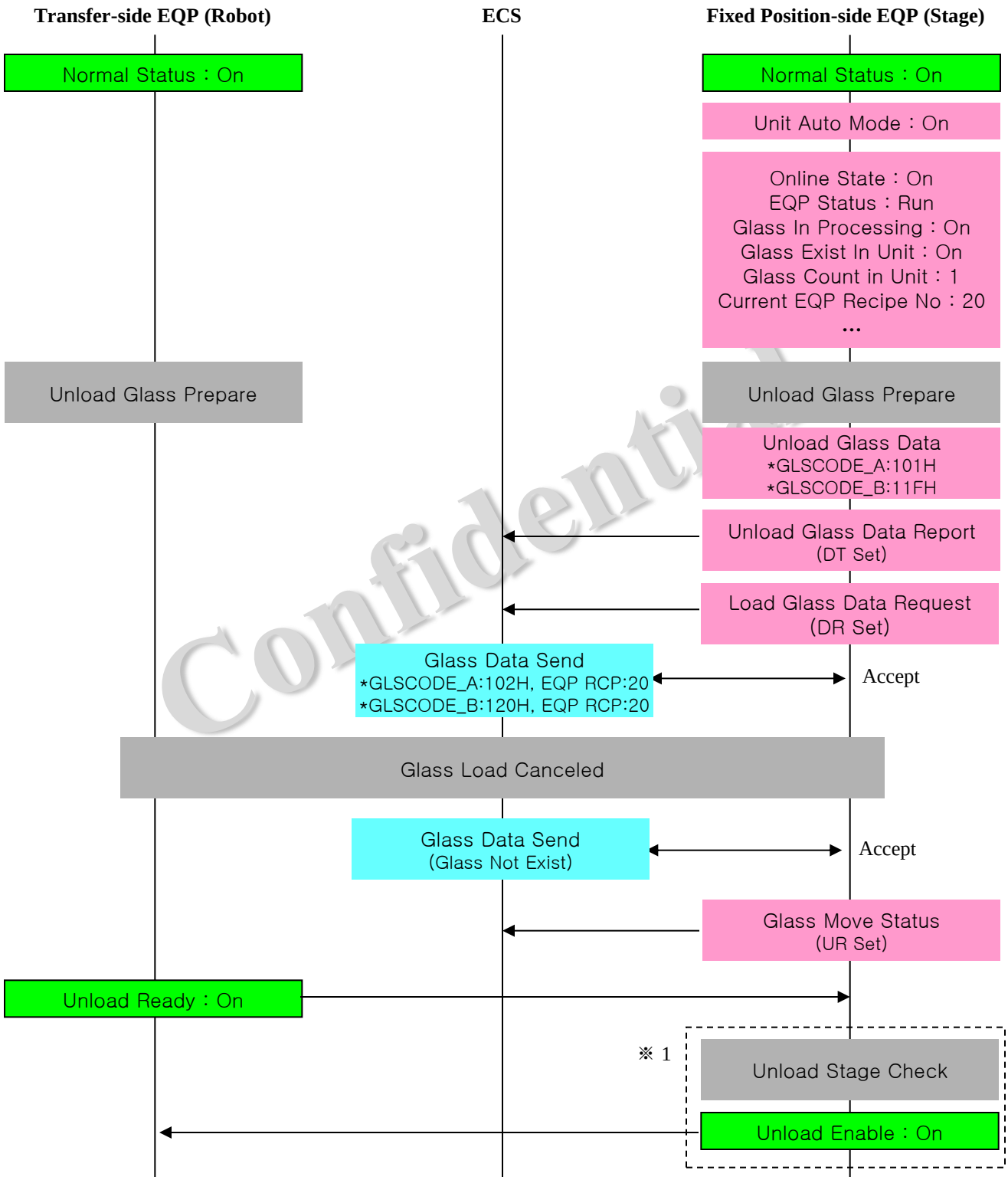
Interlock | Operator | Report | Data Set | Internal

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2) Unload Glass Exist

- Case ⓐ : Glass Data Send before 'ER'



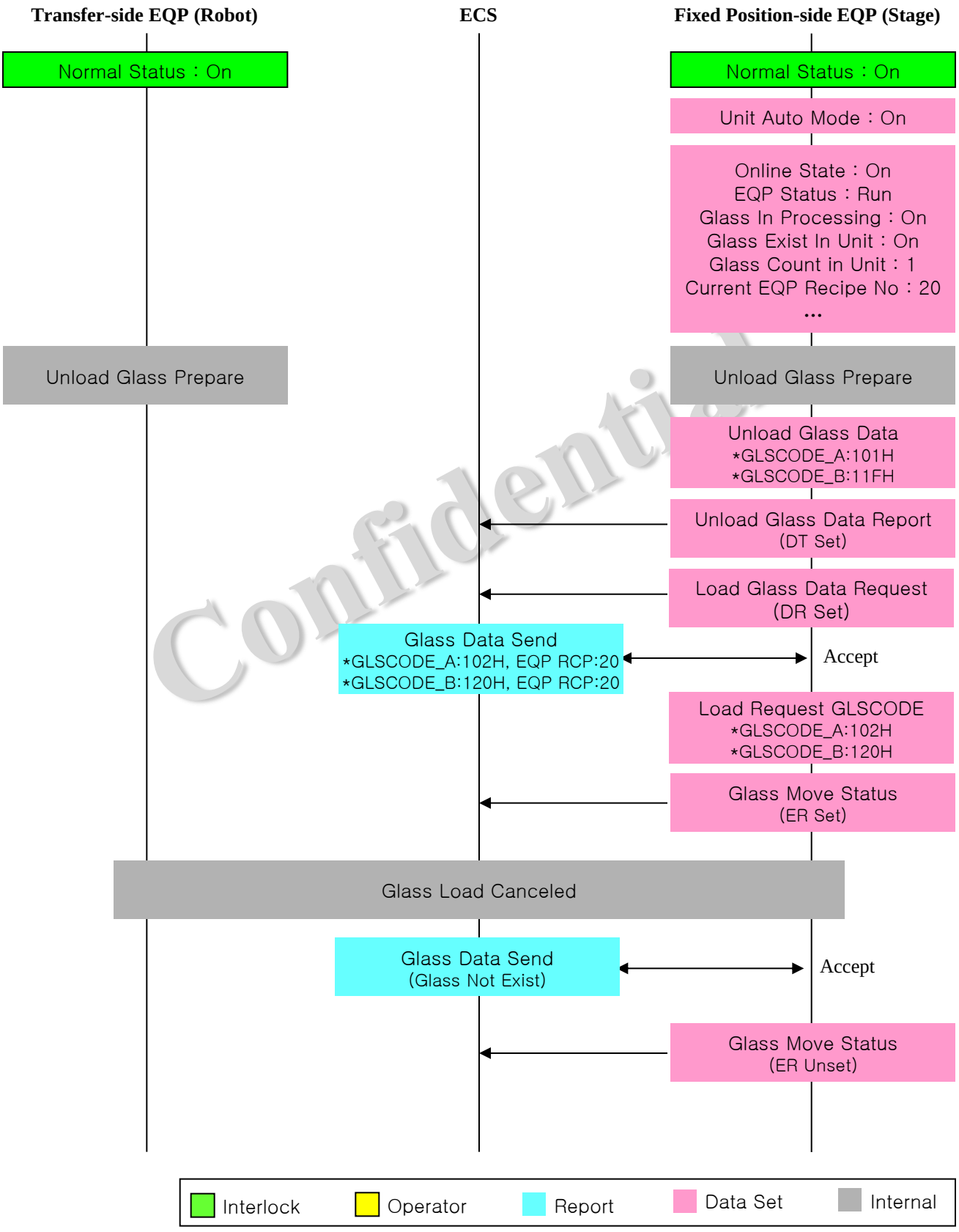
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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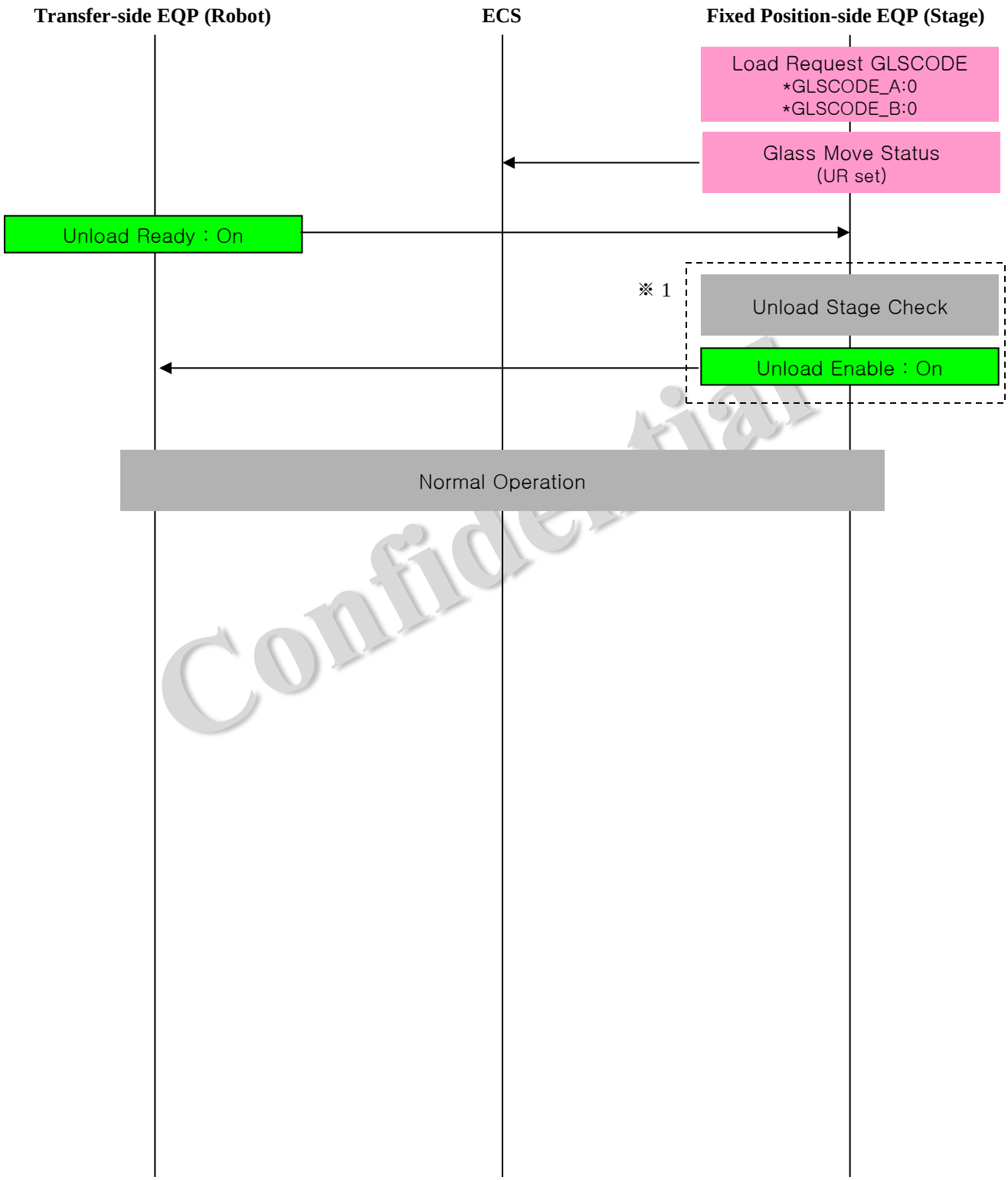
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- Case ⓑ : Glass Data Send after 'ER'



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※ 1: To set before according to a physical Interlock condition.

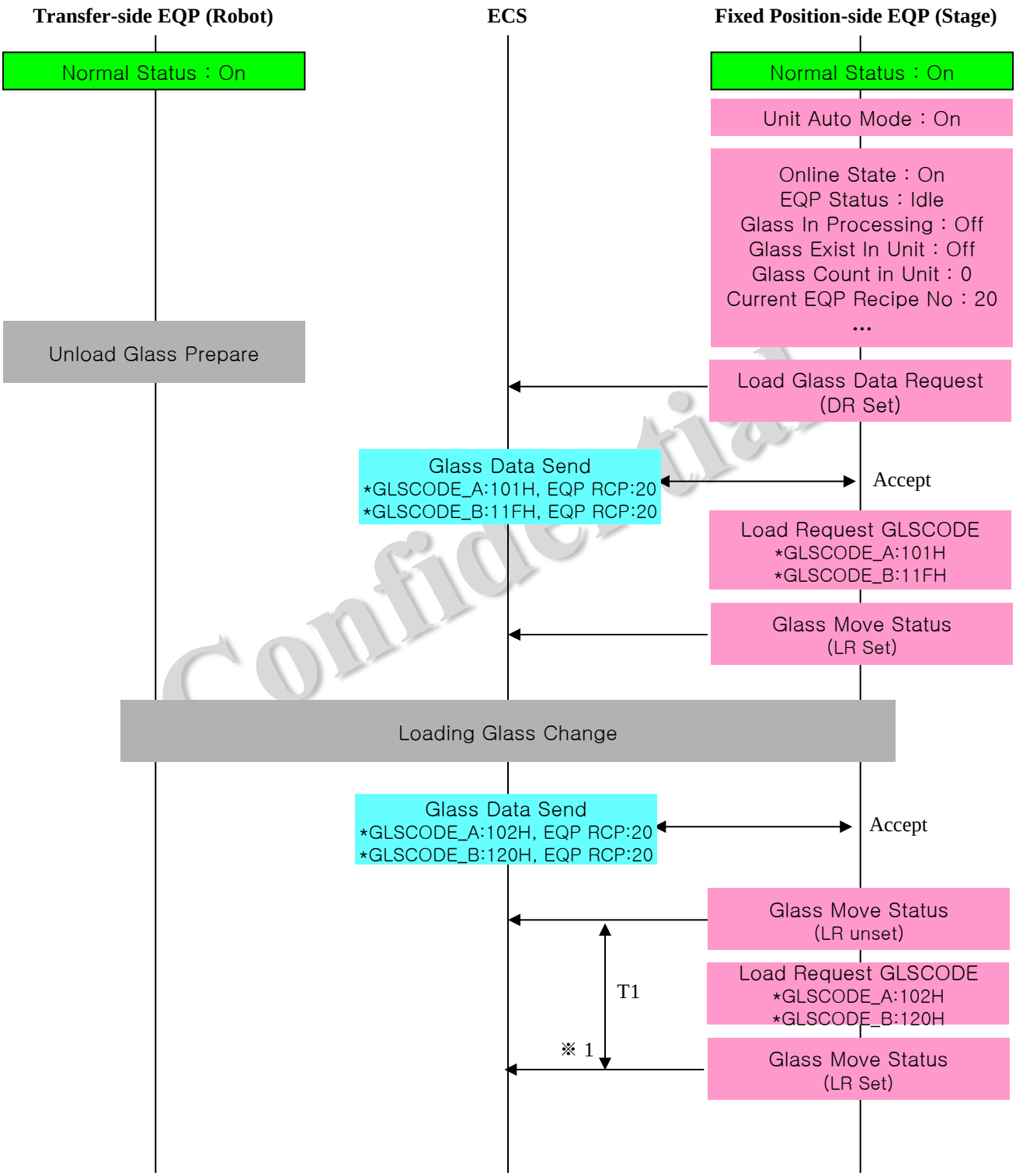
Interlock | Operator | Report | Data Set | Internal

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7.4.3 Change From Exist to Exist (Stand Alone, Robot → Stage)

1) Unload Glass Not Exist

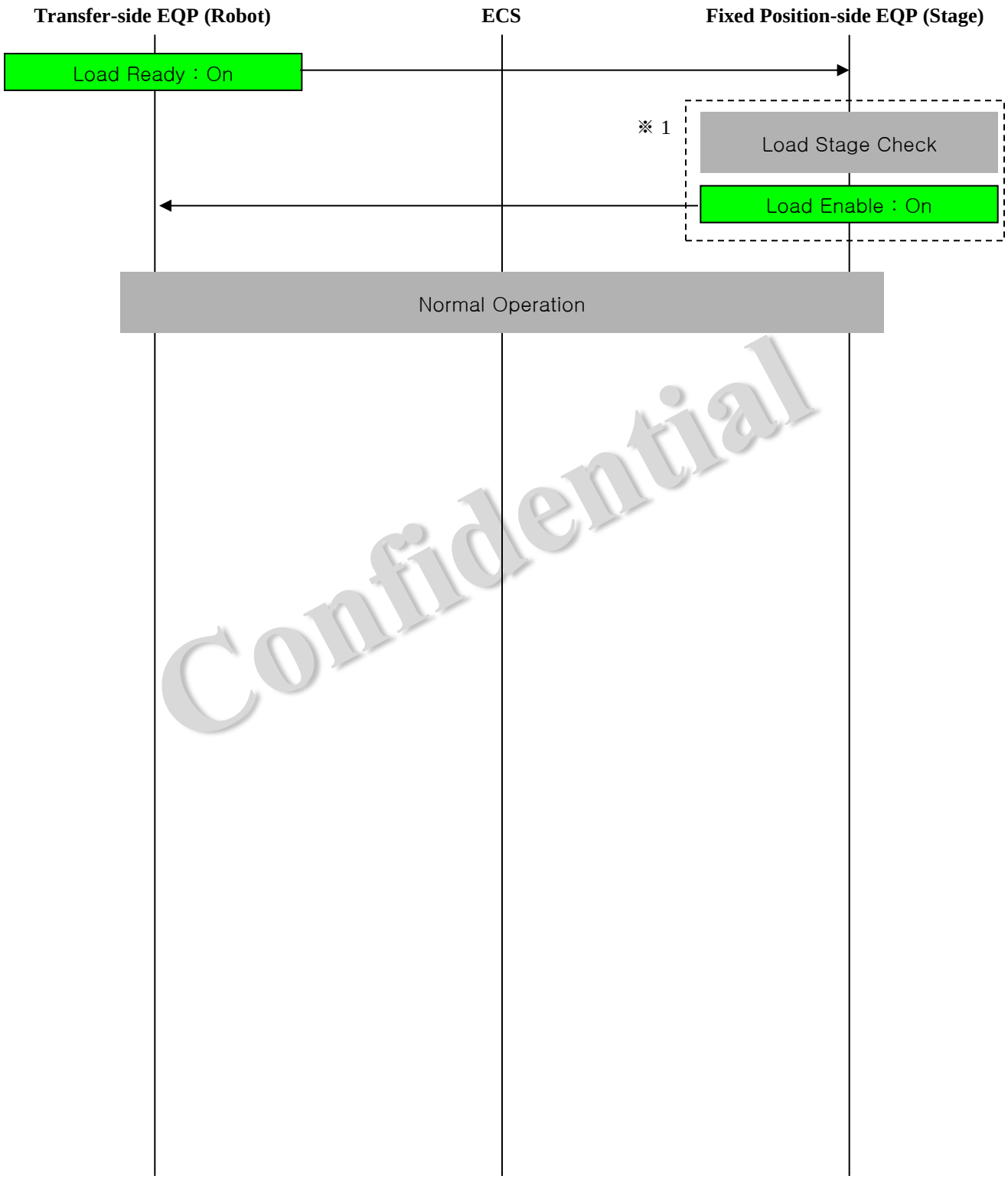


※ 1 If Glass Move Request (LR) is Unset, that signal should keep One Sec or more.
This time value should be changeable by parameter

Interlock | Operator | Report | Data Set | Internal

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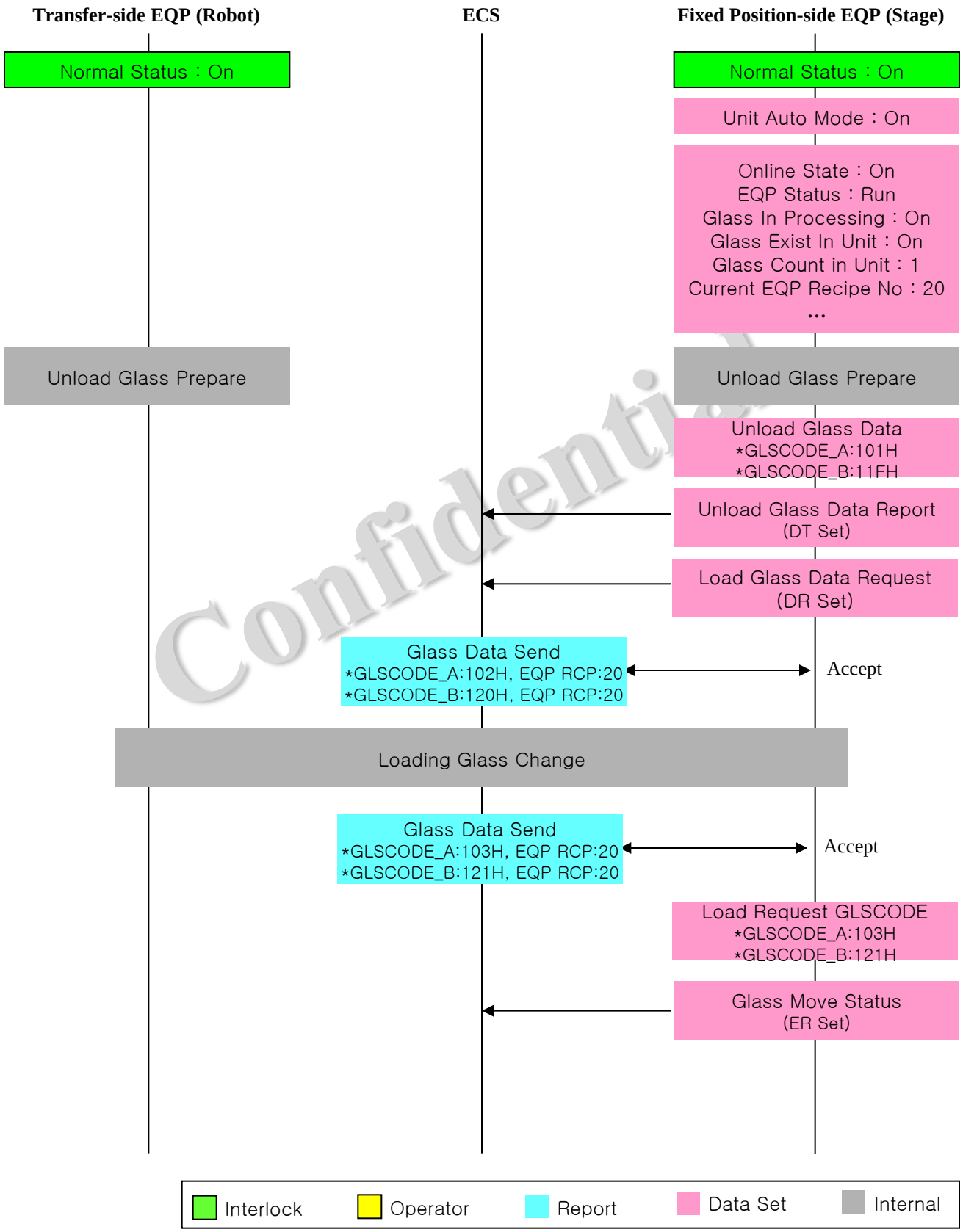
※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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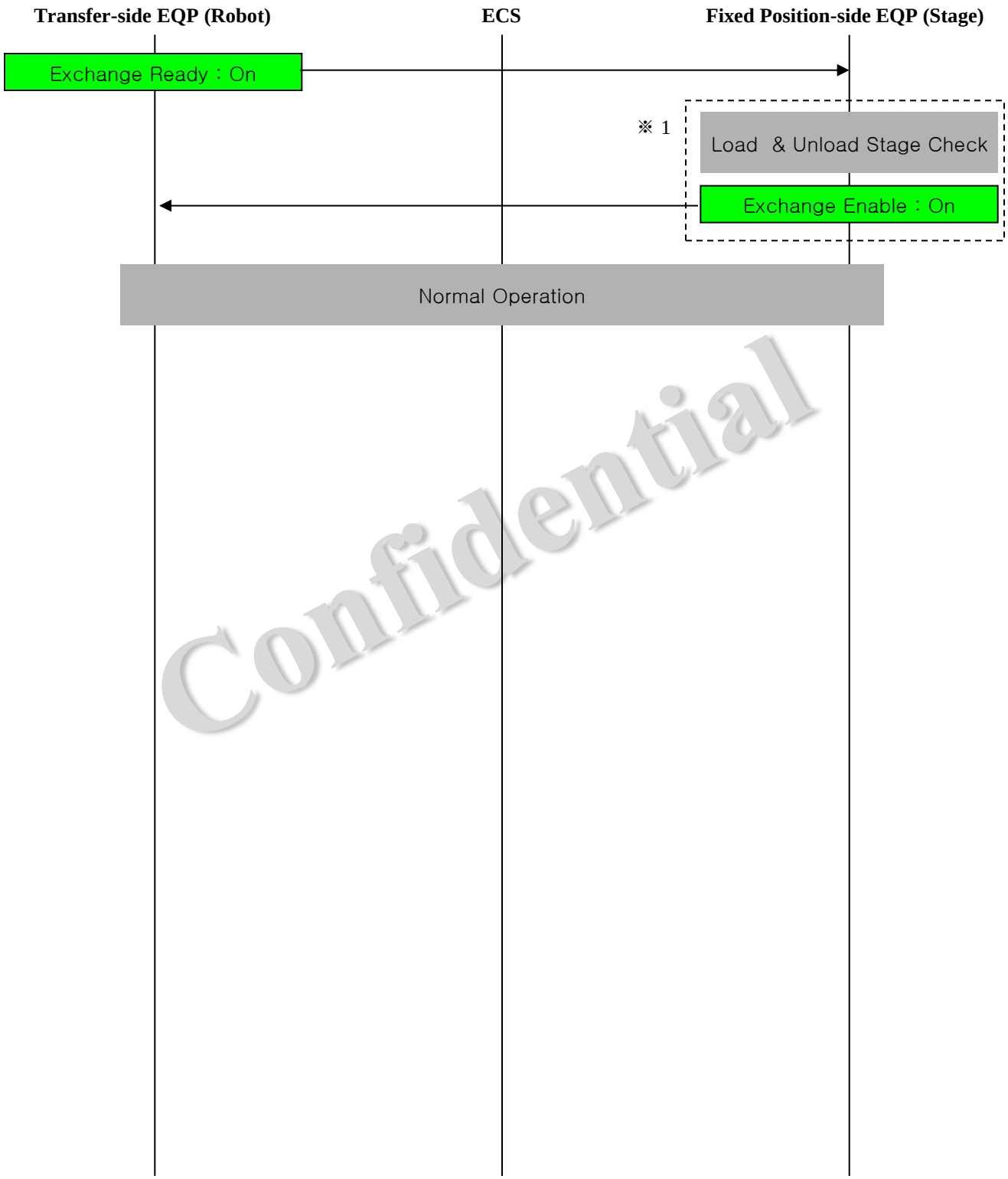
2) Unload Glass Exist

- Case ⓐ : Glass Data Send before 'ER'



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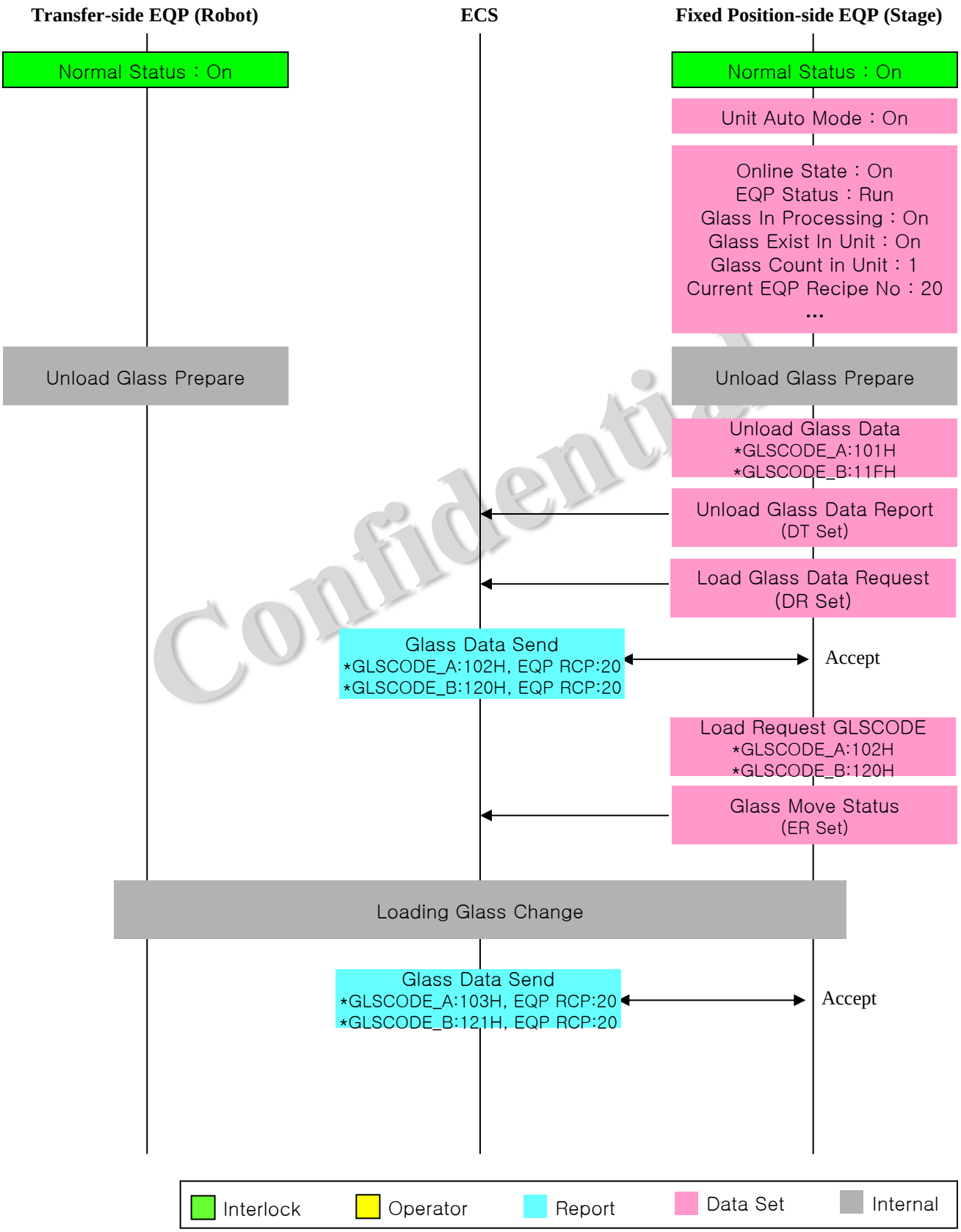
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※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

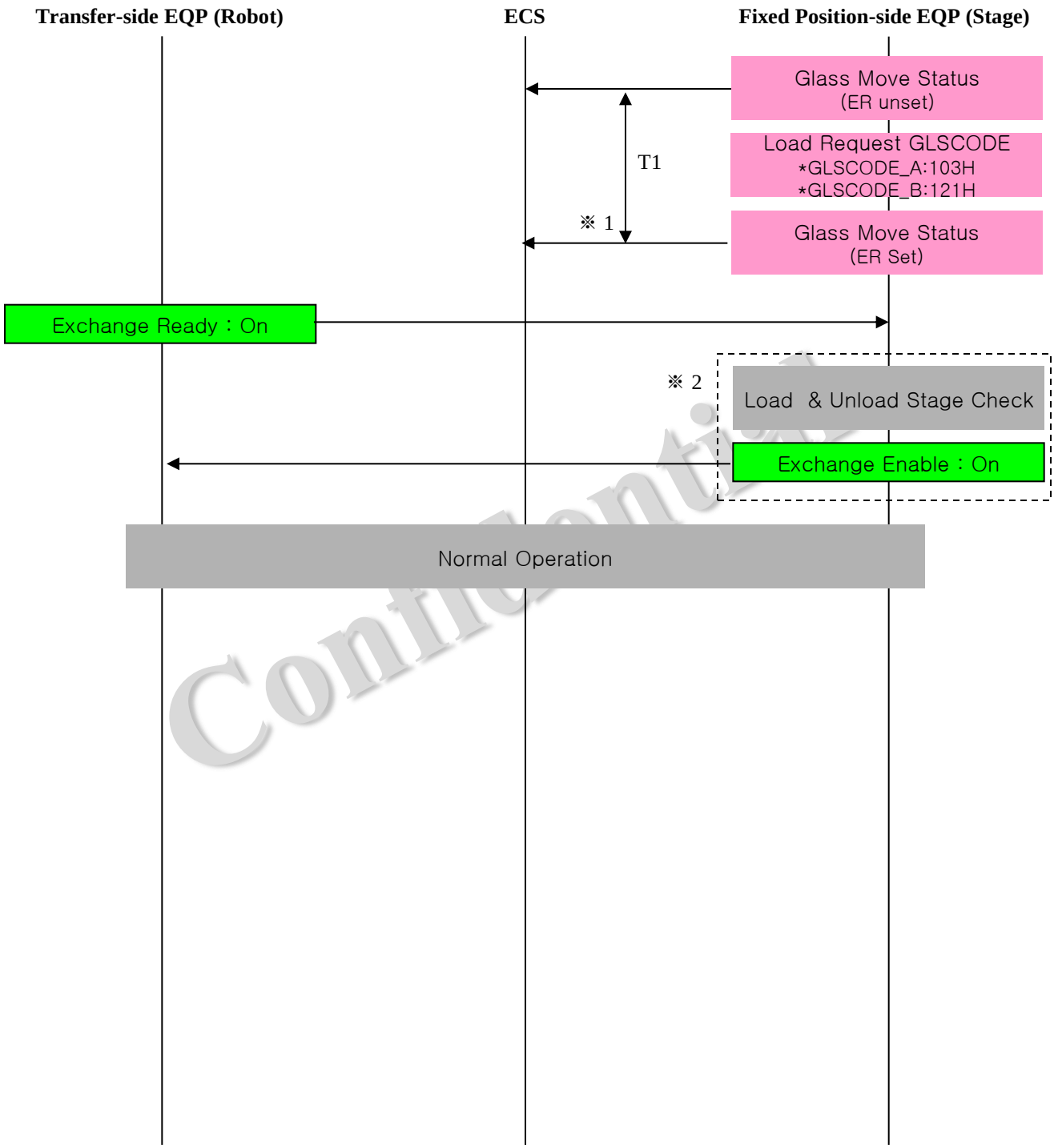
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- Case ⓑ : Glass Data Send after 'ER'



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※ 1 If Glass Move Request (ER) is Unset, that signal should keep One Sec or more.
This time value should be changeable by parameter
2: To set before according to a physical Interlock condition.

Interlock Operator Report Data Set Internal

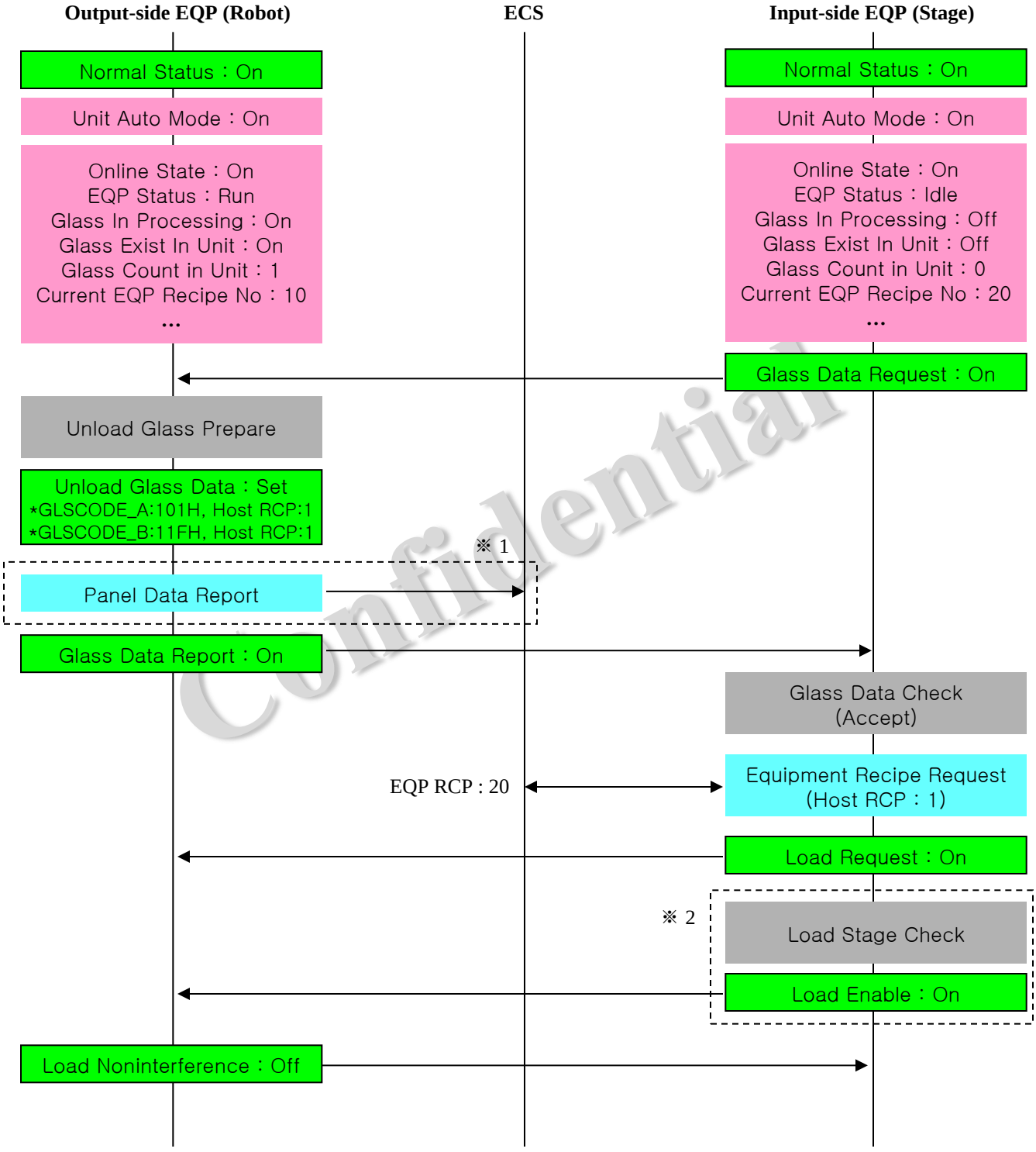
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8. Normal Operation

8.1 Glass Transfer Operation

8.1.1 Glass Transfer Operation (Process Inline, Robot → Stage)

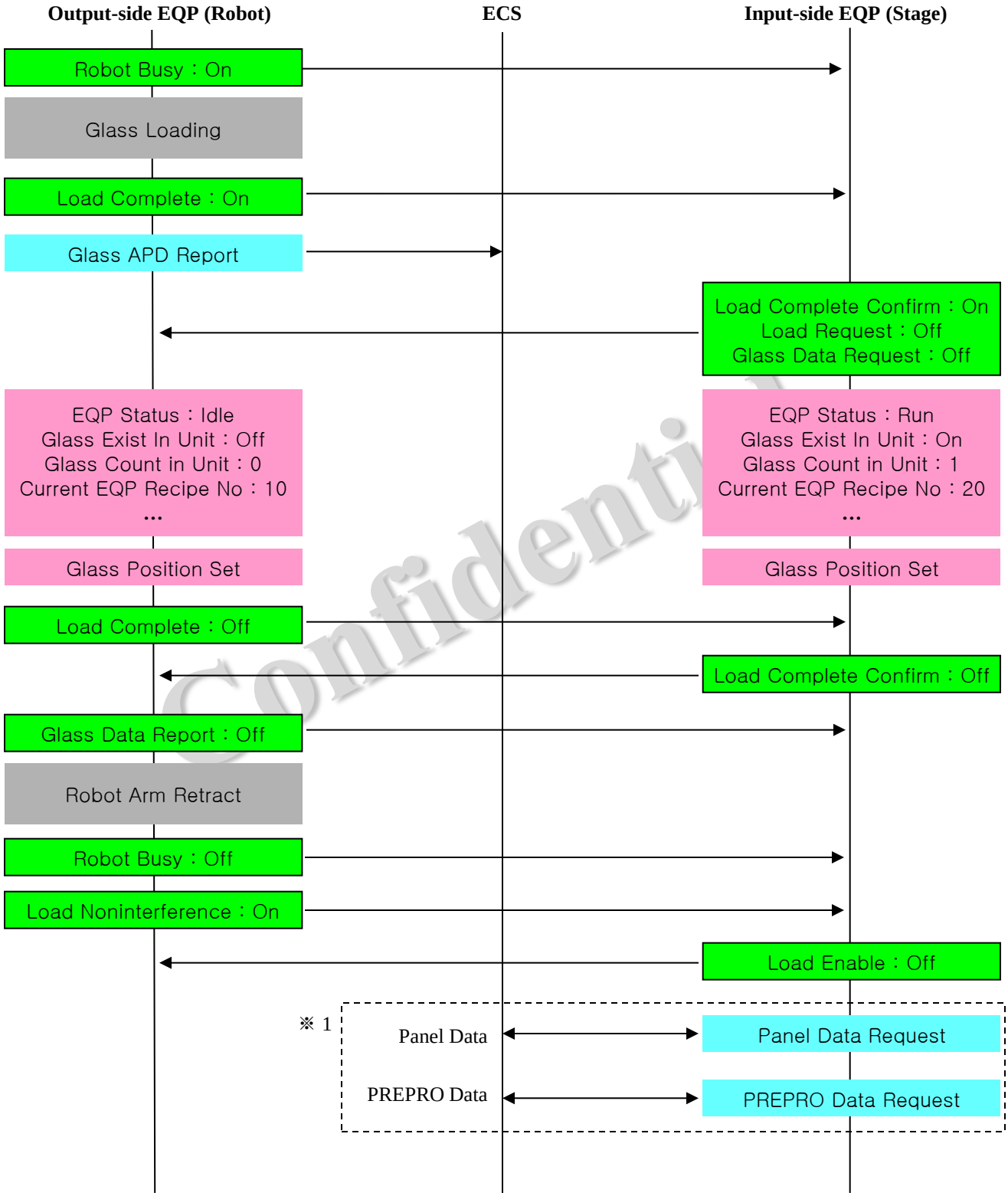


※ 1: Relative Equipment only.
※ 2: To set before according to a physical Interlock condition.



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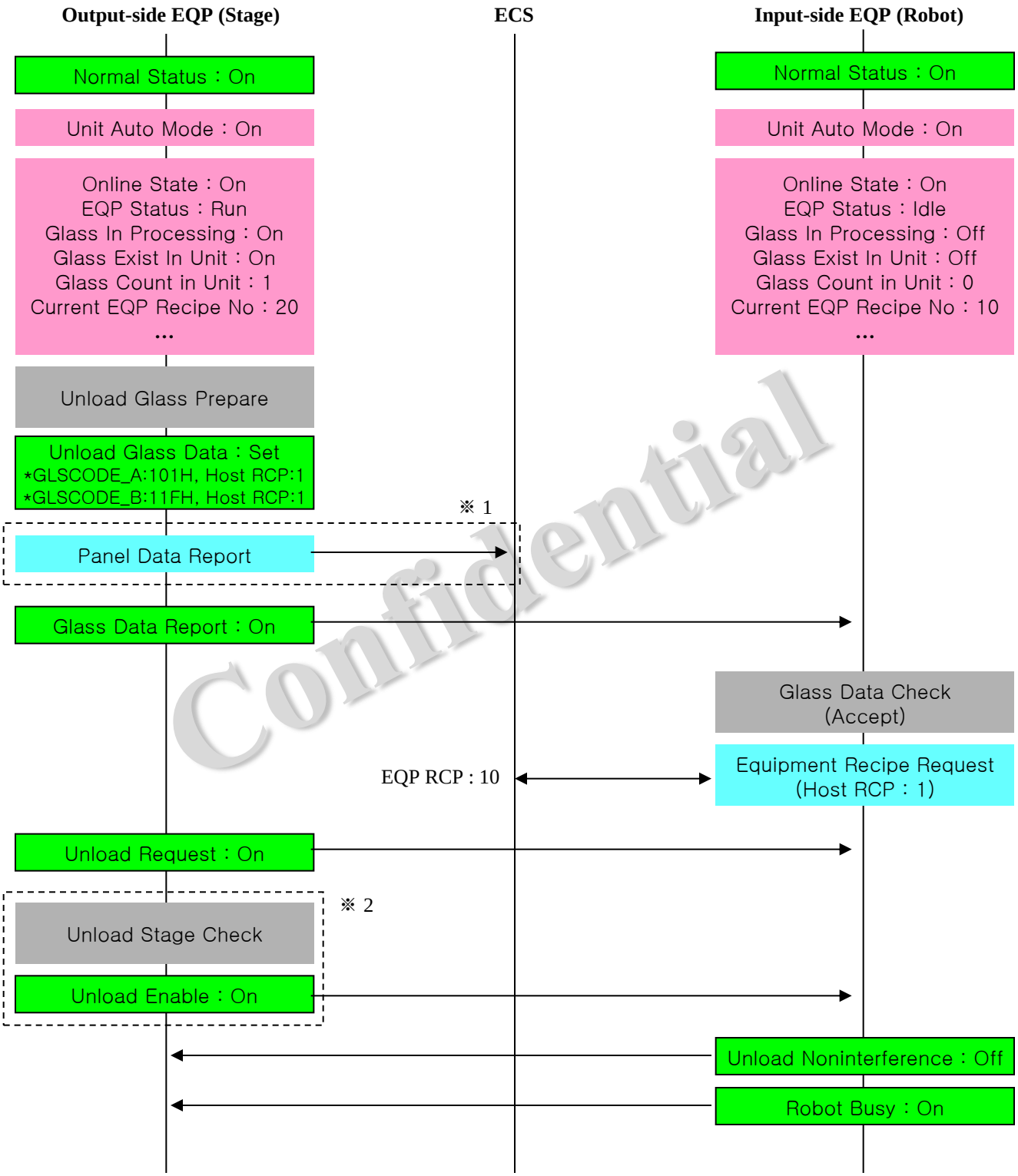
※ 1: Relative Equipment only.

Interlock | Operator | Report | Data Set | Internal

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8.1.2 Glass Transfer Operation (Process Inline, Stage → Robot)

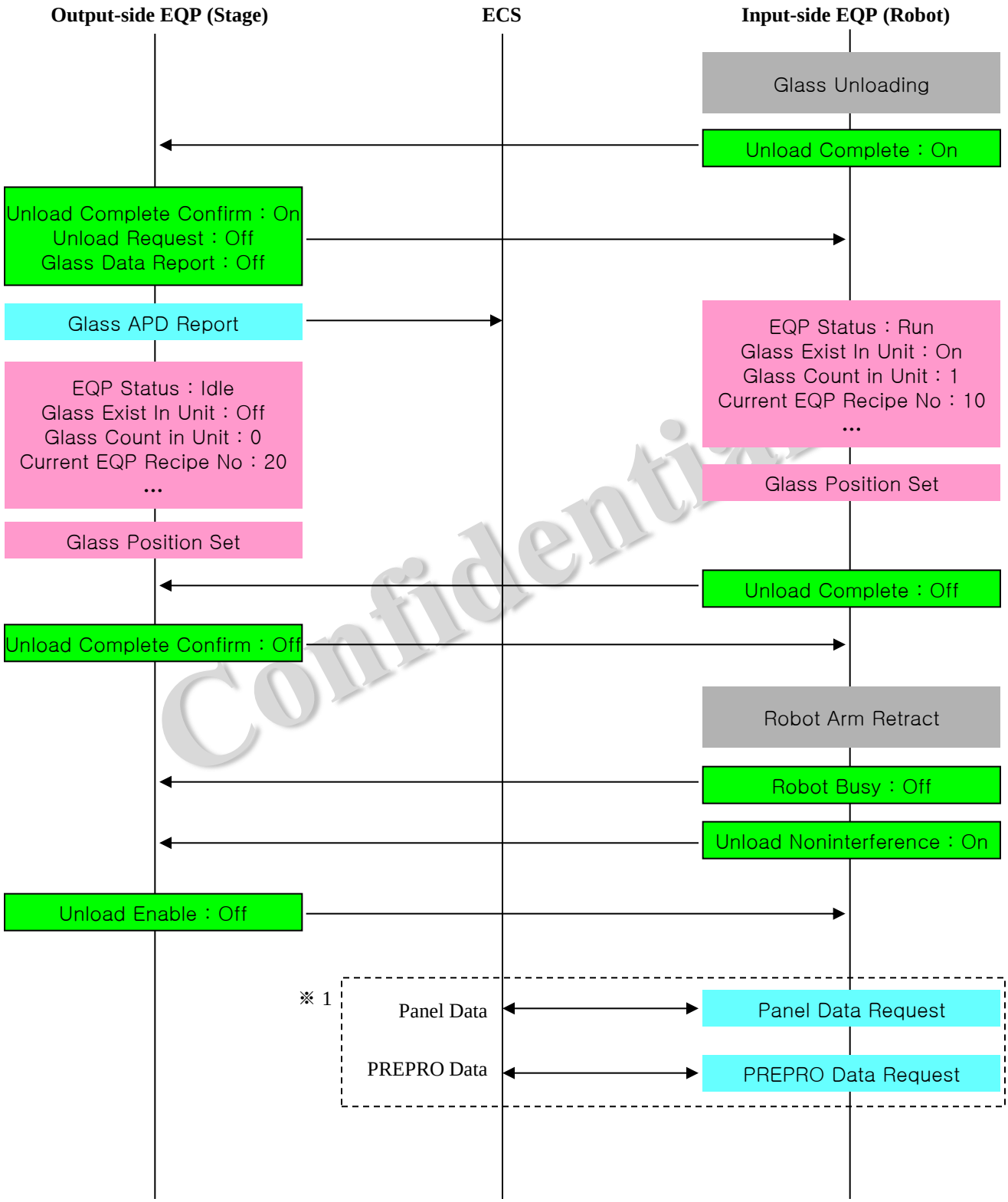


※ 1: Relative Equipment only.
※ 2: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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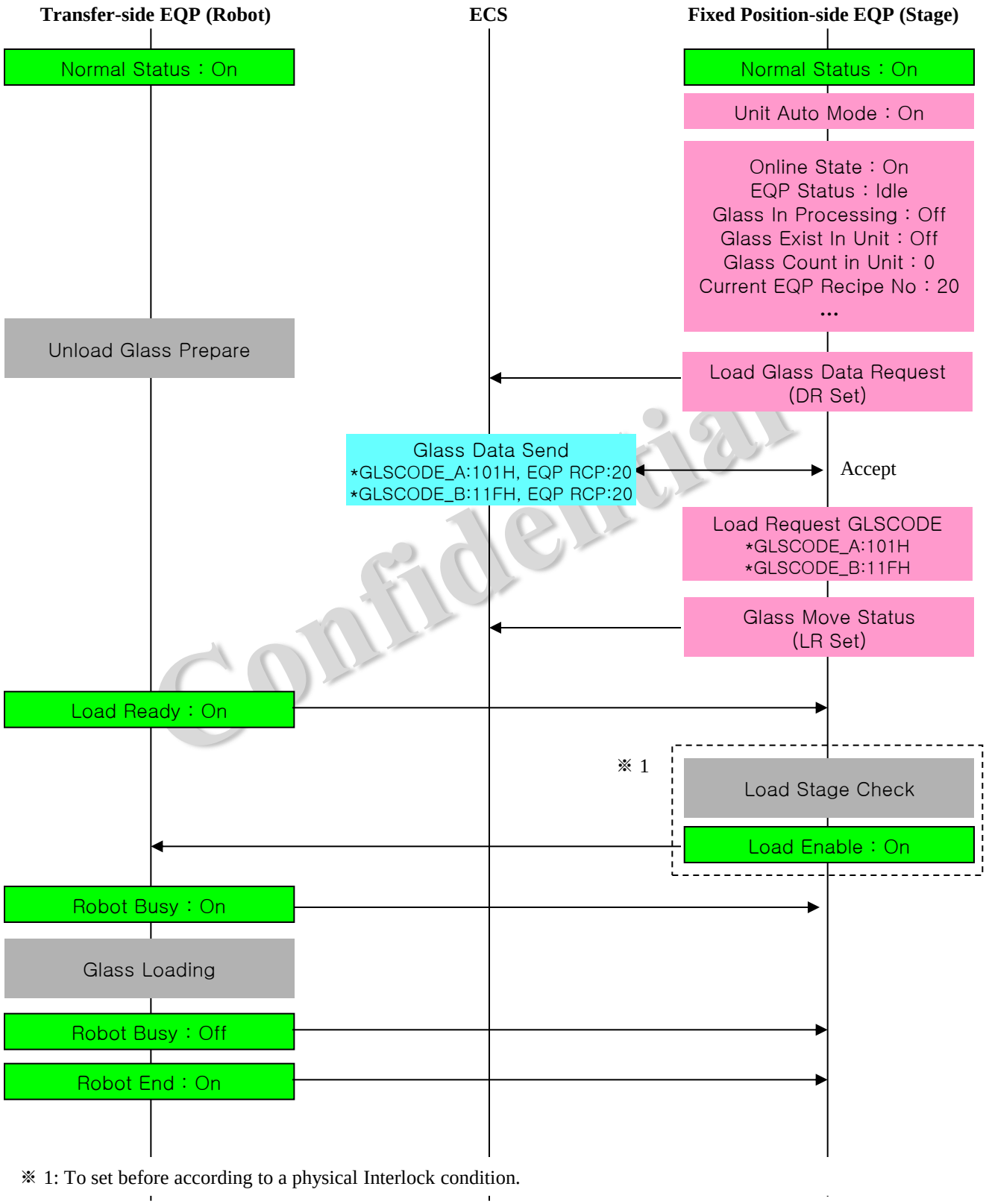
※ 1: Relative Equipment only.

Interlock | Operator | Report | Data Set | Internal

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8.1.3 Glass Transfer Operation (Stand Alone, Robot → Stage)

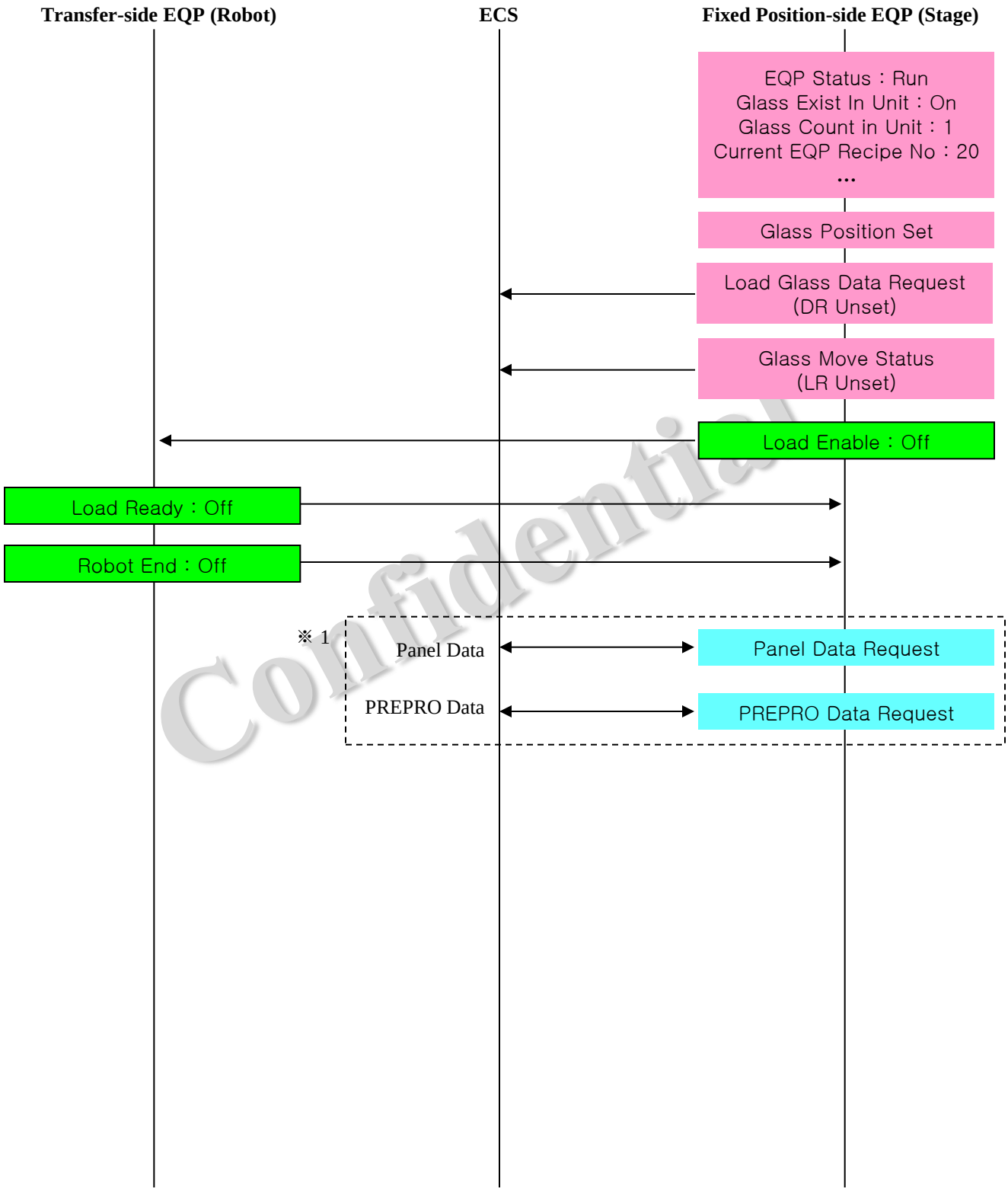


※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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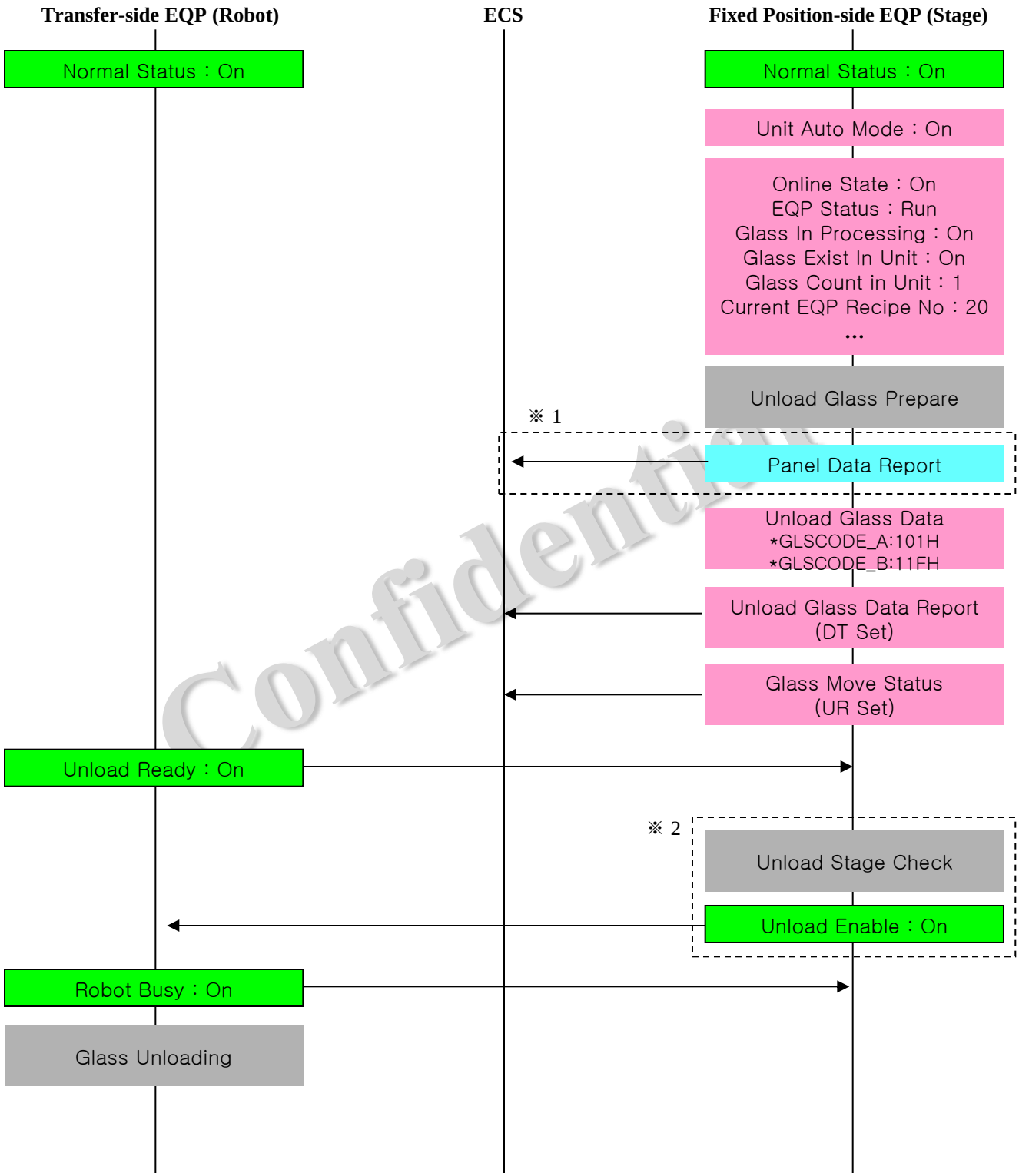
※ 1: Relative Equipment only.

Interlock | Operator | Report | Data Set | Internal

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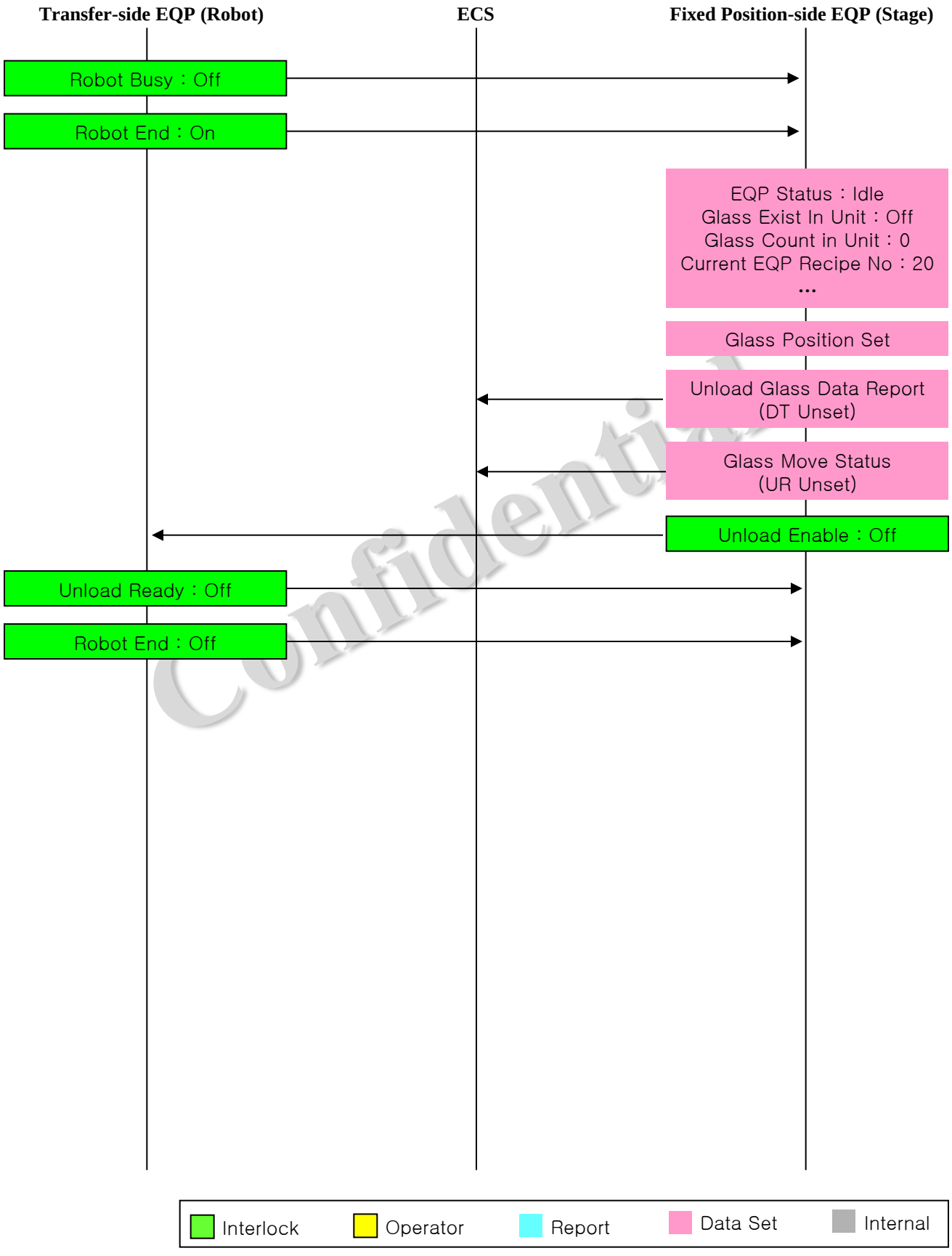
8.1.4 Glass Transfer Operation (Stand Alone, Robot ← Stage)



※ 1: Relative Equipment only.
※ 2: To set before according to a physical Interlock condition.

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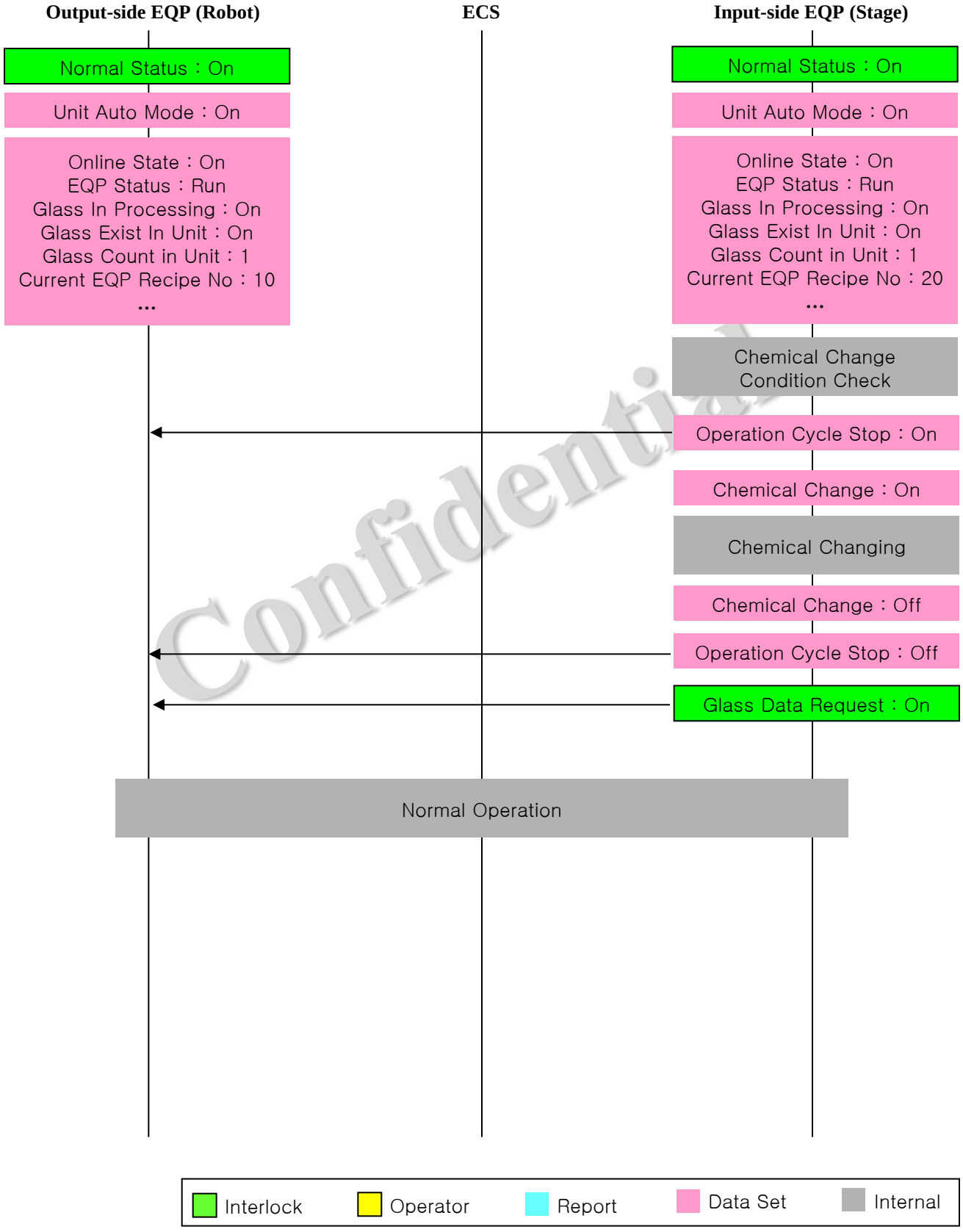


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8.2 Chemical Change Sequence

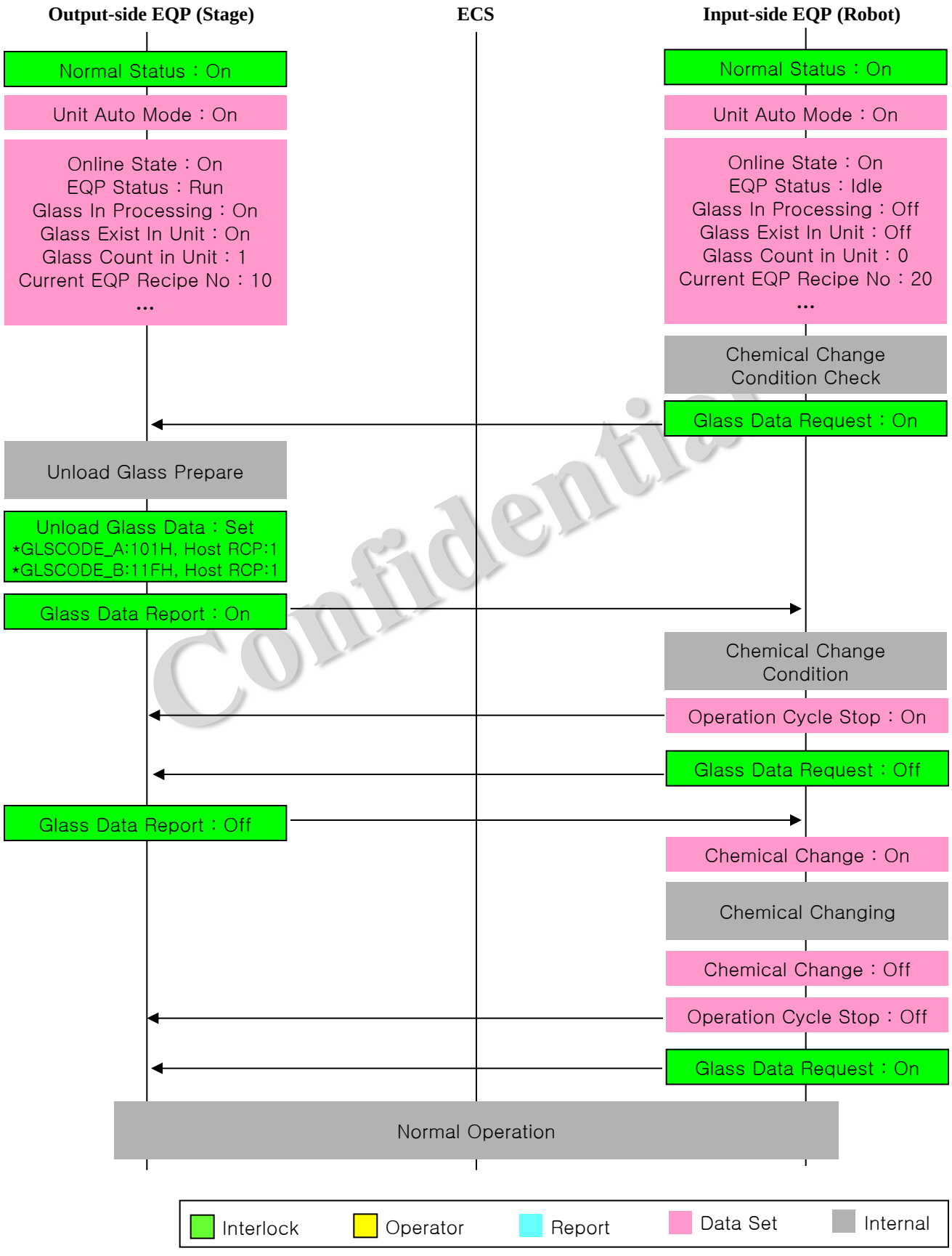
82.1 Chemical Change Before Glass Data Request (Process Inline, Robot → Stage)



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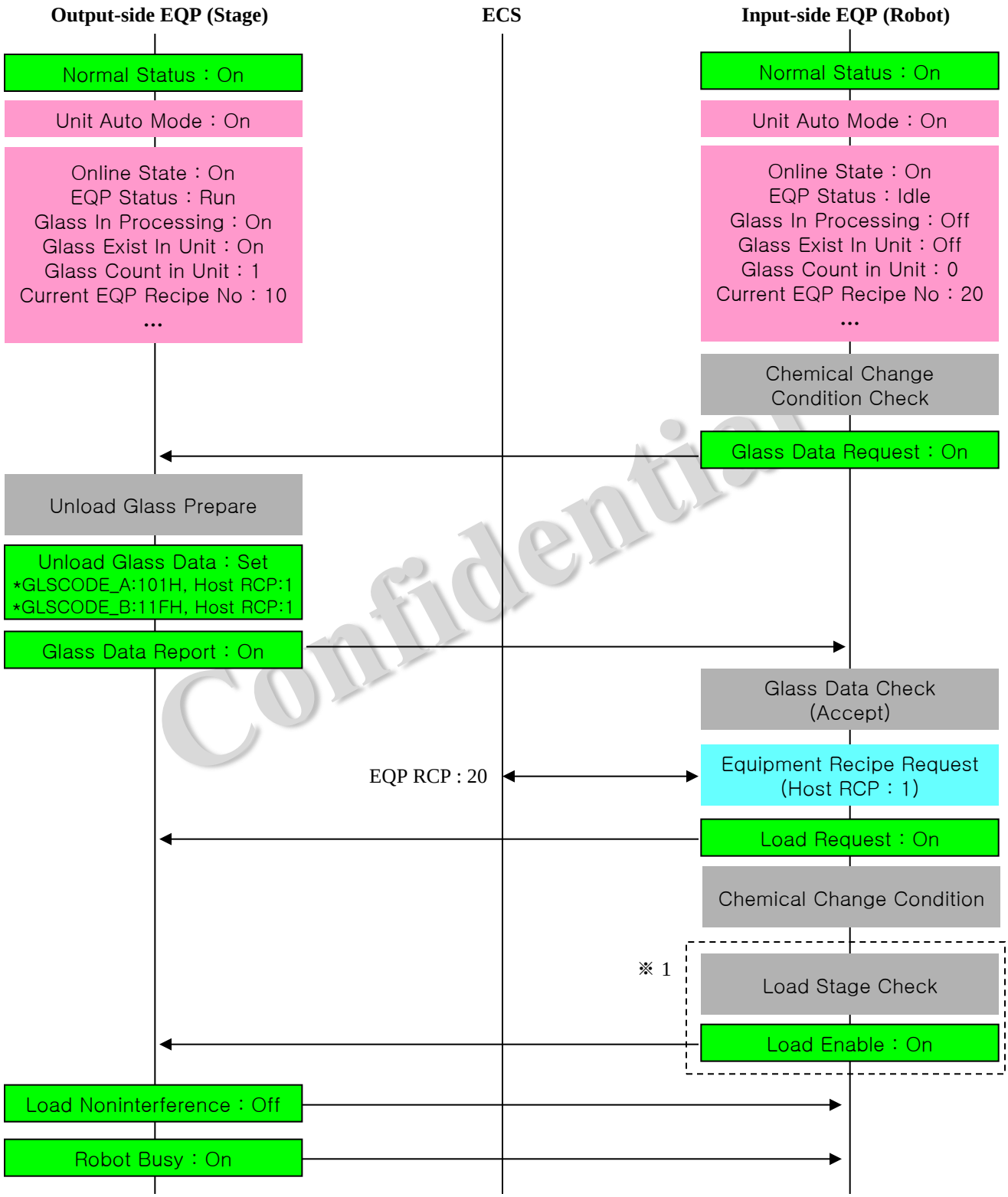
8.2.2 Chemical Change after Glass Data Request (Process Inline, Robot → Stage)



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8.2.3 Chemical Change after Load Request (Process Inline, Robot → Stage)

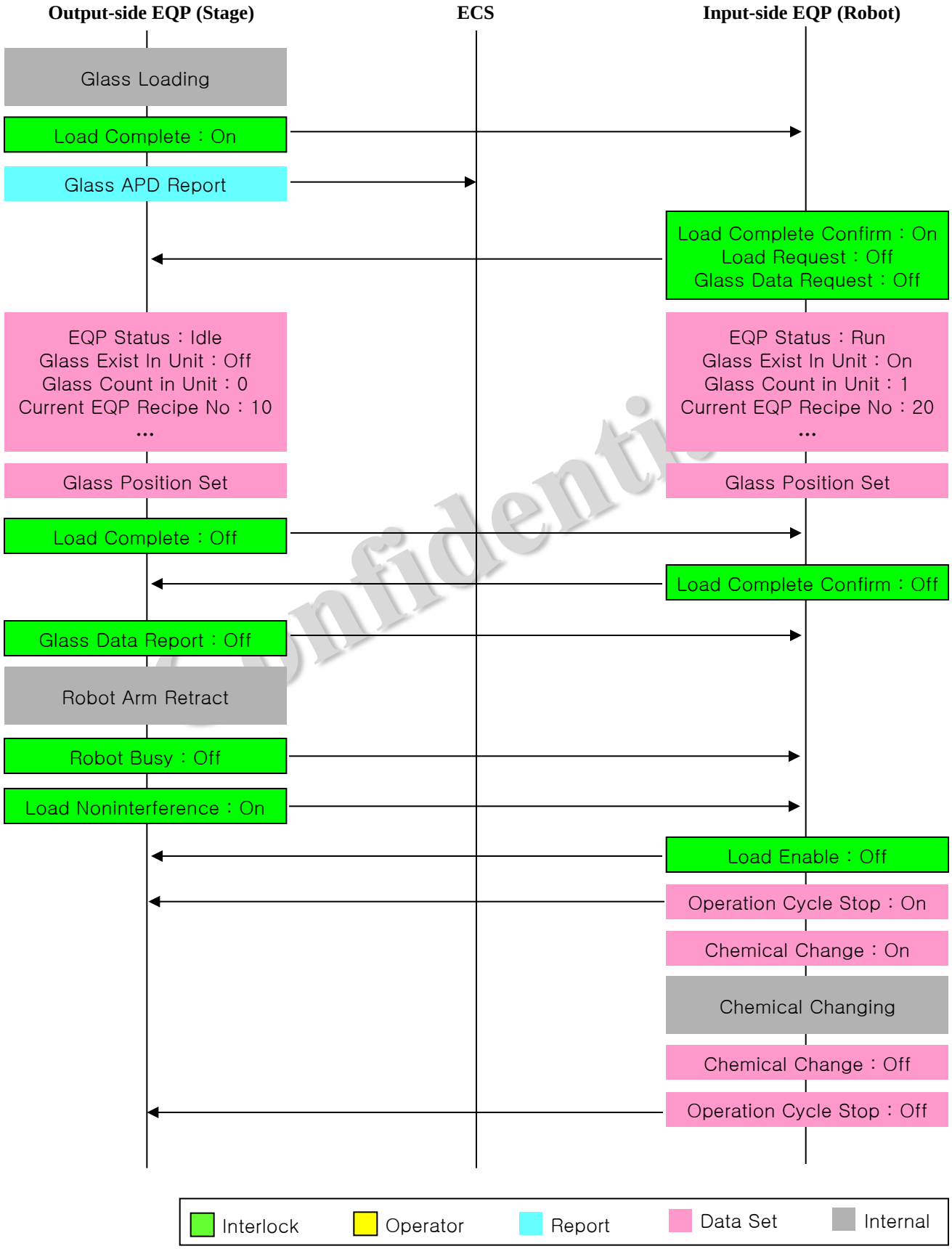


※ 1: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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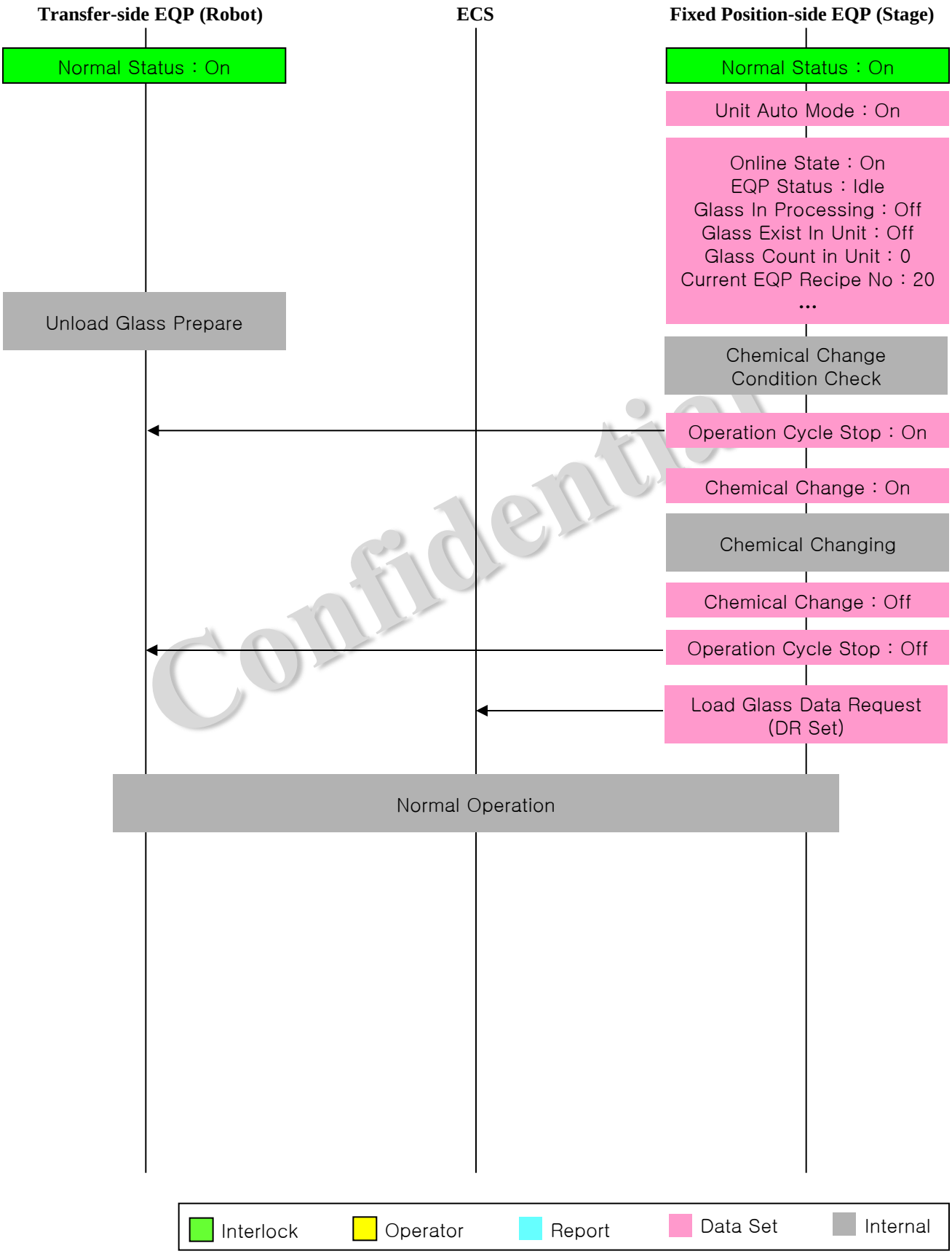
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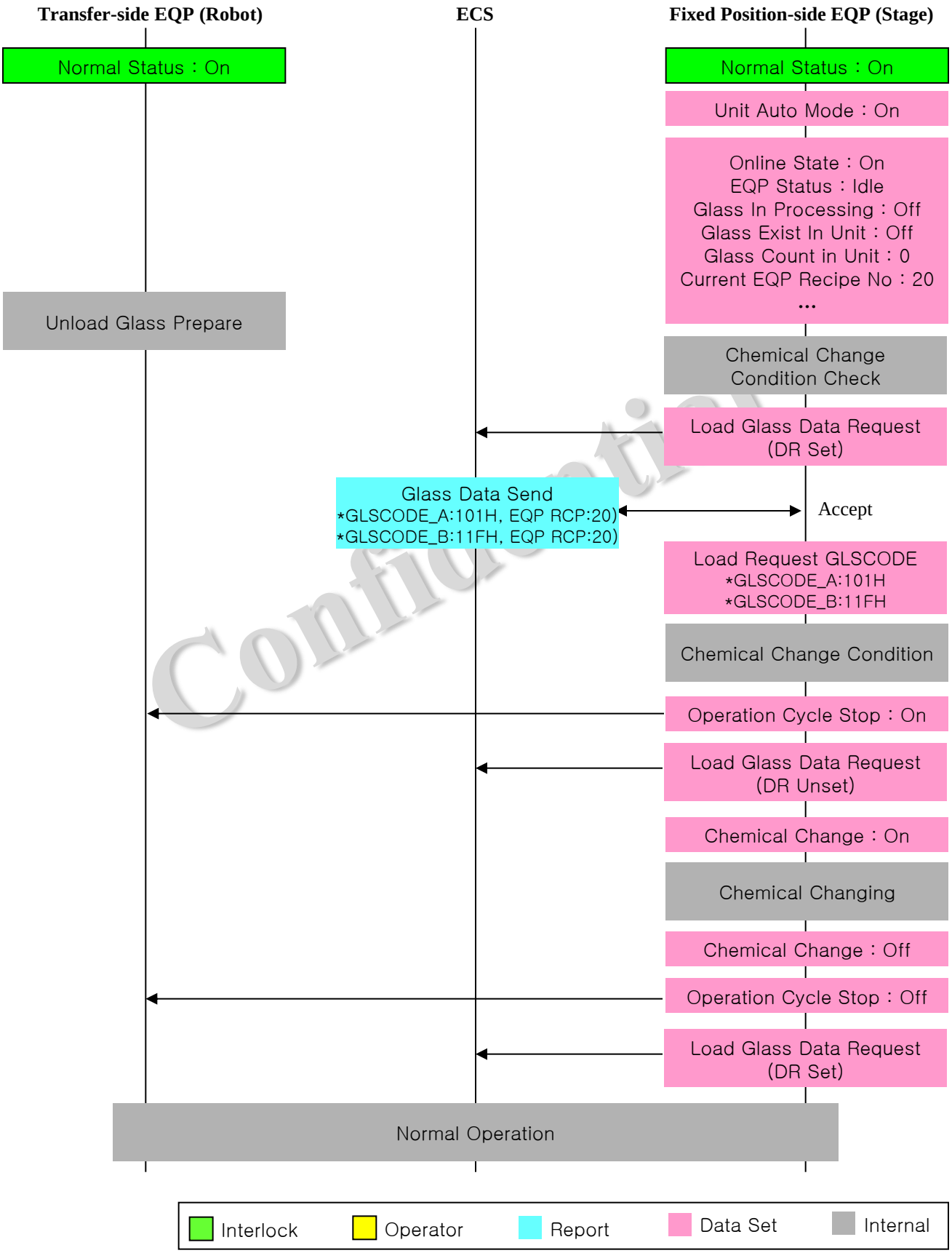
8.2.4 Chemical Change Before Glass Data Request (Stand Alone, Robot → Stage)



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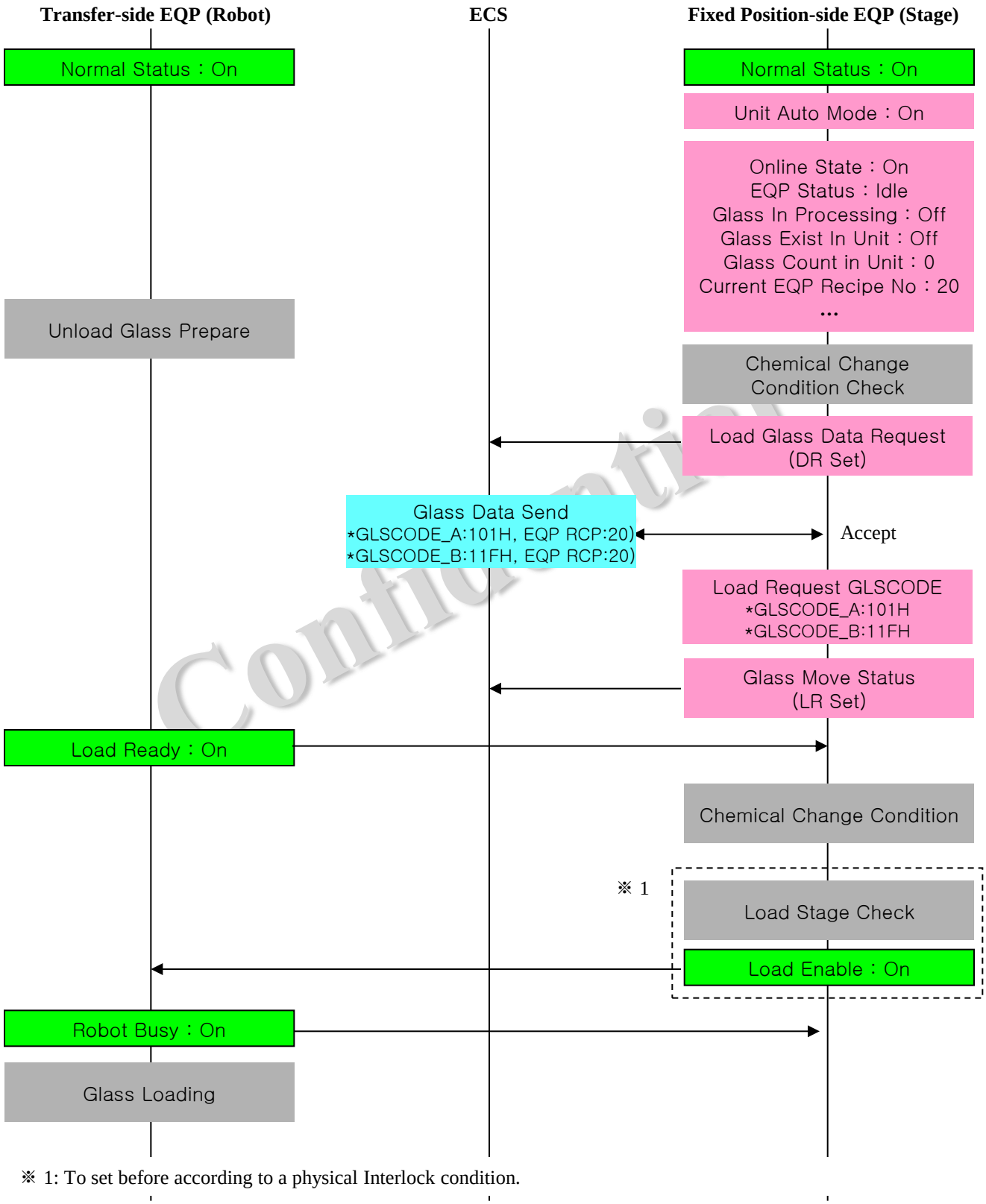
8.2.5 Chemical Change After Glass Data Request (Stand Alone, Robot → Stage)



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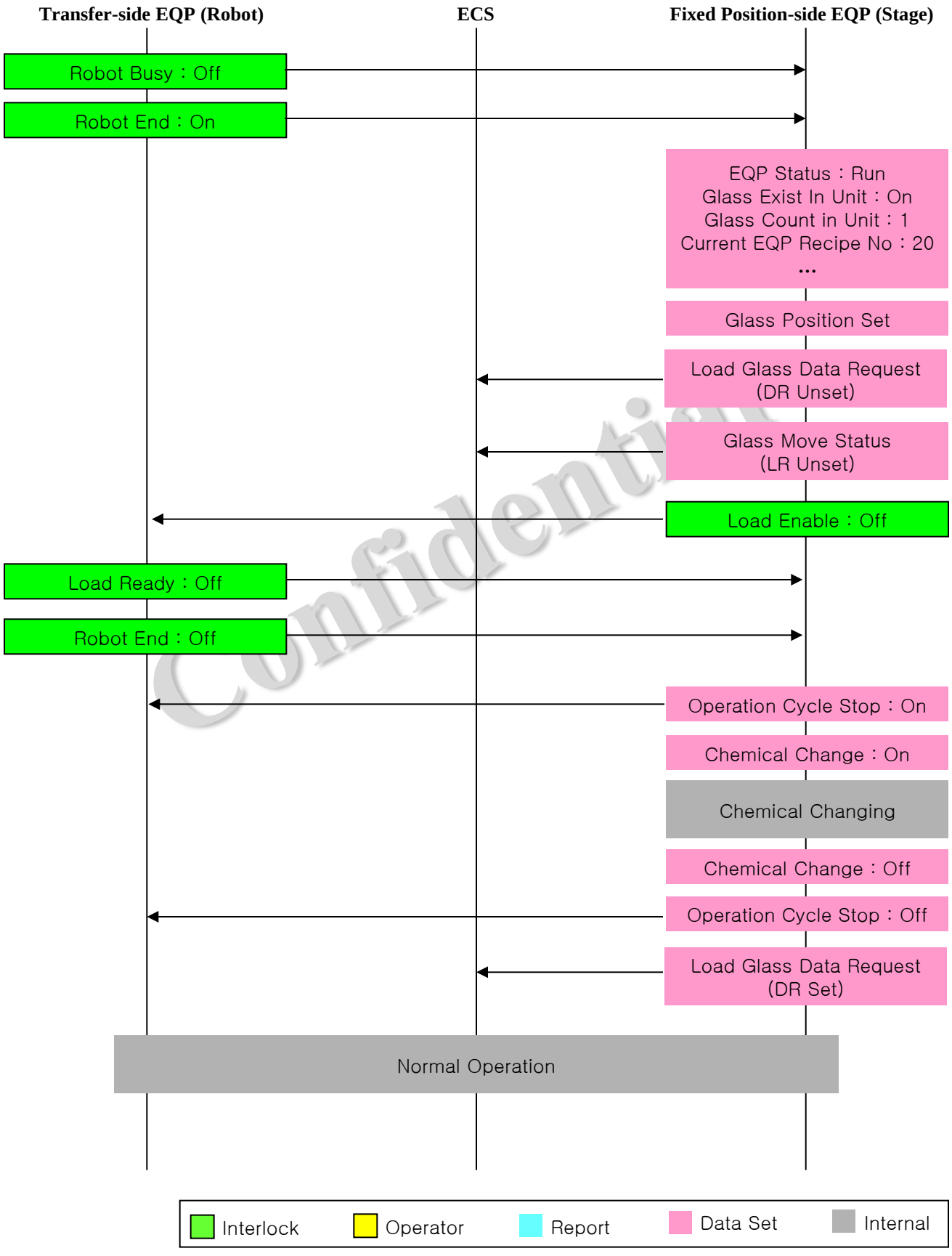
8.2.6 Chemical Change After Load Request (Stand Alone, Robot → Stage)



※ 1: To set before according to a physical Interlock condition.

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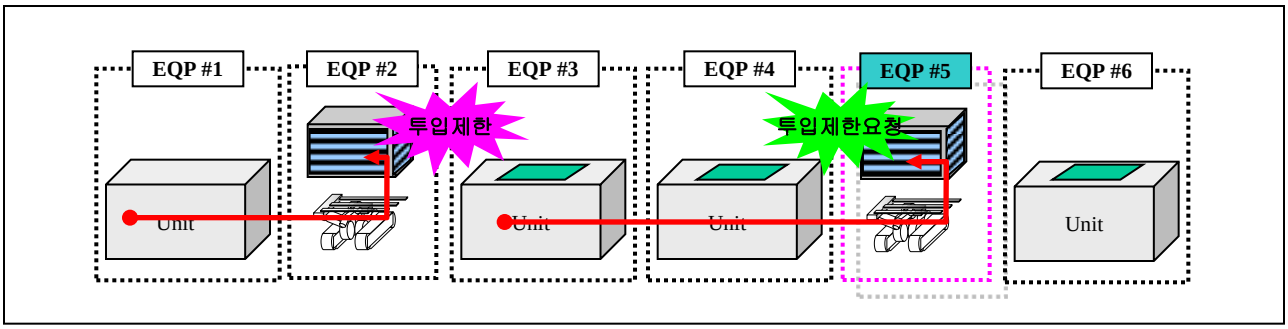


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8.3 Transfer Limit and Glass Buffering

8.3.1 하류 장비의 반송 지연



1) 투입 제한 조건 확인
- 자 장비(EQP #5)의 투입 제한 조건에 도달하였는지를 확인
- 투입 제한 요청 조건에 도달하였을 경우 Glass Transfer Enable를 Off 시킴.
- 상류 Buffering 가능 장비(EQP #2)는 자 장비(EQP #5)의 투입 제한 요청(Glass Transfer Enable Off)에 따라 하류 장비(EQP #3)로 Glass 투입을 제한함.

2) 투입 제한 해제 조건 확인
- 자 장비(EQP #5)의 투입 제한 상태에서 투입 제한 해제 조건에 도달하였는지를 확인
- 투입 제한 해제 조건에 도달하였을 경우 Glass Transfer Enable를 On 시킴.
- 상류 Buffering 가능 장비(EQP #2)는 자 장비(EQP #5)의 투입 제한 해제(Glass Transfer Enable On)에 따라 하류 장비로 Glass를 투입함.

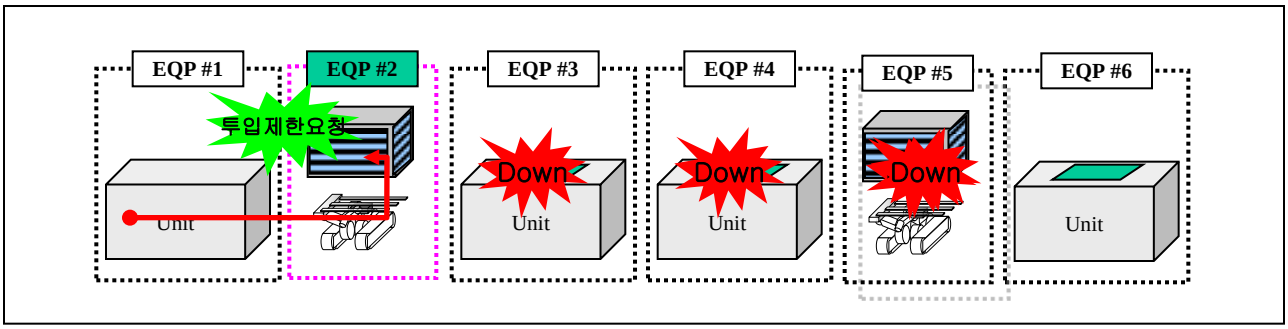
※ 투입 제한 조건
- 하류 장비의 상태(Down, 비정상적인 통신상태)
- 상류 Buffering 가능 장비 이후의 장비로부터 자 장비까지 투입된 Glass 매수가 투입 제한 매수에 도달하였을 경우
- 투입 제한 매수는 화면을 통해 입력가능 해야 함.

※ 투입 제한 해제 조건
- 하류 장비로의 반송 가능 상태
- 상류 Buffering 가능 장비 이후의 장비로부터 자 장비까지 투입된 Glass 매수가 투입 제한 해제 매수 이하로 도달 하였을 경우
- 투입 제한 해제 매수는 화면을 통해 입력 가능해야 함.

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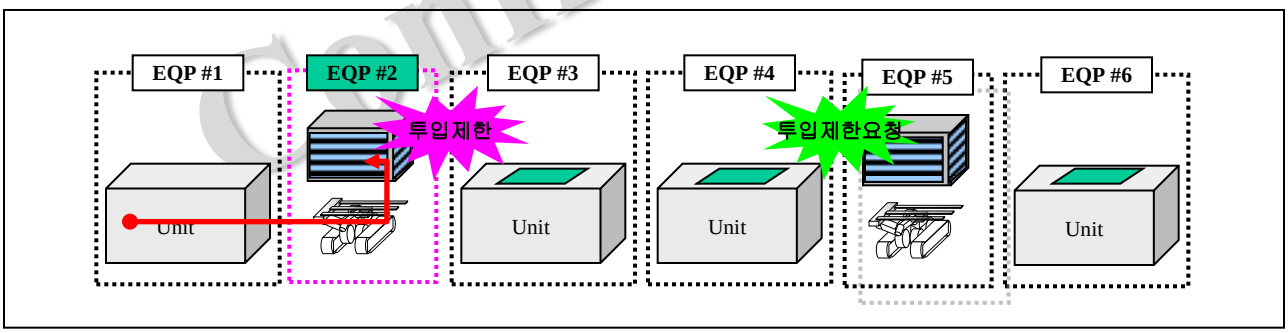
8.3.2 하류 장비의 Down 이거나 비정상적인 통신 상태일 경우



1) Glass Buffering
- 하류 장비가 Down 이거나 비정상적인 통신 상태(Abnormal baton pass Status 또는 Abnormal Cyclic Transmission Status가 2초 이상 유지)일 경우 Glass를 Buffering 한다.
- 하류 장비(EQP #3)로 투입 중(Robot Busy On 상태)일 경우 Alarm를 발생시키고 Operator를 호출하여 투입 중인 Glass를 Buffer에 수납하고 Interlock를 초기화 한다.
※ 협의에 의해 하류 장비들 중에 Down 여부를 확인할 장비를 결정한다.

2) 투입 제한 및 반송
- 8.3.1 하류 장비의 반송 지연 참조

8.3.3 하류 Buffering 가능 장비로부터의 투입 제한 요청



1) Glass Buffering
- 하류 장비(EQP #3)로 투입 중(Robot Busy On 상태)일 경우 하류 장비(EQP #3)로 Glass를 투입하며 투입(Robot Busy On) 이전일 경우 Glass를 Buffering 한다.
- 하류 Buffering 가능 장비로부터의 투입 제한 해제(EQP #5의 Glass Transfer Enable On)가 설정될 때까지 Glass를 Buffering 한다.

2) 투입 제한 및 반송
- 8.3.1 하류 장비의 반송 지연 참조

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Chapter Ⅱ. ECS-Equipment MELSEC Exception Specification

For LG Display

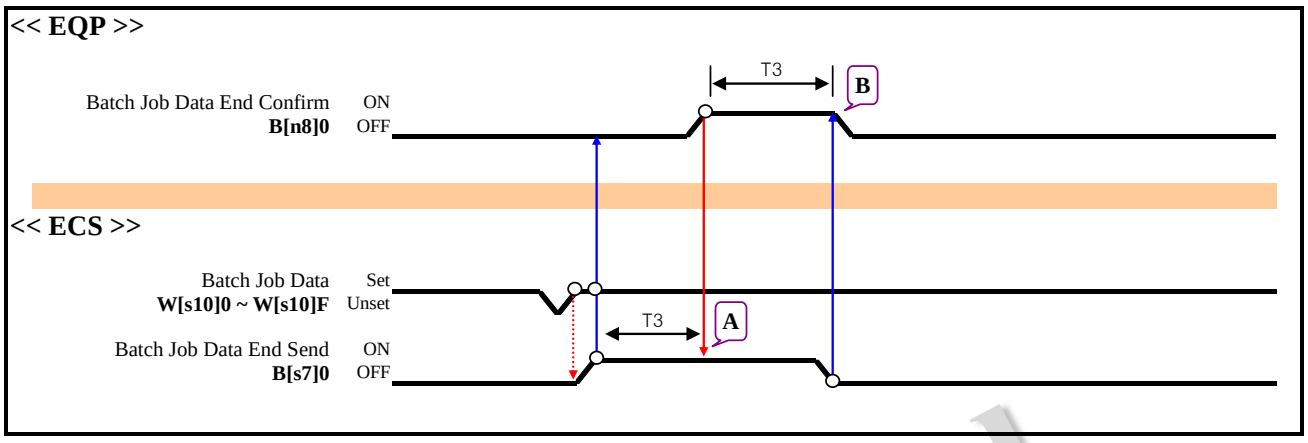
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1. Equipment Relation Sequence

1.1 Batch Job (Glass Injection) End Data Send

Batch Job의 마지막 Glass가 장비로 투입 되었을 때 전달한다.
.이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)



A : ECS가 Batch Job Data End Send 설정한 후 장비는 **3초** 이내에 Batch Job Data End Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시키며 Batch Job Data End Confirm신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

< Batch Job Data – ECS >

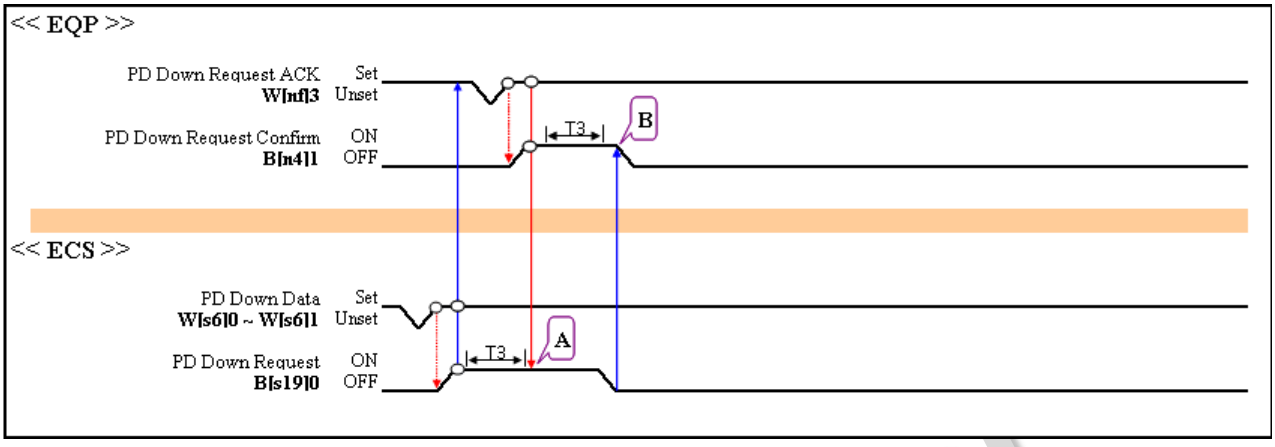
Name	Address	Format	Description
Port Number	W[s10]0	Integer	Port Number
LOT Number	W[s10]1	Integer	LOT Unique Number
LOT ID	W[s10]2 ~ W[s10]9	ASCII	LOT ID
Batch Flag	W[s10]A	ASCII	I : Injection End A : :Lot Aborted

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1.2 PD Down Request

ECS가 Host로부터 받은 ECMD 메시지를 장비에 전달하는 것으로 “PD Down Request”로 요청하며 사용여부는 합의하여 결정한다.



A : ECS가 PD Down Request 설정한 후 장비는 **3초** 이내에 PD Down Request Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.

< PD Down Request– ECS >

Name	Address	Format	Description
PD Down Unit ID	W[s6]0	Integer	-Unit ID ex) 0 : Machine PD Down (Whole Unit) n : Number #n PD Down
PD Down ECMD	W[s6]1	ASCII	- PD Down RCMD P : Process Down - ECMD (“PAUSE”) R : Resume S : Start C : Cancel M : Material Change - ECMD (“MATERIAL_CHANGE”) - EQP에서 교체 필요한 Tank 판단하여 교체 시작

< PD Down Mode ACK– EQP >

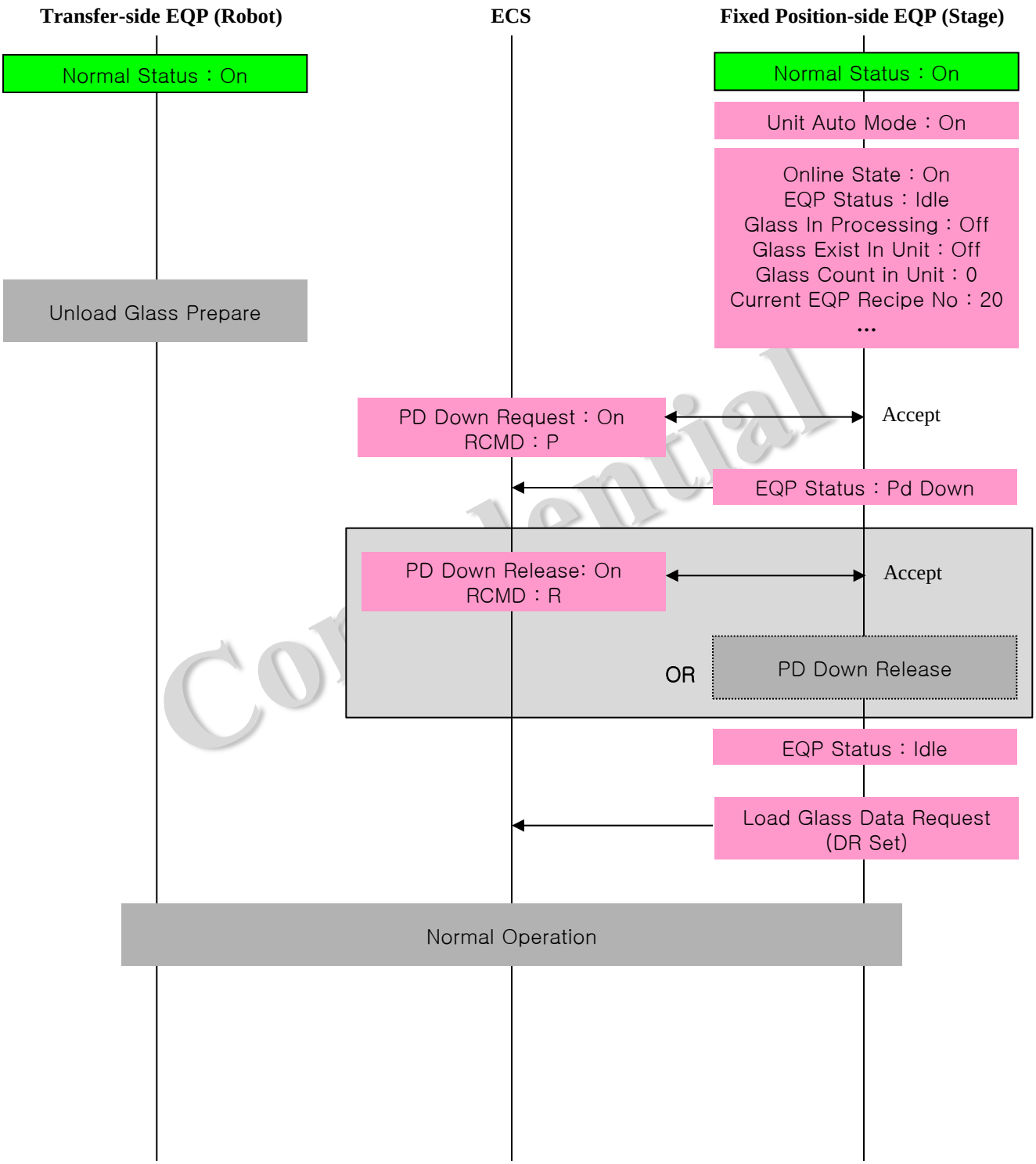
Name	Address	Format	Description
Request ACK Data	W[nf]3	Integer	0 : Accept 1 : Not Accept 2 : Already Release

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※ Reference : PD Down Request

1.2.1 PD Down Request Before Glass Data Request (Stand Alone, Robot → Stage)



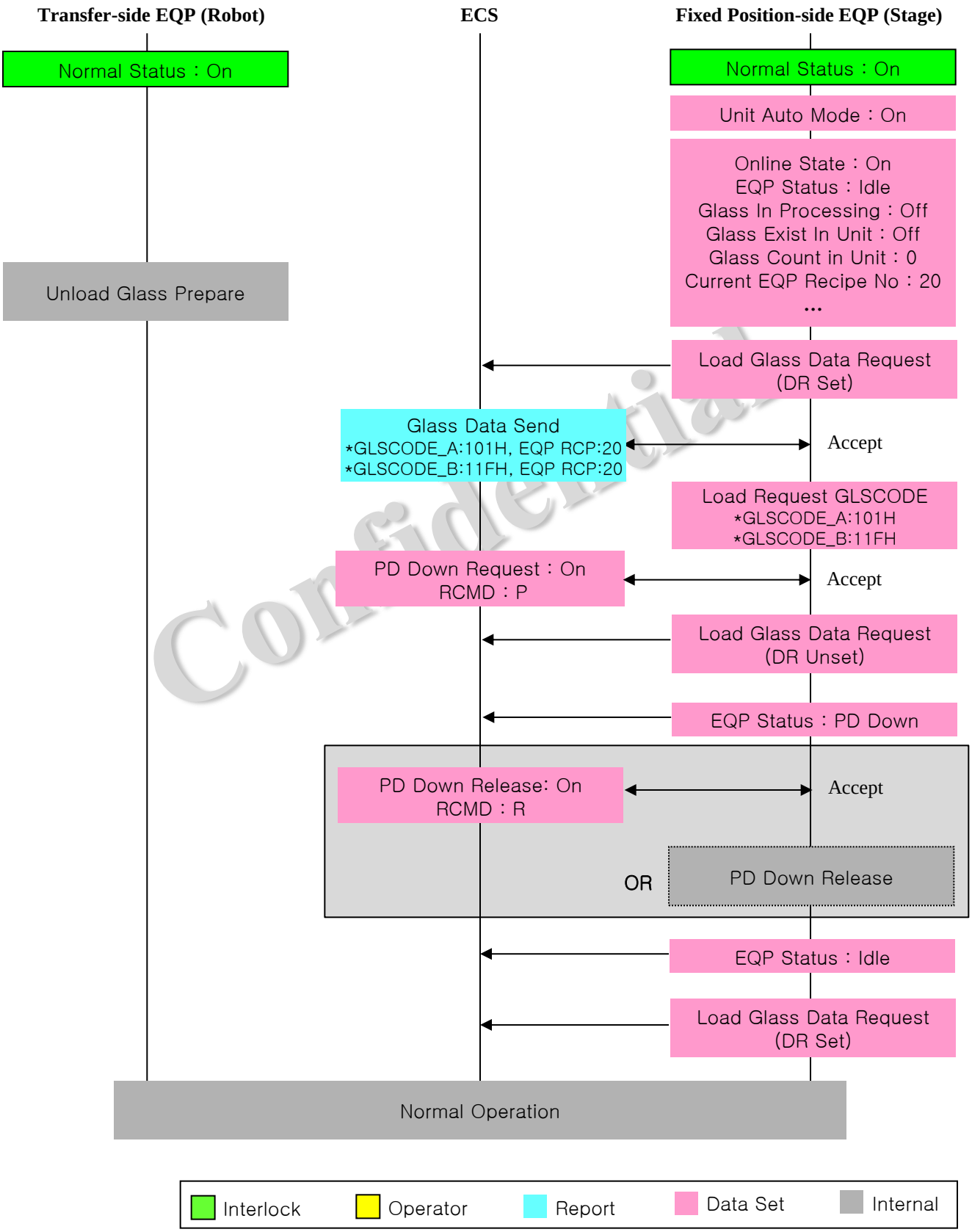
※ 1: Relative Equipment only.
※ 2: To set before according to a physical Interlock condition.

Interlock | Operator | Report | Data Set | Internal

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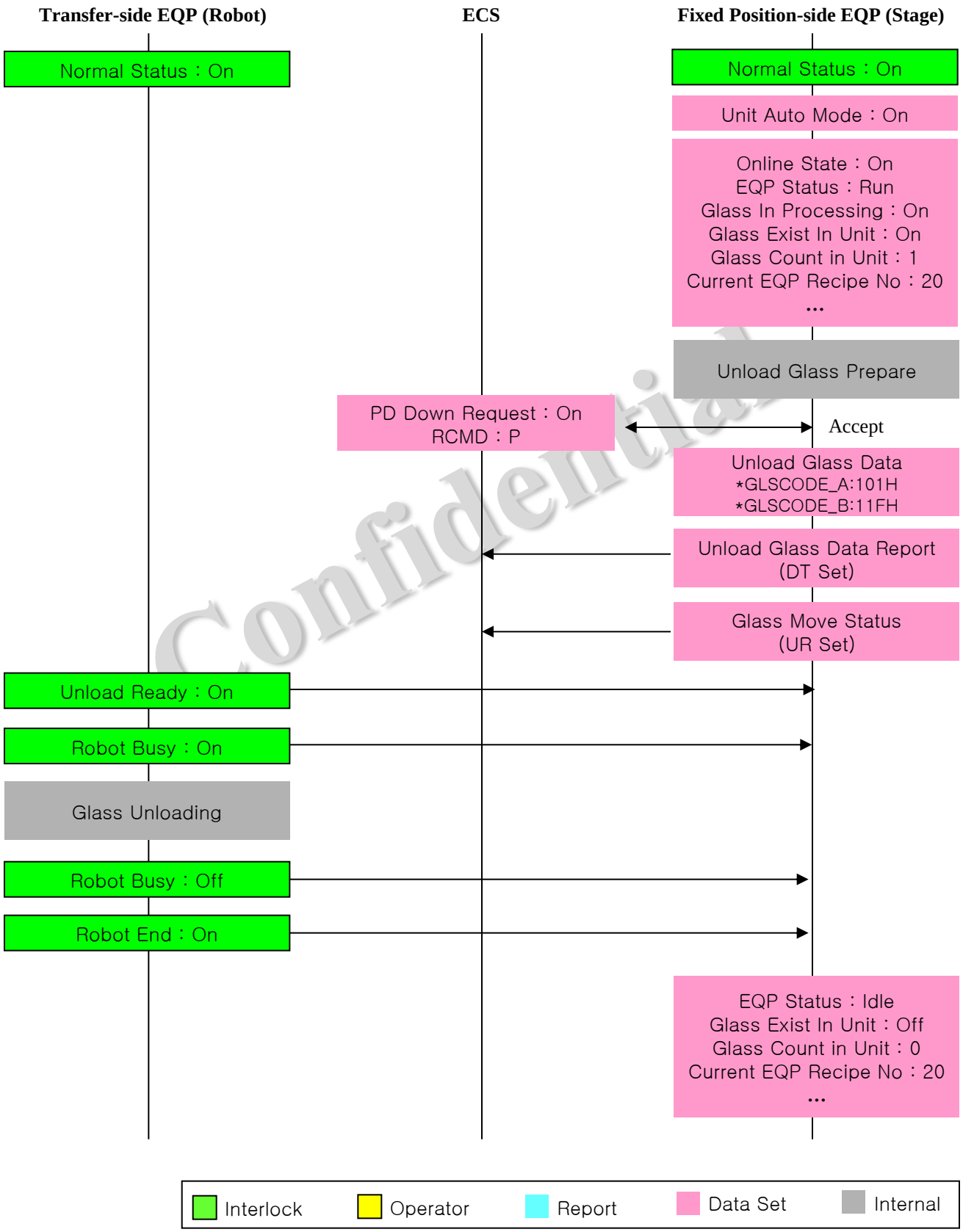
1.2.2 PD Down Request After Glass Data Request (Stand Alone, Robot → Stage)



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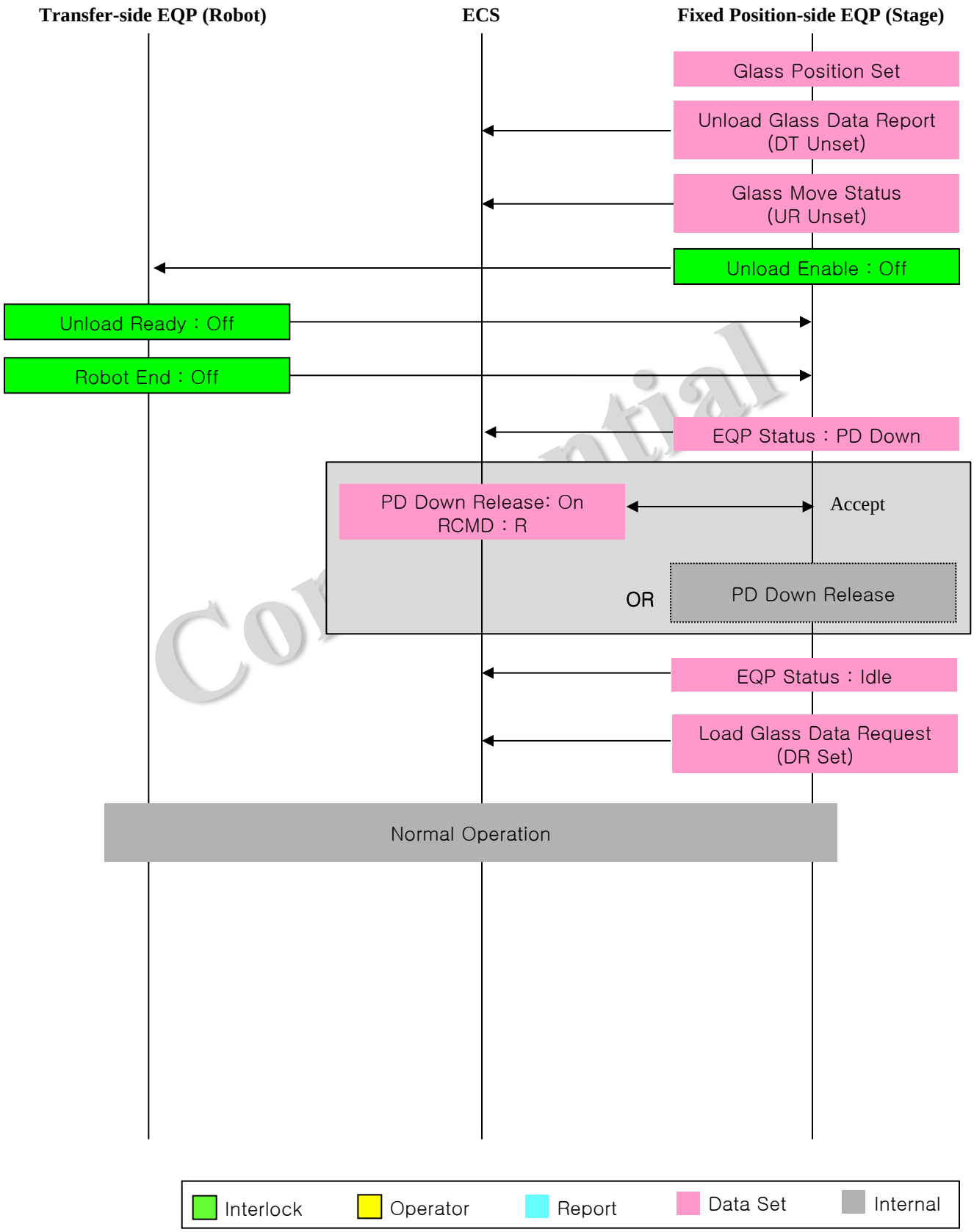
1.2.3 PD Down Request Before Glass Unloading (Stand Alone, Robot → Stage)



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1.2.3 PD Down Request Before Glass Unloading (Stand Alone, Robot → Stage)



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2. Glass Relation Sequence

2.1 Glass Move Status of Multi Stage

다수의 Stage에서 동시에 Glass Move Request가 발생하는 장비에서 사용하는 것으로 반송 장치로부터 수신(또는 배출) 완료 시 0으로 설정된다.
이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)

2.1.1 Glass Move Status (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 반송 장비로부터 완료 신호가 수신될 때까지 On 한다.

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Move Status	Stage #1 Load Request GLSCODE_A	Stage #1 Load Request GLSCODE_B	Stage #1 Unload Request GLSCODE_A	Stage #1 Unload Request GLSCODE_B	Stage #1 Unload Glass Info										

Name	Description
Glass Move Status	Bit 1 : Unload Request (UR) Bit 2 : Load Request (LR) Bit 3 : Exchange Request (ER) Bit 4 : Unload Wait Request (UW) Bit 5 : Load Wait Request (LW) Bit 6 : Pre Load Request (PR)

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2.1.2 Stage Load Request GLSCODE (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 장비 Stage로 투입(또는 대기)을 요청하는 Glass의 GLSCODE를 나타내는 것으로 투입 완료 시 0으로 설정한다

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Move Status	Stage #1 Load Request GLSCODE_A	Stage #1 Load Request GLSCODE_B	Stage #1 Unload Request GLSCODE_A	Stage #1 Unload Request GLSCODE_B	Stage #1 Unload Glass Info										

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2.1.3 Stage Unload Request GLSCODE (Stand Alone)

Stand Alone 장비에서 사용하는 것으로 장비 Stage에서 배출(또는 대기)을 요청하는 Glass의 GLSCODE를 나타내는 것으로 투입 완료 시 0으로 설정한다

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Move Status	Stage #1 Load Request GLSCODE_A	Stage #1 Load Request GLSCODE_B	Stage #1 Unload Request GLSCODE_A	Stage #1 Unload Request GLSCODE_B	Stage #1 Unload Glass Info										

2.1.4 Stage Unload Glass Info

장비의 Stage에서 배출되는 Glass의 추가 정보를 나타내는 것으로 배출 완료 시 0으로 설정한다.
사용여부 및 Glass 정보는 협의에 의해 결정한다.

W [n?]	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
Stage Move Status	Stage #1 Glass Move Status	Stage #1 Load Request GLSCODE_A	Stage #1 Load Request GLSCODE_B	Stage #1 Unload Request GLSCODE_A	Stage #1 Unload Request GLSCODE_B	Stage #1 Unload Glass Info										

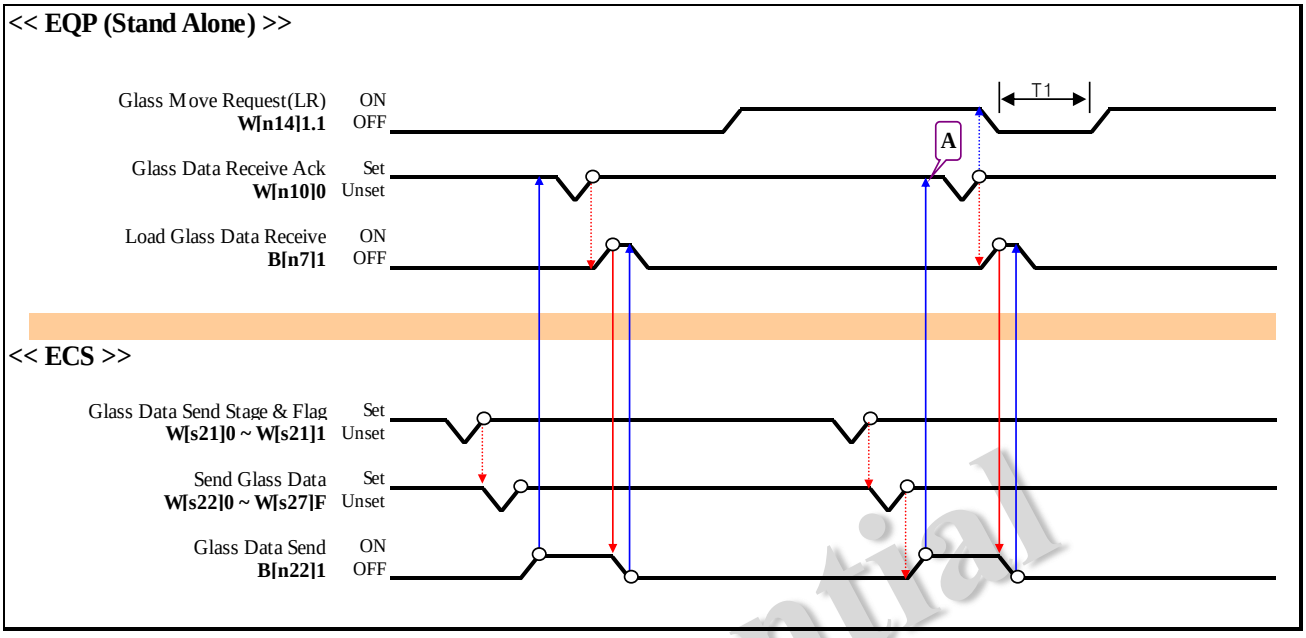
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2.2 Glass Data Send

Stand Alone 장비에서 사용하는 것으로 ECS가 장비의 특정 Stage로 투입될 Glass에 대한 정보를 전달한다.
.이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)

<< EQP (Stand Alone) >>



※ A : 장비는 Glass의 반송이 이루어지지 전(Robot Busy or Conveyor Busy)에 해당 Glass 정보가 변경되었음 보고 받았을 경우 수신 가능 여부를 확인 한 후 “Glass Move Request (LR)”를 Off시킨다.
※ Glass Move Request (LR) Unset시 1초 이상 유지하며 Parameter를 통해 변경 가능하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Send Glass Data - ECS >

Name	Address	Format	Description
Glass Data Send Flag	W[s21]0	Integer	0 : Glass Exist 1 : Glass Not Exist
Glass Data Send Stage	W[s21]1	Integer	Stage Number
Send Glass Data	W[s22]0 ~ W[s27]F	-	Glass Data (Each Glass Data of A Cut, B Cut) - Reference 4.1 Glass Data Structure

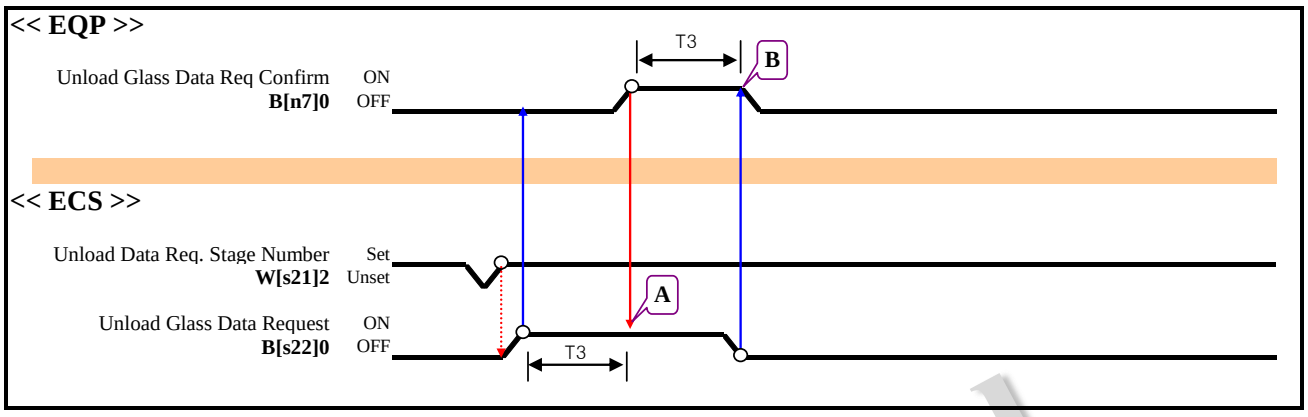
< Glass Data Receive Ack - EQP >

Name	Address	Format	Description
Glass Data Receive ACK	W[n10]0	Integer	0 : Accept 1 : Not Accept (Glass Processing) 2 : Not Accept (Unknown Recipe Number) 3 : Not Accept (Glass Data Error) Etc : Other Error

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2.3 Unload Glass Data Request

ECS가 장비의 특정 Stage의 Unload Glass Data를 요청하는 것으로 수신 시 작업자가 선택창을 통하여 Lost Glass Data Request 또는 Unload Glass Data Report을 선택 가능하게 한다.
Unload Glass Data Report 보고 시 상태가 On일 경우 1초간 Off 시킨 후 다시 On 시킨다.
이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)



A : ECS가 Unload Glass Data Request한 후 장비는 3초 이내에 Unload Glass Data Request Confirm를 보고 해야만 한다.

B : 장비는 일정시간(3초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시키며 Unload Glass Data Confirm 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

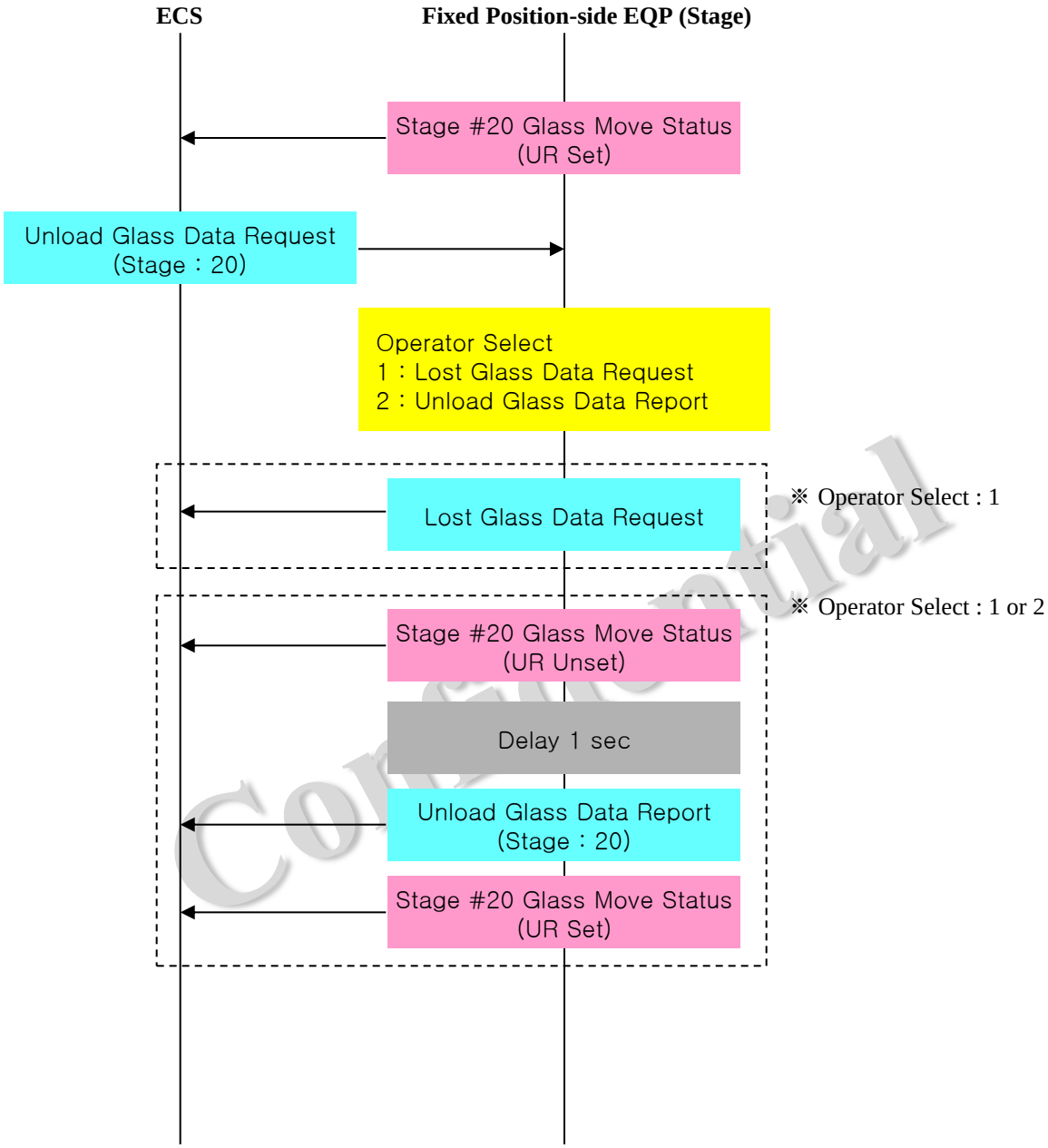
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< Unload Glass Data Request - ECS >

Name	Address	Format	Description
Unload Data Req. Stage Number	W[s21]2	Integer	Stage Number

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※ Reference : Unload Glass Data Request



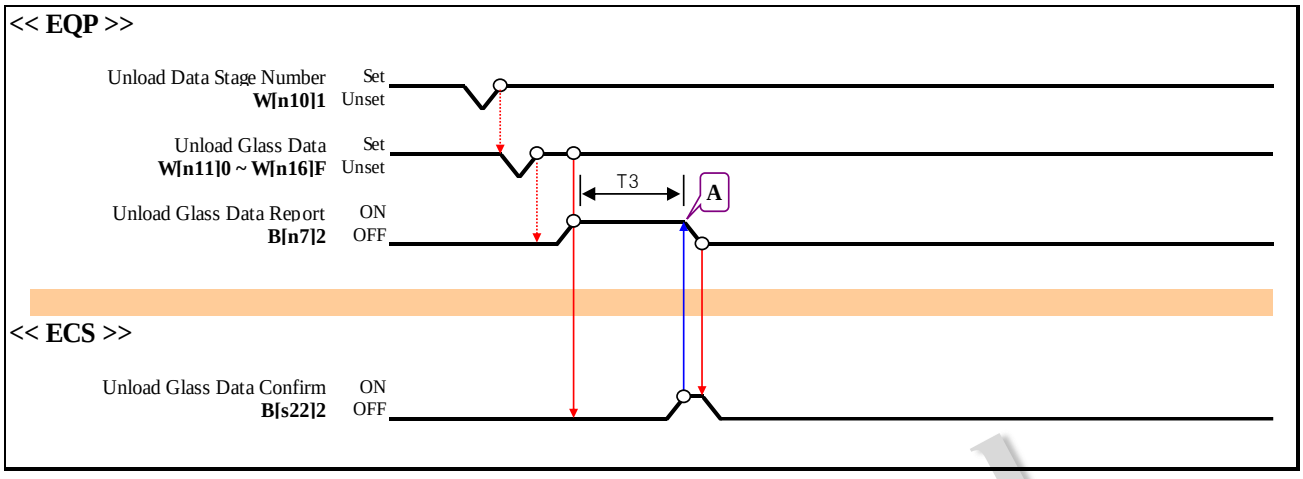
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Interlock | Operator | Report | Data Set | Internal

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2.4 Unload Glass Data Report

Stand Alone 장비에서 사용하는 것으로 장비의 특정 Stage의 배출 Glass에 대한 정보를 보고한다
이 시나리오의 사용여부는 협의하여 결정한다.(Stand Alone)



A : 장비는 일정시간(Timeout – 3초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Unload Glass Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Unload Glass Data - EQP >

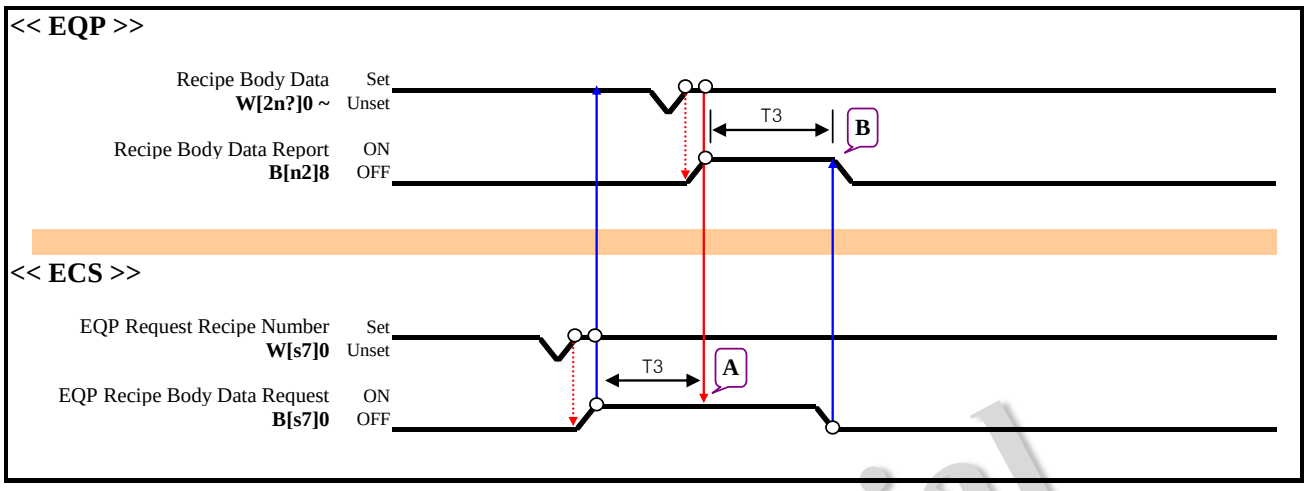
Name	Address	Format	Description
Unload Data Stage Number	W[n10]1	Integer	Stage Number
Unload Glass Data	W[n11]0 ~ W[n16]F	-	Glass Data (Each Glass Data of A Cut, B Cut) - Reference 4.1 Glass Data Structure

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3. Recipe Relation Sequence

3.1 Recipe Body Data Request

Recipe Number의 존재 여부 및 Recipe Body Data를 요청하는 것으로 Recipe Number는 존재하되 Recipe Body Data가 없을 시에는 CEID , Equipment Recipe Number 및 ACK Data 만을 보고한다.



A : ECS가 Recipe Body Data를 요구한 후 장비는 일정시간(Timeout - 3초) 이내에 Recipe Body Data를 보고 해야만 한다.

B : 장비는 일정시간(Timeout - 3 초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Recipe Body Data Request Confirm – ECS >

Name	Address	Format	Description
Request Recipe Number	W[s7]0	Integer	- Equipment Recipe Number

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< Recipe Body Data Report – EQP >

Name	Address	Format	Description
Data Report CEID	W[2n]0	Integer	3 : Recipe Body Data
Recipe Number	W[2n]1	Integer	- Equipment Recipe Number
Request Confirm ACK	W[2n]2	Integer	0 : Recipe Confirm OK 1 : Invalid Recipe Number 2 : Recipe Confirm Not Accept
Recipe Level	W[2n]3	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe)
Recipe Type	W[2n]4	ASCII	Type of recipe "M" : Machine Recipe
Recipe Version	W[2n]5 ~ W[2n]7	-	Recipe Version (The detail use is determined after discussing)
Unit #1 Equipment Specific Data V1 ~	W[2n1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Unit #1 Recipe Report V1 ~	W[2n2]0 ~	-	Recipe Body Data (The detail use is determined after discussing)
Unit #n Equipment Specific Data V1 ~	W[2n?]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Unit #n Recipe Report V1 ~	W[2n?]0 ~	-	Recipe Body Data (The detail use is determined after discussing)

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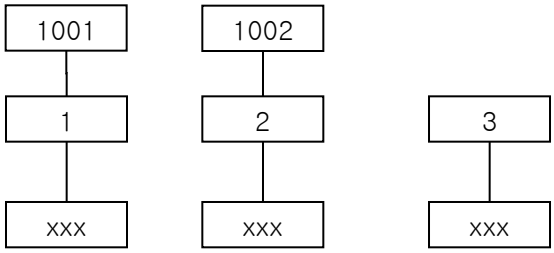
< Equipment Specific Data >

Index	Name	Length	Format	Description
1	Recipe Number	1 word	Integer	Recipe Number
2	Recipe Level	1 word	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
3	Recipe Type	1 word	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
4	Recipe Version	3 word	-	Recipe version
5	Reserve	1 word	-	Reserve
6	Unit Number	1 word	Integer	Unit Number (The detail use is determined after discussing)

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※ Example 1 (2 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Body Data Request – ECS >

Name	Value
Request Recipe Number	2

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< Recipe Body Data – EQP >

Name	Value
Recipe Report CEID	3 : Recipe Body
Recipe Number	2
Request Confirm ACK	0 : Accepted
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit #1Equipment Specific Data V1 ~	Equipment Specific
Unit #1 Recipe Report V1 ~	xxx

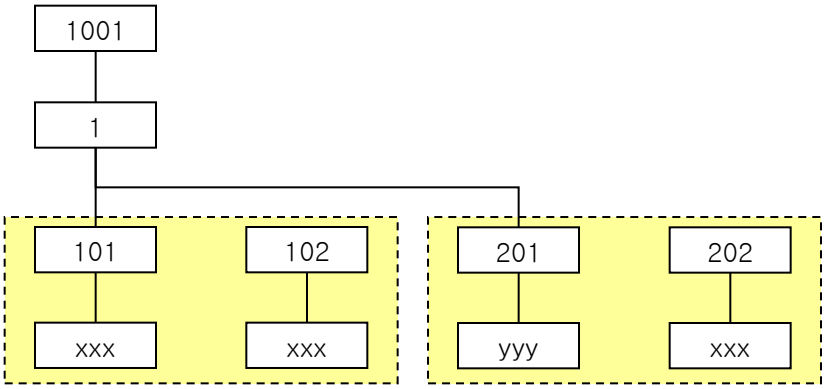
< Unit #1 Equipment Specific Data >

No	Name	Value
1	Recipe Number	2
2	Recipe Level	2 : Level 2 Recipe
3	Recipe Type	"M" : Machine Recipe
6	Unit Number	0

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※ Example 2 (3 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Body Data Request – ECS >

Name	Value
Request Recipe Number	1

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< Recipe Body Data – EQP >

Name	Value
Recipe Report CEID	3 : Recipe Body
Recipe Number	1
Request Confirm ACK	0 : Accepted
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit #1 Equipment Specific V1 ~	Equipment Specific Data
Unit #1 Recipe Report V1 ~	xxx
Unit #2 Equipment Specific V1 ~	Equipment Specific Data
Unit #2 Recipe Report V1 ~	yyy

< Unit #1 Equipment Specific Data >

No	Name	Value
1	Recipe Number	101
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe
6	Unit Number	1

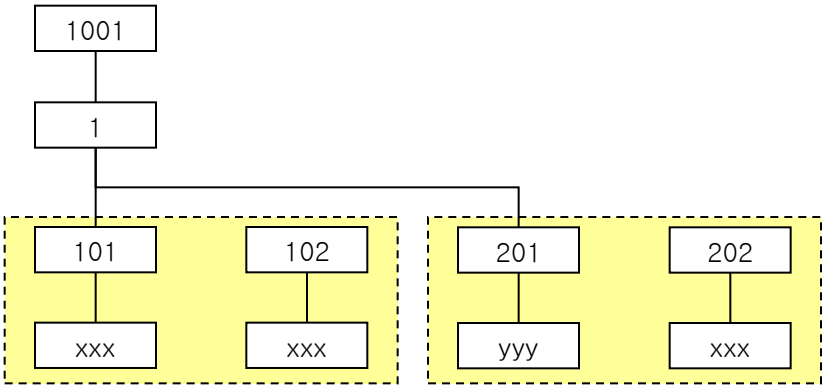
< Unit #2 Equipment Specific Data >

No	Name	Value
1	Recipe Number	201
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe
6	Unit Number	2

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※ Example 3 (3 Level Recipe, Step)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Body Data Request – ECS >

Name	Value
Request Recipe Number	1

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< Recipe Body Data – EQP >

Name	Value
Recipe Report CEID	3 : Recipe Body
Recipe Number	1
Request Confirm ACK	0 : Accepted
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit #1 Equipment Specific V1 ~	Equipment Specific Data
Unit #1 Recipe Report V1 ~	Unit Recipe Report
Unit #2 Equipment Specific V1 ~	Equipment Specific Data
Unit #2 Recipe Report V1 ~	Unit Recipe Report

< Unit #1 Equipment Specific Data >

No	Name	Value
1	Recipe Number	101
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe
6	Unit Number	1

< Unit #1 Recipe Report >

Name	Value
Step 1 Recipe Body V1 <Report Step>	1
Step 1 Recipe Body V2 ~	xxx
Step 2 Recipe Body V1 <Report Step>	3
Step 2 Recipe Body V2 ~	yyy

< Unit #2 Equipment Specific Data >

No	Name	Value
1	Recipe Number	201
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe
6	Unit Number	2

< Unit #2 Recipe Report >

Name	Value
Step 1 Recipe Body V1 <Report Step>	2
Step 1 Recipe Body V2 ~	xxx
Step 2 Recipe Body V1 <Report Step>	3
Step 2 Recipe Body V2 ~	yyy

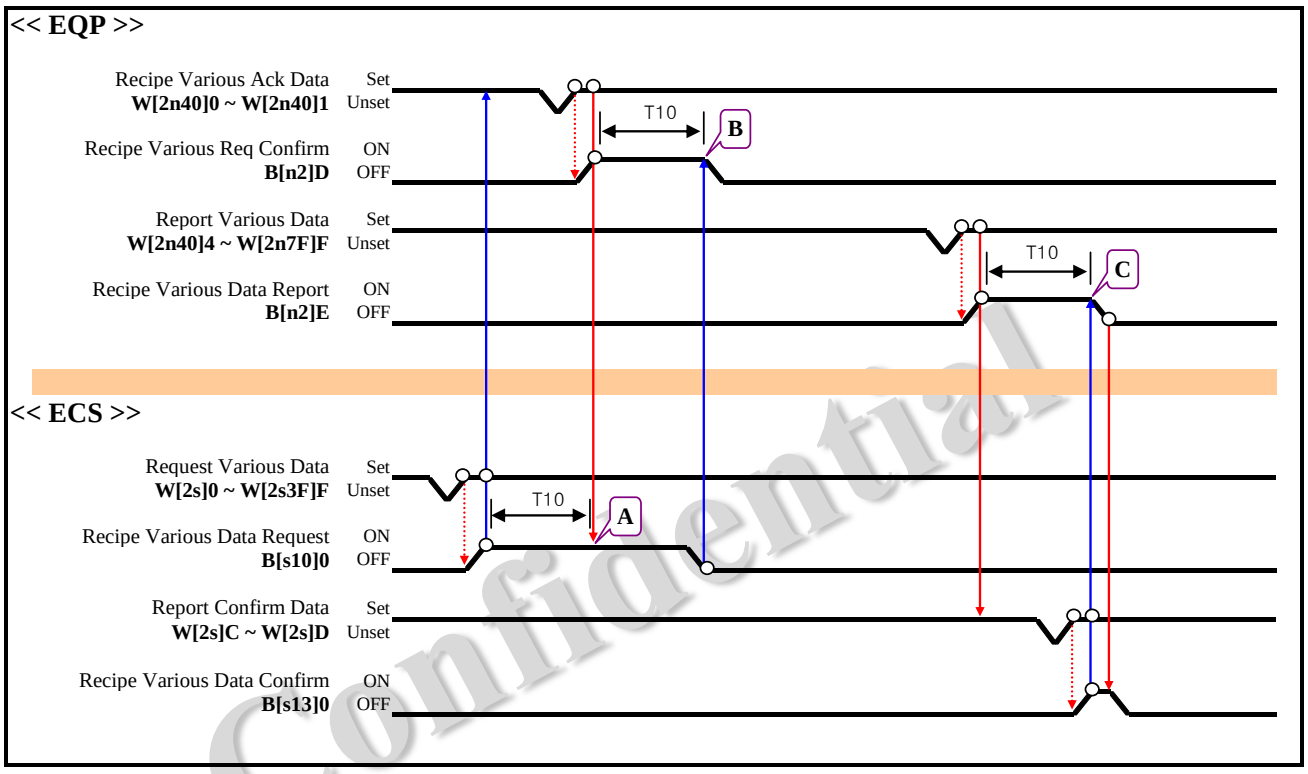
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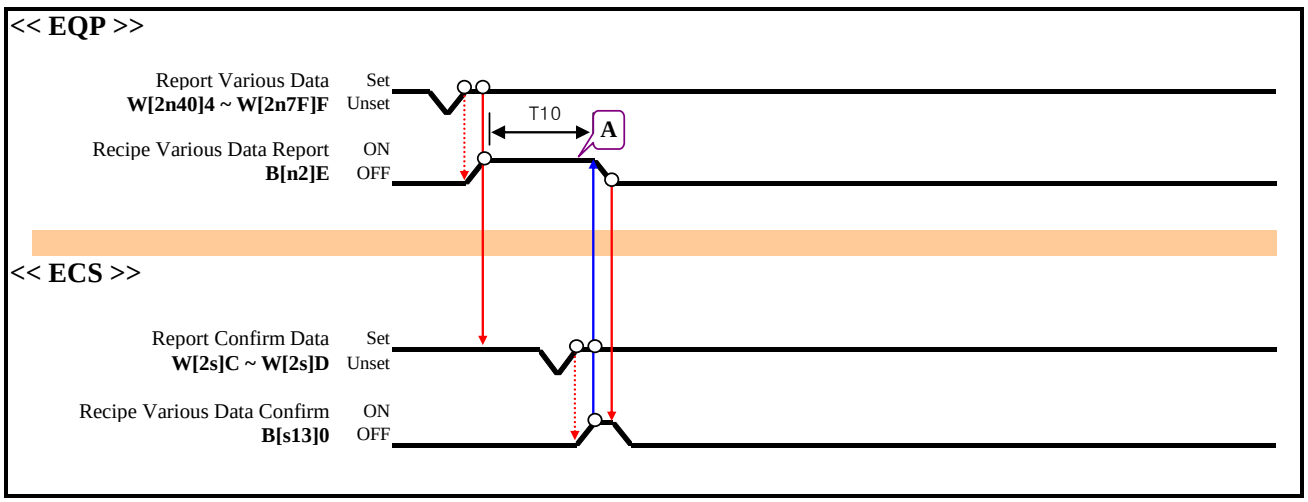
3.2 Recipe Various Data Request & Report

장비에 등록되어 있는 Recipe 정보 요청 하거나 Recipe 정보를 보고 받는 것으로 CEID에 따라 요청 및 보고 정보를 구분한다.
사용여부는 협의하여 결정한다.

※ <Recipe Various Data Request & Report> Time Chart



※ <Recipe Various Data Report> Time Chart

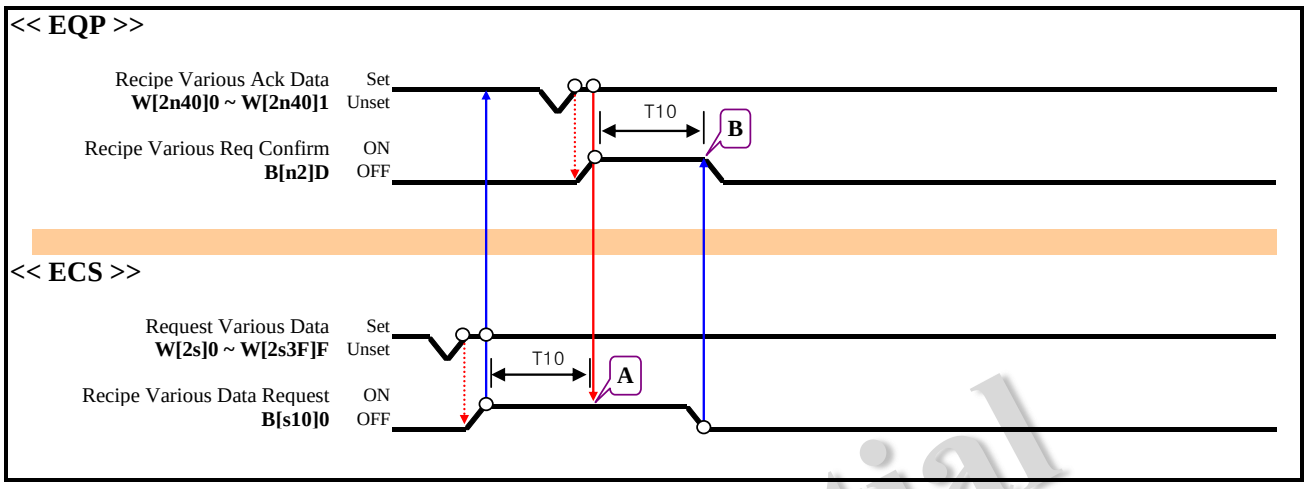


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3.2.1 Recipe Various Data Request

장비에 등록되어 있는 Recipe 정보 요청 또는 Recipe Command를 전달하는 것으로 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 10초 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 4 : Recipe Command 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 8 : EQP Constant Recipe Command
Request Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete 4 : Rename
Request Recipe Number	W[2s]2	Integer	Recipe Number
Request Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Request Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Request Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Equipment Specific Data V1 ~	W[2s1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s2]0 ~	-	- Recipe Parameter Data - Lower Recipe Number List (The detail use is determined after discussing)

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< Recipe Various ACK Data - EQP >

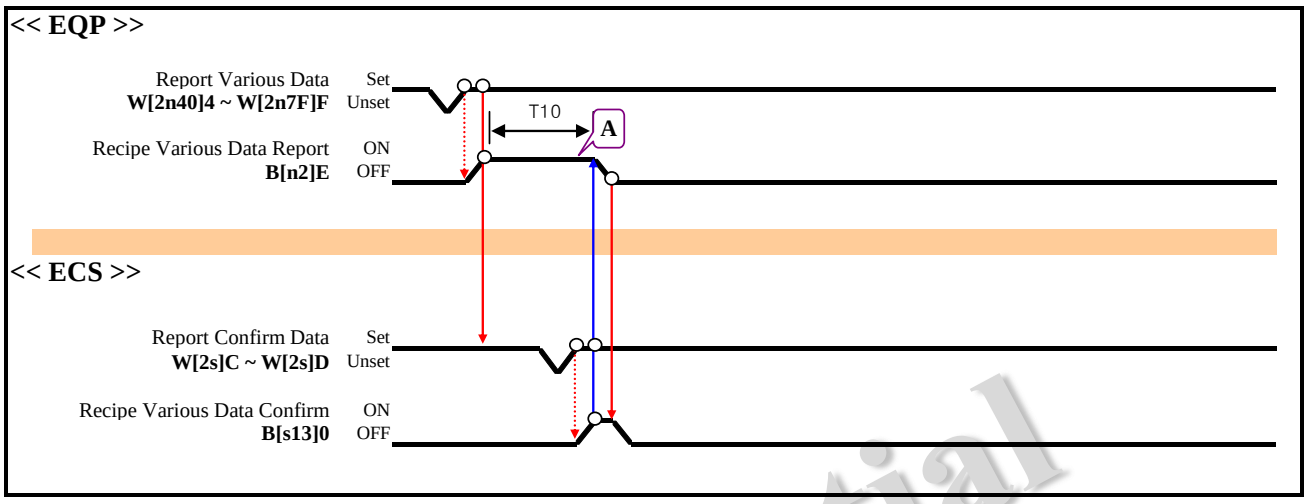
Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 4 : Recipe Command 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 8 : EQP Constant Recipe Command
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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3.2.2 Recipe Various Data Report

ECS의 요청 또는 장비에서 변경된 Recipe 정보를 보고하는 것으로 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 5 : Recipe Change Report 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 9 : EQP Constant Recipe Change Report
Recipe Report CEID Order	W[2n40]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n40]6	Integer	Recipe Total Count
Recipe Report Count	W[2n40]7	Integer	Recipe Data Structure Count
Recipe Command	W[2n40]8	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete 4 : Rename
Recipe Number	W[2n40]9	Integer	Recipe Number
Recipe Level	W[2n40]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n40]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n40]C ~ W[2n40]E	-	Recipe Version
Equipment Specific Data V1 ~	W[2n41]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Recipe Report V1 ~	W[2n42]0 ~	-	Recipe Report Data (The detail use is determined after discussing)

※ 보고하는 Recipe 데이터가 Report Various Data 영역을 초과할 경우 처음 보고 시는 CEID를 Original Value를 보고 하고 그 이후에는 10을 더한 값을 보고한다 (이 시나리오는 추후 협의에 의해 결정함)

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< Report Confirm Data – ECS >

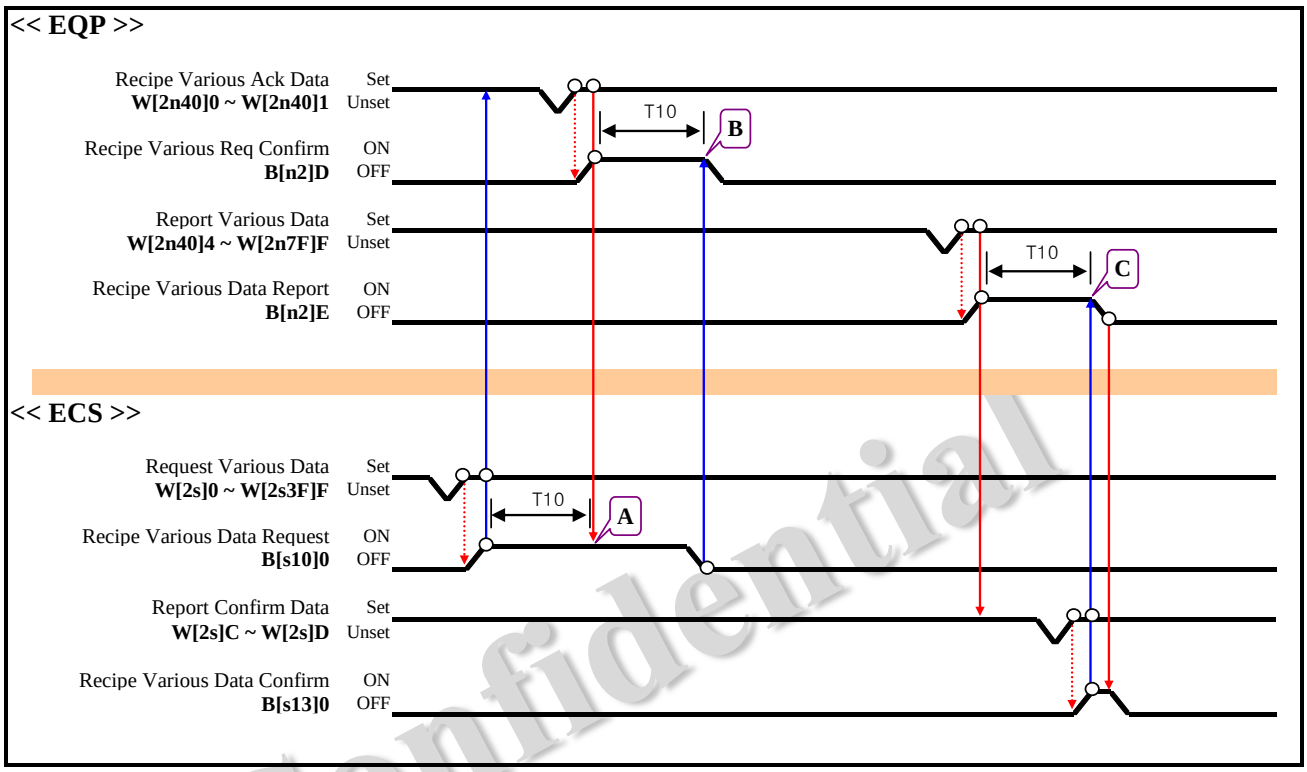
Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	1 : Recipe List 2 : Lower Recipe List 3 : Recipe Parameter 5 : Recipe Change Report 6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) 9 : EQP Constant Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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3.2.3 Recipe List Request

장비에 등록되어 있는 Recipe 정보를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout -10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	1 : Recipe List
Request Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Request Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Equipment Specific Data V1 ~	W[2s1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)

< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	1 : Recipe List
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

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< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	1 : Recipe List
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	1 : Recipe List
Recipe Report CEID Order	W[2n40]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n40]6	Integer	Recipe Total Count
Recipe Report Count	W[2n40]7	Integer	Recipe Data Structure Count
Recipe Report V1 ~	W[2n41]0 ~	-	Equipment Specific Data Recipe Report Structure Sub Recipe Number List (The detail use is determined after discussing)

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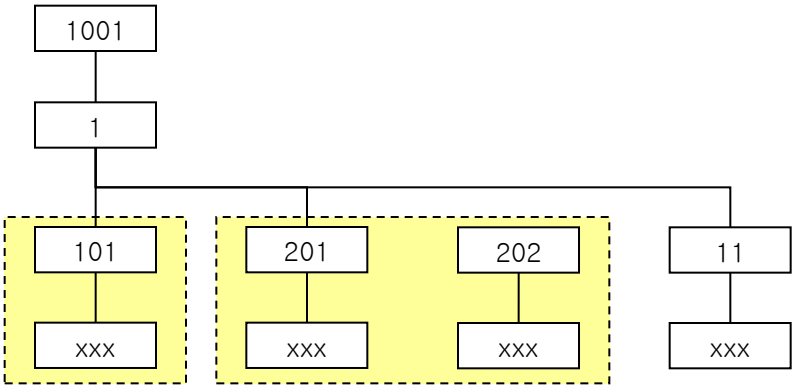
< Recipe Report Structure >

Index	Name	Length	Format	Description
1	Recipe Number	1 word	Integer	Recipe Number
2	Recipe Level	1 word	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
3	Recipe Type	1 word	ASCII	Type of recipe “M” : Machine Recipe “U” : Unit Recipe “1”, “2”, “3”, … : Special Recipe (The detail use is determined after discussing)
4	Recipe Version	3 word	-	Recipe version
5	Recipe Status	1 word	Integer	Recipe Status 1 : Running (using) 2 : Idle (Editable) 3 : Disable (locked)
6	Unit Number	1 word	Integer	Unit Number (The detail use is determined after discussing)

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※ Example 1 (Machine Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	1
Recipe Report Count	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

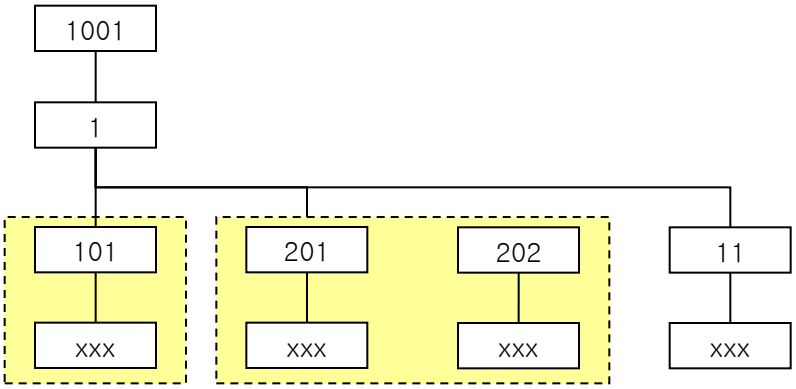
No	Name	Value
1	Recipe Number	1
2	Recipe Level	2 : Level 2 Recipe
3	Recipe Type	"M" : Machine Recipe
5	Recipe Status	2 : Idle (Editable)
6	Unit Number	0

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※ Example 2 (Special Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Unit Number	0

< Request ACK - EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	1
Recipe Report Count	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

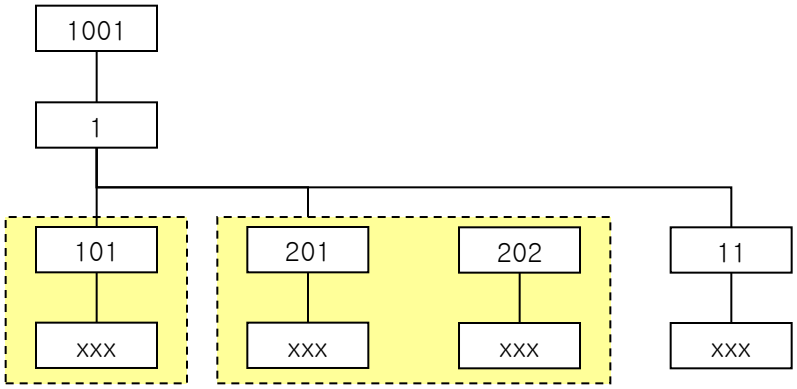
No	Name	Value 1
1	Recipe Number	11
2	Recipe Level	3 : Level 3 Recipe
3	Recipe Type	"1" : Special Recipe
5	Recipe Status	2 : Idle (Editable)
6	Unit Number	0

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※ Example 3 (Unit Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe List Request – ECS >

Name	Value
Request CEID	1 : Recipe List
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK - EQP >

Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	0 : Accepted

< Recipe List – EQP >

Name	Value
Recipe Report CEID	1 : Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	2
Recipe Report Count	2
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

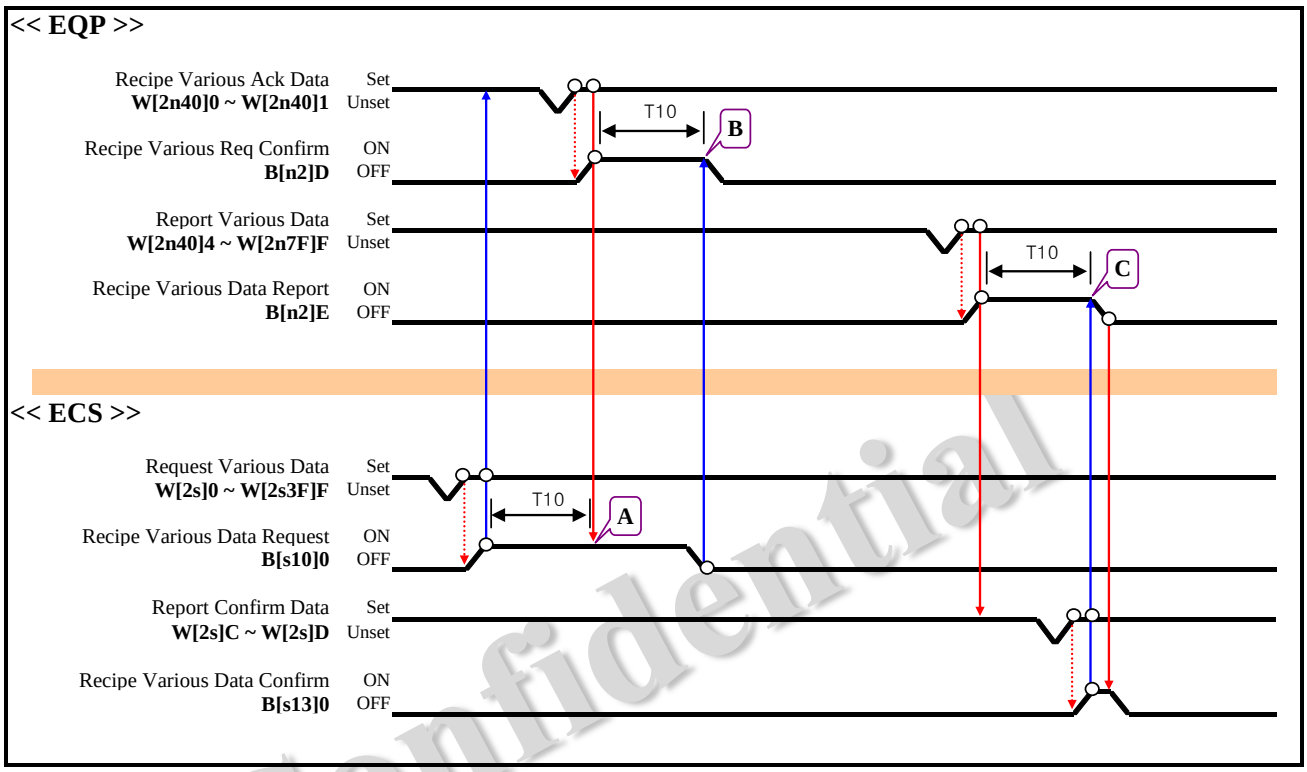
No	Name	Value 1	Value 2
1	Recipe Number	201	202
2	Recipe Level	3 : Level 3 Recipe	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe	"U" : Unit Recipe
5	Recipe Status	2 : Idle (Editable)	2 : Idle (Editable)
6	Unit Number	2	2

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3.2.4 Lower Recipe List Request

장비에 등록되어 있는 Recipe의 Lower Recipe List를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout -10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	2 : Lower Recipe List
Request Recipe Number	W[2s]2	Integer	Recipe Number
Request Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Request Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Request Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Equipment Specific Data V1 ~	W[2s1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)

< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	2 : Lower Recipe List
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	2 : Lower Recipe List
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	2 : Lower Recipe List
Recipe Report CEID Order	W[2n40]5	Integer	0 : First Report CEID 1 : CEID except the first and the last 2 : Last Report CEID
Recipe Total Count	W[2n40]6	Integer	Recipe Total Count
Recipe Report Count	W[2n40]7	Integer	Recipe Data Structure Count
Recipe Number	W[2n40]9	Integer	Requested Recipe Number
Equipment Specific Data V1 ~	W[2n41]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Recipe Report V1 ~	W[2n41]0 ~	-	Equipment Specific Data Recipe Report Structure Sub Recipe Number List (The detail use is determined after discussing)

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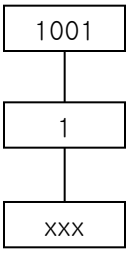
< Recipe Report Structure >

Index	Name	Length	Format	Description
1	Recipe Number	1 word	Integer	Recipe Number
2	Recipe Level	1 word	Integer	Recipe Level 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
3	Recipe Type	1 word	ASCII	Type of recipe “M” : Machine Recipe “U” : Unit Recipe “1”, “2”, “3”, … : Special Recipe (The detail use is determined after discussing)
4	Recipe Version	3 word	-	Recipe version
5	Reserve	1 word	-	Reserve
6	Unit Number	1 word	Integer	Unit Number (The detail use is determined after discussing)

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※ Example 1 (2 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Lower Recipe List Request – ECS >

Name	Value
Request CEID	2 : Lower Recipe List
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

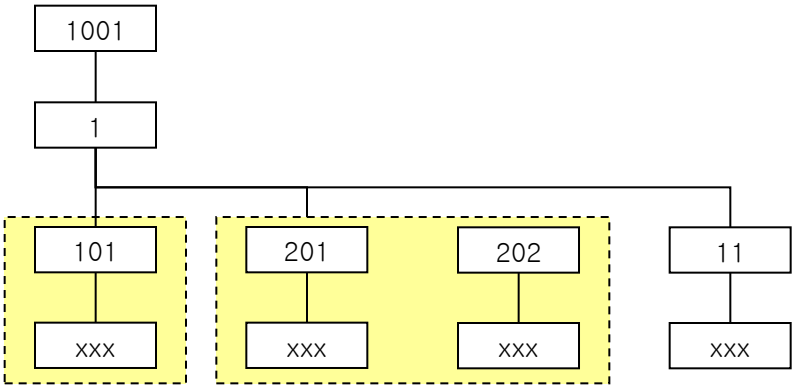
Name	Value
Request Confirm CEID	1 : Recipe List
Request Confirm ACK	5 : Recipe Number not found

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※ Example 2 (3 Level Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Lower Recipe List Request – ECS >

Name	Value
Request CEID	2 : Lower Recipe List
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	2 : Lower Recipe List
Request Confirm ACK	0 : Accepted

< Lower Recipe List – EQP >

Name	Value
Recipe Report CEID	2 : Lower Recipe List
Recipe Report CEID Order	2 : Last Report CEID
Recipe Total Count	3
Recipe Report Count	3
Recipe Number	1
Recipe Report V1 ~	Recipe Report Structure

< Recipe Report Structure >

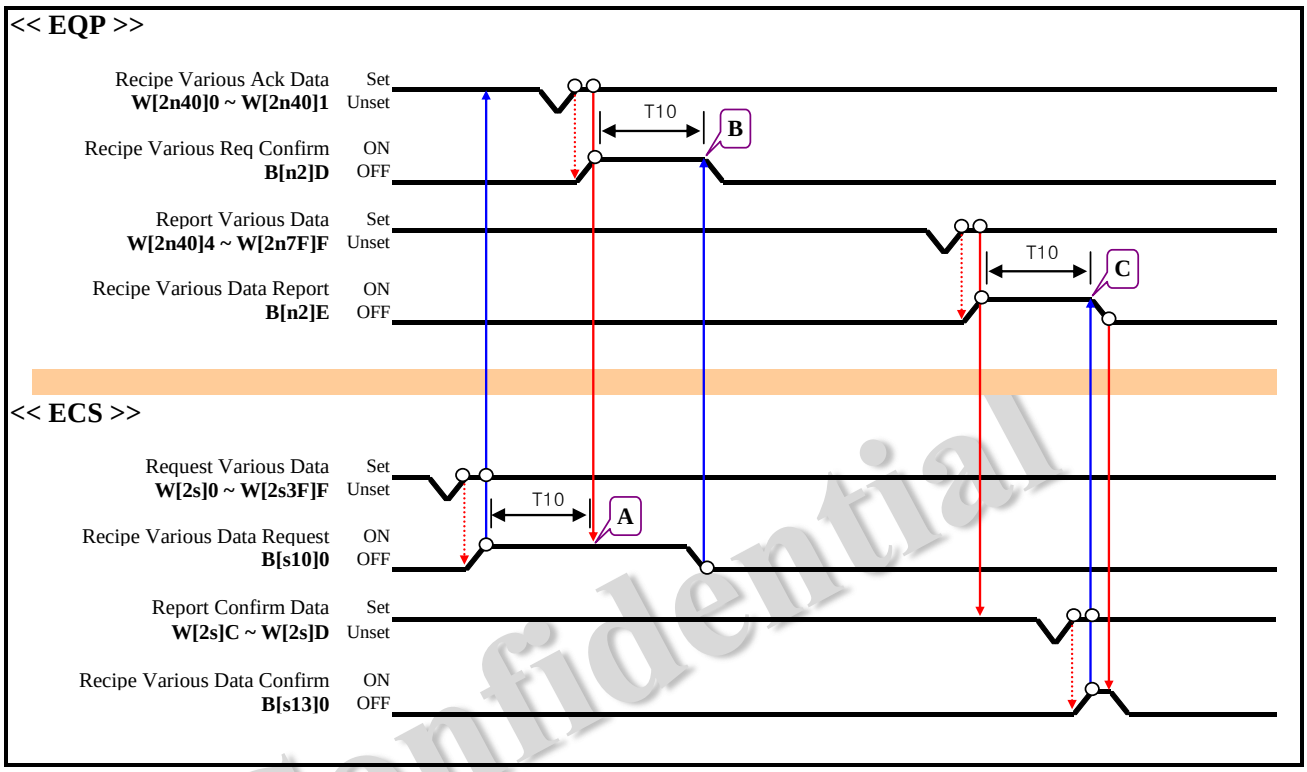
No	Name	Value 1	Value 2	Value 3
1	Recipe Number	101	201	11
2	Recipe Level	3 : Level 3 Recipe	3 : Level 3 Recipe	3 : Level 3 Recipe
3	Recipe Type	"U" : Unit Recipe	"U" : Unit Recipe	"1" : Special Recipe
6	Unit Number	1	2	0

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3.2.5 Recipe Parameter Request

장비에 등록되어 있는 Recipe의 Parameter를 요청하는 것으로 ECS는 “Recipe Various Data Request”로 요청하며 장비는 “Recipe Various Data Report”로 보고한다. (사용여부는 협의하여 결정한다)



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout -10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	3 : Recipe Parameter
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	3 : Recipe Parameter
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	3 : Recipe Parameter
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Report Various Data - EQP >

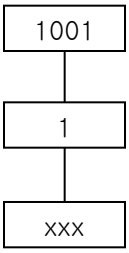
Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	3 : Recipe Parameter
Recipe Number	W[2n40]9	Integer	Recipe Number
Recipe Level	W[2n40]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n40]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n40]C ~ W[2n40]E	-	Recipe Version
Recipe Report V1 ~	W[2n42]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

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※ Example 1 (Machine Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

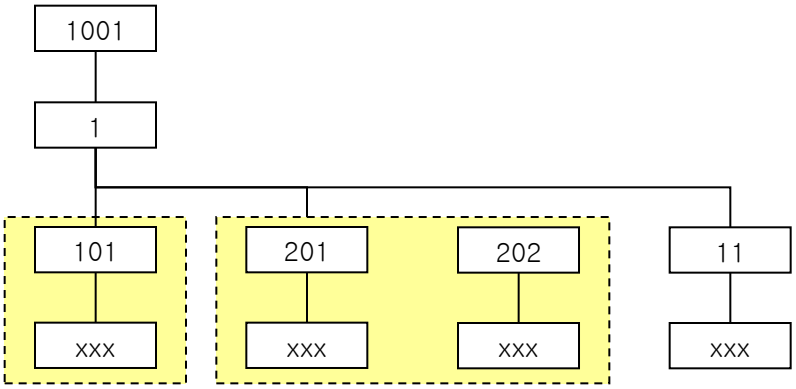
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	xxx

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※ Example 2 (Special Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	11
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

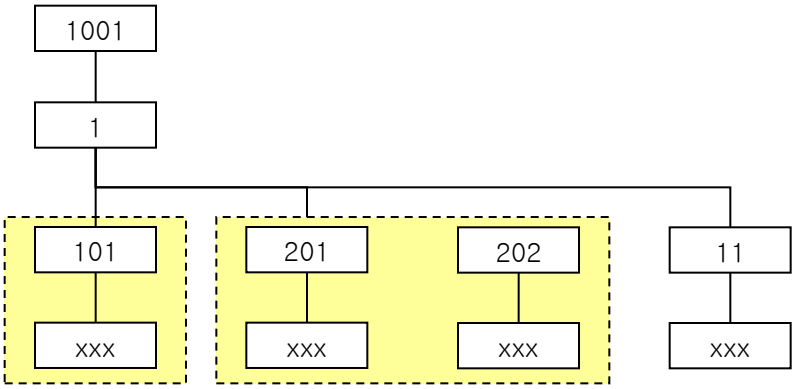
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	11
Recipe Level	3 : Level 3 Recipe
Recipe Type	"1" : Special Recipe
Recipe Report V1 ~	xxx

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※ Example 3 (Unit Recipe)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

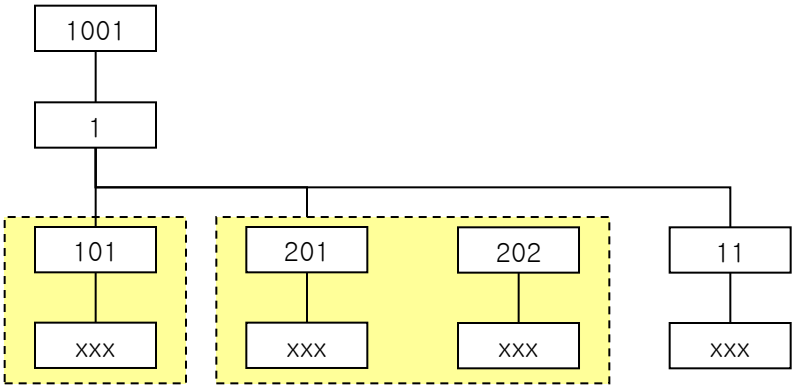
Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Equipment Specific V1	2 : Report Unit Number
Recipe Report V1 ~	xxx

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※ Example 4 (Unit Recipe, Step)

< Recipe Structure >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Level 3 : Equipment Sub-Recipe

Recipe Parameter

< Recipe Parameter Request – ECS >

Name	Value
Request CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	3 : Recipe Parameter
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

Name	Value
Recipe Report CEID	3 : Recipe Parameter
Recipe Number	201
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Equipment Specific V1	2 : Report Unit Number
Recipe Report V1 ~	Unit Recipe Report

< Unit Recipe Report >

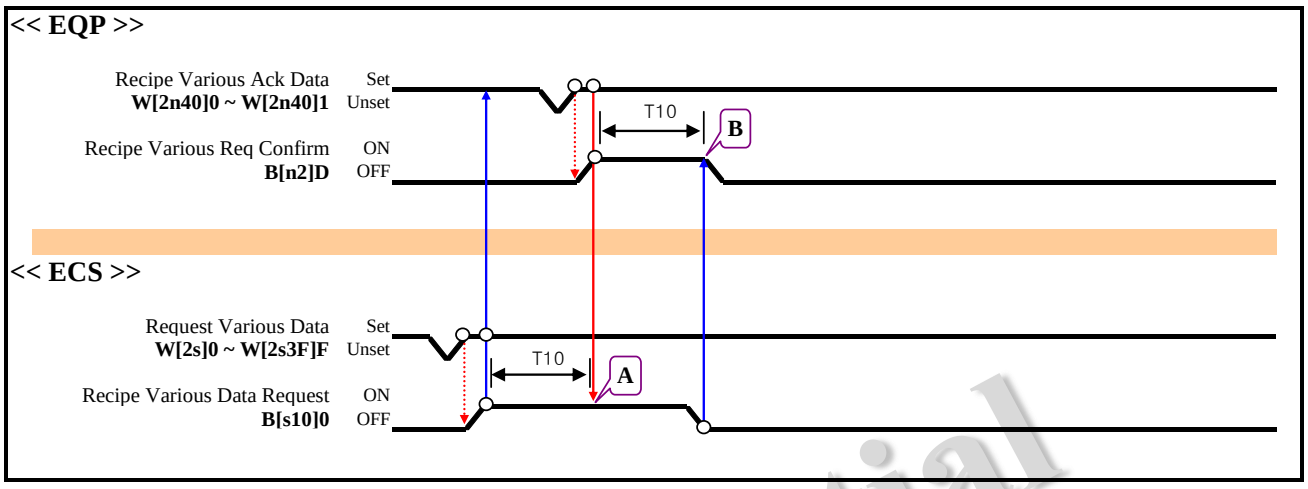
Name	Value
Step 1 Recipe Body V1 <Report Step>	1
Step 1 Recipe Body V2 ~	xxx
Step 2 Recipe Body V1 <Report Step>	2
Step 2 Recipe Body V2 ~	yyy
Step 3 Recipe Body V1 <Report Step>	3
Step 3 Recipe Body V2 ~	zzz

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3.2.6 Recipe Command Send

장비에 Recipe의 Command를 전달하는 것으로 ECS는 “Recipe Various Data Request”로 요청 하며 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete 4 : Rename
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Change Request CEID	W[2s]9	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2s1]0 ~	-	Sub Recipe Number List
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	4 : Recipe Command
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

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< Built Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 1 : Built
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Change Request CEID	W[2s]9	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2s1]0 ~	-	Sub Recipe Number List
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

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< Change Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 2 : Change
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe “M” : Machine Recipe “U” : Unit Recipe “1”, “2”, “3”, … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Change Request CEID	W[2s]9	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2s1]0 ~	-	Sub Recipe Number List
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

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< Delete Case – ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 3 : Delete
Recipe Number	W[2s]2	Integer	Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Change Request CEID	W[2s]9	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2s1]0 ~	-	Sub Recipe Number List
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

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< Rename Case – ECS >

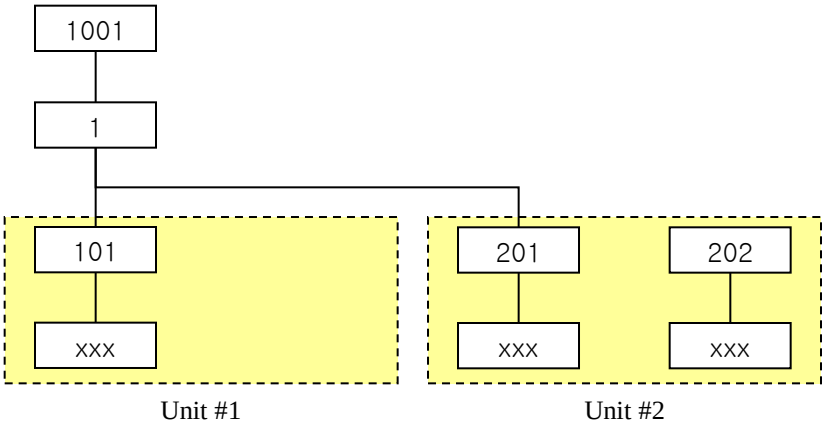
Name	Address	Format	Description
Request CEID	W[2s]0	Integer	4 : Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 4 : Rename
Recipe Number	W[2s]2	Integer	New Recipe Number
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Recipe Version
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Recipe Change Request CEID	W[2s]9	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2s1]0 ~	-	Sub Recipe Number List
Recipe Parameter V1 ~	W[2s2]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

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※ Example 1 (Equipment Sub-Recipe Built)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

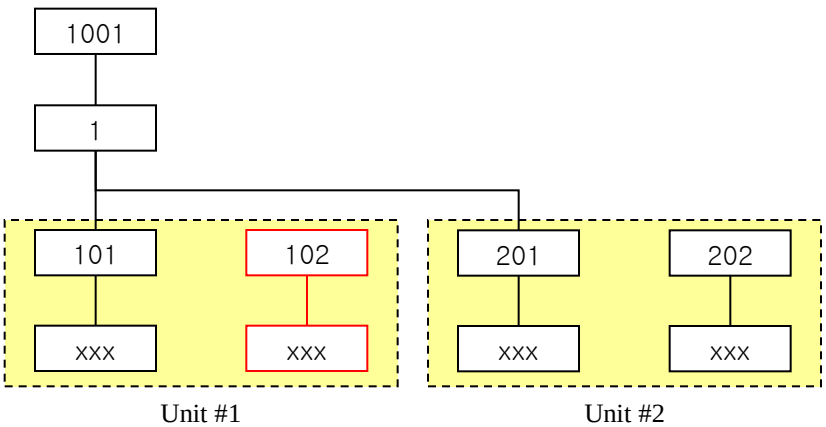
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	102
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U": Unit Recipe
Unit Number	1
Recipe Parameter V1 ~	xxx

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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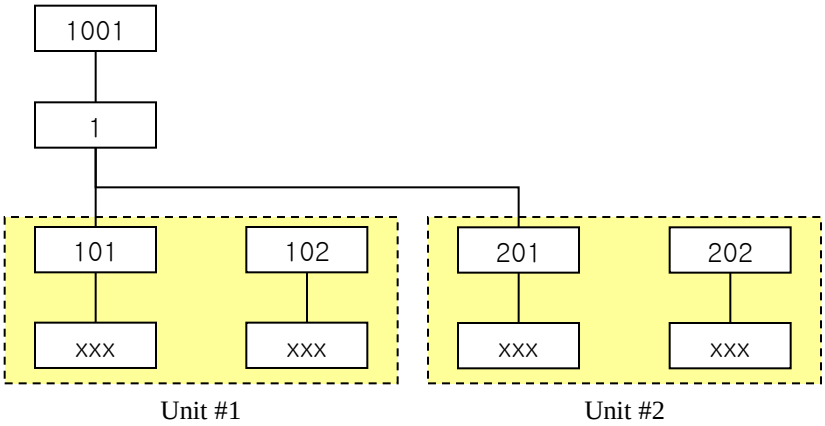
< Recipe Structure (After) >



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※ Example 2 (Equipment Recipe Built)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

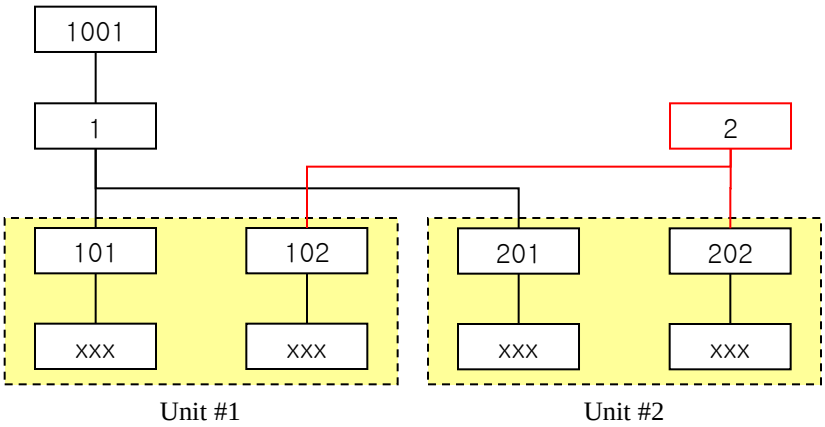
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	2
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Equipment Specific V1	102 : Unit #1 Sub-Recipe
Equipment Specific V2	202 : Unit #2 Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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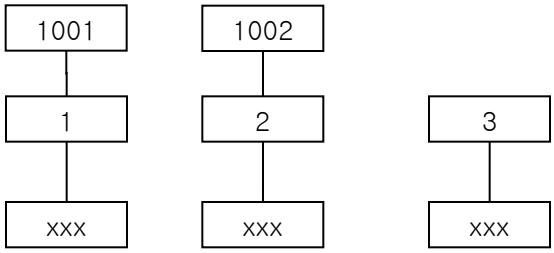
< Recipe Structure (After) >



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※ Example 3 (Equipment Recipe Built)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

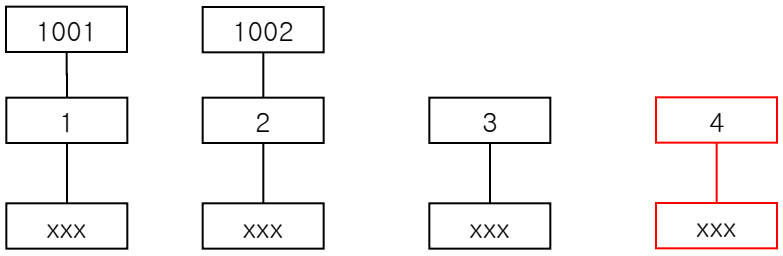
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	1 : Built
Recipe Number	4
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	xxx

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

< Recipe Structure (After) >



Level 1 : Host Recipe

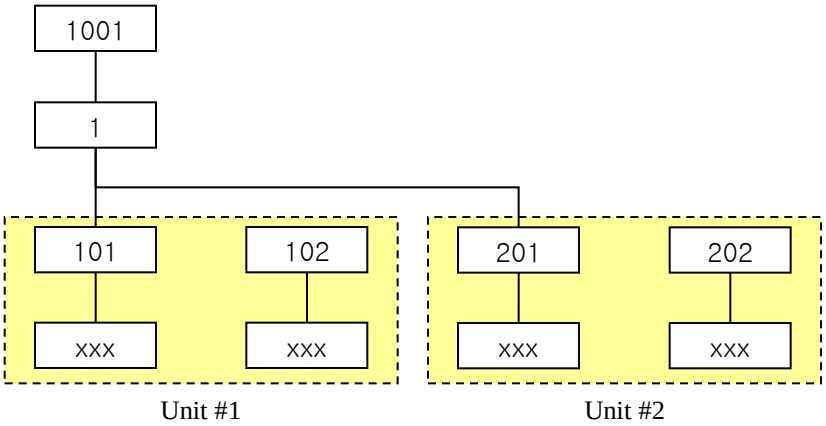
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 4 (Equipment Sub-Recipe Change)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

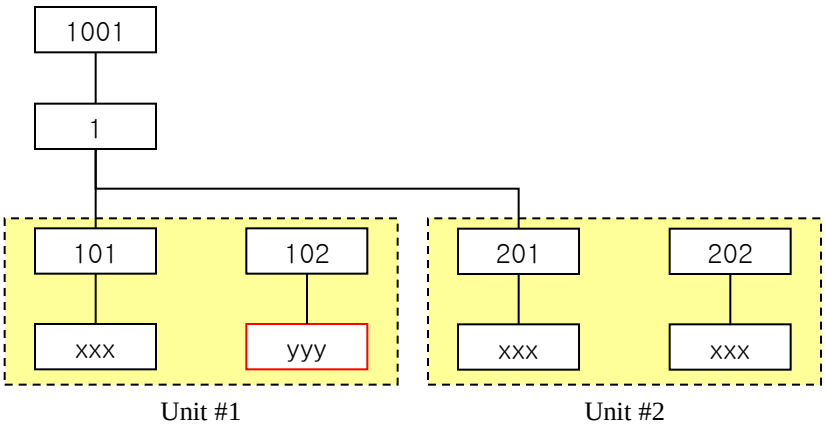
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	102
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" :Unit Recipe
Unit Number	1
Recipe Parameter V1 ~	yyy

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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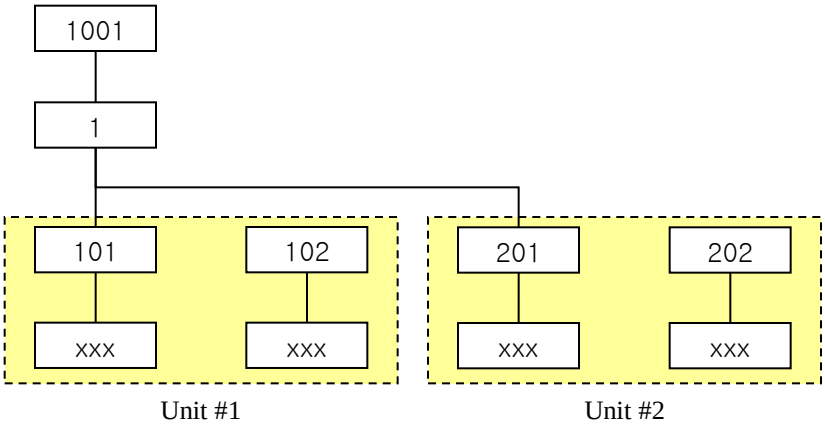
< Recipe Structure (After) >



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※ Example 5 (Equipment Recipe Change)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

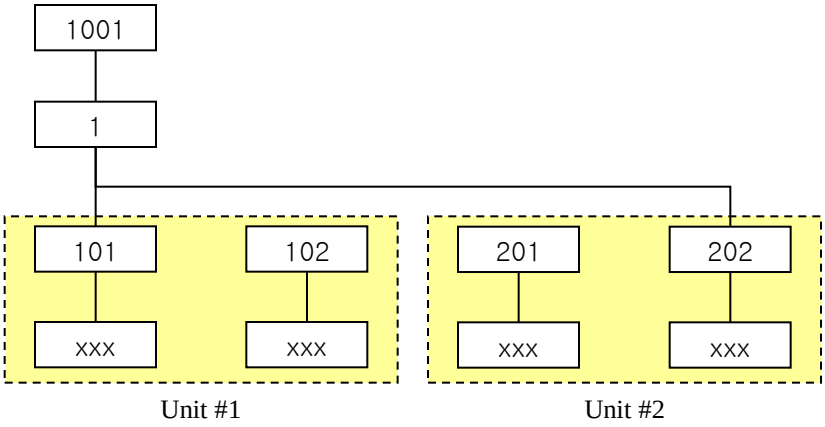
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	1
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Equipment Specific V1	101 : Unit #1 Sub-Recipe
Equipment Specific V2	202 : Unit #2 Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

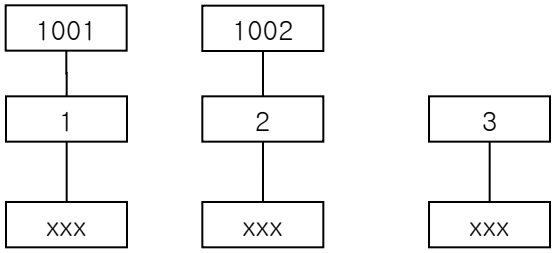
< Recipe Structure (After) >



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※ Example 6 (Equipment Recipe Change)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

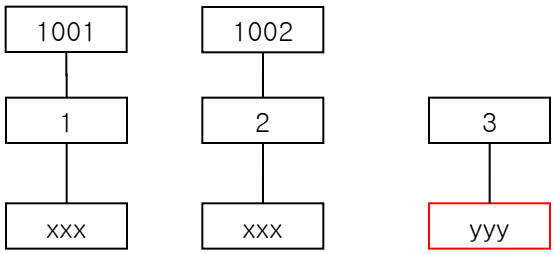
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	2 : Change
Recipe Number	3
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	yyy

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

Confidential

< Recipe Structure (After) >



Level 1 : Host Recipe

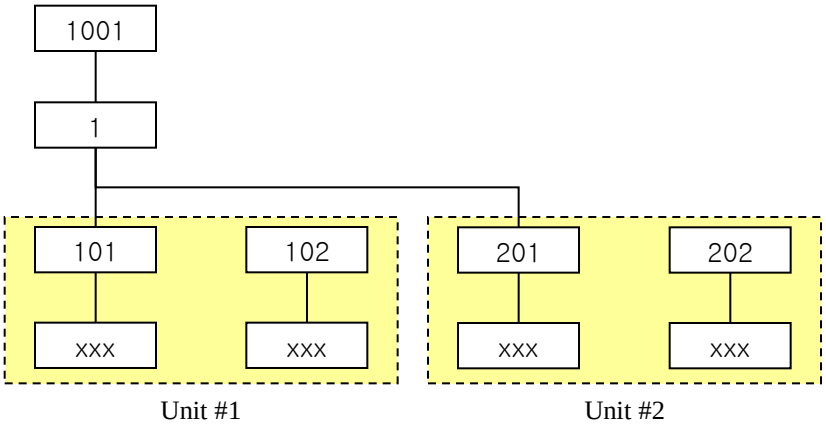
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 7 (Equipment Sub-Recipe Delete)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

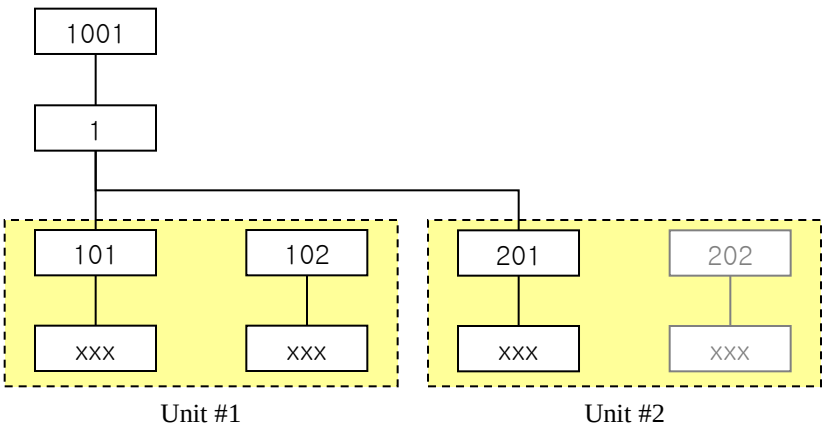
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	3 : Delete
Recipe Number	202
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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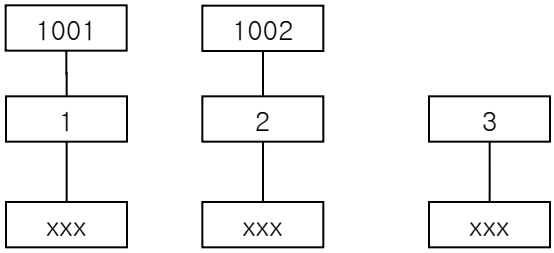
< Recipe Structure (After) >



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※ Example 8 (Equipment Recipe Delete)

< Recipe Structure (Before) >



Level 1 : Host Recipe

Level 2 : Equipment Recipe

Recipe Parameter

< Recipe Command Send – ECS >

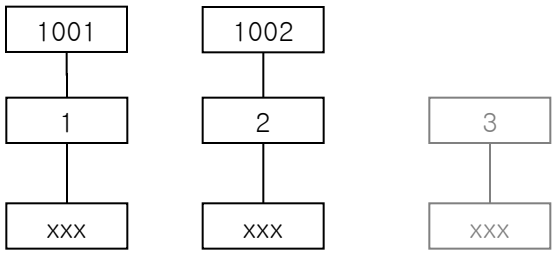
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	3 : Delete
Recipe Number	3
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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< Recipe Structure (After) >



Level 1 : Host Recipe

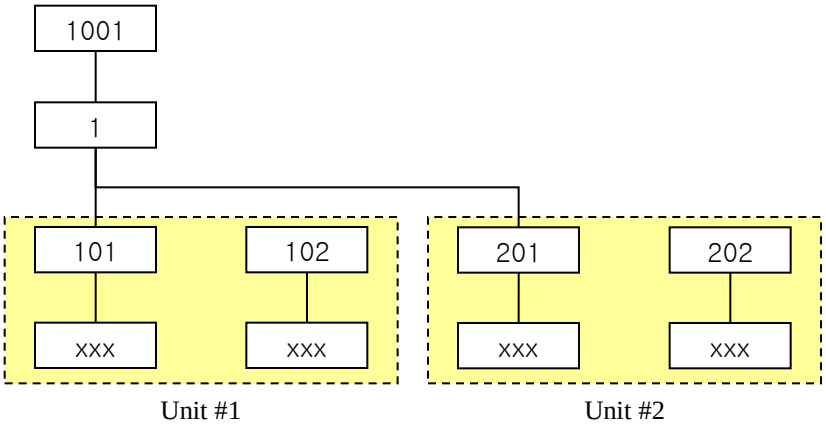
Level 2 : Equipment Recipe

Recipe Parameter

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※ Example 9 (Equipment Sub-Recipe Rename)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

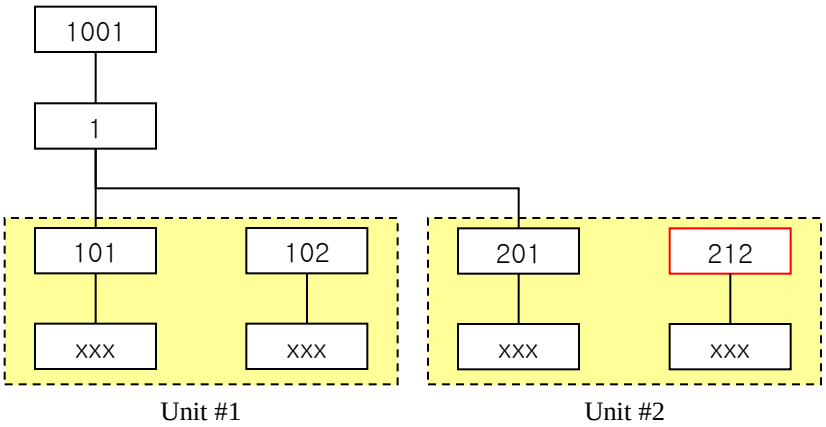
Name	Value
Request CEID	4 : Recipe Command
Recipe Command	4 : Rename
Recipe Number	212
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2
Equipment Specific V1	202 : From Sub-Recipe

< Request ACK – EQP >

Name	Value
Request Confirm CEID	4 : Recipe Command
Request Confirm ACK	0 : Accepted

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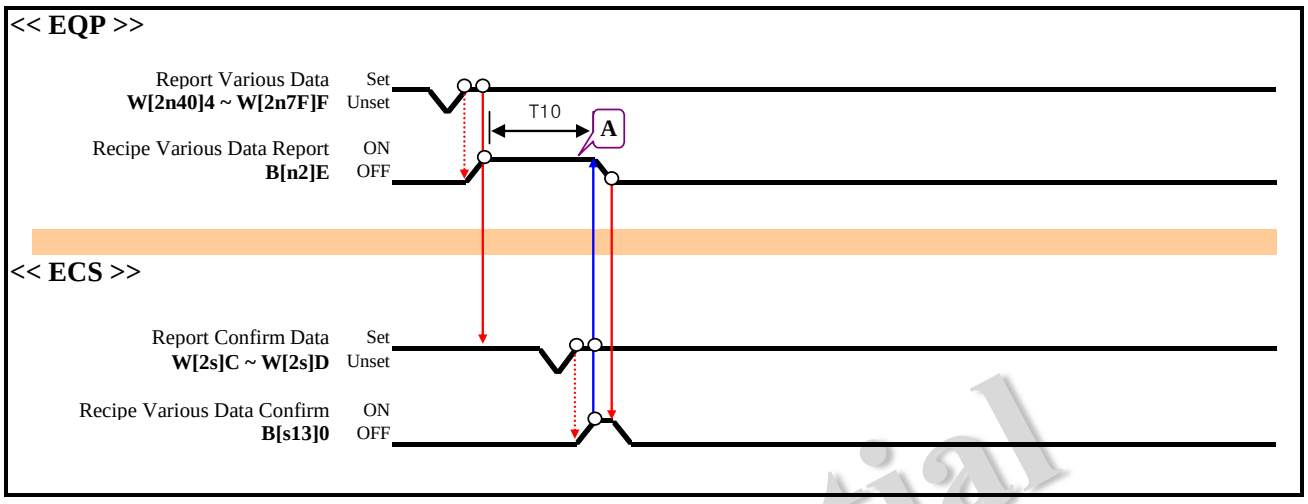
< Recipe Structure (After) >



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3.2.7 Recipe Change Report

작업자에 의해 장비에서 Recipe가 변경되었을 경우 보고하는 것으로 장비는 “Recipe Various Data Report” 로 보고 하며 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	5 : Recipe Change Report
Recipe Command	W[2n40]8	Integer	Recipe Command 1 : Built 2 : Change 3 : Delete 4 : Rename
Recipe Number	W[2n40]9	Integer	Recipe Number
Recipe Level	W[2n40]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n40]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n40]C ~ W[2n40]E	-	Recipe Version
Recipe Change Report CEID	W[2n40]F	Integer	Recipe Change Report CEID 1 : Sub Recipe List 2 : Recipe Parameter 3 : Sub Recipe List & Recipe Parameter
Sub Recipe Number	W[2n41]0 ~	-	Sub Recipe Number List
Recipe Report V1 ~	W[2n42]0 ~	-	Equipment Specific Data Recipe Parameter Data (The detail use is determined after discussing)

< Report Confirm Data – ECS >

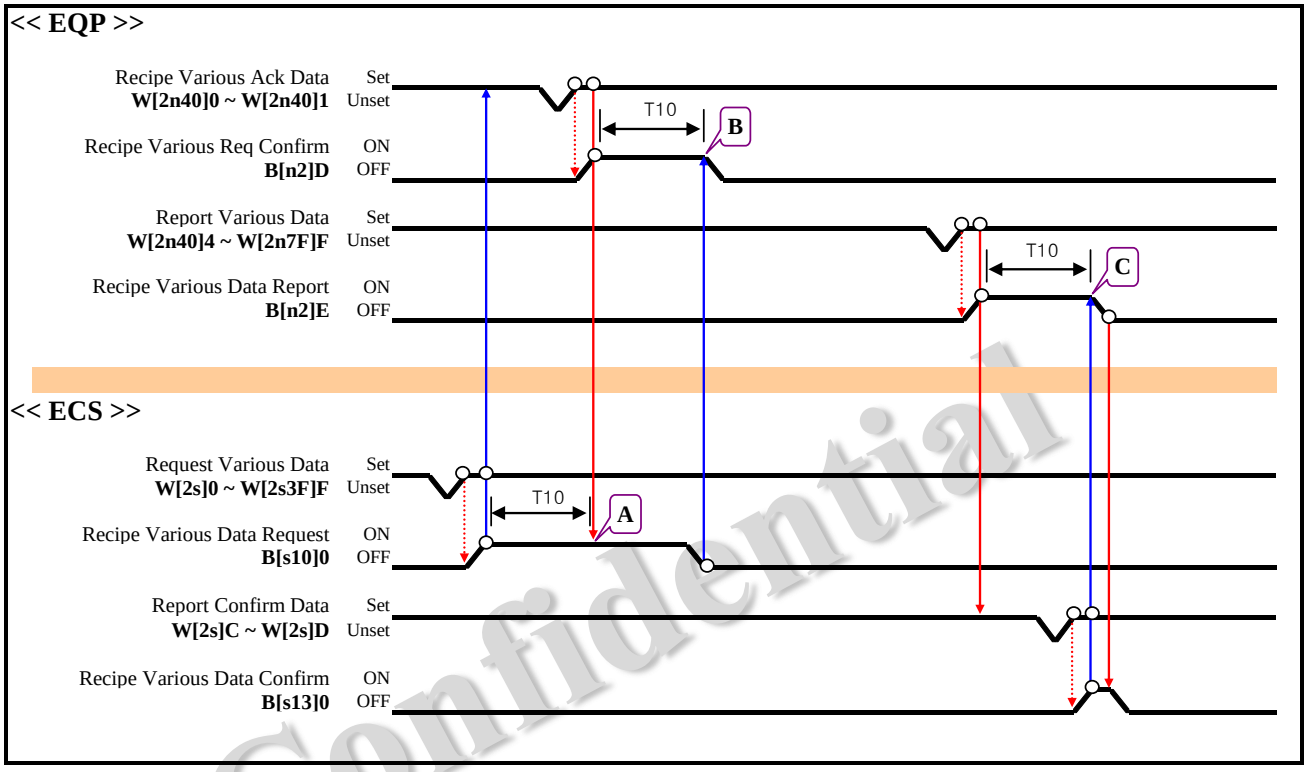
Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	5 : Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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3.2.8 Equipment Constant Recipe Parameter Request

장비에 등록되어 있는 Equipment Constant Recipe의 Parameter를 요청하는 것으로 ECS는 "Recipe Various Data Request"로 요청하며 장비는 "Recipe Various Data Report"로 보고한다.
(사용여부는 협의하여 결정한다)



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A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

C : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Recipe Number	W[2s]2	Integer	Not Used
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Not Used
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Equipment Specific Data V1 ~	W[2s1]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)

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< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Report Confirm ACK	W[2s]D	Integer	0 : Accepted

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< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 2 : Recipe Number Using 3 : Equipment not responding 4 : Version not found 5 : Recipe Number not found 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist

< Report Various Data - EQP >

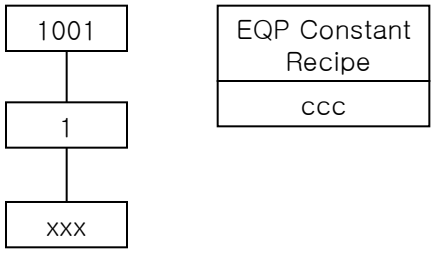
Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	6 : EQP Constant Recipe Parameter (As Host Online) 7 : EQP Constant Recipe Parameter (As Host Request) (The detail use is determined after discussing)
Recipe Number	W[2n40]9	Integer	Not Used
Recipe Level	W[2n40]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n40]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n40]C ~ W[2n40]E	-	Not Used
Equipment Specific Data V1 ~	W[2n41]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Recipe Report V1 ~	W[2n42]0 ~	-	Recipe Parameter Data (The detail use is determined after discussing)

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※ Example 1 (As Host Online)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

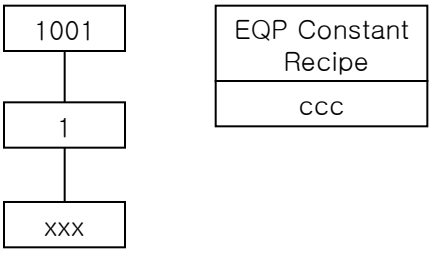
Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 2 (As Host Request)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

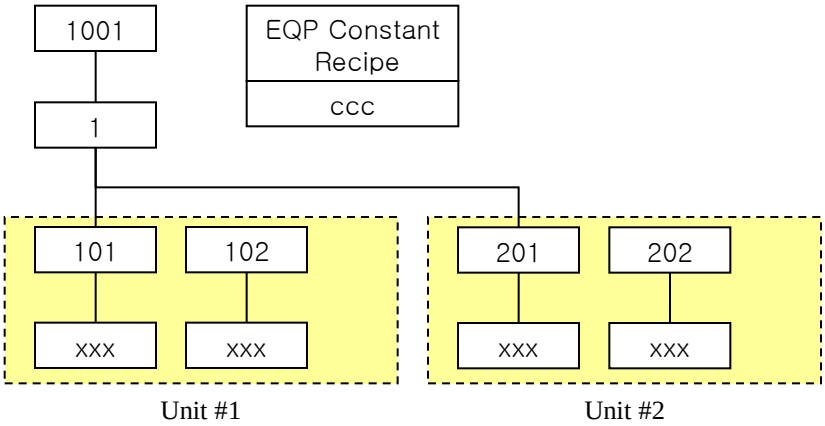
Name	Value
Recipe Report CEID	7 : EQP Constant Recipe Parameter (As Host Request)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 3 (Machine Recipe)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

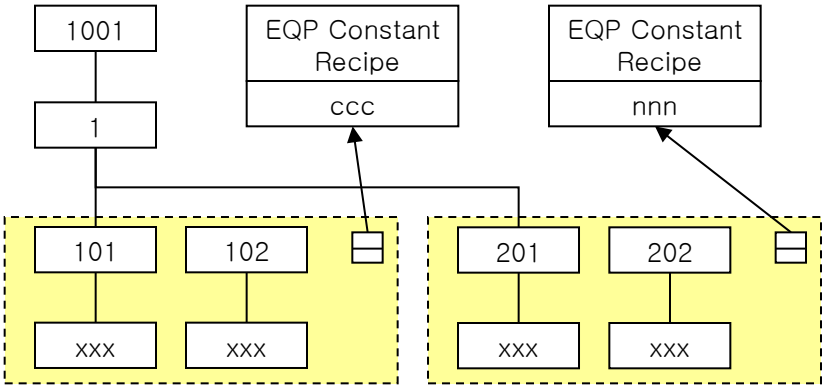
Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Recipe Report V1 ~	ccc

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※ Example 4 (Unit Recipe)

< Recipe Structure >



< Recipe Parameter Request – ECS >

Name	Value
Request CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2

< Request ACK – EQP >

Name	Value
Request Confirm CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Request Confirm ACK	0 : Accepted

< Recipe Parameter – EQP >

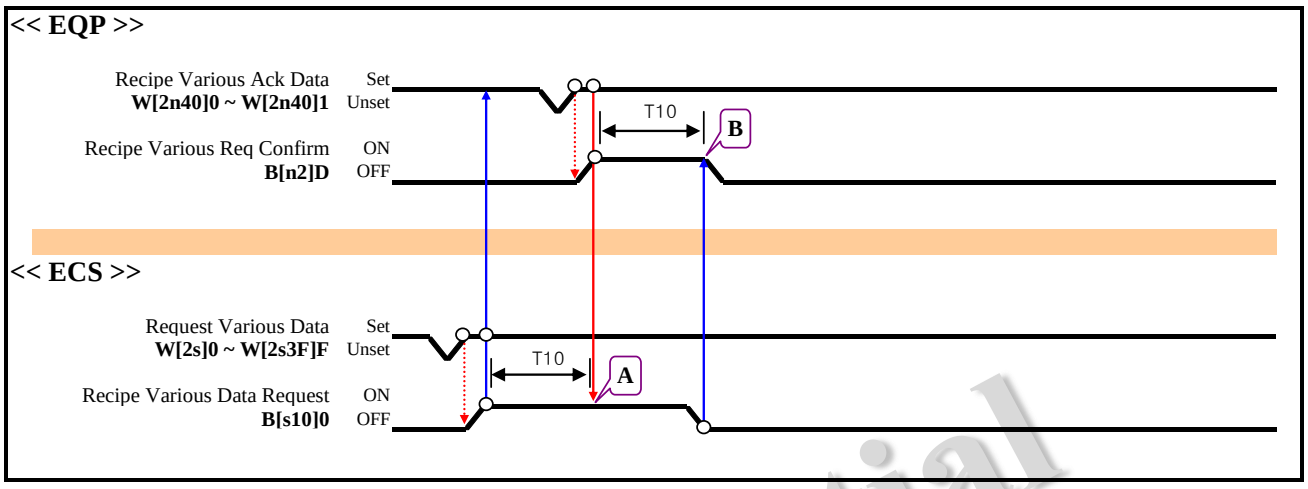
Name	Value
Recipe Report CEID	6 : EQP Constant Recipe Parameter (As Host Online)
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Equipment Specific V1	2 : Report Unit Number
Recipe Report V1 ~	nnn

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3.2.9 Equipment Constant Recipe Command Send

장비에 Equipment Constant Recipe의 Command를 전달하는 것으로 ECS는 “Recipe Various Data Request”로 요청 하며 사용여부는 협의하여 결정한다.



A : ECS가 Recipe Various Request 설정한 후 장비는 **10초** 이내에 Recipe Various Req. Confirm를 보고 해야만 한다.

B : 장비는 일정시간(timeout - 10초)안에 ECS의 Trigger가 Off되지 않을 경우 Light Alarm를 발생시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Request Various Data - ECS >

Name	Address	Format	Description
Request CEID	W[2s]0	Integer	8 : EQP Constant Recipe Command
Recipe Command	W[2s]1	Integer	Recipe Command 2 : Change
Recipe Number	W[2s]2	Integer	Not Used
Recipe Level	W[2s]3	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2s]4	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2s]5 ~ W[2s]7	-	Not Used
Request Unit Number	W[2s]8	Integer	Unit Number (The detail use is determined after discussing)
Equipment Specific Data V1 ~	W[2s1]0 ~	-	- Equipment Specific Data (The detail use is determined after discussing)
Recipe Parameter V1 ~	W[2s2]0 ~	-	Recipe Parameter Data (The detail use is determined after discussing)

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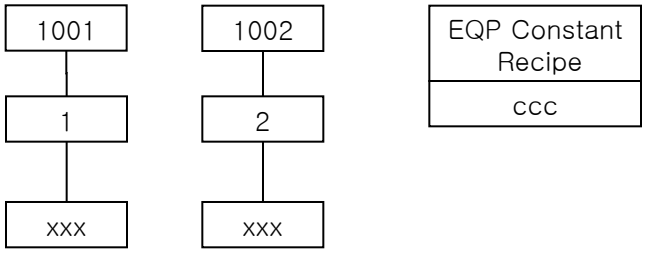
< Recipe Various ACK Data - EQP >

Name	Address	Format	Description
Request Confirm CEID	W[2n40]0	Integer	8 : EQP Constant Recipe Command
Request Confirm ACK	W[2n40]1	Integer	0 : Accepted 1 : Permission not granted 3 : Equipment not responding 6 : Length error 7 : Parameter range error 8 : Same name of recipes exist 9 : Invalid parameter 10 : Equipment in Off-line Mode

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※ Example 1 (Machine Recipe)

< Recipe Structure (Before) >



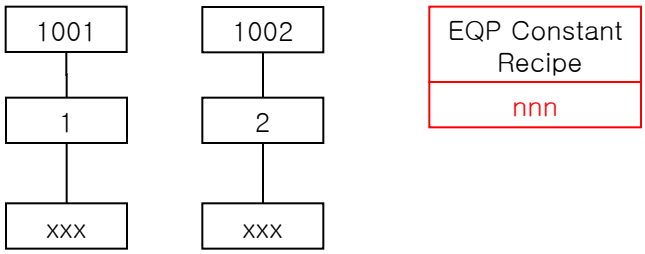
< Recipe Command Send – ECS >

Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

< Recipe Structure (After) >

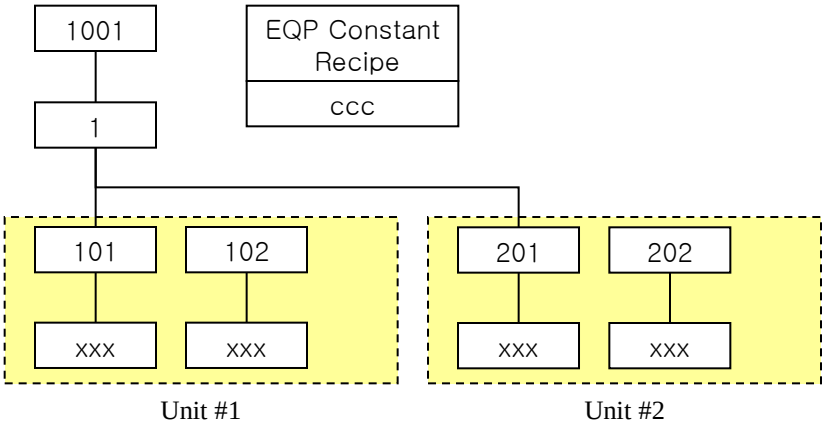


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※ Example 2 (Machine Recipe)

< Recipe Structure (Before) >



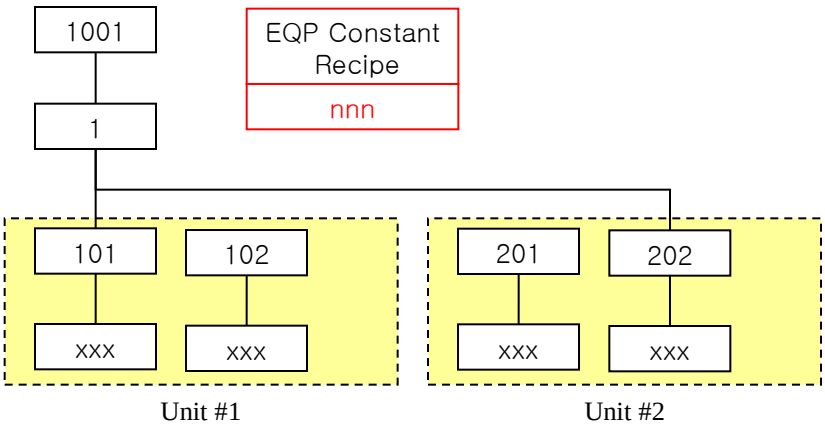
< Recipe Command Send – ECS >

Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	2 : Level 2 Recipe
Recipe Type	"M" : Machine Recipe
Unit Number	0
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

< Recipe Structure (After) >

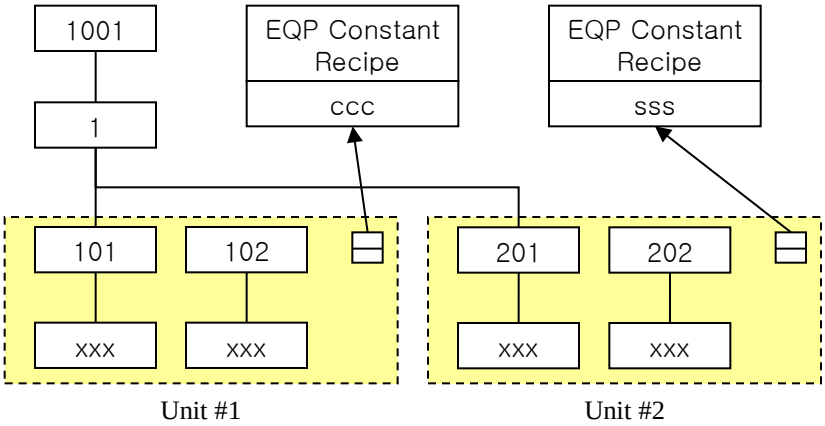


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※ Example 3 (Unit Recipe)

< Recipe Structure (Before) >



< Recipe Command Send – ECS >

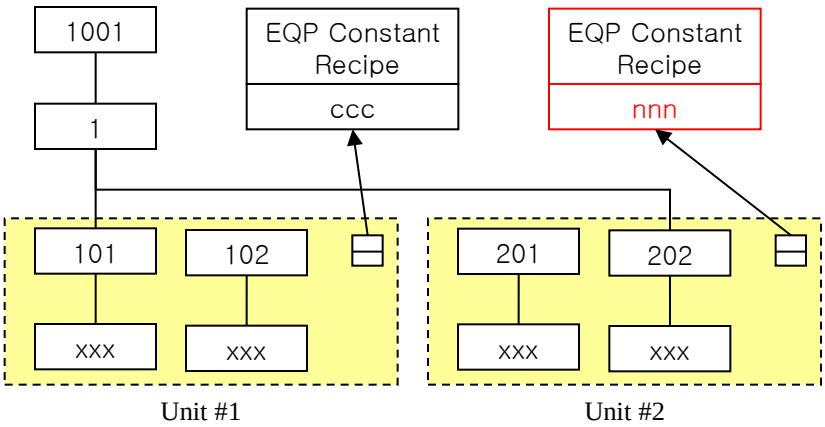
Name	Value
Request CEID	8 : EQP Constant Recipe Command
Recipe Command	2 : Change
Recipe Number	0
Recipe Level	3 : Level 3 Recipe
Recipe Type	"U" : Unit Recipe
Unit Number	2
Recipe Parameter V1 ~	nnn

< Request ACK – EQP >

Name	Value
Request Confirm CEID	8 : EQP Constant Recipe Command
Request Confirm ACK	0 : Accepted

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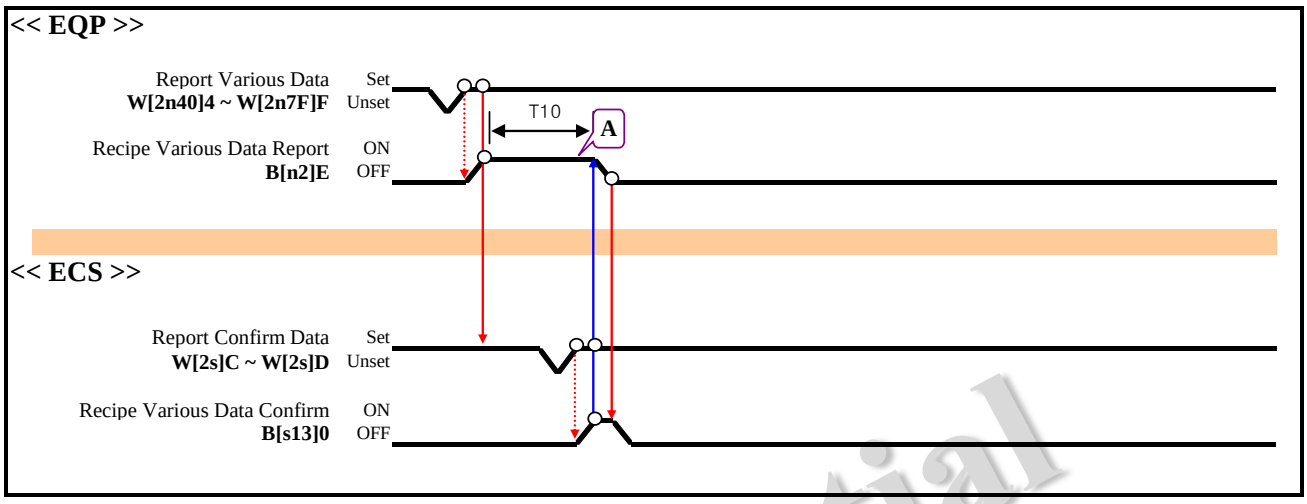
< Recipe Structure (After) >



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3.2.10 Equipment Constant Recipe Change Report

작업자에 의해 장비에서 Equipment Constant Recipe가 변경되었을 경우 보고하는 것으로 장비는 "Recipe Various Data Report"로 보고 하며 사용여부는 협의하여 결정한다.



A : 장비는 일정시간(Timeout - 10초)안에 ECS의 Trigger가 On되지 않을 경우 Light Alarm를 발생시키며 Recipe Various Data Report 신호를 Off 시킨다.

※ Timeout 은 Parameter를 통해 변경 가능 하도록 해야만 한다.
※ 각 장비 별로 Handshake 또는 Data address가 다를 수 있다.

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< Report Various Data - EQP >

Name	Address	Format	Description
Recipe Report CEID	W[2n40]4	Integer	9 : EQP Constant Recipe Change Report
Recipe Command	W[2n40]8	Integer	Recipe Command 2 : Change
Recipe Number	W[2n40]9	Integer	Not Used
Recipe Level	W[2n40]A	Integer	Level of recipe 2 : Level 2 Recipe (Equipment Recipe) 3 : Level 3 Recipe (Equipment Sub-Recipe)
Recipe Type	W[2n40]B	ASCII	Type of recipe "M" : Machine Recipe "U" : Unit Recipe "1", "2", "3", … : Special Recipe (The detail use is determined after discussing)
Recipe Version	W[2n40]C ~ W[2n40]E	-	Not Used
Equipment Specific Data V1 ~	W[2n41]0 ~	-	Equipment Specific Data (The detail use is determined after discussing)
Recipe Report V1 ~	W[2n42]0 ~	-	Recipe Parameter Data (The detail use is determined after discussing)

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< Report Confirm Data – ECS >

Name	Address	Format	Description
Report Confirm CEID	W[2s]C	Integer	9 : EQP Constant Recipe Change Report
Report Confirm ACK	W[2s]D	Integer	0 : Accepted 1 : Permission not granted 2 : Length error 3 : Recipe not found 4 : Parameter range error 5 : Saved for additional use

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